Tedtronics — perfboard revision 5

7×10 cm stock · two banks · six interchangeable positions · three active capsules

ENGINEERING PROTOTYPE — NOT RELEASED FOR SOLDERING

Software checks: **pass**. Actual hardware, module envelopes, timing and electrical performance remain pending. Read the release gates before soldering.

Architecture

- ESP1 owns routing, rail enable and fault reset.
- ESP2 owns smart modes, voltage scanning and joystick ADC1 inputs.
- Each bank shares an ADS1115 voltage scanner and buffered selector.
- Every position retains independent output clamps and series-path isolation.
- Pi and its peripherals have a separate supply. RC522 uses the purchased MIFARE tags.
- Socket through-hole ICs. Carrier interfaces are labelled pigtails; their connector numbering is not IC numbering.

Construction comparison

Metric	Legacy	Revision 5
Boards	17	27
Functional parts	551	457
Adapter assemblies	47	23
Local wire segments	1511	1392
Board solder estimate	5224	4946

The legacy design used larger boards. Revision 5 also has 341 distribution posts and 279 harness conductors. Counts exclude carrier internals and adapter/socket preparation. This revision reduces adapters; it is not a claim of fewer boards or a finished minimum-work optimum.

Electrical bounds and release gates

Calculated design bounds

- Voltage input impedance: ≥1.98 MΩ with 1% divider resistors.
- Worst divided input at 5.25 V: 2.919 V.
- Static source/sink clamp maximum: 11.08 mA.
- Current shunt: 1 Ω ±1%; 10 mV shunt burden at 10 mA, plus switch and wiring resistance.
- Bank scan target: ≤20 ms. Wait for conversion-ready, discard after switching and keep timestamps.
- Watchdog expiry target: 200–350 ms; actual HC123 timing must be measured before permanent installation.
- 5 V guard static trip: 5.053–5.227 V. Approximately 23 mV margin below the contact ceiling does not establish safe fault transients.

These calculations do not establish loop stability, transient limits or measured accuracy.

Pending physical evidence

- Actual stock grid and plating
- Pogo pin diameter, compressed height and mounting fit
- DevKit pinout and body measurements
- Chosen ADS1115/INA226 carrier interfaces and envelopes
- Bench qualification of output clamps, supply guards and partial-power states
- Firmware implementation and measured acceptance

Verify the exact carrier pads, actual stock grid, socket bodies, compressed pogo heights and inter-board mounting offsets.

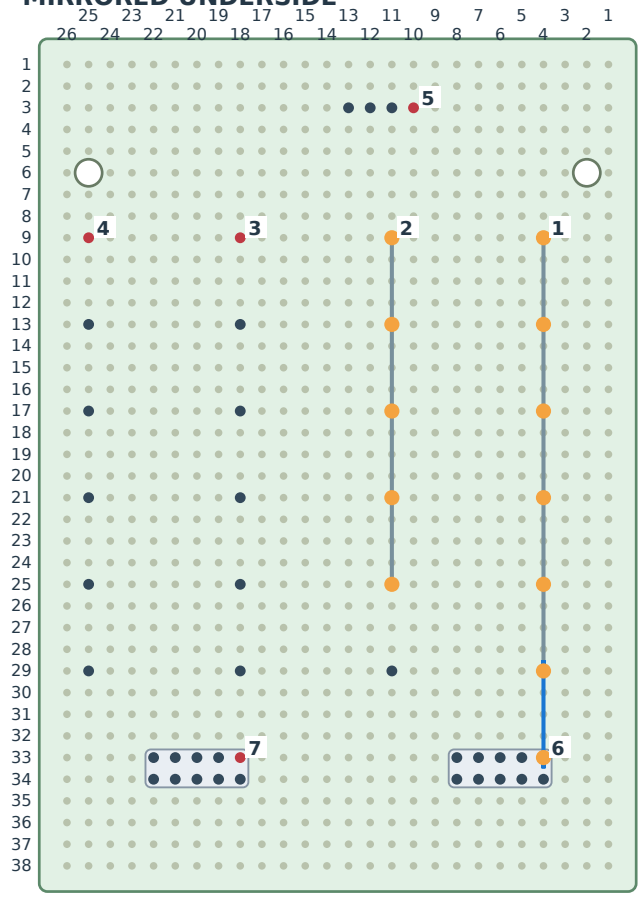
Current contact geometry needs **156 pogo pins**. The inventory lists 150, leaving a six-pin shortage before spares.

The 12 V adapter provides a nominal 24 W budget. Measure fan, light, controller and buck input loads. Do not include the Pi in this budget.

CP00 — a1

Install these wires after their endpoint components; inspect each net with power isolated.

a1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

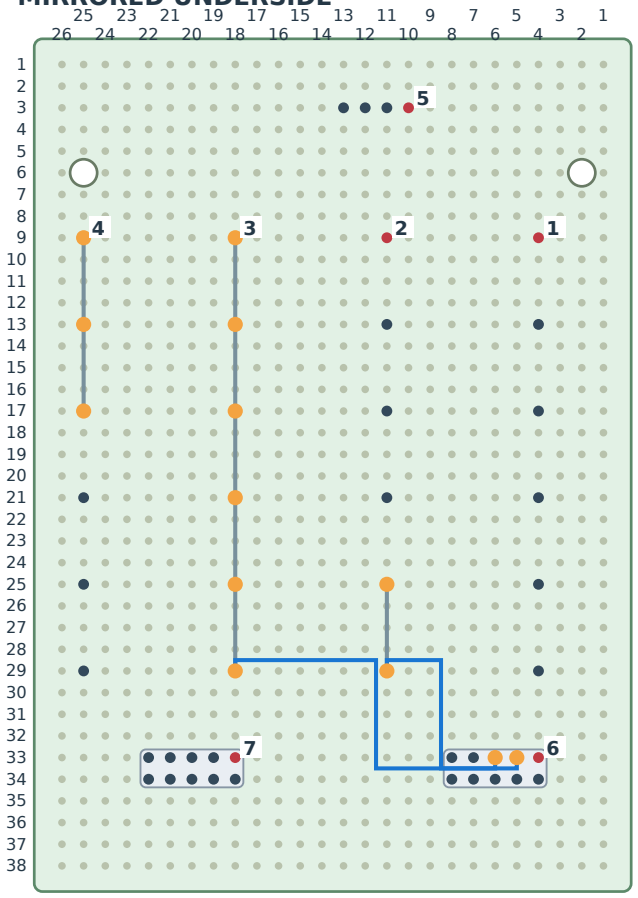
ID / net	From → to	TOP holes
W7 / NODE00	J_NODE00.1 → J_NODE00.2	(4,9) → (4,13)
W8 / NODE00	J_NODE00.2 → J_NODE00.3	(4,13) → (4,17)
W9 / NODE00	J_NODE00.3 → J_NODE00.4	(4,17) → (4,21)
W10 / NODE00	J_NODE00.4 → J_NODE00.5	(4,21) → (4,25)
W11 / NODE00	J_NODE00.5 → J_NODE00.6	(4,25) → (4,29)
W12 / NODE00	J_NODE00.6 → J_NODE.1	(4,29) → (4,33)
W14 / NODE01	J_NODE01.1 → J_NODE01.2	(11,9) → (11,13)
W15 / NODE01	J_NODE01.2 → J_NODE01.3	(11,13) → (11,17)
W16 / NODE01	J_NODE01.3 → J_NODE01.4	(11,17) → (11,21)
W17 / NODE01	J_NODE01.4 → J_NODE01.5	(11,21) → (11,25)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP00 — a1

Install these wires after their endpoint components; inspect each net with power isolated.

a1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

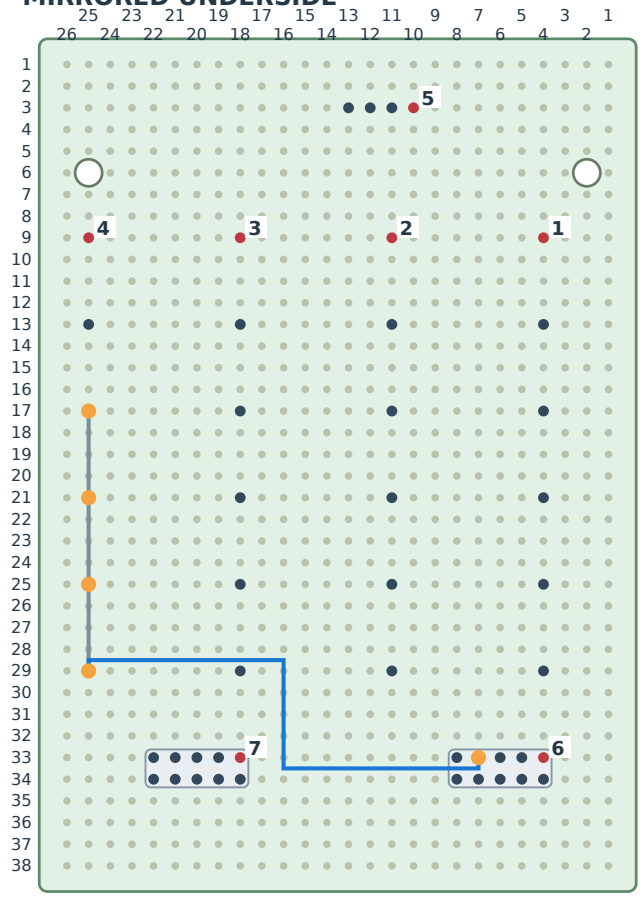
ID / net	From → to	TOP holes
W18 / NODE01	J_NODE01.5 → J_NODE01.6	(11,25) → (11,29)
W19 / NODE01	J_NODE01.6 → J_NODE.3	(11,29) → (5,33)
W21 / NODE02	J_NODE02.1 → J_NODE02.2	(18,9) → (18,13)
W22 / NODE02	J_NODE02.2 → J_NODE02.3	(18,13) → (18,17)
W23 / NODE02	J_NODE02.3 → J_NODE02.4	(18,17) → (18,21)
W24 / NODE02	J_NODE02.4 → J_NODE02.5	(18,21) → (18,25)
W25 / NODE02	J_NODE02.5 → J_NODE02.6	(18,25) → (18,29)
W26 / NODE02	J_NODE02.6 → J_NODE.5	(18,29) → (6,33)
W28 / NODE03	J_NODE03.1 → J_NODE03.2	(25,9) → (25,13)
W29 / NODE03	J_NODE03.2 → J_NODE03.3	(25,13) → (25,17)

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CP00 — a1

Install these wires after their endpoint components; inspect each net with power isolated.

a1 — MIRRORED UNDERSIDE



ID / net	From → to	TOP holes
W30 / NODE03	J_NODE03.3 → J_NODE03.4	(25,17) → (25,21)
W31 / NODE03	J_NODE03.4 → J_NODE03.5	(25,21) → (25,25)
W32 / NODE03	J_NODE03.5 → J_NODE03.6	(25,25) → (25,29)
W33 / NODE03	J_NODE03.6 → J_NODE.7	(25,29) → (7,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

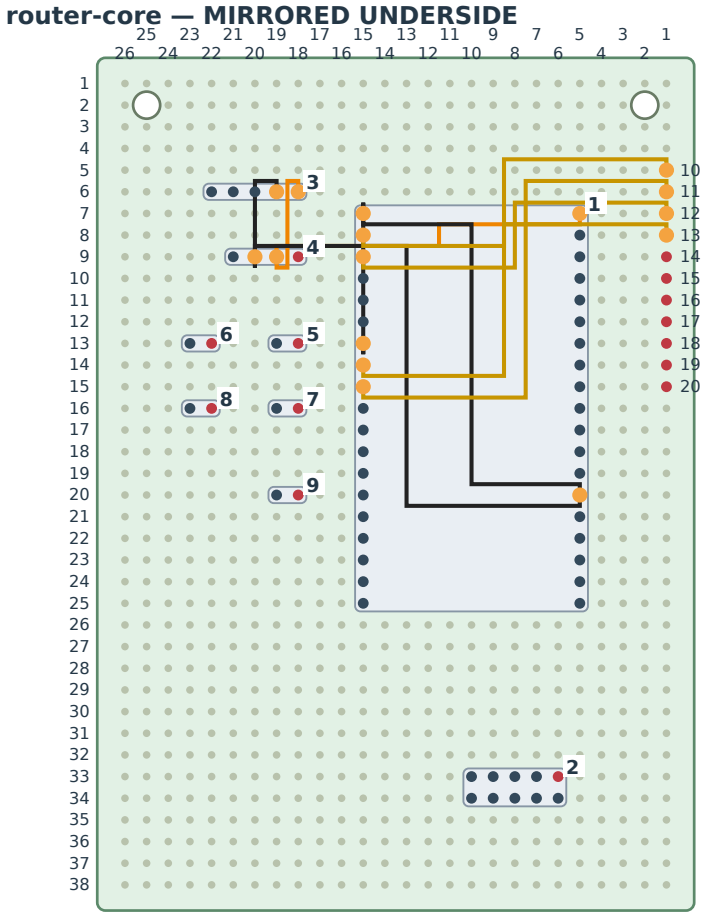
70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP00 — router-core

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

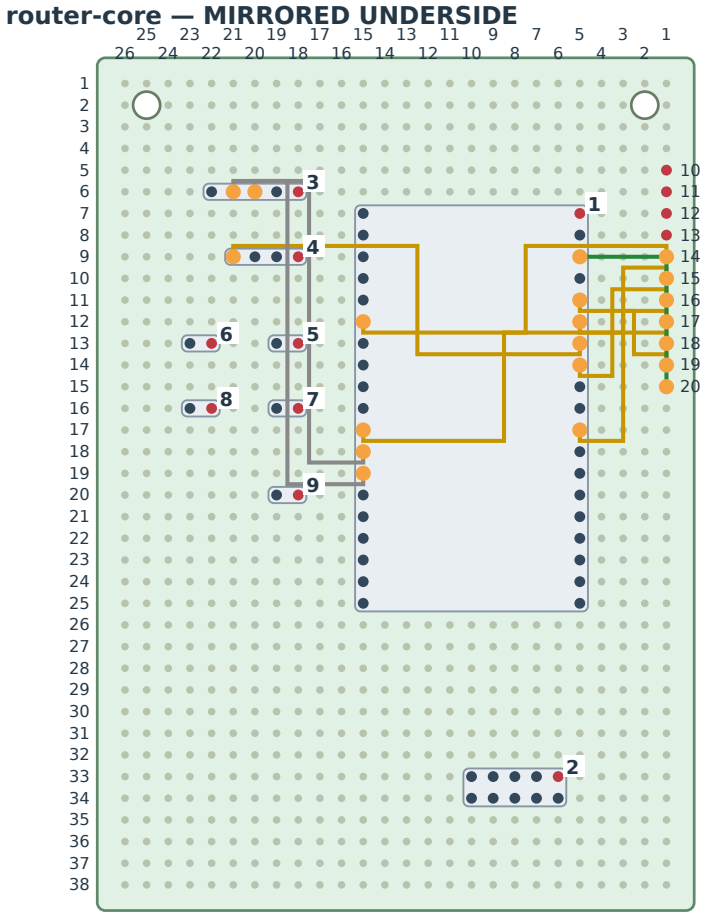
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / USB_ROUTER_3V3	J_FIXTURE.1 → J_PROTOTYPE.2	(18,6) → (19,9)
W2 / USB_ROUTER_3V3	J_PROTOTYPE.2 → U_ESP1.1	(19,9) → (5,7)
W3 / GND	J_FIXTURE.2 → J_PROTOTYPE.3	(19,6) → (20,9)
W4 / GND	J_PROTOTYPE.3 → U_ESP1.14	(20,9) → (5,20)
W5 / GND	U_ESP1.14 → U_ESP1.20	(5,20) → (15,7)
W6 / GND	U_ESP1.20 → U_ESP1.26	(15,7) → (15,13)
W13 / XBAR_DAT	J_LINK_1.1 → U_ESP1.21	(1,5) → (15,8)
W14 / XBAR_CLK	J_LINK_2.1 → U_ESP1.28	(1,6) → (15,15)
W15 / XBAR_LATCH	J_LINK_3.1 → U_ESP1.22	(1,7) → (15,9)
W16 / STB_A_IN	J_LINK_4.1 → U_ESP1.27	(1,8) → (15,14)

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CP00 — router-core

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

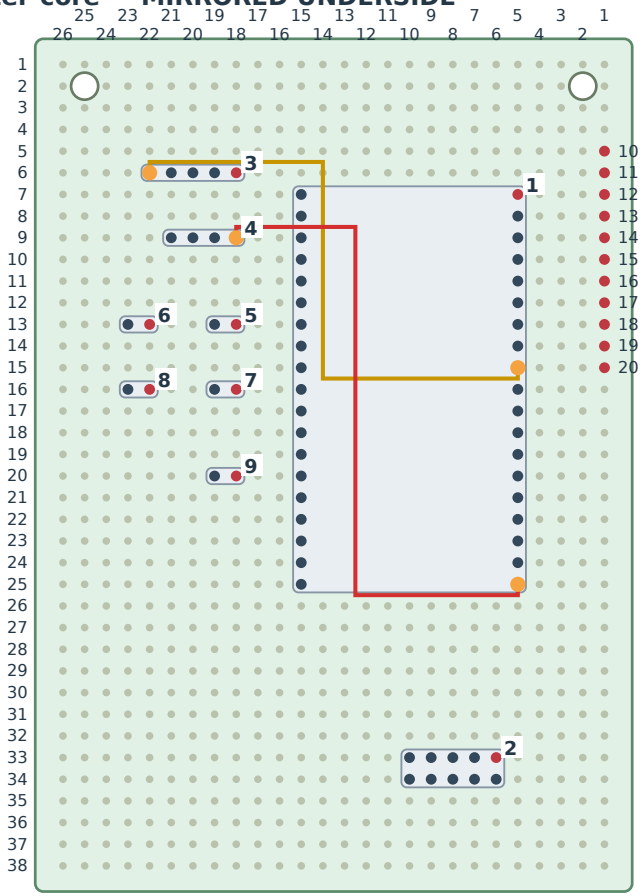
ID / net	From → to	TOP holes
W17 / STB_B_IN	J_LINK_5.1 → U_ESP1.25	(1,9) → (15,12)
W18 / RST_RELEASE	J_LINK_6.1 → U_ESP1.11	(1,10) → (5,17)
W19 / HEARTBEAT	J_LINK_7.1 → U_ESP1.8	(1,11) → (5,14)
W20 / EDU_PWR_EN	J_PROTOTYPE.4 → U_ESP1.7	(21,9) → (5,13)
W22 / FAULT_USB	J_LINK_8.1 → U_ESP1.6	(1,12) → (5,12)
W23 / EDU_FAULT_RESET	J_LINK_9.1 → U_ESP1.30	(1,13) → (15,17)
W25 / CURRENT_USB	J_LINK_10.1 → U_ESP1.5	(1,14) → (5,11)
W26 / DIAG_ADC	J_LINK_11.1 → U_ESP1.3	(1,15) → (5,9)
W27 / FIX_SDA	J_FIXTURE.3 → U_ESP1.32	(20,6) → (15,19)
W28 / FIX_SCL	J_FIXTURE.4 → U_ESP1.31	(21,6) → (15,18)

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CP00 — router-core

Install these wires after their endpoint components; inspect each net with power isolated.

router-core — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

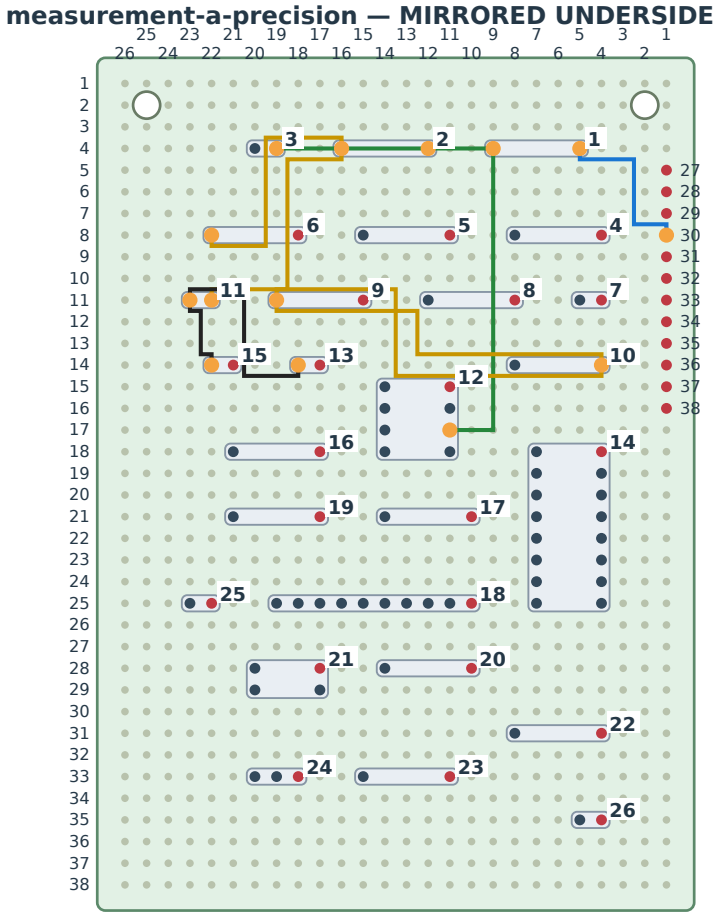
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W29 / FIX_STIM	J_FIXTURE.5 → U_ESP1.9	(22,6) → (5,15)
W30 / USB_5V	J_PROTOTYPE.1 → U_ESP1.19	(18,9) → (5,25)

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CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



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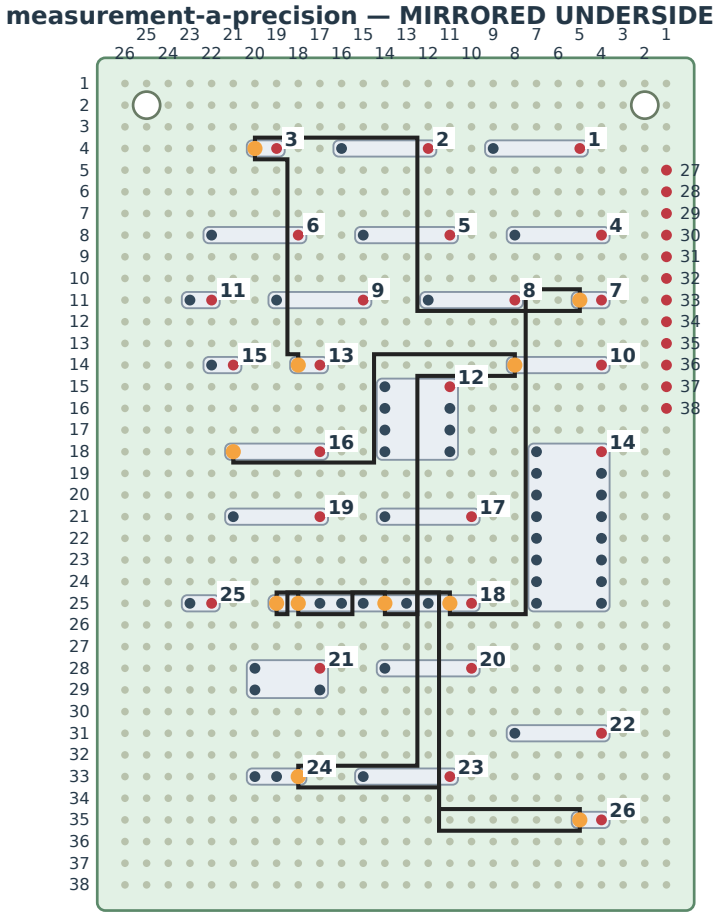
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / NODE00	J_LINK_4.1 → R_P0_1_TOP.1	(1,8) → (5,4)
W2 / P0_1_DIV	C_P0_1.1 → R_P0_1_BOTTOM.1	(19,4) → (12,4)
W3 / P0_1_DIV	R_P0_1_BOTTOM.1 → R_P0_1_TOP.2	(12,4) → (9,4)
W4 / P0_1_DIV	R_P0_1_TOP.2 → U_A_PRECISION.3	(9,4) → (11,17)
W5 / A_PREC_BIAS	C_A_PREC_BIAS.1 → R_A_PREC_BIAS_BOTTOM.1	(22,11) → (4,14)
W6 / A_PREC_BIAS	R_A_PREC_BIAS_BOTTOM.1 → R_A_PREC_BIAS_TOP.2	(4,14) → (19,11)
W7 / A_PREC_BIAS	R_A_PREC_BIAS_TOP.2 → R_P0_1_BOTTOM.2	(19,11) → (16,4)
W8 / A_PREC_BIAS	R_P0_1_BOTTOM.2 → R_P0_2_BOTTOM.2	(16,4) → (22,8)
W9 / GND	C_A_MUX.2 → C_A_PREC_BIAS.2	(22,14) → (23,11)
W10 / GND	C_A_PREC_BIAS.2 → C_A_PRECISION.2	(23,11) → (18,14)

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CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

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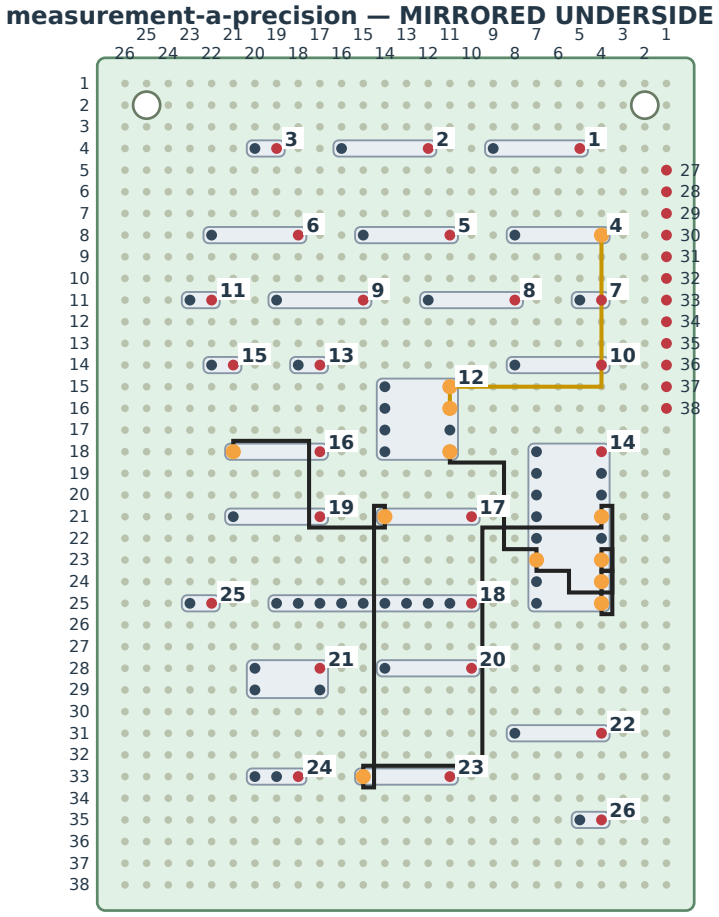
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	C_A_PRECISION.2 → C_P0_1.2	(18,14) → (20,4)
W12 / GND	C_P0_1.2 → C_P0_2.2	(20,4) → (5,11)
W13 / GND	C_P0_2.2 → J_A_ADC_MODULE.2	(5,11) → (11,25)
W14 / GND	J_A_ADC_MODULE.2 → J_A_ADC_MODULE.5	(11,25) → (14,25)
W15 / GND	J_A_ADC_MODULE.5 → J_A_ADC_MODULE.9	(14,25) → (18,25)
W16 / GND	J_A_ADC_MODULE.9 → J_A_ADC_MODULE.10	(18,25) → (19,25)
W17 / GND	J_A_ADC_MODULE.10 → JP_RETURN.2	(19,25) → (5,35)
W18 / GND	JP_RETURN.2 → Q_STIM.1	(5,35) → (18,33)
W19 / GND	Q_STIM.1 → R_A_PREC_BIAS_BOTTOM.2	(18,33) → (8,14)
W20 / GND	R_A_PREC_BIAS_BOTTOM.2 → R_A_SEL_A.2	(8,14) → (21,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

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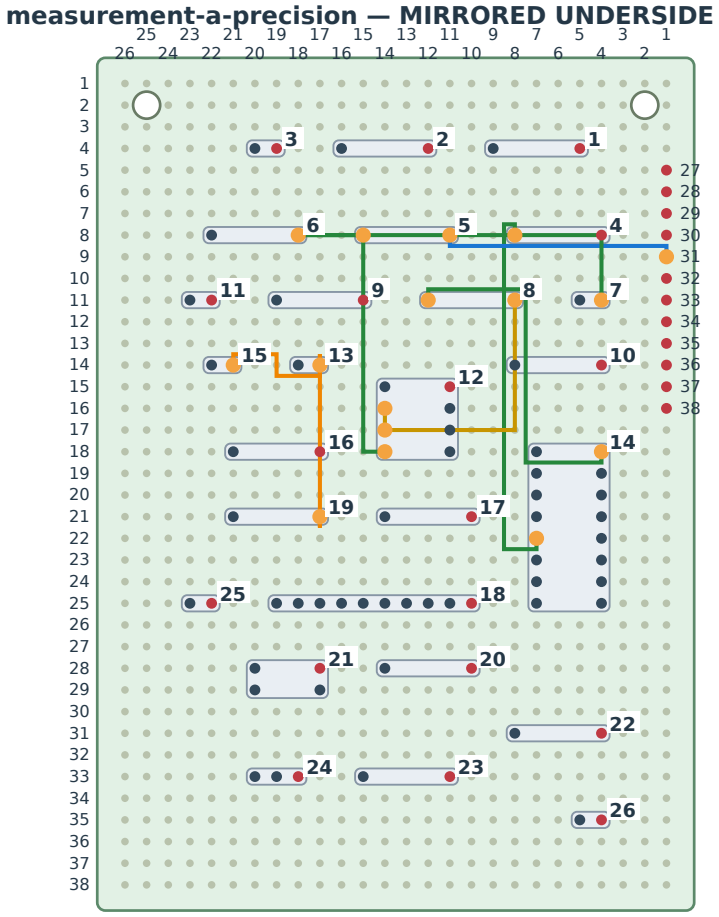
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / GND	R_A_SEL_A.2 → R_A_SEL_B.2	(21,18) → (14,21)
W22 / GND	R_A_SEL_B.2 → R_OFF.2	(14,21) → (15,33)
W23 / GND	R_OFF.2 → U_A_MUX.4	(15,33) → (4,21)
W24 / GND	U_A_MUX.4 → U_A_MUX.6	(4,21) → (4,23)
W25 / GND	U_A_MUX.6 → U_A_MUX.7	(4,23) → (4,24)
W26 / GND	U_A_MUX.7 → U_A_MUX.8	(4,24) → (4,25)
W27 / GND	U_A_MUX.8 → U_A_MUX.11	(4,25) → (7,23)
W28 / GND	U_A_MUX.11 → U_A_PRECISION.4	(7,23) → (11,18)
W29 / P0_1_BUF	R_P0_1_ADC.1 → U_A_PRECISION.1	(4,8) → (11,15)
W30 / P0_1_BUF	U_A_PRECISION.1 → U_A_PRECISION.2	(11,15) → (11,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

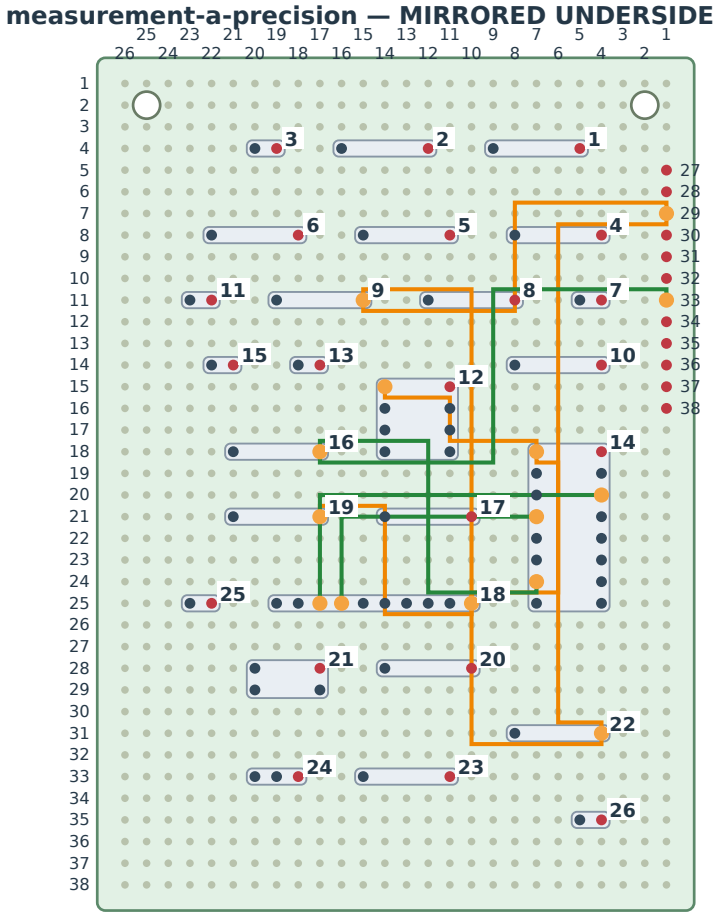
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / P0_1_MUX	R_P0_1_ADC.2 → U_A_MUX.12	(8,8) → (7,22)
W32 / NODE01	J_LINK_5.1 → R_P0_2_TOP.1	(1,9) → (11,8)
W33 / P0_2_DIV	C_P0_2.1 → R_P0_2_BOTTOM.1	(4,11) → (18,8)
W34 / P0_2_DIV	R_P0_2_BOTTOM.1 → R_P0_2_TOP.2	(18,8) → (15,8)
W35 / P0_2_DIV	R_P0_2_TOP.2 → U_A_PRECISION.5	(15,8) → (14,18)
W36 / P0_2_BUF	R_P0_2_ADC.1 → U_A_PRECISION.6	(8,11) → (14,17)
W37 / P0_2_BUF	U_A_PRECISION.6 → U_A_PRECISION.7	(14,17) → (14,16)
W38 / P0_2_MUX	R_P0_2_ADC.2 → U_A_MUX.1	(12,11) → (4,18)
W39 / 3V3_SYS	C_A_MUX.1 → C_A_PRECISION.1	(21,14) → (17,14)
W40 / 3V3_SYS	C_A_PRECISION.1 → D_STIM.1	(17,14) → (17,21)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

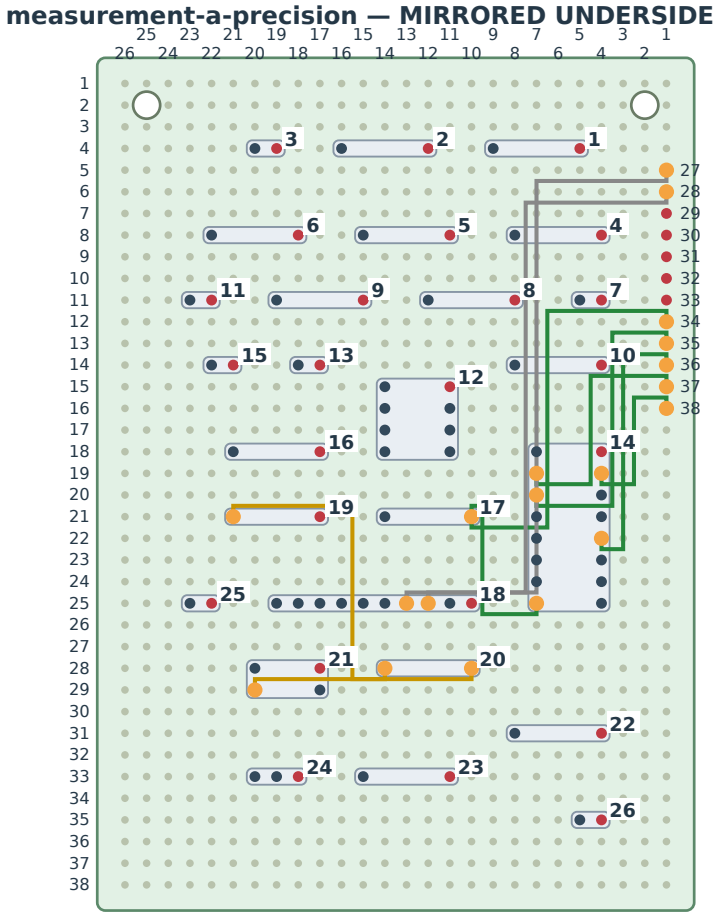
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / 3V3_SYS	D_STIM.1 → J_A_ADC_MODULE.1	(17,21) → (10,25)
W42 / 3V3_SYS	J_A_ADC_MODULE.1 → J_LINK_3.1	(10,25) → (1,7)
W43 / 3V3_SYS	J_LINK_3.1 → R_A_PREC_BIAS_TOP.1	(1,7) → (15,11)
W44 / 3V3_SYS	R_A_PREC_BIAS_TOP.1 → R_LED.1	(15,11) → (4,31)
W45 / 3V3_SYS	R_LED.1 → U_A_MUX.16	(4,31) → (7,18)
W46 / 3V3_SYS	U_A_MUX.16 → U_A_PRECISION.8	(7,18) → (14,15)
W47 / ADC_A_P2	J_A_ADC_MODULE.8 → U_A_MUX.3	(17,25) → (4,20)
W48 / ADC_A_P1	J_A_ADC_MODULE.7 → U_A_MUX.13	(16,25) → (7,21)
W49 / MUX_A_A	J_LINK_7.1 → R_A_SEL_A.1	(1,11) → (17,18)
W50 / MUX_A_A	R_A_SEL_A.1 → U_A_MUX.10	(17,18) → (7,24)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

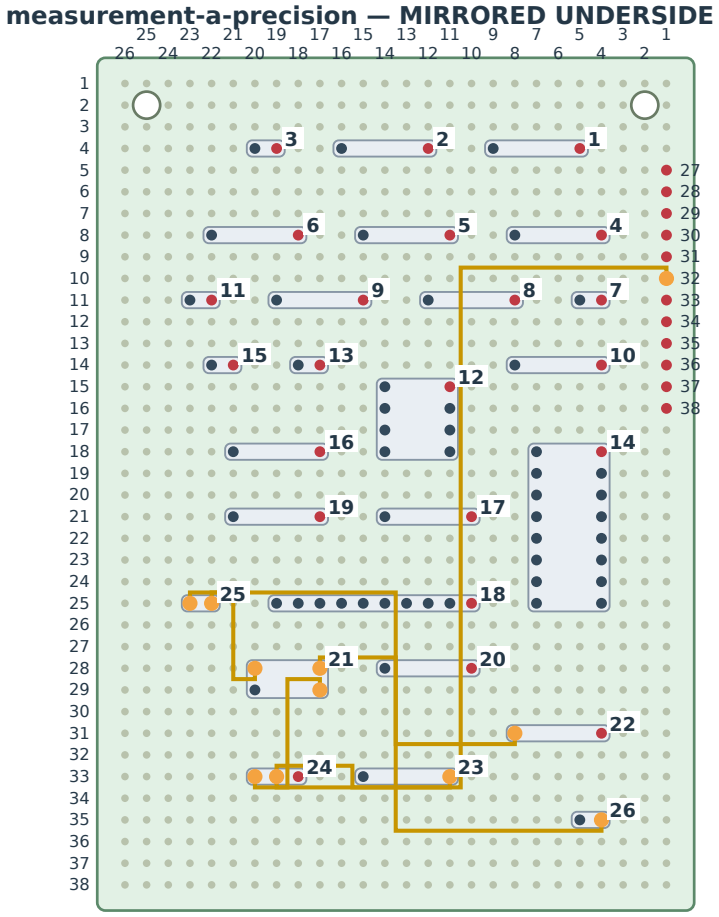
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W51 / MUX_A_B	J_LINK_8.1 → R_A_SEL_B.1	(1,12) → (10,21)
W52 / MUX_A_B	R_A_SEL_B.1 → U_A_MUX.9	(10,21) → (7,25)
W53 / P1_1_MUX	J_LINK_9.1 → U_A_MUX.14	(1,13) → (7,20)
W54 / P1_2_MUX	J_LINK_10.1 → U_A_MUX.5	(1,14) → (4,22)
W55 / P2_1_MUX	J_LINK_11.1 → U_A_MUX.15	(1,15) → (7,19)
W56 / P2_2_MUX	J_LINK_12.1 → U_A_MUX.2	(1,16) → (4,19)
W57 / SCL_POSITION_0	J_A_ADC_MODULE.3 → J_LINK_1.1	(12,25) → (1,5)
W58 / SDA_POSITION_0	J_A_ADC_MODULE.4 → J_LINK_2.1	(13,25) → (1,6)
W59 / STIM_FEED	D_STIM.2 → R_STIM.1	(21,21) → (10,28)
W60 / STIM_LIMIT	K_STIM.3 → R_STIM.2	(20,29) → (14,28)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — measurement-a-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

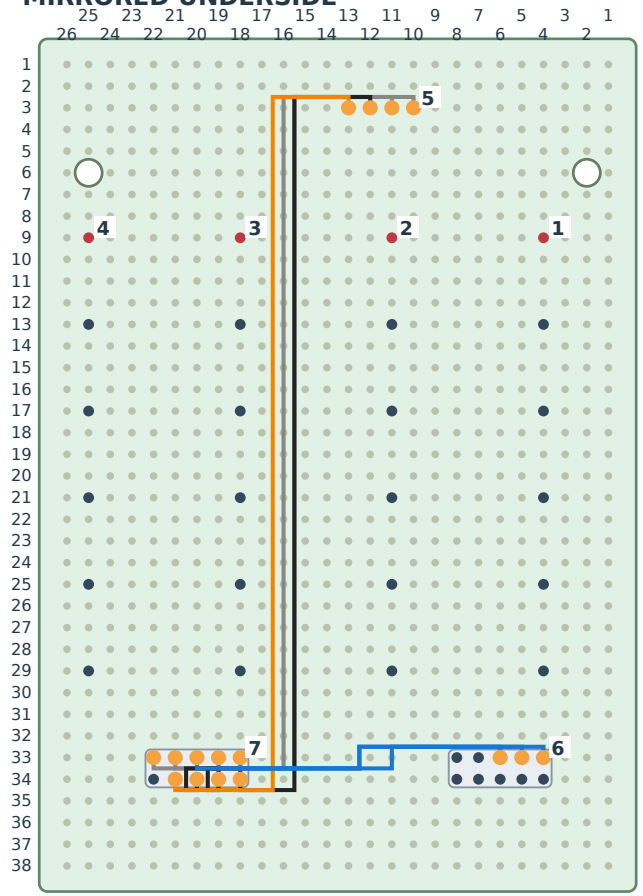
ID / net	From → to	TOP holes
W61 / STIM_LED_A	K_STIM.1 → R_LED.2	(17,28) → (8,31)
W62 / STIM_LED_K	K_STIM.2 → Q_STIM.3	(17,29) → (20,33)
W63 / TEST_A	J_DIAGNOSTIC.1 → K_STIM.4	(22,25) → (20,28)
W64 / FIX_STIM	J_LINK_6.1 → Q_STIM.2	(1,10) → (19,33)
W65 / FIX_STIM	Q_STIM.2 → R_OFF.1	(19,33) → (11,33)
W66 / TEST_B	J_DIAGNOSTIC.2 → JP_RETURN.1	(23,25) → (4,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP06 — a1

Install these wires after their endpoint components; inspect each net with power isolated.

a1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

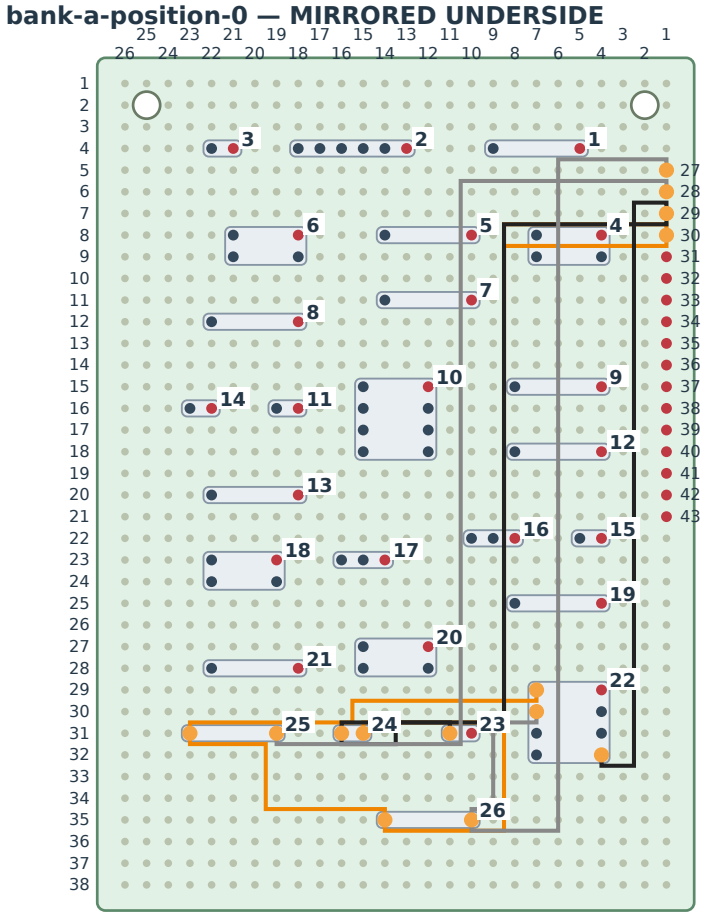
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / SCL_POSITION_0	J_SERVICE.1 → J_SMART.9	(10,3) → (22,33)
W2 / SDA_POSITION_0	J_SERVICE.2 → J_SMART.7	(11,3) → (21,33)
W3 / GND	J_SERVICE.3 → J_SMART.2	(12,3) → (18,34)
W4 / GND	J_SMART.2 → J_SMART.4	(18,34) → (19,34)
W5 / GND	J_SMART.4 → J_SMART.6	(19,34) → (20,34)
W6 / 3V3_SYS	J_SERVICE.4 → J_SMART.8	(13,3) → (21,34)
W13 / NODE00	J_NODE.1 → J_SMART.1	(4,33) → (18,33)
W20 / NODE01	J_NODE.3 → J_SMART.3	(5,33) → (19,33)
W27 / NODE02	J_NODE.5 → J_SMART.5	(6,33) → (20,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP06 — bank-a-position-0

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

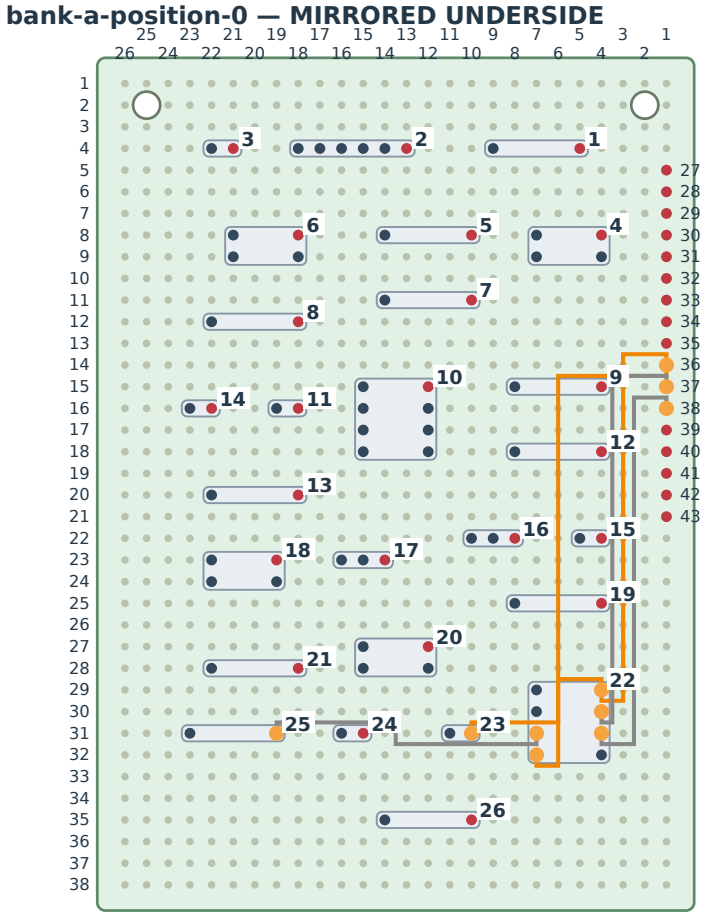
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W5 / 3V3_SYS	C_P0_BUS_B.1 → J_LINK_4.1	(15,31) → (1,8)
W6 / 3V3_SYS	J_LINK_4.1 → R_P0_SCL.2	(1,8) → (14,35)
W7 / 3V3_SYS	R_P0_SCL.2 → R_P0_SDA.2	(14,35) → (23,31)
W8 / 3V3_SYS	R_P0_SDA.2 → U_P0_BUS.8	(23,31) → (7,29)
W11 / GND	C_P0_BUS_A.2 → C_P0_BUS_B.2	(11,31) → (16,31)
W12 / GND	C_P0_BUS_B.2 → J_LINK_3.1	(16,31) → (1,7)
W13 / GND	J_LINK_3.1 → U_P0_BUS.4	(1,7) → (4,32)
W19 / SCL_POSITION_0	J_LINK_1.1 → R_P0_SCL.1	(1,5) → (10,35)
W20 / SCL_POSITION_0	R_P0_SCL.1 → U_P0_BUS.7	(10,35) → (7,30)
W22 / SDA_POSITION_0	J_LINK_2.1 → R_P0_SDA.1	(1,6) → (19,31)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP06 — bank-a-position-0

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

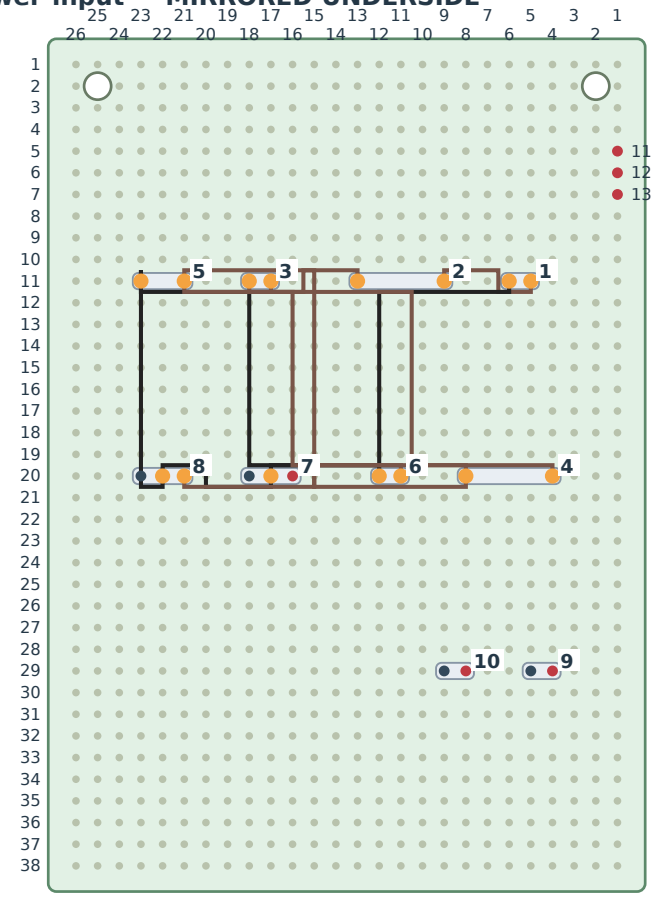
ID / net	From → to	TOP holes
W23 / SDA_POSITION_0	R_P0_SDA.1 → U_P0_BUS.6	(19,31) → (7,31)
W60 / USB_SMART_3V3	C_P0_BUS_A.1 → J_LINK_10.1	(10,31) → (1,14)
W61 / USB_SMART_3V3	J_LINK_10.1 → U_P0_BUS.1	(1,14) → (4,29)
W62 / USB_SMART_3V3	U_P0_BUS.1 → U_P0_BUS.5	(4,29) → (7,32)
W63 / SCL_A	J_LINK_11.1 → U_P0_BUS.2	(1,15) → (4,30)
W64 / SDA_0	J_LINK_12.1 → U_P0_BUS.3	(1,16) → (4,31)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP09 — power-input

Install these wires after their endpoint components; inspect each net with power isolated.

power-input — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

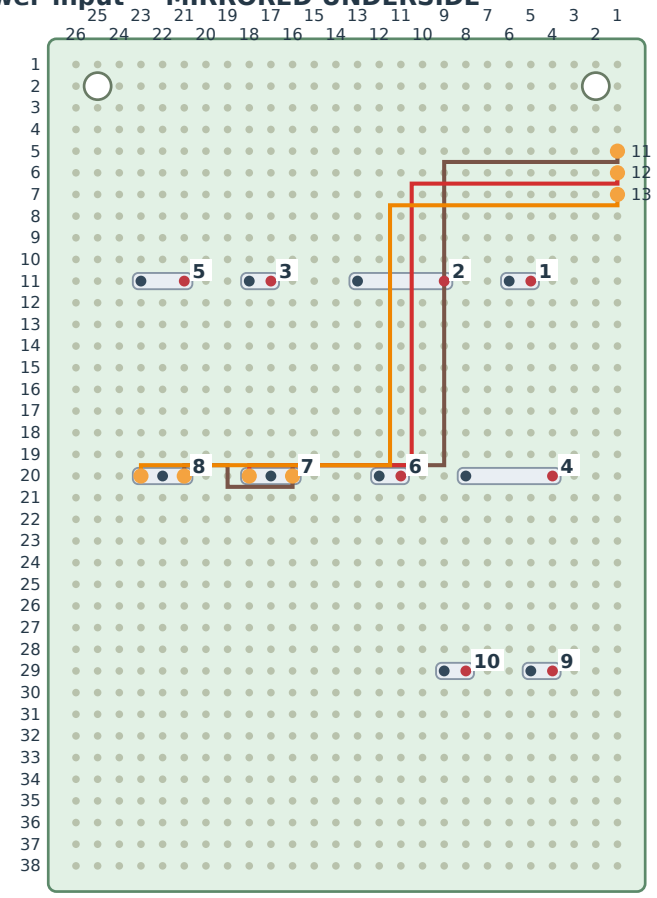
ID / net	From → to	TOP holes
W1 / 12V_IN	F_INPUT.1 → J_INPUT.1	(9,11) → (5,11)
W2 / GND	C_INPUT_HF.2 → C_INPUT.2	(12,20) → (23,11)
W3 / GND	C_INPUT.2 → J_BUCK_3V3.2	(23,11) → (22,20)
W4 / GND	J_BUCK_3V3.2 → J_BUCK_5V.2	(22,20) → (17,20)
W5 / GND	J_BUCK_5V.2 → J_INPUT.2	(17,20) → (6,11)
W6 / 12V_FUSED	F_INPUT.2 → J_SW_MAIN.1	(13,11) → (17,11)
W7 / 12V_SWITCHED	D_REVERSE.1 → J_SW_MAIN.2	(4,20) → (18,11)
W8 / 12V_SYS	C_INPUT_HF.1 → C_INPUT.1	(11,20) → (21,11)
W9 / 12V_SYS	C_INPUT.1 → D_REVERSE.2	(21,11) → (8,20)
W10 / 12V_SYS	D_REVERSE.2 → J_BUCK_3V3.1	(8,20) → (21,20)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP09 — power-input

Install these wires after their endpoint components; inspect each net with power isolated.

power-input — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

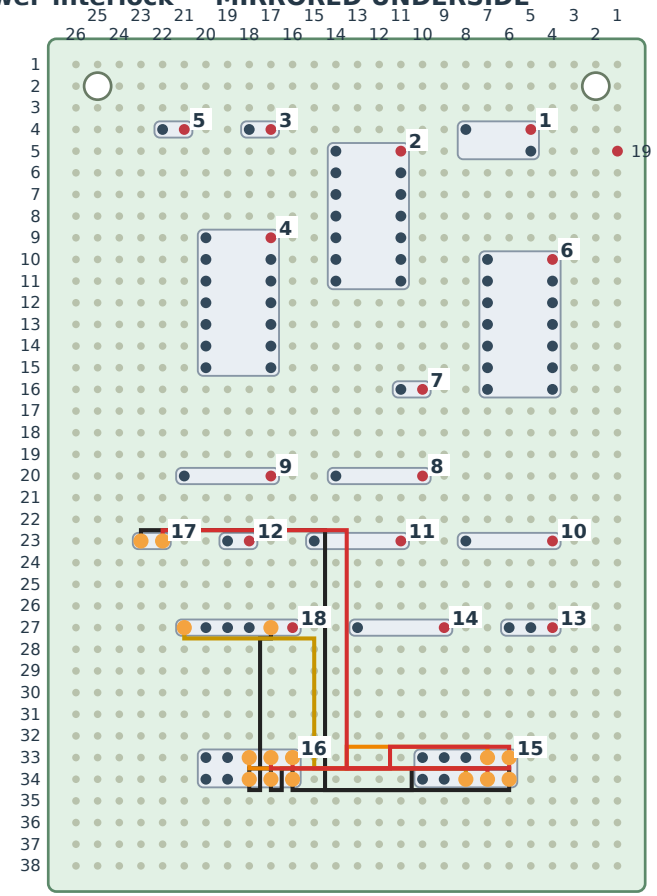
ID / net	From → to	TOP holes
W11 / 12V_SYS	J_BUCK_3V3.1 → J_BUCK_5V.1	(21,20) → (16,20)
W12 / 12V_SYS	J_BUCK_5V.1 → J_LINK_1.1	(16,20) → (1,5)
W13 / BUCK_5V	J_BUCK_5V.3 → J_LINK_2.1	(18,20) → (1,6)
W14 / BUCK_3V3	J_BUCK_3V3.3 → J_LINK_3.1	(23,20) → (1,7)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP09 — power-interlock

Install these wires after their endpoint components; inspect each net with power isolated.

power-interlock — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

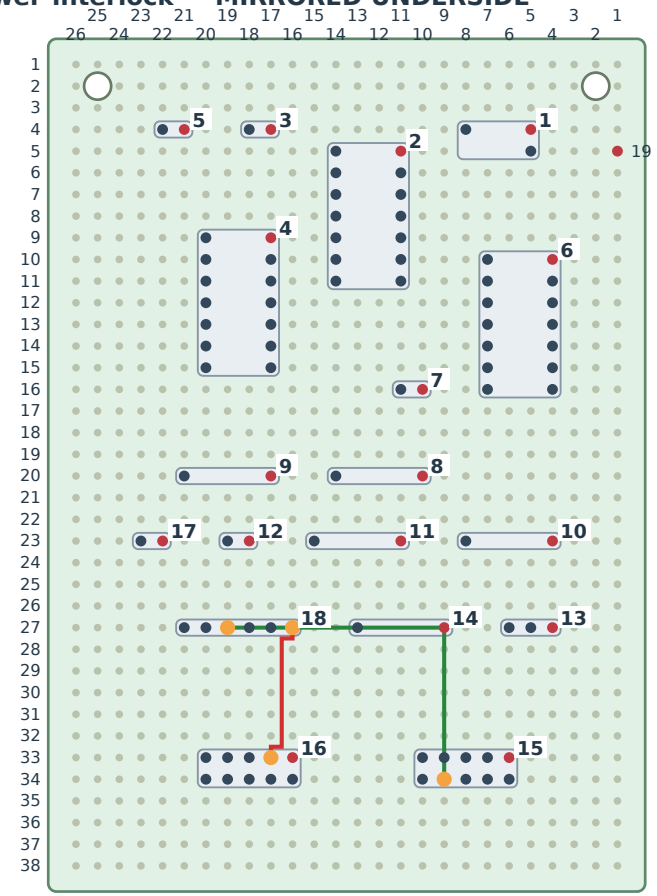
ID / net	From → to	TOP holes
W1 / GND	J_FIXED_BATTERY.2 → J_MATRIX.2	(23,23) → (6,34)
W2 / GND	J_MATRIX.2 → J_SMART.2	(6,34) → (16,34)
W3 / GND	J_SMART.2 → J_SMART.4	(16,34) → (17,34)
W4 / GND	J_SMART.4 → J_SMART.6	(17,34) → (18,34)
W5 / GND	J_SMART.6 → J_TEST.2	(18,34) → (17,27)
W35 / 3V3_SYS	J_MATRIX.3 → J_SMART.5	(7,33) → (18,33)
W46 / EDU_FAULT	J_MATRIX.6 → J_TEST.6	(8,34) → (21,27)
W57 / 5V_SYS	J_MATRIX.4 → J_SMART.1	(7,34) → (16,33)
W63 / 5V_EDU	J_FIXED_BATTERY.1 → J_MATRIX.1	(22,23) → (6,33)
W64 / 5V_EDU	J_MATRIX.1 → J_SMART.3	(6,33) → (17,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP09 — power-interlock

Install these wires after their endpoint components; inspect each net with power isolated.

power-interlock — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

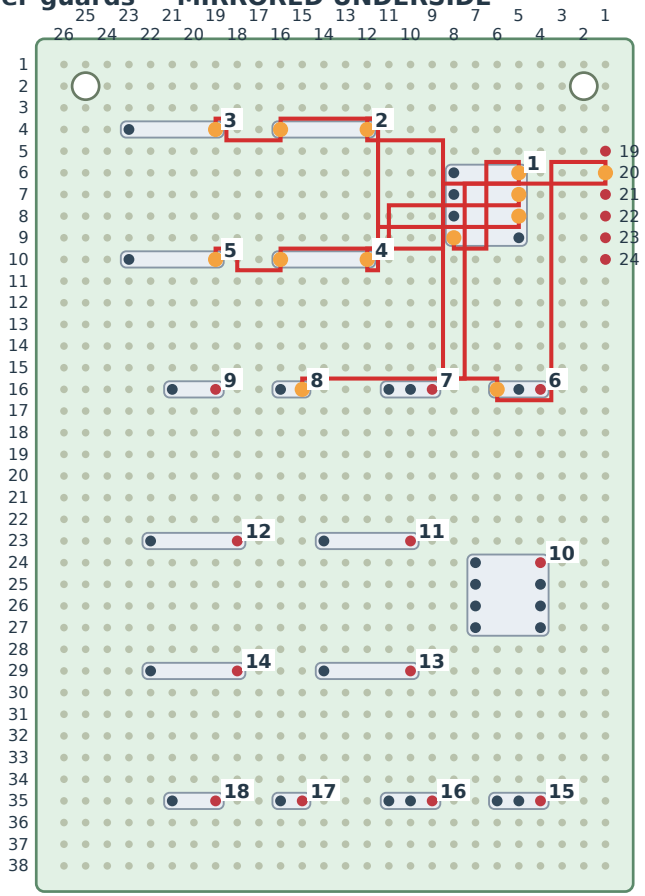
ID / net	From → to	TOP holes
W65 / 5V_EDU	J_SMART.3 → J_TEST.1	(17,33) → (16,27)
W66 / CURRENT_SENSE	J_MATRIX.8 → J_TEST.4	(9,34) → (19,27)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP10 — power-guards

Install these wires after their endpoint components; inspect each net with power isolated.

power-guards — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

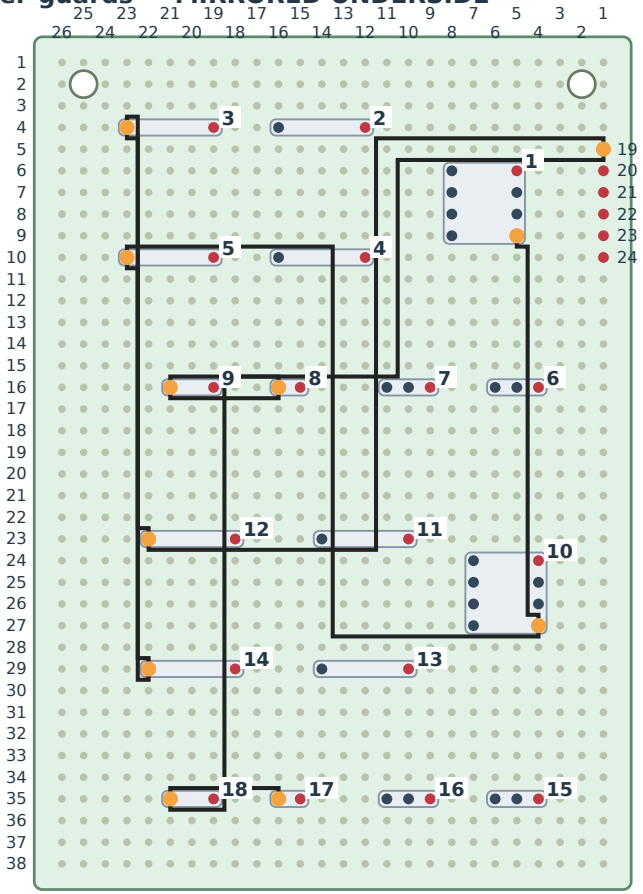
ID / net	From → to	TOP holes
W1 / BUCK_5V	C_OV_5V_IN.1 → J_LINK_2.1	(15,16) → (1,6)
W2 / BUCK_5V	J_LINK_2.1 → Q_OV_5V_IN.3	(1,6) → (6,16)
W3 / BUCK_5V	Q_OV_5V_IN.3 → R_OV_5V_TOP.1	(6,16) → (12,4)
W4 / BUCK_5V	R_OV_5V_TOP.1 → R_UV_5V_TOP.1	(12,4) → (12,10)
W5 / BUCK_5V	R_UV_5V_TOP.1 → U_OV_5V.1	(12,10) → (5,6)
W6 / BUCK_5V	U_OV_5V.1 → U_OV_5V.5	(5,6) → (8,9)
W7 / UV_5V	R_UV_5V_BOTTOM.1 → R_UV_5V_TOP.2	(19,10) → (16,10)
W8 / UV_5V	R_UV_5V_TOP.2 → U_OV_5V.2	(16,10) → (5,7)
W9 / OV_5V	R_OV_5V_BOTTOM.1 → R_OV_5V_TOP.2	(19,4) → (16,4)
W10 / OV_5V	R_OV_5V_TOP.2 → U_OV_5V.3	(16,4) → (5,8)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP10 — power-guards

Install these wires after their endpoint components; inspect each net with power isolated.

power-guards — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

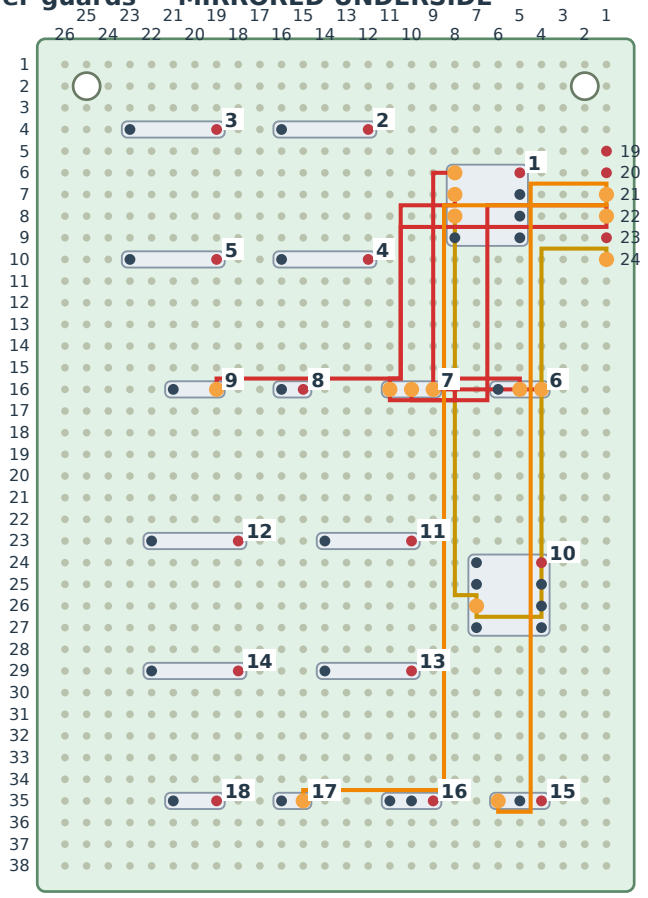
ID / net	From → to	TOP holes
W11 / GND	C_OV_3V3_IN.2 → C_OV_3V3_OUT.2	(16,35) → (21,35)
W12 / GND	C_OV_3V3_OUT.2 → C_OV_5V_IN.2	(21,35) → (16,16)
W13 / GND	C_OV_5V_IN.2 → C_OV_5V_OUT.2	(16,16) → (21,16)
W14 / GND	C_OV_5V_OUT.2 → J_LINK_1.1	(21,16) → (1,5)
W15 / GND	J_LINK_1.1 → R_OV_3V3_BOTTOM.2	(1,5) → (22,23)
W16 / GND	R_OV_3V3_BOTTOM.2 → R_OV_5V_BOTTOM.2	(22,23) → (23,4)
W17 / GND	R_OV_5V_BOTTOM.2 → R_UV_3V3_BOTTOM.2	(23,4) → (22,29)
W18 / GND	R_UV_3V3_BOTTOM.2 → R_UV_5V_BOTTOM.2	(22,29) → (23,10)
W19 / GND	R_UV_5V_BOTTOM.2 → U_OV_3V3.4	(23,10) → (4,27)
W20 / GND	U_OV_3V3.4 → U_OV_5V.4	(4,27) → (5,9)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP10 — power-guards

Install these wires after their endpoint components; inspect each net with power isolated.

power-guards — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

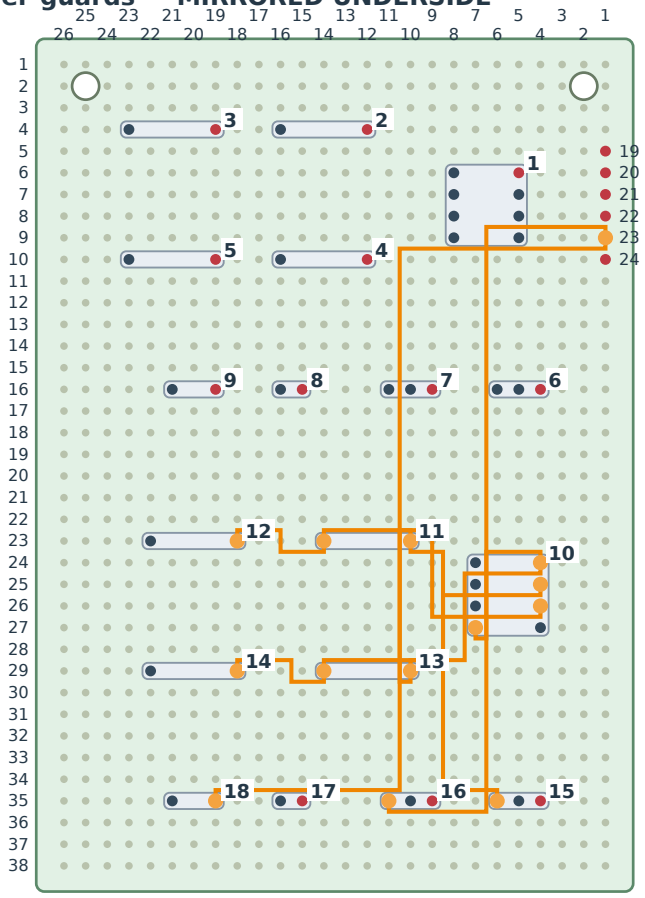
ID / net	From → to	TOP holes
W21 / PRESET_N	J_LINK_6.1 → U_OV_3V3.6	(1,10) → (7,26)
W22 / PRESET_N	U_OV_3V3.6 → U_OV_5V.6	(7,26) → (8,8)
W23 / SAFE_5V	C_OV_5V_OUT.1 → J_LINK_4.1	(19,16) → (1,8)
W24 / SAFE_5V	J_LINK_4.1 → Q_OV_5V_OUT.3	(1,8) → (11,16)
W25 / SAFE_5V	Q_OV_5V_OUT.3 → U_OV_5V.7	(11,16) → (8,7)
W26 / OV_GATE_5V	Q_OV_5V_IN.1 → Q_OV_5V_OUT.1	(4,16) → (9,16)
W27 / OV_GATE_5V	Q_OV_5V_OUT.1 → U_OV_5V.8	(9,16) → (8,6)
W28 / OV_MID_5V	Q_OV_5V_IN.2 → Q_OV_5V_OUT.2	(5,16) → (10,16)
W29 / BUCK_3V3	C_OV_3V3_IN.1 → J_LINK_3.1	(15,35) → (1,7)
W30 / BUCK_3V3	J_LINK_3.1 → Q_OV_3V3_IN.3	(1,7) → (6,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP10 — power-guards

Install these wires after their endpoint components; inspect each net with power isolated.

power-guards — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

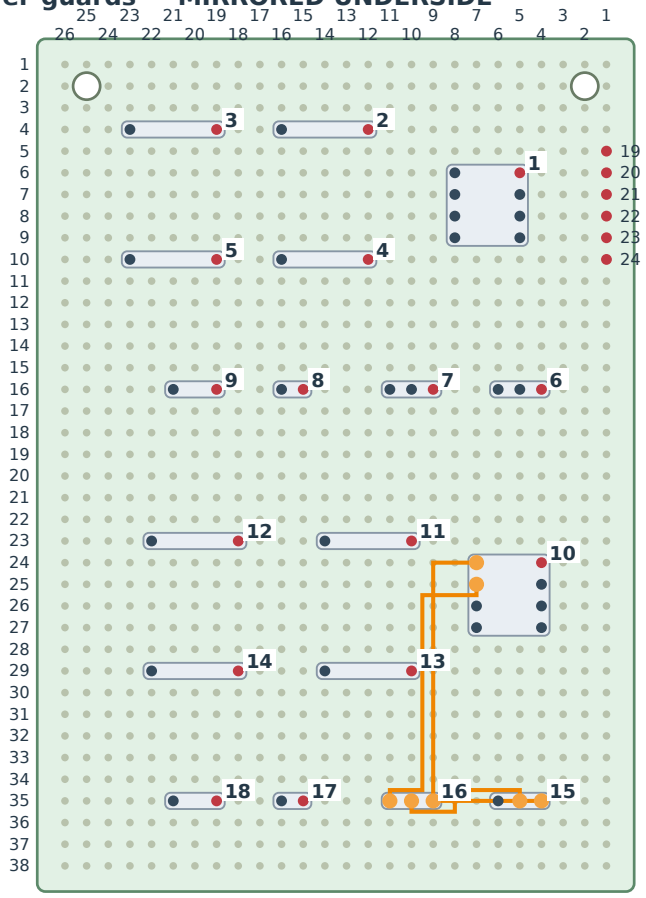
ID / net	From → to	TOP holes
W31 / BUCK_3V3	Q_OV_3V3_IN.3 → R_OV_3V3_TOP.1	(6,35) → (10,23)
W32 / BUCK_3V3	R_OV_3V3_TOP.1 → R_UV_3V3_TOP.1	(10,23) → (10,29)
W33 / BUCK_3V3	R_UV_3V3_TOP.1 → U_OV_3V3.1	(10,29) → (4,24)
W34 / BUCK_3V3	U_OV_3V3.1 → U_OV_3V3.5	(4,24) → (7,27)
W35 / UV_3V3	R_UV_3V3_BOTTOM.1 → R_UV_3V3_TOP.2	(18,29) → (14,29)
W36 / UV_3V3	R_UV_3V3_TOP.2 → U_OV_3V3.2	(14,29) → (4,25)
W37 / OV_3V3	R_OV_3V3_BOTTOM.1 → R_OV_3V3_TOP.2	(18,23) → (14,23)
W38 / OV_3V3	R_OV_3V3_TOP.2 → U_OV_3V3.3	(14,23) → (4,26)
W39 / SAFE_3V3	C_OV_3V3_OUT.1 → J_LINK_5.1	(19,35) → (1,9)
W40 / SAFE_3V3	J_LINK_5.1 → Q_OV_3V3_OUT.3	(1,9) → (11,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP10 — power-guards

Install these wires after their endpoint components; inspect each net with power isolated.

power-guards — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

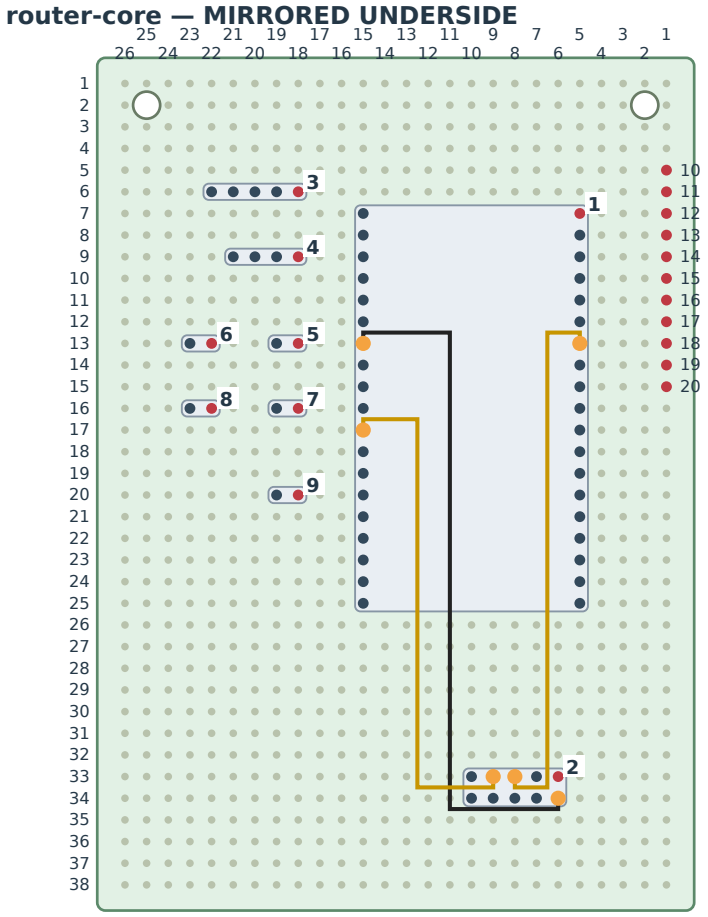
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / SAFE_3V3	Q_OV_3V3_OUT.3 → U_OV_3V3.7	(11,35) → (7,25)
W42 / OV_GATE_3V3	Q_OV_3V3_IN.1 → Q_OV_3V3_OUT.1	(4,35) → (9,35)
W43 / OV_GATE_3V3	Q_OV_3V3_OUT.1 → U_OV_3V3.8	(9,35) → (7,24)
W44 / OV_MID_3V3	Q_OV_3V3_IN.2 → Q_OV_3V3_OUT.2	(5,35) → (10,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — router-core

Install these wires after their endpoint components; inspect each net with power isolated.



ID / net	From → to	TOP holes
W7 / GND	U_ESP1.26 → J_POWER.2	(15,13) → (6,34)
W21 / EDU_PWR_EN	U_ESP1.7 → J_POWER.5	(5,13) → (8,33)
W24 / EDU_FAULT_RESET	U_ESP1.30 → J_POWER.7	(15,17) → (9,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

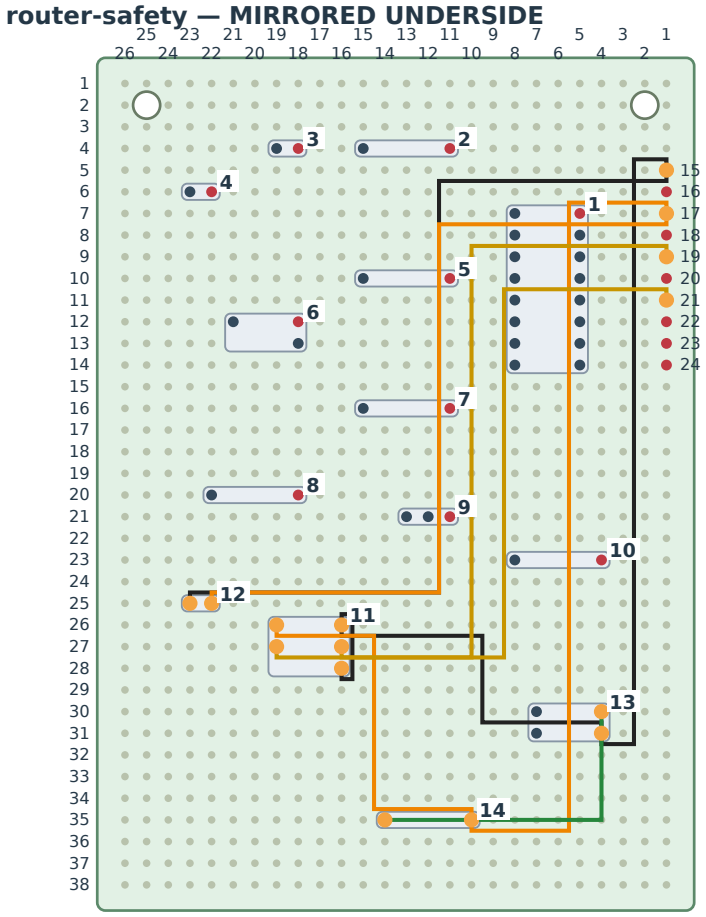
70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP11 — router-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

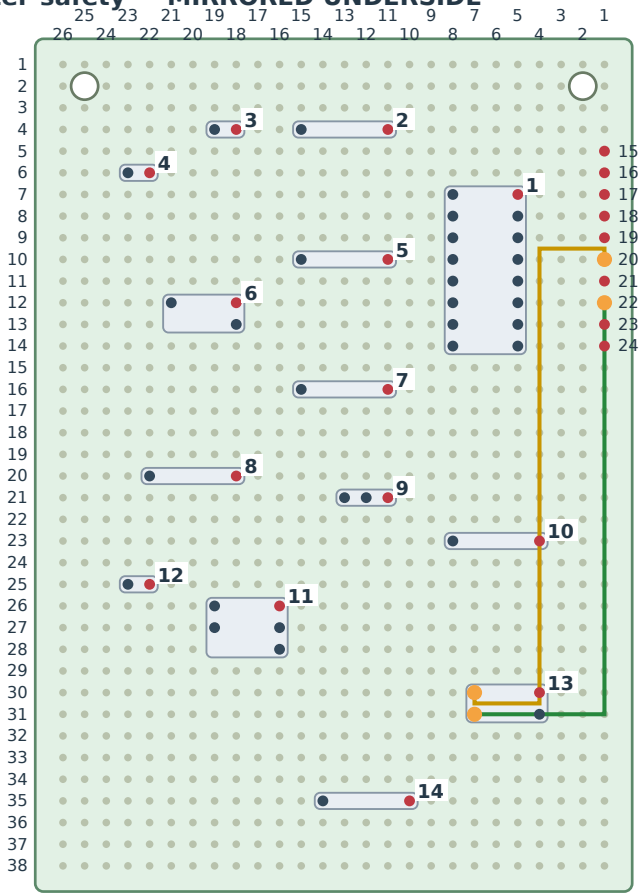
ID / net	From → to	TOP holes
W1 / GND	C_RETURN.2 → J_LINK_1.1	(23,25) → (1,5)
W2 / GND	J_LINK_1.1 → K_ADC.2	(1,5) → (4,31)
W3 / GND	K_ADC.2 → U_FAULT_RETURN.1	(4,31) → (16,26)
W4 / GND	U_FAULT_RETURN.1 → U_FAULT_RETURN.3	(16,26) → (16,28)
W25 / USB_ROUTER_3V3	C_RETURN.1 → J_LINK_3.1	(22,25) → (1,7)
W26 / USB_ROUTER_3V3	J_LINK_3.1 → R_KADC_LED.1	(1,7) → (10,35)
W27 / USB_ROUTER_3V3	R_KADC_LED.1 → U_FAULT_RETURN.5	(10,35) → (19,26)
W36 / EDU_FAULT	J_LINK_7.1 → U_FAULT_RETURN.2	(1,11) → (16,27)
W37 / FAULT_USB	J_LINK_5.1 → U_FAULT_RETURN.4	(1,9) → (19,27)
W38 / K_ADC_LED	K_ADC.1 → R_KADC_LED.2	(4,30) → (14,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — router-safety

Install these wires after their endpoint components; inspect each net with power isolated.

router-safety — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

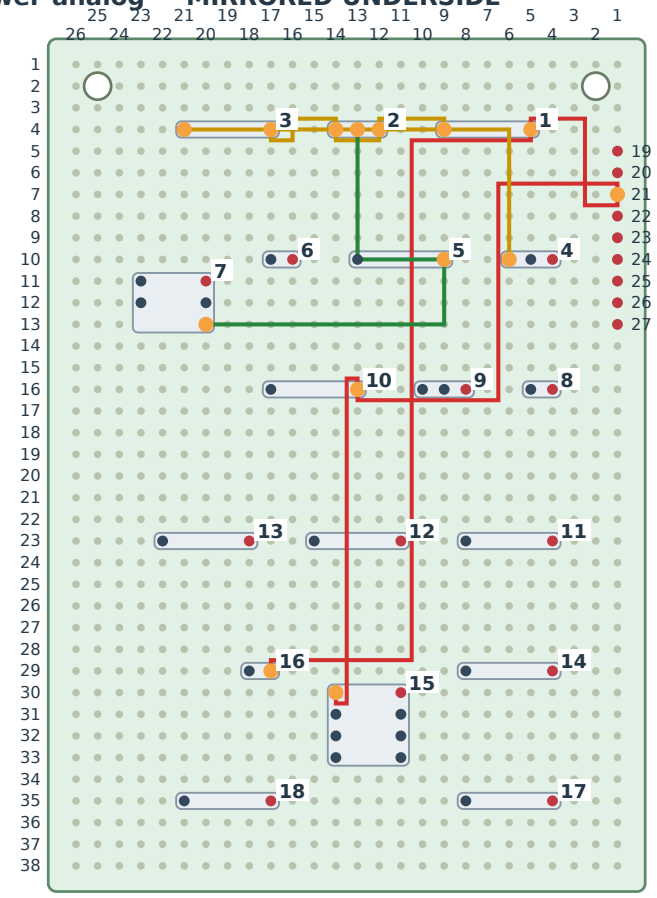
ID / net	From → to	TOP holes
W39 / CURRENT_SENSE	J_LINK_8.1 → K_ADC.3	(1,12) → (7,31)
W40 / CURRENT_USB	J_LINK_6.1 → K_ADC.4	(1,10) → (7,30)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-analog

Install these wires after their endpoint components; inspect each net with power isolated.

power-analog — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

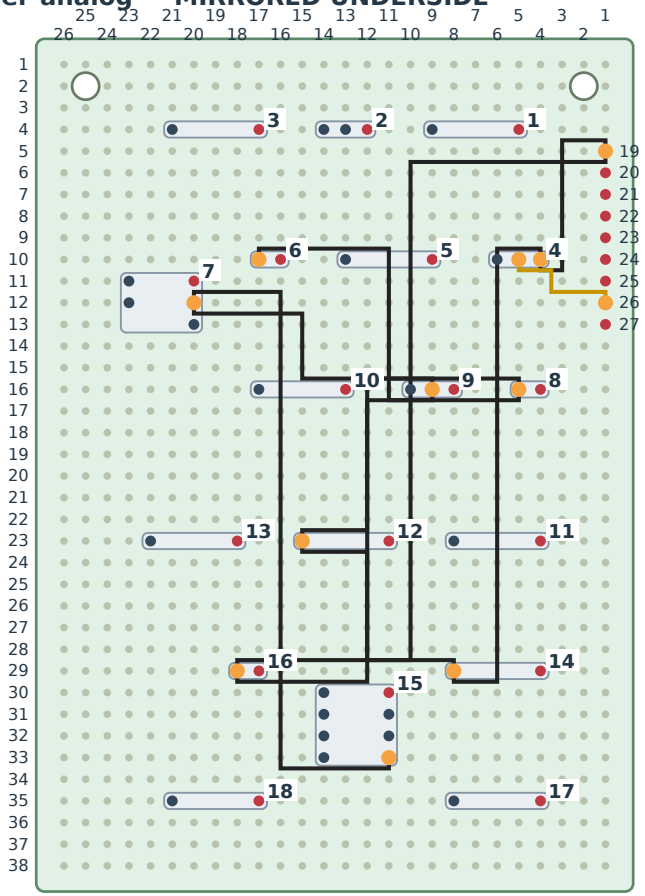
ID / net	From → to	TOP holes
W1 / 5V_SYS	C_TRIP.1 → F_EDU.1	(17,29) → (5,4)
W2 / 5V_SYS	F_EDU.1 → J_LINK_3.1	(5,4) → (1,7)
W3 / 5V_SYS	J_LINK_3.1 → R_REF_FEED.1	(1,7) → (13,16)
W4 / 5V_SYS	R_REF_FEED.1 → U_TRIP.8	(13,16) → (14,30)
W5 / PTC_OUT	F_EDU.2 → Q_PASS.3	(9,4) → (14,4)
W6 / PTC_OUT	Q_PASS.3 → R_GATE_OFF.1	(14,4) → (17,4)
W7 / PASS_GATE	Q_ENABLE.3 → Q_PASS.1	(6,10) → (12,4)
W8 / PASS_GATE	Q_PASS.1 → R_GATE_OFF.2	(12,4) → (21,4)
W9 / SHUNT_P	Q_PASS.2 → R_SHUNT.1	(13,4) → (9,10)
W10 / SHUNT_P	R_SHUNT.1 → U_SENSE.3	(9,10) → (20,13)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-analog

Install these wires after their endpoint components; inspect each net with power isolated.

power-analog — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

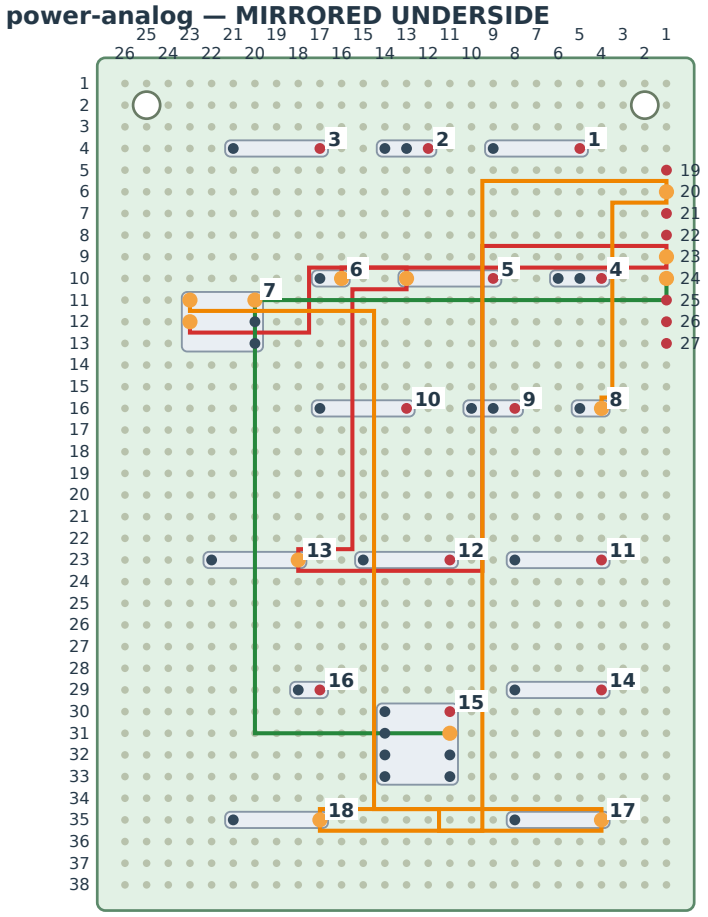
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	C_EDU.2 → C_SENSE.2	(17,10) → (5,16)
W12 / GND	C_SENSE.2 → C_TRIP.2	(5,16) → (18,29)
W13 / GND	C_TRIP.2 → J_LINK_1.1	(18,29) → (1,5)
W14 / GND	J_LINK_1.1 → Q_ENABLE.1	(1,5) → (4,10)
W15 / GND	Q_ENABLE.1 → R_EDU_BOTTOM.2	(4,10) → (8,29)
W16 / GND	R_EDU_BOTTOM.2 → R_TRIP_BOTTOM.2	(8,29) → (15,23)
W17 / GND	R_TRIP_BOTTOM.2 → U_REF.2	(15,23) → (9,16)
W18 / GND	U_REF.2 → U_SENSE.2	(9,16) → (20,12)
W19 / GND	U_SENSE.2 → U_TRIP.4	(20,12) → (11,33)
W20 / ALLOW_RAIL	J_LINK_8.1 → Q_ENABLE.2	(1,12) → (5,10)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-analog

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

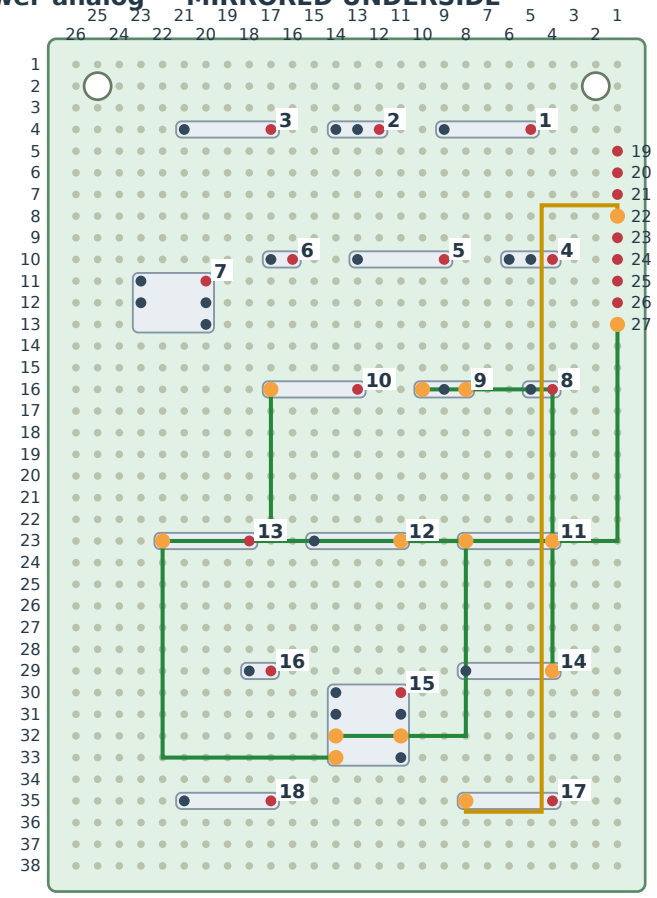
ID / net	From → to	TOP holes
W21 / 5V_EDU	C_EDU.1 → J_LINK_5.1	(16,10) → (1,9)
W22 / 5V_EDU	J_LINK_5.1 → R_EDU_TOP.1	(1,9) → (18,23)
W23 / 5V_EDU	R_EDU_TOP.1 → R_SHUNT.2	(18,23) → (13,10)
W24 / 5V_EDU	R_SHUNT.2 → U_SENSE.4	(13,10) → (23,12)
W25 / CURRENT_SENSE	J_LINK_6.1 → U_SENSE.1	(1,10) → (20,11)
W26 / CURRENT_SENSE	U_SENSE.1 → U_TRIP.2	(20,11) → (11,31)
W27 / 3V3_SYS	C_SENSE.1 → J_LINK_2.1	(4,16) → (1,6)
W28 / 3V3_SYS	J_LINK_2.1 → R_EDU_ON.1	(1,6) → (17,35)
W29 / 3V3_SYS	R_EDU_ON.1 → R_PRESET.1	(17,35) → (4,35)
W30 / 3V3_SYS	R_PRESET.1 → U_SENSE.5	(4,35) → (23,11)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-analog

Install these wires after their endpoint components; inspect each net with power isolated.

power-analog — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

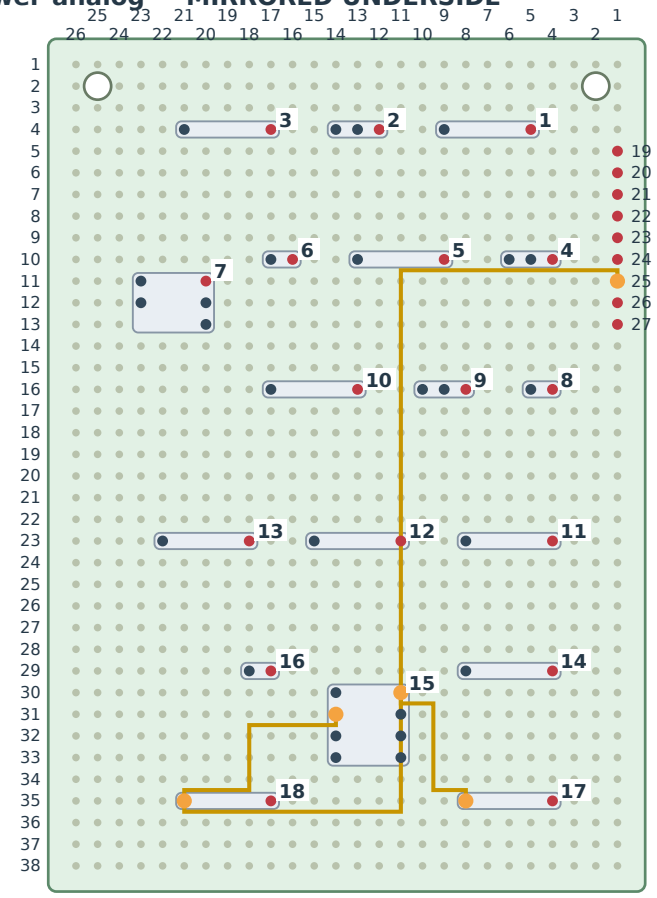
ID / net	From → to	TOP holes
W31 / REF_2V495	R_REF_FEED.2 → R_TRIP_TOP.1	(17,16) → (4,23)
W32 / REF_2V495	R_TRIP_TOP.1 → U_REF.1	(4,23) → (8,16)
W33 / REF_2V495	U_REF.1 → U_REF.3	(8,16) → (10,16)
W34 / REF_TRIP	J_LINK_9.1 → R_TRIP_BOTTOM.1	(1,13) → (11,23)
W35 / REF_TRIP	R_TRIP_BOTTOM.1 → R_TRIP_TOP.2	(11,23) → (8,23)
W36 / REF_TRIP	R_TRIP_TOP.2 → U_TRIP.3	(8,23) → (11,32)
W37 / REF_TRIP	U_TRIP.3 → U_TRIP.6	(11,32) → (14,32)
W38 / EDU_DIV	R_EDU_BOTTOM.1 → R_EDU_TOP.2	(4,29) → (22,23)
W39 / EDU_DIV	R_EDU_TOP.2 → U_TRIP.5	(22,23) → (14,33)
W40 / PRESET_N	J_LINK_4.1 → R_PRESET.2	(1,8) → (8,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-analog

Install these wires after their endpoint components; inspect each net with power isolated.

power-analog — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

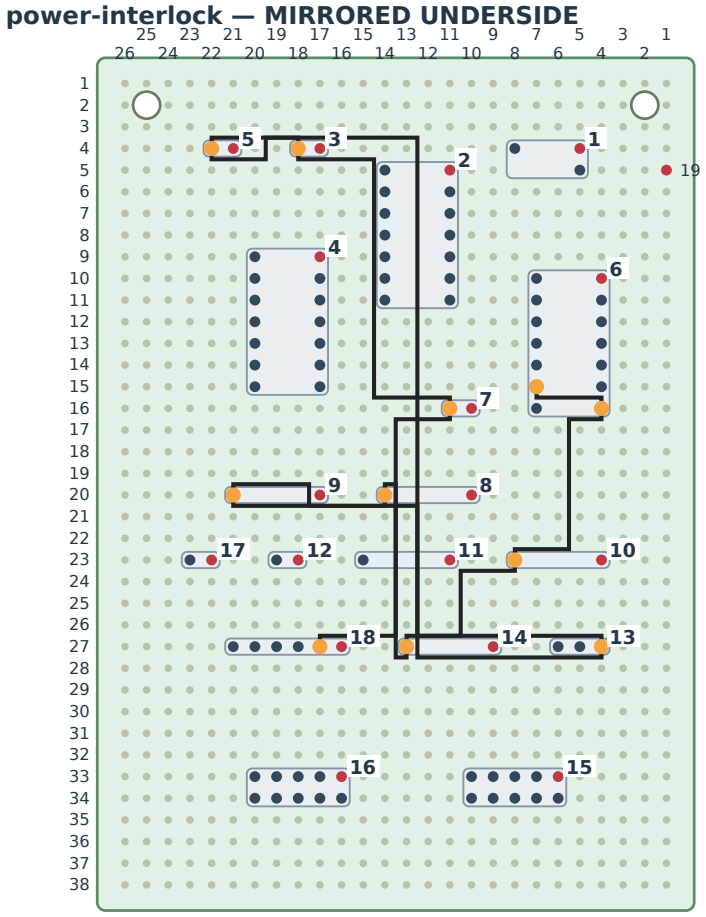
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / PRESET_N	R_PRESET.2 → U_TRIP.1	(8,35) → (11,30)
W42 / EDU_ON	J_LINK_7.1 → R_EDU_ON.2	(1,11) → (21,35)
W43 / EDU_ON	R_EDU_ON.2 → U_TRIP.7	(21,35) → (14,31)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-interlock

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

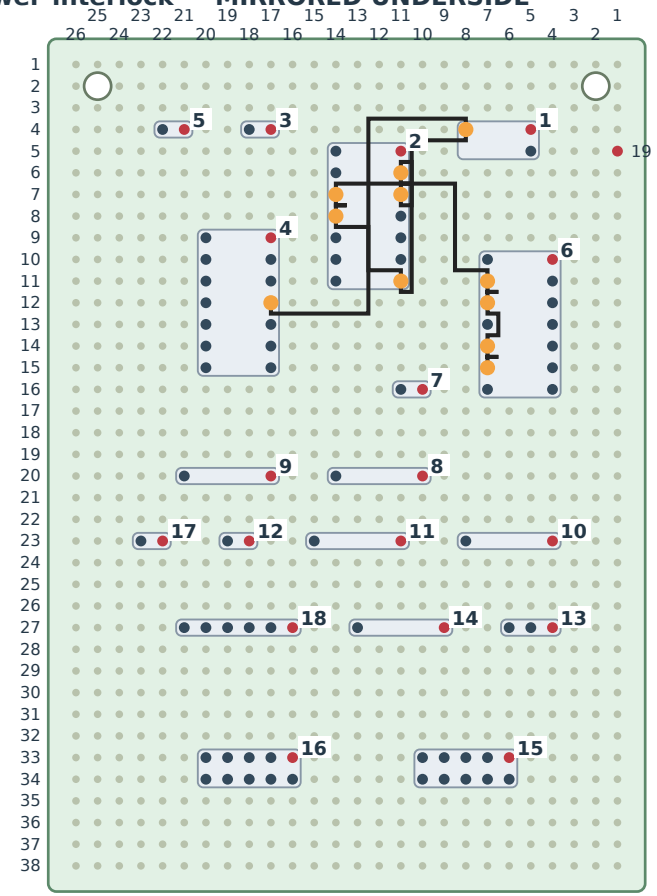
ID / net	From → to	TOP holes
W6 / GND	J_TEST.2 → C_ALLOW.2	(17,27) → (11,16)
W7 / GND	C_ALLOW.2 → C_FAULT.2	(11,16) → (18,4)
W8 / GND	C_FAULT.2 → C_RESET.2	(18,4) → (22,4)
W9 / GND	C_RESET.2 → Q_BATTERY_LED.1	(22,4) → (4,27)
W10 / GND	Q_BATTERY_LED.1 → R_EDU_FAULT_RESET_OFF.2	(4,27) → (21,20)
W11 / GND	R_EDU_FAULT_RESET_OFF.2 → R_EDU_PWR_EN_OFF.2	(21,20) → (14,20)
W12 / GND	R_EDU_PWR_EN_OFF.2 → R_LED_GATE_OFF.2	(14,20) → (13,27)
W13 / GND	R_LED_GATE_OFF.2 → R_ROUTER_ALIVE_OFF.2	(13,27) → (8,23)
W14 / GND	R_ROUTER_ALIVE_OFF.2 → U_ALLOW.7	(8,23) → (4,16)
W15 / GND	U_ALLOW.7 → U_ALLOW.9	(4,16) → (7,15)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-interlock

Install these wires after their endpoint components; inspect each net with power isolated.

power-interlock — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

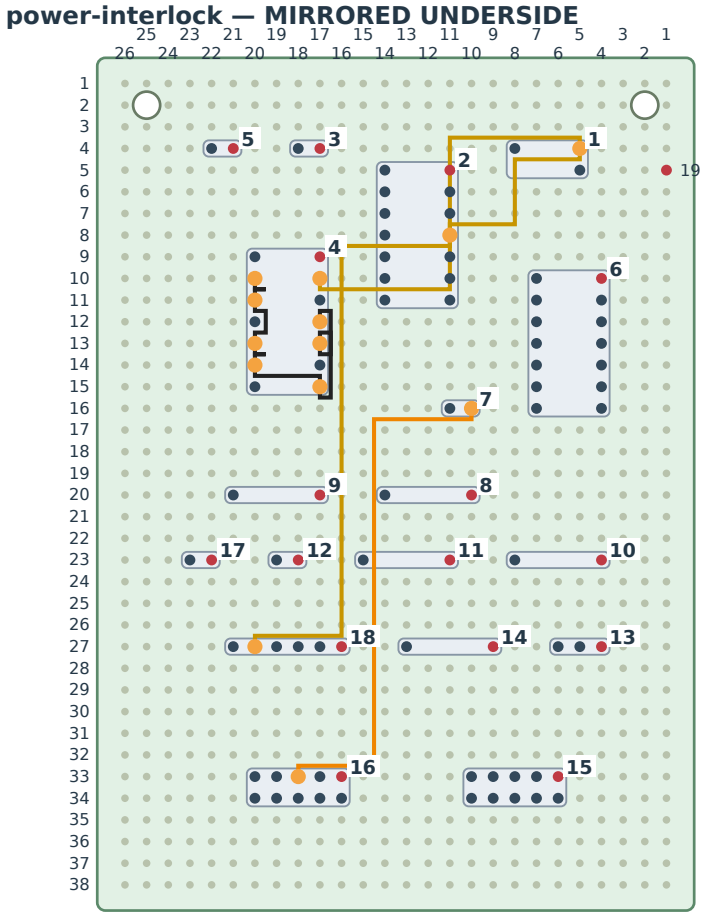
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W16 / GND	U_ALLOW.9 → U_ALLOW.10	(7,15) → (7,14)
W17 / GND	U_ALLOW.10 → U_ALLOW.12	(7,14) → (7,12)
W18 / GND	U_ALLOW.12 → U_ALLOW.13	(7,12) → (7,11)
W19 / GND	U_ALLOW.13 → U_FAULT.2	(7,11) → (11,6)
W20 / GND	U_FAULT.2 → U_FAULT.3	(11,6) → (11,7)
W21 / GND	U_FAULT.3 → U_FAULT.7	(11,7) → (11,11)
W22 / GND	U_FAULT.7 → U_FAULT.11	(11,11) → (14,8)
W23 / GND	U_FAULT.11 → U_FAULT.12	(14,8) → (14,7)
W24 / GND	U_FAULT.12 → U_POR.3	(14,7) → (8,4)
W25 / GND	U_POR.3 → U_RESET.4	(8,4) → (17,12)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-interlock

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

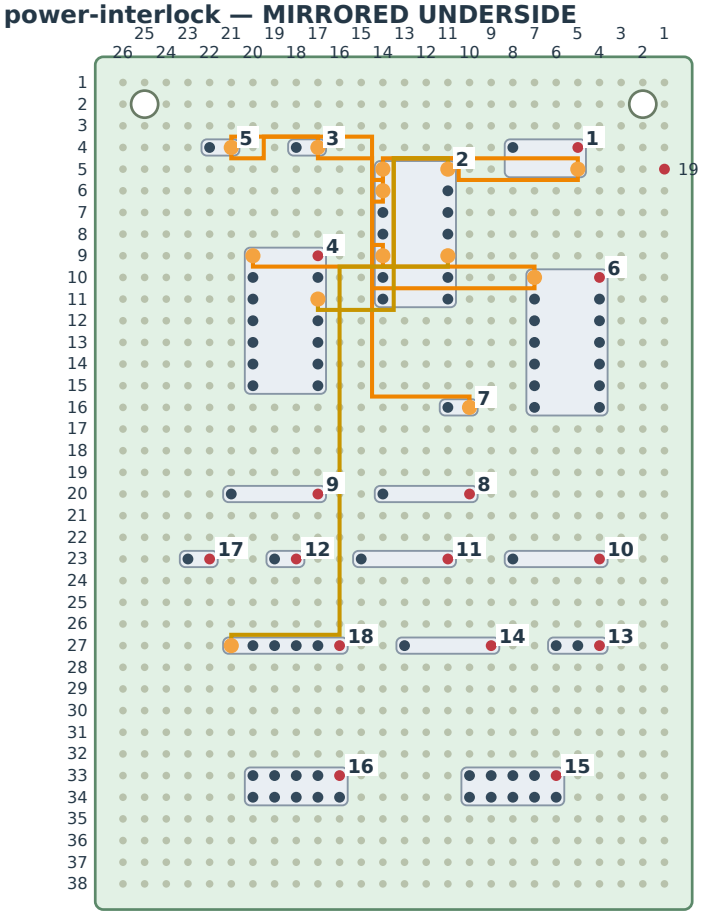
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W26 / GND	U_RESET.4 → U_RESET.5	(17,12) → (17,13)
W27 / GND	U_RESET.5 → U_RESET.7	(17,13) → (17,15)
W28 / GND	U_RESET.7 → U_RESET.9	(17,15) → (20,14)
W29 / GND	U_RESET.9 → U_RESET.10	(20,14) → (20,13)
W30 / GND	U_RESET.10 → U_RESET.12	(20,13) → (20,11)
W31 / GND	U_RESET.12 → U_RESET.13	(20,11) → (20,10)
W32 / PRESET_N	J_TEST.5 → U_FAULT.4	(20,27) → (11,8)
W33 / PRESET_N	U_FAULT.4 → U_POR.1	(11,8) → (5,4)
W34 / PRESET_N	U_POR.1 → U_RESET.2	(5,4) → (17,10)
W36 / 3V3_SYS	J_SMART.5 → C_ALLOW.1	(18,33) → (10,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-interlock

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

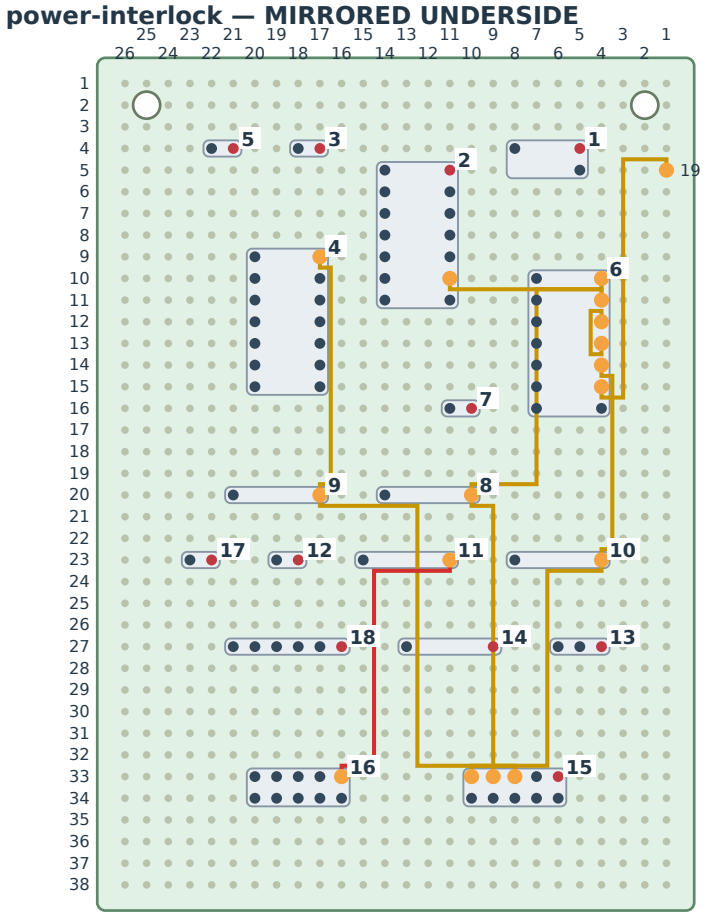
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W37 / 3V3_SYS	C_ALLOW.1 → C_FAULT.1	(10,16) → (17,4)
W38 / 3V3_SYS	C_FAULT.1 → C_RESET.1	(17,4) → (21,4)
W39 / 3V3_SYS	C_RESET.1 → U_ALLOW.14	(21,4) → (7,10)
W40 / 3V3_SYS	U_ALLOW.14 → U_FAULT.10	(7,10) → (14,9)
W41 / 3V3_SYS	U_FAULT.10 → U_FAULT.13	(14,9) → (14,6)
W42 / 3V3_SYS	U_FAULT.13 → U_FAULT.14	(14,6) → (14,5)
W43 / 3V3_SYS	U_FAULT.14 → U_POR.2	(14,5) → (5,5)
W44 / 3V3_SYS	U_POR.2 → U_RESET.14	(5,5) → (20,9)
W45 / CLEAR_N	U_FAULT.1 → U_RESET.3	(11,5) → (17,11)
W47 / EDU_FAULT	J_TEST.6 → U_FAULT.5	(21,27) → (11,9)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-interlock

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

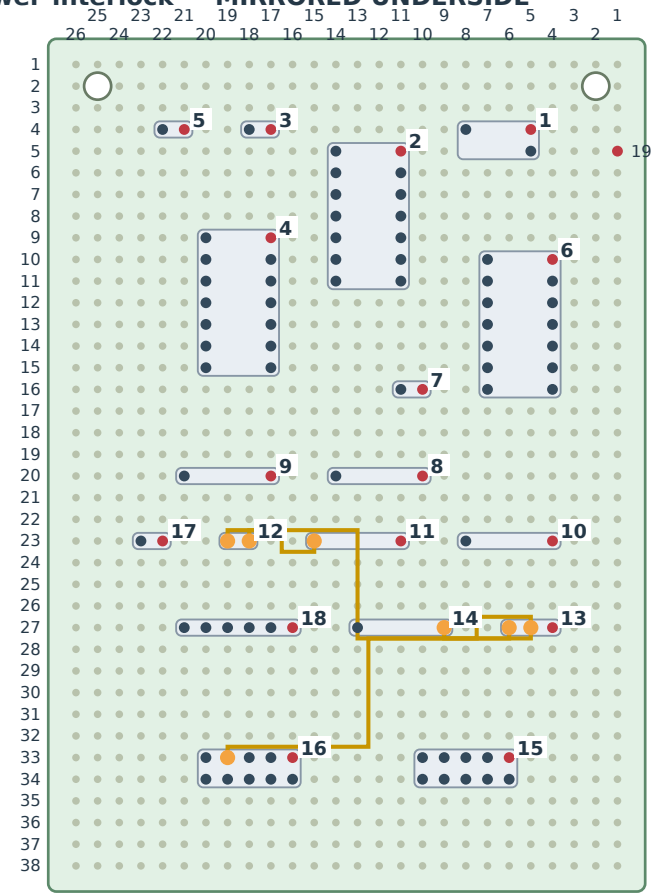
ID / net	From → to	TOP holes
W48 / NOT_FAULT	U_ALLOW.2 → U_FAULT.6	(4,11) → (11,10)
W49 / EDU_FAULT_RESET	J_MATRIX.7 → R_EDU_FAULT_RESET_OFF.1	(9,33) → (17,20)
W50 / EDU_FAULT_RESET	R_EDU_FAULT_RESET_OFF.1 → U_RESET.1	(17,20) → (17,9)
W51 / EDU_PWR_EN	J_MATRIX.5 → R_EDU_PWR_EN_OFF.1	(8,33) → (10,20)
W52 / EDU_PWR_EN	R_EDU_PWR_EN_OFF.1 → U_ALLOW.1	(10,20) → (4,10)
W53 / ENABLE_NO_FAULT	U_ALLOW.3 → U_ALLOW.4	(4,12) → (4,13)
W54 / ROUTER_ALIVE	J_MATRIX.9 → R_ROUTER_ALIVE_OFF.1	(10,33) → (4,23)
W55 / ROUTER_ALIVE	R_ROUTER_ALIVE_OFF.1 → U_ALLOW.5	(4,23) → (4,14)
W56 / ALLOW_RAIL	J_LINK_1.1 → U_ALLOW.6	(1,5) → (4,15)
W58 / 5V_SYS	J_SMART.1 → R_BATTERY_LED.1	(16,33) → (11,23)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP11 — power-interlock

Install these wires after their endpoint components; inspect each net with power isolated.

power-interlock — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

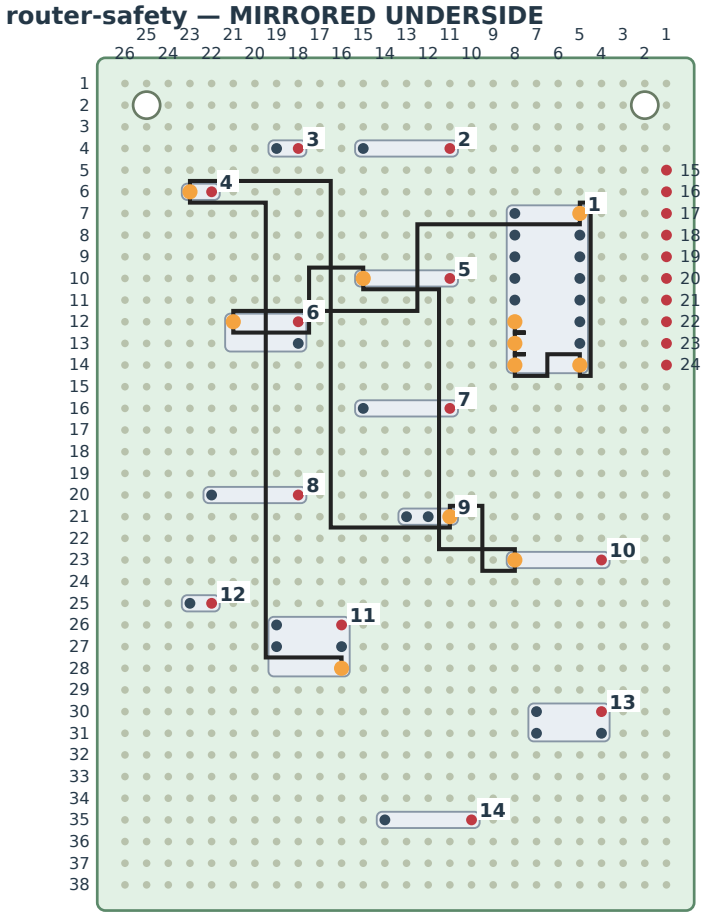
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W59 / BATTERY_LED_A	LED_BATTERY.1 → R_BATTERY_LED.2	(18,23) → (15,23)
W60 / BATTERY_LED_K	LED_BATTERY.2 → Q_BATTERY_LED.3	(19,23) → (6,27)
W61 / EDU_ON	J_SMART.7 → Q_BATTERY_LED.2	(19,33) → (5,27)
W62 / EDU_ON	Q_BATTERY_LED.2 → R_LED_GATE_OFF.1	(5,27) → (9,27)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP13 — router-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

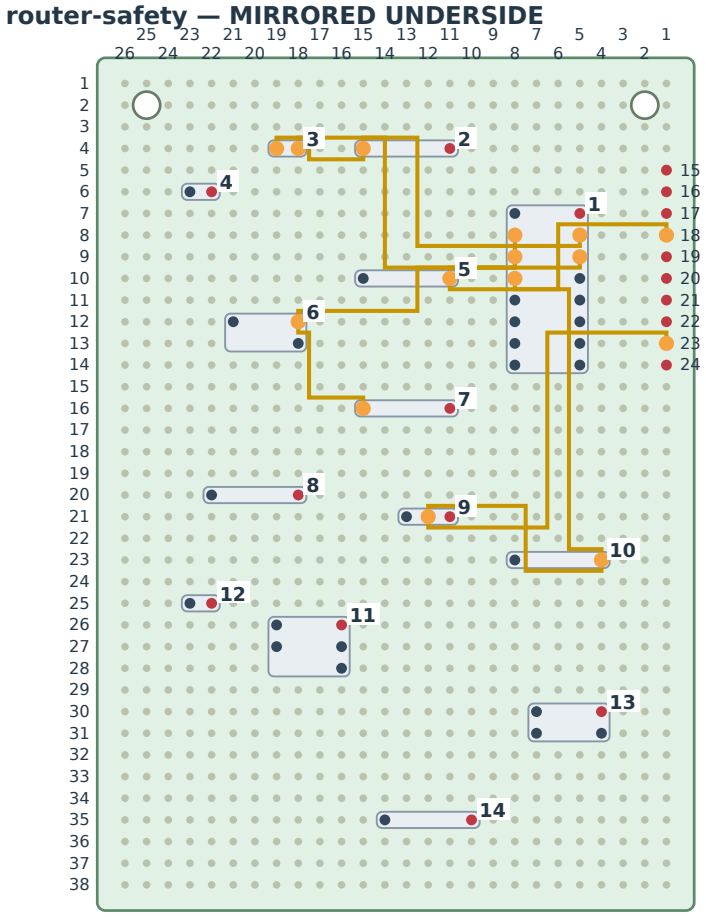
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W5 / GND	U_FAULT_RETURN.3 → C_WATCH_VCC.2	(16,28) → (23,6)
W6 / GND	C_WATCH_VCC.2 → Q_BUF.1	(23,6) → (11,21)
W7 / GND	Q_BUF.1 → R_ALIVE_OFF.2	(11,21) → (8,23)
W8 / GND	R_ALIVE_OFF.2 → R_HEART.2	(8,23) → (15,10)
W9 / GND	R_HEART.2 → U_USB_POR.3	(15,10) → (21,12)
W10 / GND	U_USB_POR.3 → U_WATCH.1	(21,12) → (5,7)
W11 / GND	U_WATCH.1 → U_WATCH.8	(5,7) → (5,14)
W12 / GND	U_WATCH.8 → U_WATCH.9	(5,14) → (8,14)
W13 / GND	U_WATCH.9 → U_WATCH.10	(8,14) → (8,13)
W14 / GND	U_WATCH.10 → U_WATCH.11	(8,13) → (8,12)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP13 — router-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

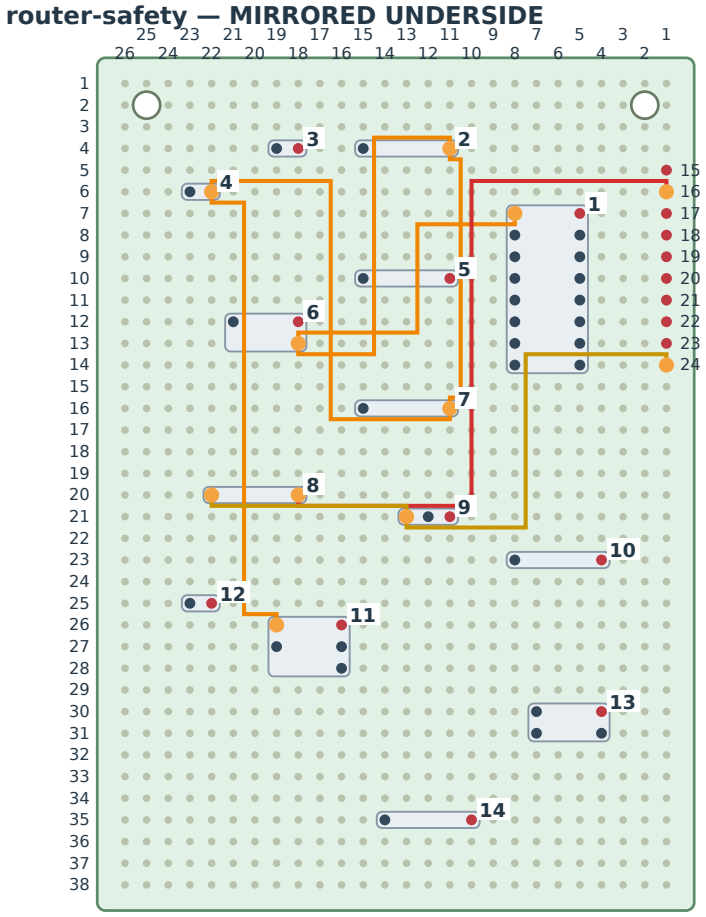
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W15 / HEARTBEAT	J_LINK_4.1 → R_HEART.1	(1,8) → (11,10)
W16 / HEARTBEAT	R_HEART.1 → U_WATCH.2	(11,10) → (5,8)
W17 / USB_RESET_N	R_USB_POR.2 → U_USB_POR.1	(15,16) → (18,12)
W18 / USB_RESET_N	U_USB_POR.1 → U_WATCH.3	(18,12) → (5,9)
W19 / ROUTER_ALIVE	J_LINK_9.1 → Q_BUF.2	(1,13) → (12,21)
W20 / ROUTER_ALIVE	Q_BUF.2 → R_ALIVE_OFF.1	(12,21) → (4,23)
W21 / ROUTER_ALIVE	R_ALIVE_OFF.1 → U_WATCH.13	(4,23) → (8,10)
W22 / WATCH_CX	C_WATCH.2 → U_WATCH.14	(19,4) → (8,9)
W23 / WATCH_RX	C_WATCH.1 → R_WATCH.2	(18,4) → (15,4)
W24 / WATCH_RX	R_WATCH.2 → U_WATCH.15	(15,4) → (8,8)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP13 — router-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

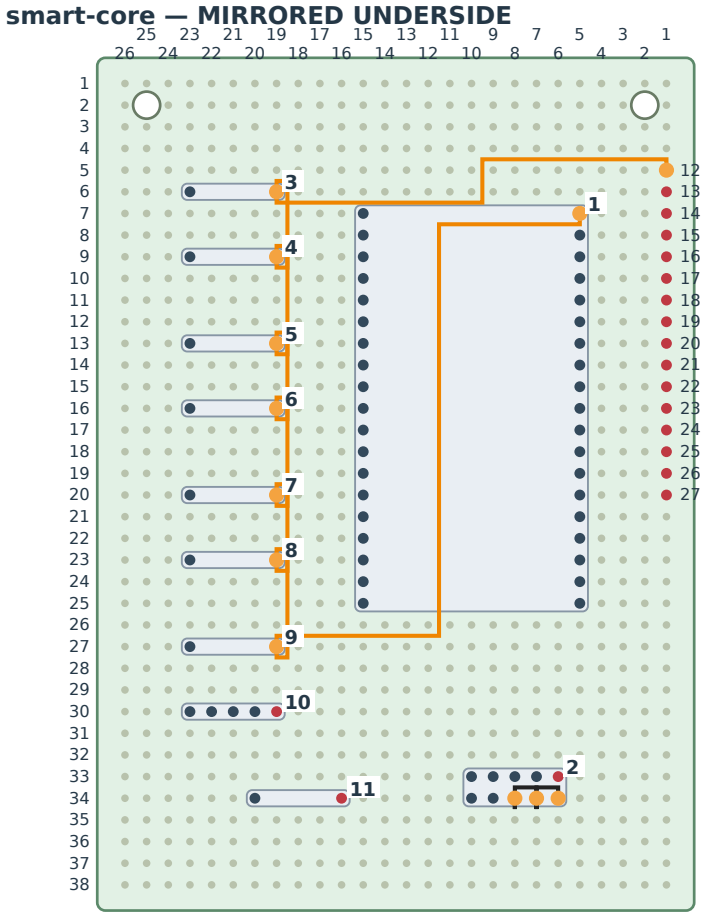
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W28 / USB_ROUTER_3V3	U_FAULT_RETURN.5 → C_WATCH_VCC.1	(19,26) → (22,6)
W29 / USB_ROUTER_3V3	C_WATCH_VCC.1 → R_USB_POR.1	(22,6) → (11,16)
W30 / USB_ROUTER_3V3	R_USB_POR.1 → R_WATCH.1	(11,16) → (11,4)
W31 / USB_ROUTER_3V3	R_WATCH.1 → U_USB_POR.2	(11,4) → (18,13)
W32 / USB_ROUTER_3V3	U_USB_POR.2 → U_WATCH.16	(18,13) → (8,7)
W33 / 5V_SYS	J_LINK_2.1 → R_BUF_OE.1	(1,6) → (18,20)
W34 / XBAR_OE_N	J_LINK_10.1 → Q_BUF.3	(1,14) → (13,21)
W35 / XBAR_OE_N	Q_BUF.3 → R_BUF_OE.2	(13,21) → (22,20)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-core

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

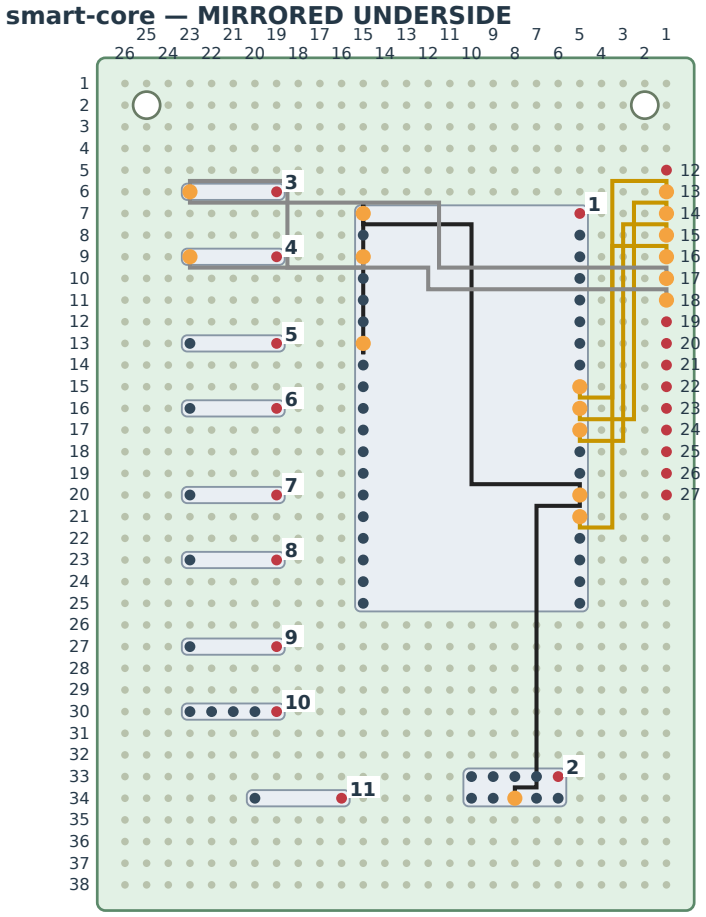
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / USB_SMART_3V3	J_LINK_1.1 → R_SCL_A.1	(1,5) → (19,6)
W2 / USB_SMART_3V3	R_SCL_A.1 → R_SDA_0.1	(19,6) → (19,9)
W3 / USB_SMART_3V3	R_SDA_0.1 → R_SDA_1.1	(19,9) → (19,13)
W4 / USB_SMART_3V3	R_SDA_1.1 → R_SDA_2.1	(19,13) → (19,16)
W5 / USB_SMART_3V3	R_SDA_2.1 → R_SDA_3.1	(19,16) → (19,20)
W6 / USB_SMART_3V3	R_SDA_3.1 → R_SDA_4.1	(19,20) → (19,23)
W7 / USB_SMART_3V3	R_SDA_4.1 → R_SDA_5.1	(19,23) → (19,27)
W8 / USB_SMART_3V3	R_SDA_5.1 → U_ESP2.1	(19,27) → (5,7)
W11 / GND	J_POWER.2 → J_POWER.4	(6,34) → (7,34)
W12 / GND	J_POWER.4 → J_POWER.6	(7,34) → (8,34)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-core

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

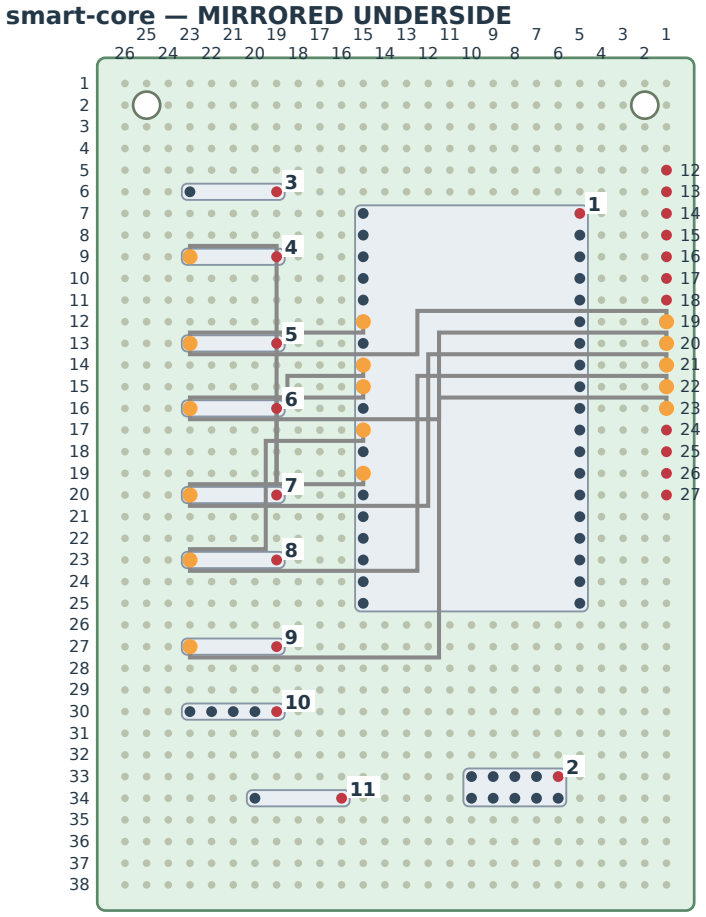
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W13 / GND	J_POWER.6 → U_ESP2.14	(8,34) → (5,20)
W14 / GND	U_ESP2.14 → U_ESP2.20	(5,20) → (15,7)
W15 / GND	U_ESP2.20 → U_ESP2.26	(15,7) → (15,13)
W17 / SR_SRCLK	J_LINK_2.1 → U_ESP2.9	(1,6) → (5,15)
W18 / SR_RCLK	J_LINK_3.1 → U_ESP2.10	(1,7) → (5,16)
W19 / SR_SER	J_LINK_4.1 → U_ESP2.11	(1,8) → (5,17)
W20 / SMART_HEARTBEAT	J_LINK_5.1 → U_ESP2.15	(1,9) → (5,21)
W21 / SCL_A	J_LINK_6.1 → R_SCL_A.2	(1,10) → (23,6)
W22 / SCL_A	R_SCL_A.2 → U_ESP2.22	(23,6) → (15,9)
W23 / SDA_0	J_LINK_7.1 → R_SDA_0.2	(1,11) → (23,9)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-core

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

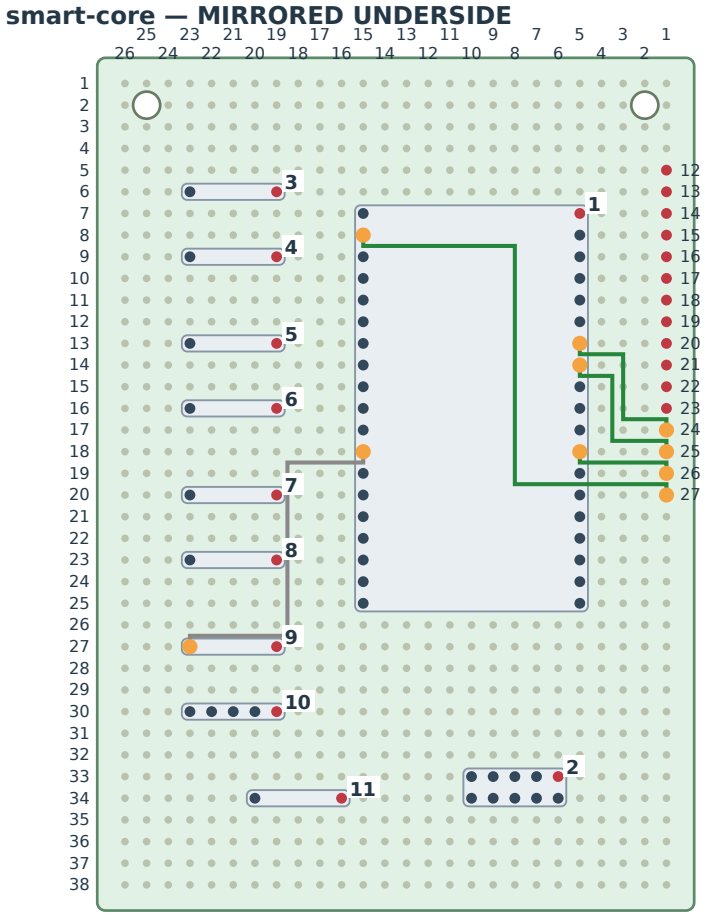
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W24 / SDA_0	R_SDA_0.2 → U_ESP2.32	(23,9) → (15,19)
W25 / SDA_1	J_LINK_8.1 → R_SDA_1.2	(1,12) → (23,13)
W26 / SDA_1	R_SDA_1.2 → U_ESP2.25	(23,13) → (15,12)
W27 / SDA_2	J_LINK_9.1 → R_SDA_2.2	(1,13) → (23,16)
W28 / SDA_2	R_SDA_2.2 → U_ESP2.27	(23,16) → (15,14)
W29 / SDA_3	J_LINK_10.1 → R_SDA_3.2	(1,14) → (23,20)
W30 / SDA_3	R_SDA_3.2 → U_ESP2.28	(23,20) → (15,15)
W31 / SDA_4	J_LINK_11.1 → R_SDA_4.2	(1,15) → (23,23)
W32 / SDA_4	R_SDA_4.2 → U_ESP2.30	(23,23) → (15,17)
W33 / SDA_5	J_LINK_12.1 → R_SDA_5.2	(1,16) → (23,27)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-core

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

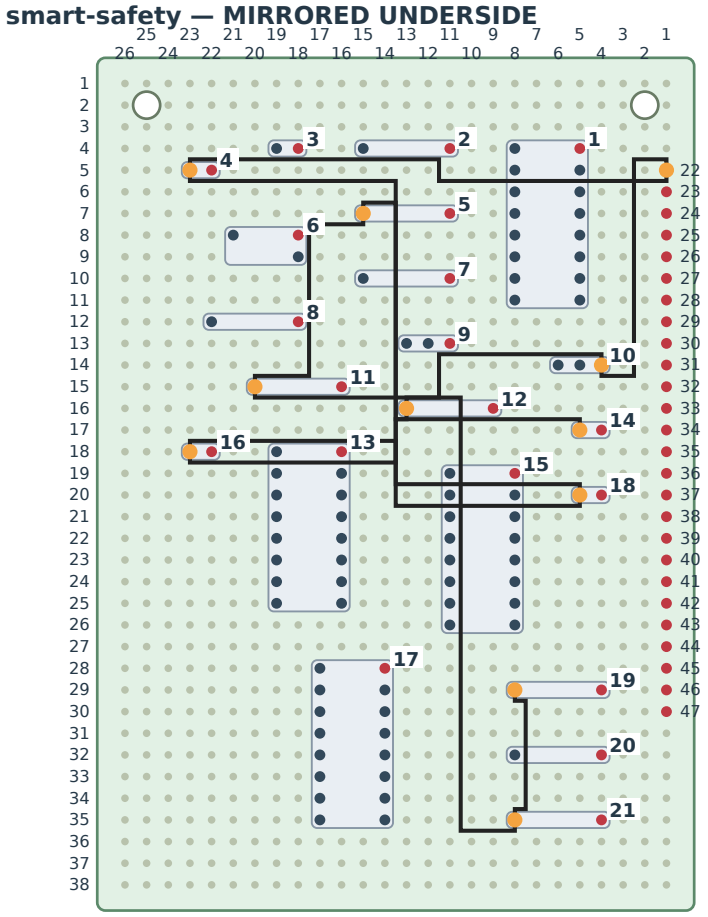
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W34 / SDA_5	R_SDA_5.2 → U_ESP2.31	(23,27) → (15,18)
W35 / MUX_A_A	J_LINK_13.1 → U_ESP2.7	(1,17) → (5,13)
W36 / MUX_A_B	J_LINK_14.1 → U_ESP2.8	(1,18) → (5,14)
W37 / MUX_B_A	J_LINK_15.1 → U_ESP2.12	(1,19) → (5,18)
W38 / MUX_B_B	J_LINK_16.1 → U_ESP2.21	(1,20) → (15,8)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

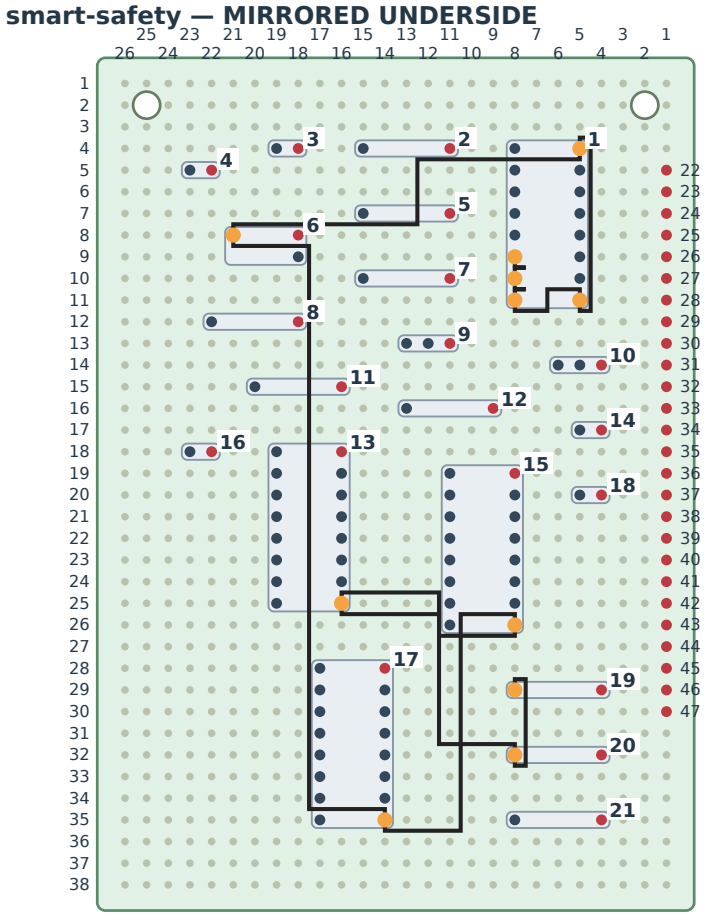
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / GND	C_SR0.2 → C_SR1.2	(5,17) → (23,18)
W2 / GND	C_SR1.2 → C_SR2.2	(23,18) → (5,20)
W3 / GND	C_SR2.2 → C_WATCH_VCC.2	(5,20) → (23,5)
W4 / GND	C_WATCH_VCC.2 → J_LINK_1.1	(23,5) → (1,5)
W5 / GND	J_LINK_1.1 → Q_RAIL.1	(1,5) → (4,14)
W6 / GND	Q_RAIL.1 → R_ALIVE_OFF.2	(4,14) → (20,15)
W7 / GND	R_ALIVE_OFF.2 → R_HEART.2	(20,15) → (15,7)
W8 / GND	R_HEART.2 → R_RAIL_OFF.2	(15,7) → (13,16)
W9 / GND	R_RAIL_OFF.2 → R_SR_RCLK_OFF.2	(13,16) → (8,35)
W10 / GND	R_SR_RCLK_OFF.2 → R_SR_SER_OFF.2	(8,35) → (8,29)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

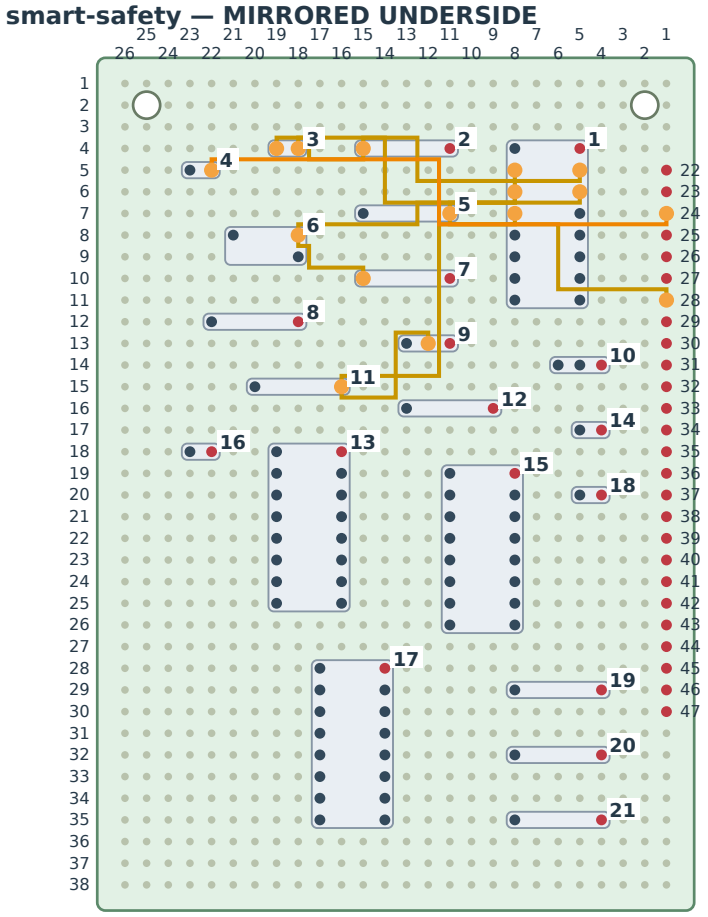
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	R_SR_SER_OFF.2 → R_SR_SRCLK_OFF.2	(8,29) → (8,32)
W12 / GND	R_SR_SRCLK_OFF.2 → U_SR0.8	(8,32) → (16,25)
W13 / GND	U_SR0.8 → U_SR1.8	(16,25) → (8,26)
W14 / GND	U_SR1.8 → U_SR2.8	(8,26) → (14,35)
W15 / GND	U_SR2.8 → U_USB_POR.3	(14,35) → (21,8)
W16 / GND	U_USB_POR.3 → U_WATCH.1	(21,8) → (5,4)
W17 / GND	U_WATCH.1 → U_WATCH.8	(5,4) → (5,11)
W18 / GND	U_WATCH.8 → U_WATCH.9	(5,11) → (8,11)
W19 / GND	U_WATCH.9 → U_WATCH.10	(8,11) → (8,10)
W20 / GND	U_WATCH.10 → U_WATCH.11	(8,10) → (8,9)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

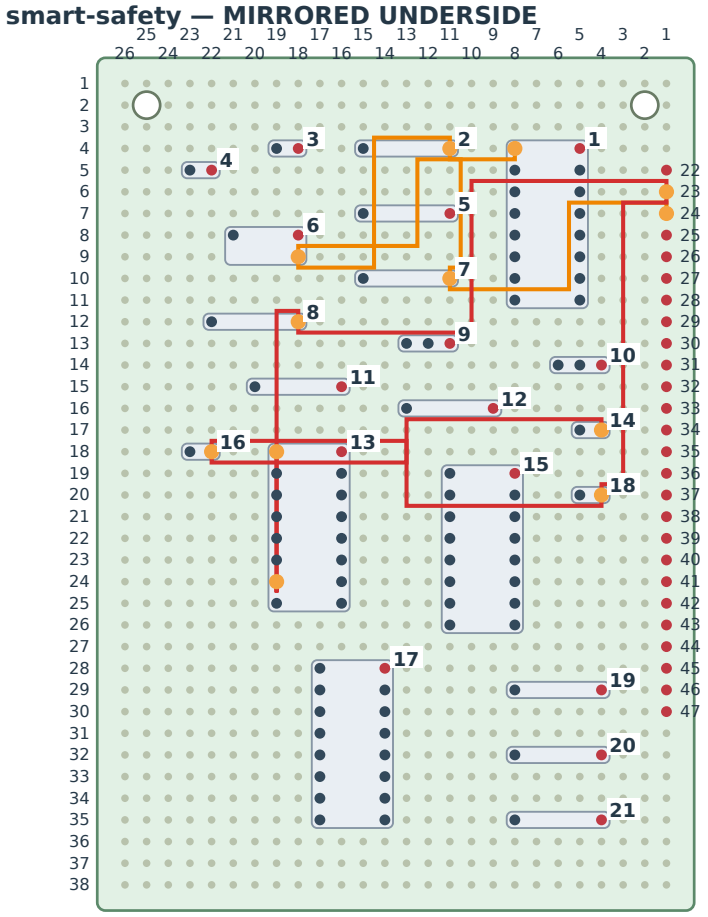
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / SMART_HEARTBEAT	J_LINK_7.1 → R_HEART.1	(1,11) → (11,7)
W22 / SMART_HEARTBEAT	R_HEART.1 → U_WATCH.2	(11,7) → (5,5)
W23 / SMART_USB_RESET_N	R_USB_POR.2 → U_USB_POR.1	(15,10) → (18,8)
W24 / SMART_USB_RESET_N	U_USB_POR.1 → U_WATCH.3	(18,8) → (5,6)
W25 / SMART_ALIVE	Q_ALIVE.2 → R_ALIVE_OFF.1	(12,13) → (16,15)
W26 / SMART_ALIVE	R_ALIVE_OFF.1 → U_WATCH.13	(16,15) → (8,7)
W27 / SMART_WATCH_CX	C_WATCH.2 → U_WATCH.14	(19,4) → (8,6)
W28 / SMART_WATCH_RX	C_WATCH.1 → R_WATCH.2	(18,4) → (15,4)
W29 / SMART_WATCH_RX	R_WATCH.2 → U_WATCH.15	(15,4) → (8,5)
W30 / USB_SMART_3V3	C_WATCH_VCC.1 → J_LINK_3.1	(22,5) → (1,7)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

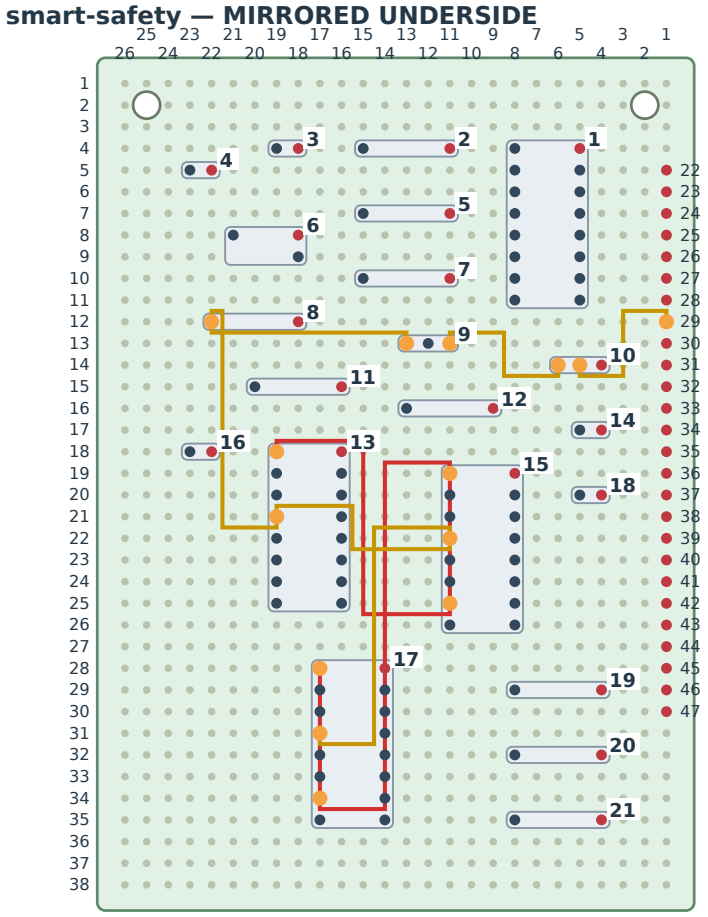
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / USB_SMART_3V3	J_LINK_3.1 → R_USB_POR.1	(1,7) → (11,10)
W32 / USB_SMART_3V3	R_USB_POR.1 → R_WATCH.1	(11,10) → (11,4)
W33 / USB_SMART_3V3	R_WATCH.1 → U_USB_POR.2	(11,4) → (18,9)
W34 / USB_SMART_3V3	U_USB_POR.2 → U_WATCH.16	(18,9) → (8,4)
W35 / 5V_SYS	C_SR0.1 → C_SR1.1	(4,17) → (22,18)
W36 / 5V_SYS	C_SR1.1 → C_SR2.1	(22,18) → (4,20)
W37 / 5V_SYS	C_SR2.1 → J_LINK_2.1	(4,20) → (1,6)
W38 / 5V_SYS	J_LINK_2.1 → R_OE.1	(1,6) → (18,12)
W39 / 5V_SYS	R_OE.1 → U_SR0.10	(18,12) → (19,24)
W40 / 5V_SYS	U_SR0.10 → U_SR0.16	(19,24) → (19,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

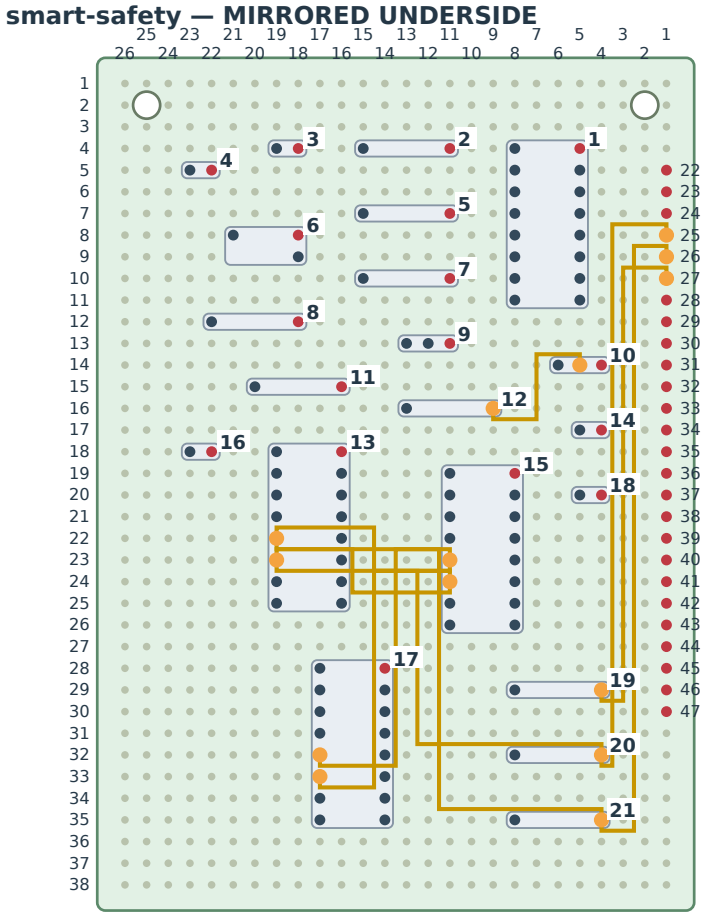
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / 5V_SYS	U_SR0.16 → U_SR1.10	(19,18) → (11,25)
W42 / 5V_SYS	U_SR1.10 → U_SR1.16	(11,25) → (11,19)
W43 / 5V_SYS	U_SR1.16 → U_SR2.10	(11,19) → (17,34)
W44 / 5V_SYS	U_SR2.10 → U_SR2.16	(17,34) → (17,28)
W45 / SR_OE_N	Q_ALIVE.3 → R_OE.2	(13,13) → (22,12)
W46 / SR_OE_N	R_OE.2 → U_SR0.13	(22,12) → (19,21)
W47 / SR_OE_N	U_SR0.13 → U_SR1.13	(19,21) → (11,22)
W48 / SR_OE_N	U_SR1.13 → U_SR2.13	(11,22) → (17,31)
W49 / OE_MID	Q_ALIVE.1 → Q_RAIL.3	(11,13) → (6,14)
W50 / EDU_ON	J_LINK_8.1 → Q_RAIL.2	(1,12) → (5,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

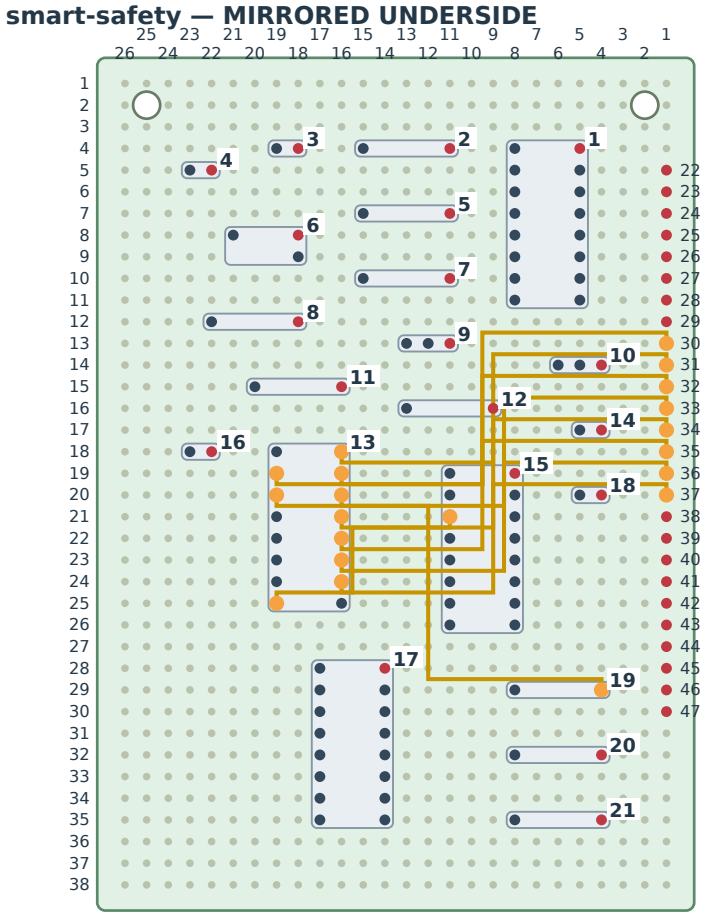
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W51 / EDU_ON	Q_RAIL.2 → R_RAIL_OFF.1	(5,14) → (9,16)
W52 / SR_SRCLK	J_LINK_4.1 → R_SR_SRCLK_OFF.1	(1,8) → (4,32)
W53 / SR_SRCLK	R_SR_SRCLK_OFF.1 → U_SR0.11	(4,32) → (19,23)
W54 / SR_SRCLK	U_SR0.11 → U_SR1.11	(19,23) → (11,24)
W55 / SR_SRCLK	U_SR1.11 → U_SR2.11	(11,24) → (17,33)
W56 / SR_RCLK	J_LINK_5.1 → R_SR_RCLK_OFF.1	(1,9) → (4,35)
W57 / SR_RCLK	R_SR_RCLK_OFF.1 → U_SR0.12	(4,35) → (19,22)
W58 / SR_RCLK	U_SR0.12 → U_SR1.12	(19,22) → (11,23)
W59 / SR_RCLK	U_SR1.12 → U_SR2.12	(11,23) → (17,32)
W60 / SR_SER	J_LINK_6.1 → R_SR_SER_OFF.1	(1,10) → (4,29)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

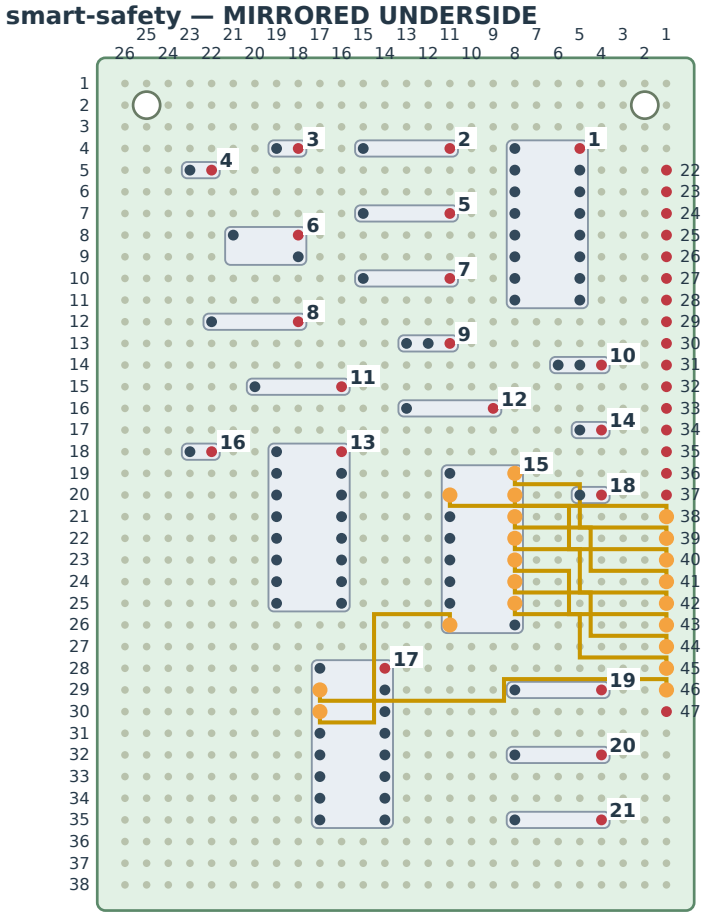
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W61 / SR_SER	R_SR_SER_OFF.1 → U_SR0.14	(4,29) → (19,20)
W62 / SR_CHAIN_1	U_SR0.9 → U_SR1.14	(19,25) → (11,21)
W63 / RELAY_0	J_LINK_9.1 → U_SR0.15	(1,13) → (19,19)
W64 / RELAY_1	J_LINK_10.1 → U_SR0.1	(1,14) → (16,18)
W65 / RELAY_2	J_LINK_11.1 → U_SR0.2	(1,15) → (16,19)
W66 / RELAY_3	J_LINK_12.1 → U_SR0.3	(1,16) → (16,20)
W67 / RELAY_4	J_LINK_13.1 → U_SR0.4	(1,17) → (16,21)
W68 / RELAY_5	J_LINK_14.1 → U_SR0.5	(1,18) → (16,22)
W69 / RELAY_6	J_LINK_15.1 → U_SR0.6	(1,19) → (16,23)
W70 / RELAY_7	J_LINK_16.1 → U_SR0.7	(1,20) → (16,24)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-safety

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

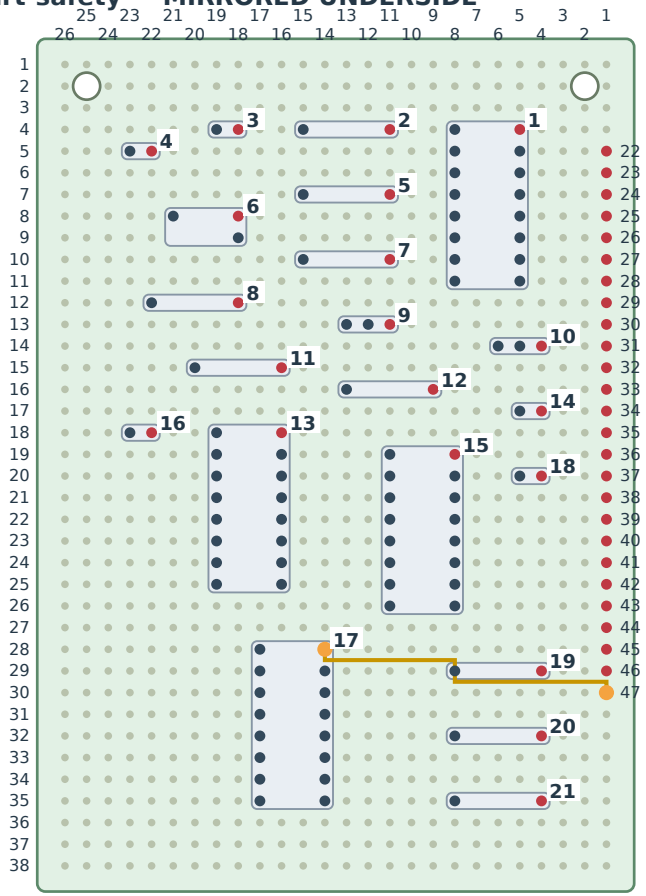
ID / net	From → to	TOP holes
W71 / SR_CHAIN_2	U_SR1.9 → U_SR2.14	(11,26) → (17,30)
W72 / RELAY_8	J_LINK_17.1 → U_SR1.15	(1,21) → (11,20)
W73 / RELAY_9	J_LINK_18.1 → U_SR1.1	(1,22) → (8,19)
W74 / RELAY_10	J_LINK_19.1 → U_SR1.2	(1,23) → (8,20)
W75 / RELAY_11	J_LINK_20.1 → U_SR1.3	(1,24) → (8,21)
W76 / RELAY_12	J_LINK_21.1 → U_SR1.4	(1,25) → (8,22)
W77 / RELAY_13	J_LINK_22.1 → U_SR1.5	(1,26) → (8,23)
W78 / RELAY_14	J_LINK_23.1 → U_SR1.6	(1,27) → (8,24)
W79 / RELAY_15	J_LINK_24.1 → U_SR1.7	(1,28) → (8,25)
W80 / RELAY_16	J_LINK_25.1 → U_SR2.15	(1,29) → (17,29)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — smart-safety

Install these wires after their endpoint components; inspect each net with power isolated.

smart-safety — MIRRORED UNDERSIDE



ID / net	From → to	TOP holes
W81 / RELAY_17	J_LINK_26.1 → U_SR2.1	(1,30) → (14,28)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

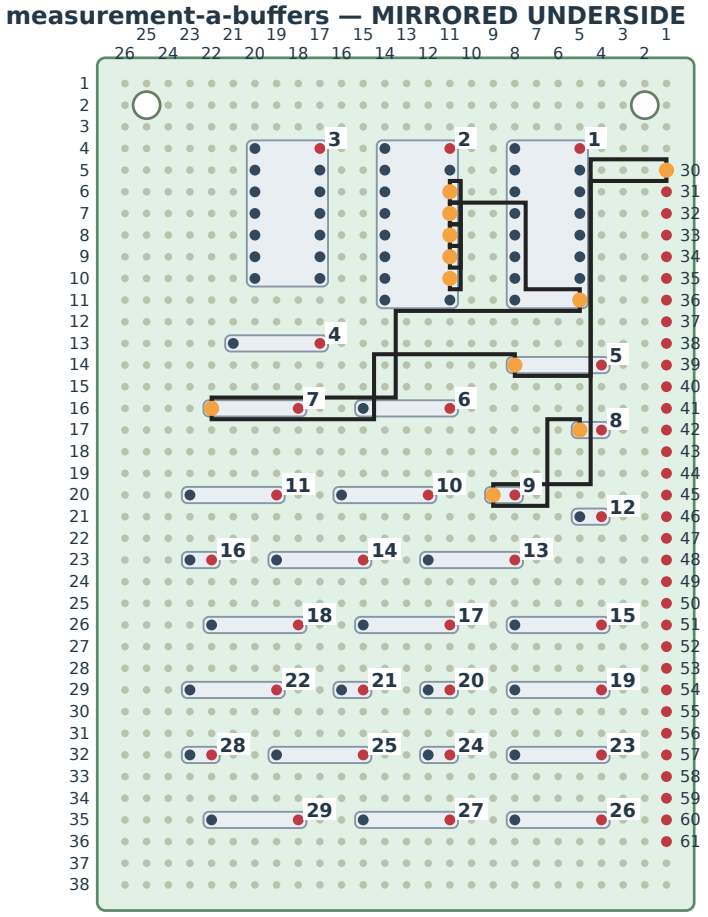
70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP15 — measurement-a-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

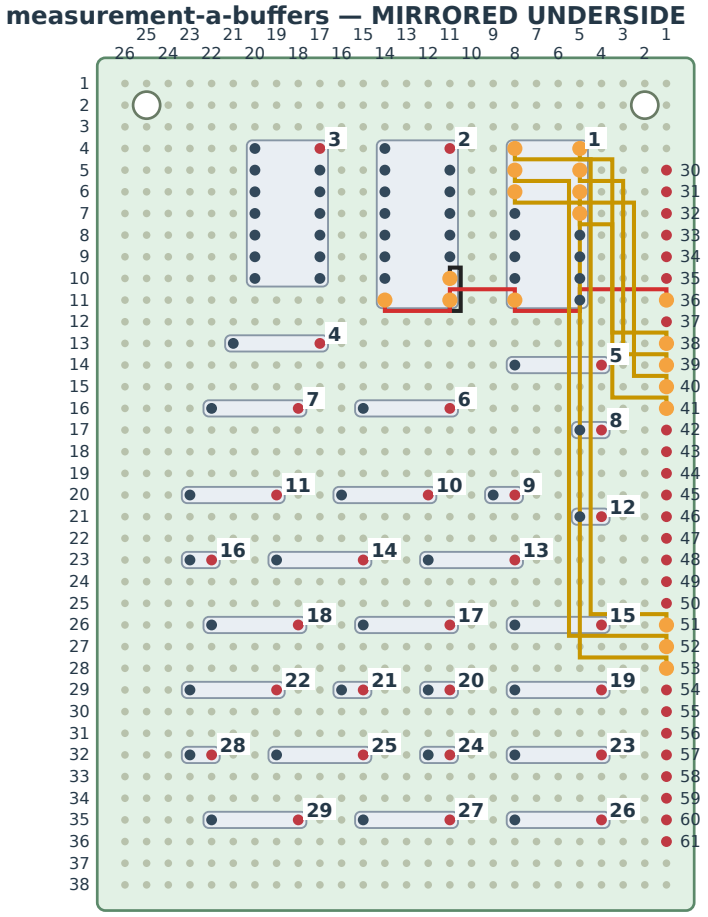
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / GND	C_A_REFHI.2 → C_A_REFLO.2	(5,17) → (9,20)
W2 / GND	C_A_REFLO.2 → J_LINK_1.1	(9,20) → (1,5)
W3 / GND	J_LINK_1.1 → R_A_HI_BOTTOM.2	(1,5) → (8,14)
W4 / GND	R_A_HI_BOTTOM.2 → R_A_LO_BOTTOM.2	(8,14) → (22,16)
W5 / GND	R_A_LO_BOTTOM.2 → U_A_LED_DRIVER_0.8	(22,16) → (5,11)
W6 / GND	U_A_LED_DRIVER_0.8 → U_A_LED_DRIVER_1.3	(5,11) → (11,6)
W7 / GND	U_A_LED_DRIVER_1.3 → U_A_LED_DRIVER_1.4	(11,6) → (11,7)
W8 / GND	U_A_LED_DRIVER_1.4 → U_A_LED_DRIVER_1.5	(11,7) → (11,8)
W9 / GND	U_A_LED_DRIVER_1.5 → U_A_LED_DRIVER_1.6	(11,8) → (11,9)
W10 / GND	U_A_LED_DRIVER_1.6 → U_A_LED_DRIVER_1.7	(11,9) → (11,10)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — measurement-a-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

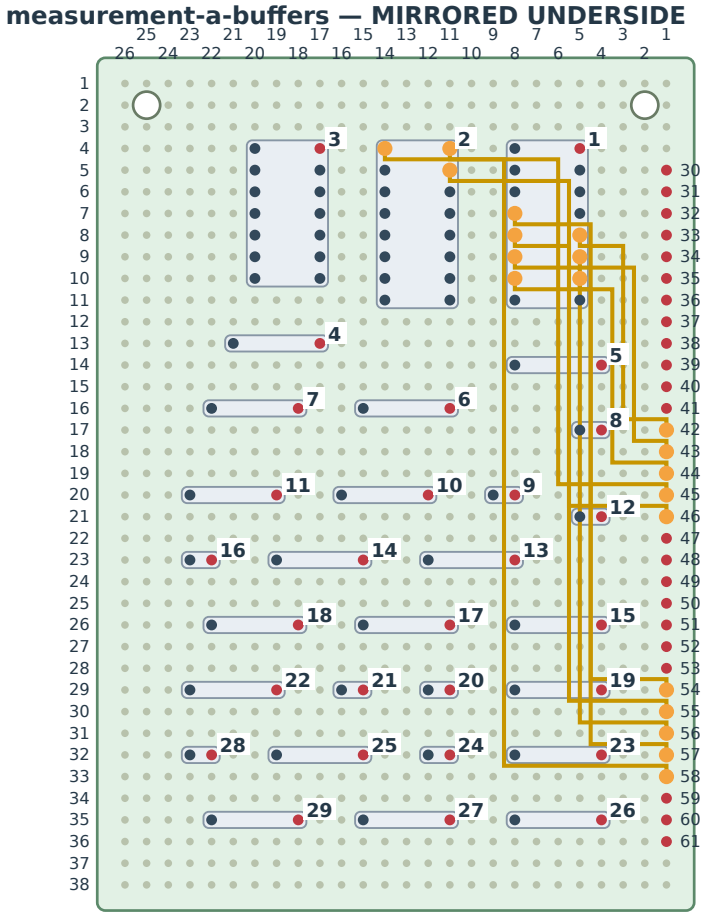
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	U_A_LED_DRIVER_1.7 → U_A_LED_DRIVER_1.8	(11,10) → (11,11)
W20 / 5V_SYS	J_LINK_7.1 → U_A_LED_DRIVER_0.9	(1,11) → (8,11)
W21 / 5V_SYS	U_A_LED_DRIVER_0.9 → U_A_LED_DRIVER_1.9	(8,11) → (14,11)
W22 / RELAY_0	J_LINK_9.1 → U_A_LED_DRIVER_0.1	(1,13) → (5,4)
W23 / P0_K_HIGH_CATHODE	J_LINK_22.1 → U_A_LED_DRIVER_0.16	(1,26) → (8,4)
W24 / RELAY_1	J_LINK_10.1 → U_A_LED_DRIVER_0.2	(1,14) → (5,5)
W25 / P0_K_LOW_CATHODE	J_LINK_23.1 → U_A_LED_DRIVER_0.15	(1,27) → (8,5)
W26 / RELAY_2	J_LINK_11.1 → U_A_LED_DRIVER_0.3	(1,15) → (5,6)
W27 / P0_AMM_CATHODE	J_LINK_24.1 → U_A_LED_DRIVER_0.14	(1,28) → (8,6)
W28 / RELAY_3	J_LINK_12.1 → U_A_LED_DRIVER_0.4	(1,16) → (5,7)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — measurement-a-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

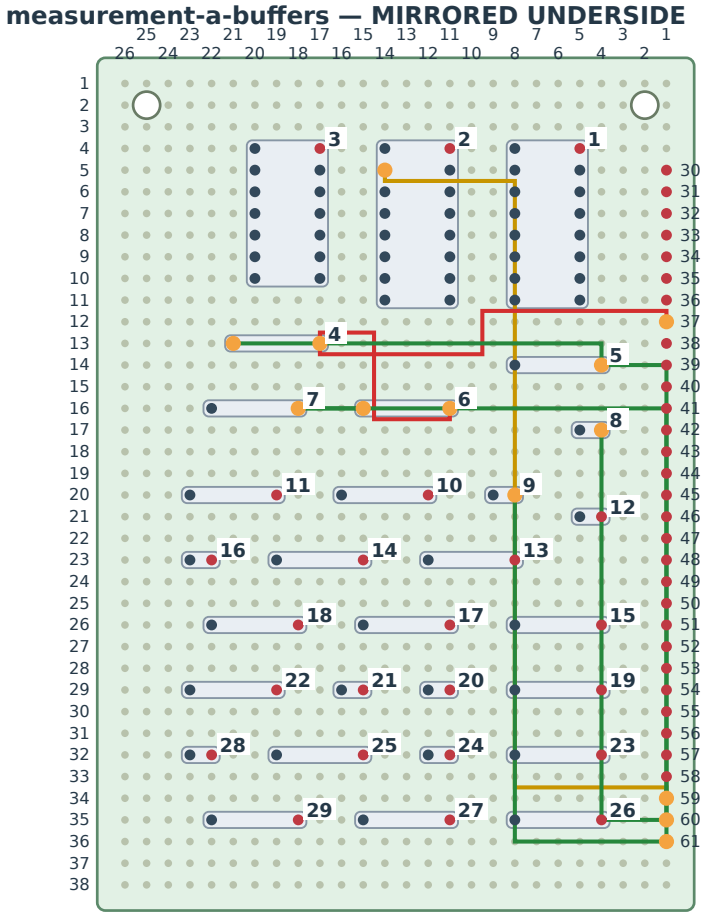
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W29 / P1_K_HIGH_CATHODE	J_LINK_25.1 → U_A_LED_DRIVER_0.13	(1,29) → (8,7)
W30 / RELAY_4	J_LINK_13.1 → U_A_LED_DRIVER_0.5	(1,17) → (5,8)
W31 / P1_K_LOW_CATHODE	J_LINK_26.1 → U_A_LED_DRIVER_0.12	(1,30) → (8,8)
W32 / RELAY_5	J_LINK_14.1 → U_A_LED_DRIVER_0.6	(1,18) → (5,9)
W33 / P1_AMM_CATHODE	J_LINK_27.1 → U_A_LED_DRIVER_0.11	(1,31) → (8,9)
W34 / RELAY_6	J_LINK_15.1 → U_A_LED_DRIVER_0.7	(1,19) → (5,10)
W35 / P2_K_HIGH_CATHODE	J_LINK_28.1 → U_A_LED_DRIVER_0.10	(1,32) → (8,10)
W36 / RELAY_7	J_LINK_16.1 → U_A_LED_DRIVER_1.1	(1,20) → (11,4)
W37 / P2_K_LOW_CATHODE	J_LINK_29.1 → U_A_LED_DRIVER_1.16	(1,33) → (14,4)
W38 / RELAY_8	J_LINK_17.1 → U_A_LED_DRIVER_1.2	(1,21) → (11,5)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — measurement-a-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

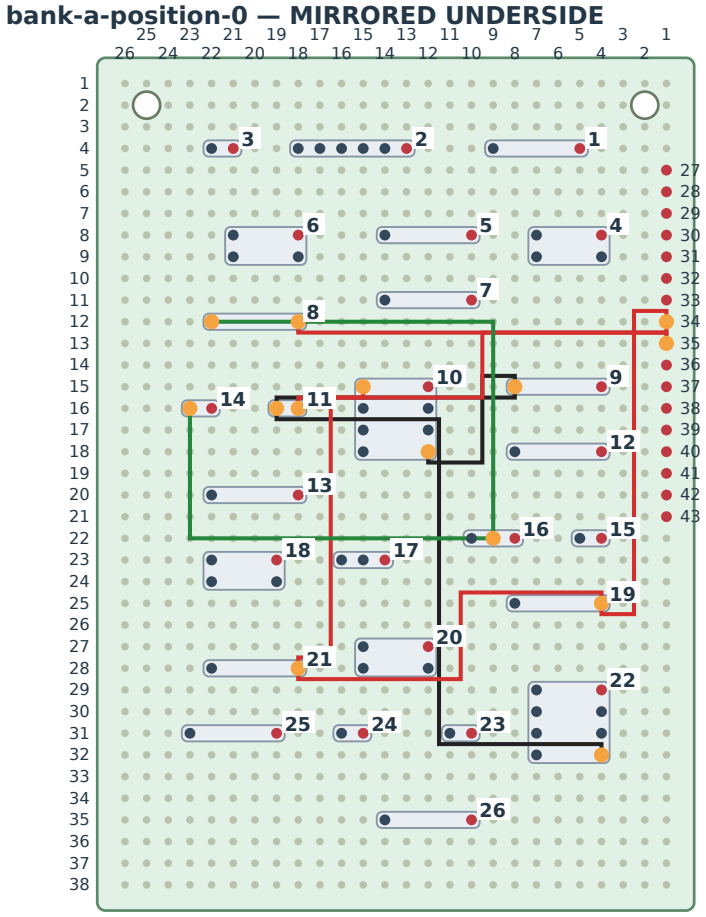
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W39 / P2_AMM_CATHODE	J_LINK_30.1 → U_A_LED_DRIVER_1.15	(1,34) → (14,5)
W63 / 5V_EDU	J_LINK_8.1 → R_A_HI_TOP.1	(1,12) → (17,13)
W64 / 5V_EDU	R_A_HI_TOP.1 → R_A_LO_TOP.1	(17,13) → (11,16)
W65 / A_REF_HI	C_A_REFHI.1 → J_LINK_31.1	(4,17) → (1,35)
W66 / A_REF_HI	J_LINK_31.1 → R_A_HI_BOTTOM.1	(1,35) → (4,14)
W67 / A_REF_HI	R_A_HI_BOTTOM.1 → R_A_HI_TOP.2	(4,14) → (21,13)
W68 / A_REF_LO	C_A_REFLO.1 → J_LINK_32.1	(8,20) → (1,36)
W69 / A_REF_LO	J_LINK_32.1 → R_A_LO_BOTTOM.1	(1,36) → (18,16)
W70 / A_REF_LO	R_A_LO_BOTTOM.1 → R_A_LO_TOP.2	(18,16) → (15,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — bank-a-position-0

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

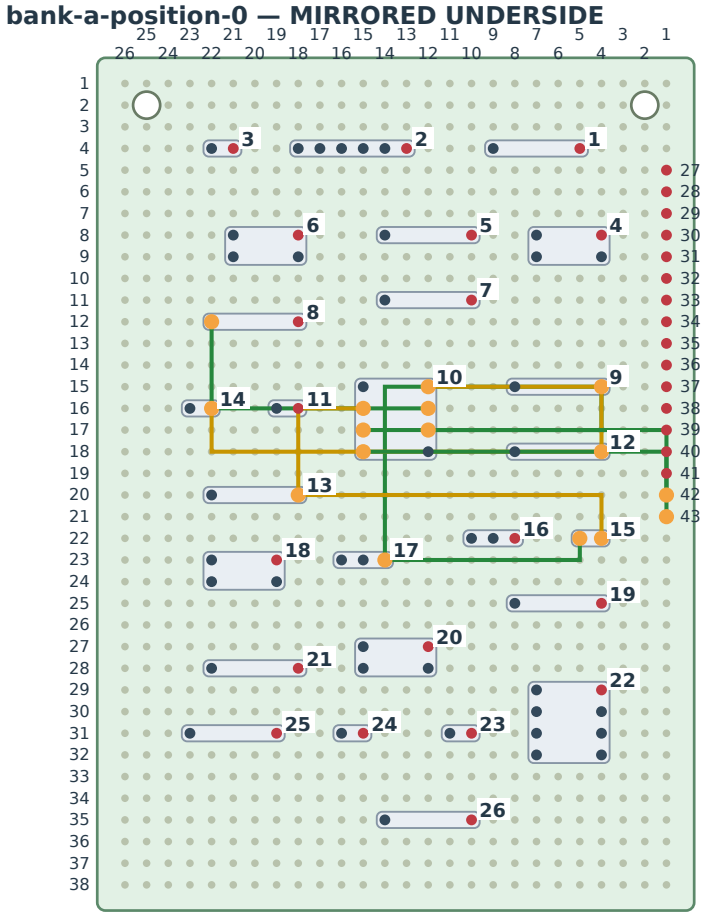
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W14 / GND	U_P0_BUS.4 → C_P0_LIMIT.2	(4,32) → (19,16)
W15 / GND	C_P0_LIMIT.2 → R_P0_SINK_SENSE.2	(19,16) → (8,15)
W16 / GND	R_P0_SINK_SENSE.2 → U_P0_LIMIT.4	(8,15) → (12,18)
W29 / 5V_SYS	C_P0_LIMIT.1 → J_LINK_8.1	(18,16) → (1,12)
W30 / 5V_SYS	J_LINK_8.1 → R_P0_K_HIGH_LED.1	(1,12) → (4,25)
W31 / 5V_SYS	R_P0_K_HIGH_LED.1 → R_P0_K_LOW_LED.1	(4,25) → (18,28)
W32 / 5V_SYS	R_P0_K_LOW_LED.1 → U_P0_LIMIT.8	(18,28) → (15,15)
W37 / 5V_EDU	J_LINK_9.1 → R_P0_SOURCE_SENSE.1	(1,13) → (18,12)
W38 / P0_SOURCE_SENSE	C_P0_HI_COMP.2 → Q_P0_HIGH.2	(23,16) → (9,22)
W39 / P0_SOURCE_SENSE	Q_P0_HIGH.2 → R_P0_SOURCE_SENSE.2	(9,22) → (22,12)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — bank-a-position-0

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

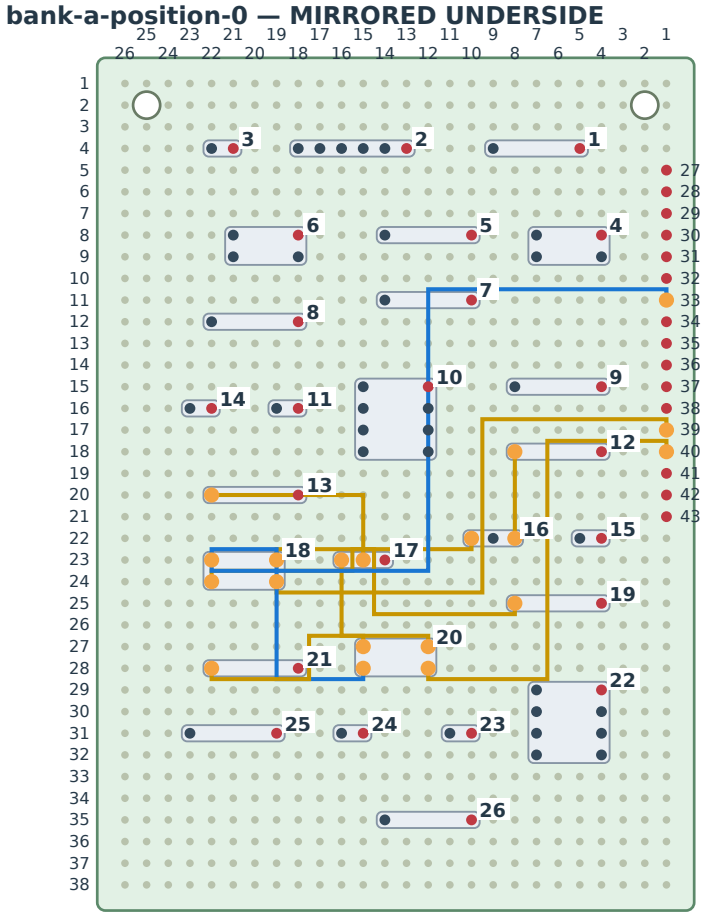
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W40 / P0_SOURCE_SENSE	R_P0_SOURCE_SENSE.2 → U_P0_LIMIT.2	(22,12) → (12,16)
W41 / P0_SINK_SENSE	C_P0_LO_COMP.2 → Q_P0_LOW.1	(5,22) → (14,23)
W42 / P0_SINK_SENSE	Q_P0_LOW.1 → R_P0_SINK_SENSE.1	(14,23) → (4,15)
W43 / P0_SINK_SENSE	R_P0_SINK_SENSE.1 → U_P0_LIMIT.6	(4,15) → (15,17)
W44 / P0_HI_GATE_DRIVE	C_P0_HI_COMP.1 → R_P0_HI_GATE.1	(22,16) → (4,18)
W45 / P0_HI_GATE_DRIVE	R_P0_HI_GATE.1 → U_P0_LIMIT.1	(4,18) → (12,15)
W46 / A_REF_HI	J_LINK_16.1 → U_P0_LIMIT.3	(1,20) → (12,17)
W47 / A_REF_LO	J_LINK_17.1 → U_P0_LIMIT.5	(1,21) → (15,18)
W48 / P0_LO_GATE_DRIVE	C_P0_LO_COMP.1 → R_P0_LO_GATE.1	(4,22) → (18,20)
W49 / P0_LO_GATE_DRIVE	R_P0_LO_GATE.1 → U_P0_LIMIT.7	(18,20) → (15,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP15 — bank-a-position-0

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

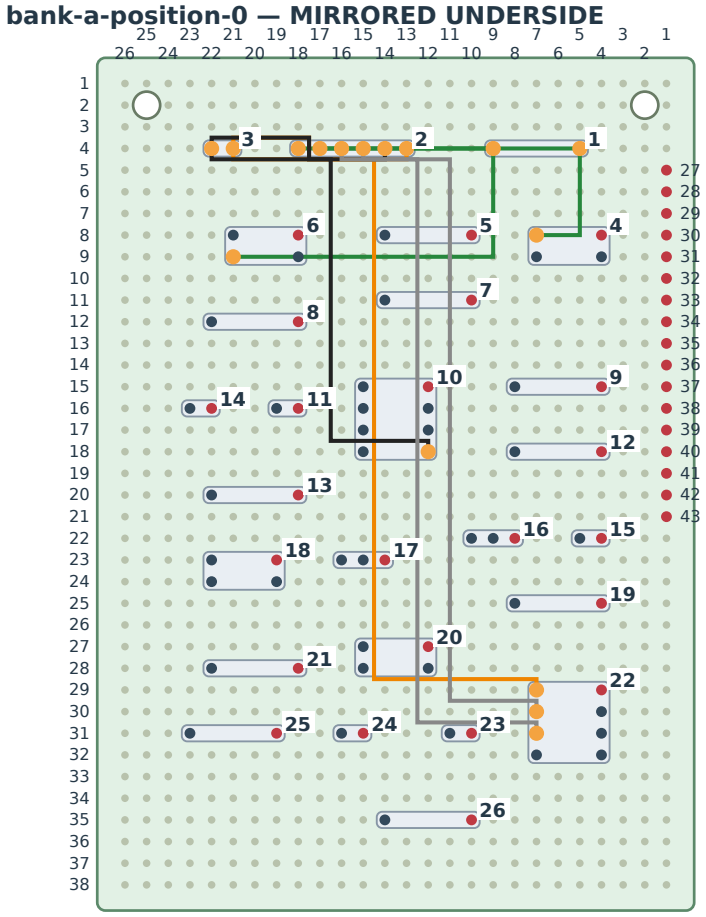
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W50 / P0_HI_GATE	Q_P0_HIGH.1 → R_P0_HI_GATE.2	(8,22) → (8,18)
W51 / P0_LO_GATE	Q_P0_LOW.2 → R_P0_LO_GATE.2	(15,23) → (22,20)
W52 / P0_SOURCE_OUT	K_P0_HIGH.3 → Q_P0_HIGH.3	(22,24) → (10,22)
W53 / P0_SINK_IN	K_P0_LOW.4 → Q_P0_LOW.3	(15,27) → (16,23)
W54 / P0_K_HIGH_LED	K_P0_HIGH.1 → R_P0_K_HIGH_LED.2	(19,23) → (8,25)
W55 / P0_K_HIGH_CATHODE	J_LINK_13.1 → K_P0_HIGH.2	(1,17) → (19,24)
W56 / NODE02	J_LINK_7.1 → K_P0_HIGH.4	(1,11) → (22,23)
W57 / NODE02	K_P0_HIGH.4 → K_P0_LOW.3	(22,23) → (15,28)
W58 / P0_K_LOW_LED	K_P0_LOW.1 → R_P0_K_LOW_LED.2	(12,27) → (22,28)
W59 / P0_K_LOW_CATHODE	J_LINK_14.1 → K_P0_LOW.2	(1,18) → (12,28)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP22 — bank-a-position-0

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

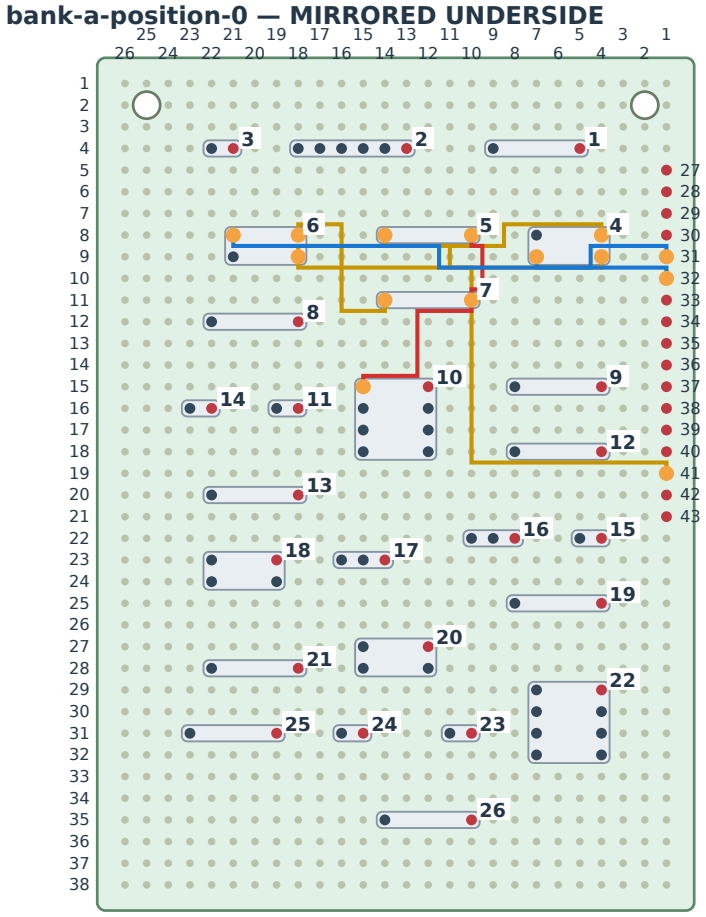
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / P0_SHUNT_P	R_P0_SHUNT.1 → J_P0_CURRENT_MODULE.5	(5,4) → (17,4)
W2 / P0_SHUNT_P	R_P0_SHUNT.1 → K_P0_AMM_P.4	(5,4) → (7,8)
W3 / P0_SHUNT_N	R_P0_SHUNT.2 → J_P0_CURRENT_MODULE.6	(9,4) → (18,4)
W4 / P0_SHUNT_N	R_P0_SHUNT.2 → K_P0_AMM_N.3	(9,4) → (21,9)
W9 / 3V3_SYS	U_P0_BUS.8 → C_P0_CURRENT.1	(7,29) → (21,4)
W10 / 3V3_SYS	C_P0_CURRENT.1 → J_P0_CURRENT_MODULE.1	(21,4) → (13,4)
W17 / GND	U_P0_LIMIT.4 → C_P0_CURRENT.2	(12,18) → (22,4)
W18 / GND	C_P0_CURRENT.2 → J_P0_CURRENT_MODULE.2	(22,4) → (14,4)
W21 / SCL_POSITION_0	U_P0_BUS.7 → J_P0_CURRENT_MODULE.3	(7,30) → (15,4)
W24 / SDA_POSITION_0	U_P0_BUS.6 → J_P0_CURRENT_MODULE.4	(7,31) → (16,4)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP22 — bank-a-position-0

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

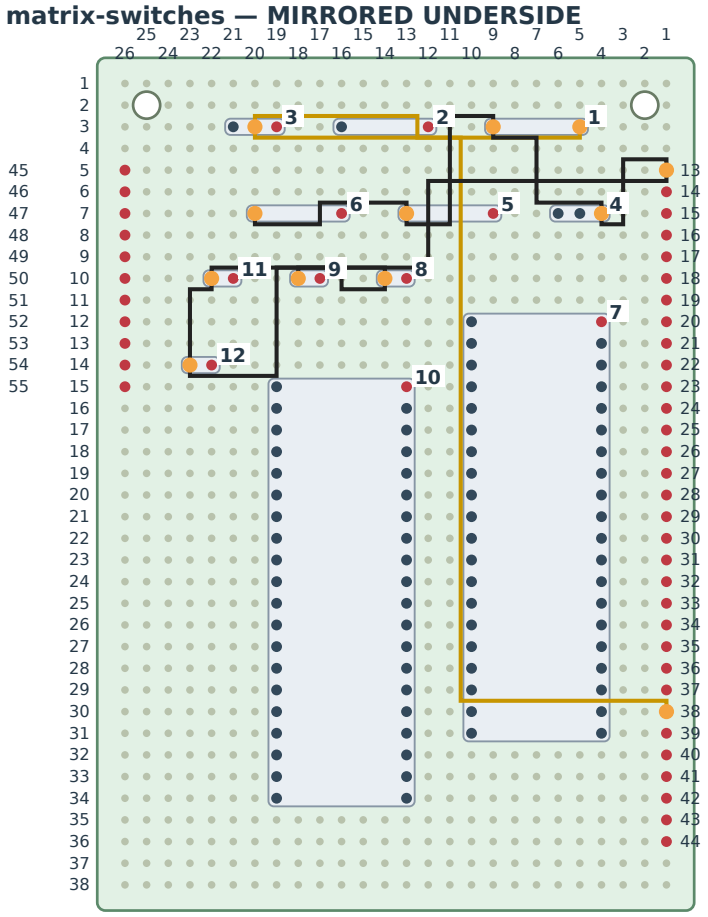
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W25 / P0_K_AMM_P_LED	K_P0_AMM_P.1 → R_P0_K_AMM_P_LED.2	(4,8) → (14,8)
W26 / P0_AMM_CATHODE	J_LINK_15.1 → K_P0_AMM_N.2	(1,19) → (18,9)
W27 / P0_AMM_CATHODE	K_P0_AMM_N.2 → K_P0_AMM_P.2	(18,9) → (4,9)
W28 / NODE00	J_LINK_5.1 → K_P0_AMM_P.3	(1,9) → (7,9)
W33 / 5V_SYS	U_P0_LIMIT.8 → R_P0_K_AMM_N_LED.1	(15,15) → (10,11)
W34 / 5V_SYS	R_P0_K_AMM_N_LED.1 → R_P0_K_AMM_P_LED.1	(10,11) → (10,8)
W35 / P0_K_AMM_N_LED	K_P0_AMM_N.1 → R_P0_K_AMM_N_LED.2	(18,8) → (14,11)
W36 / NODE01	J_LINK_6.1 → K_P0_AMM_N.4	(1,10) → (21,8)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP24 — matrix-switches

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

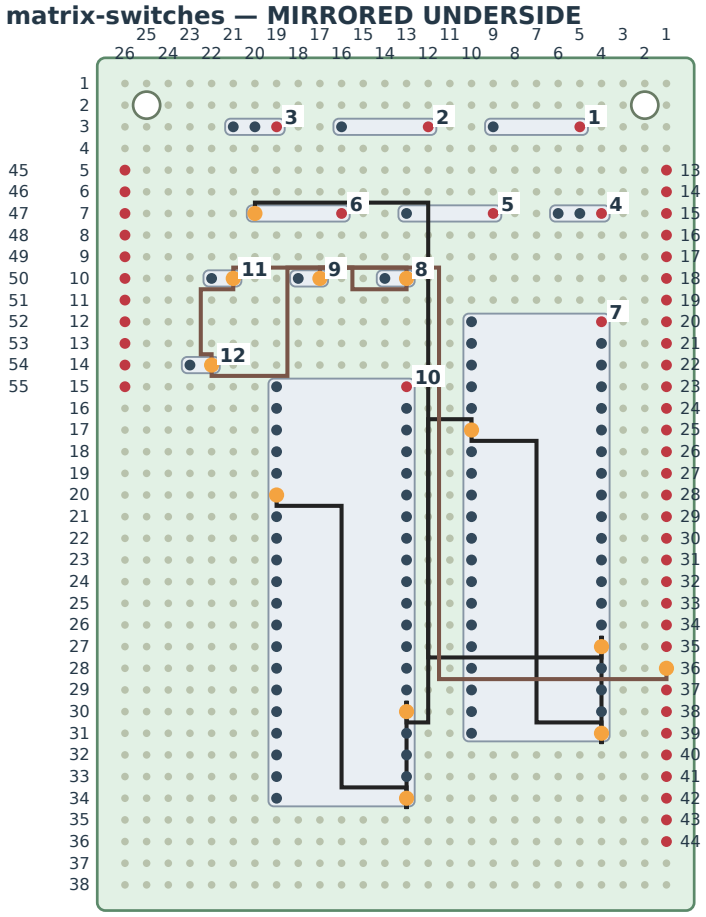
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / RST_RELEASE	J_LINK_26.1 → Q_RESET_RELEASE.2	(1,30) → (20,3)
W2 / RST_RELEASE	Q_RESET_RELEASE.2 → R_RST_RELEASE_OFF.1	(20,3) → (5,3)
W3 / GND	C_MT0_BULK.2 → C_MT0.2	(18,10) → (14,10)
W4 / GND	C_MT0.2 → C_MT1_BULK.2	(14,10) → (23,14)
W5 / GND	C_MT1_BULK.2 → C_MT1.2	(23,14) → (22,10)
W6 / GND	C_MT1.2 → J_LINK_1.1	(22,10) → (1,5)
W7 / GND	J_LINK_1.1 → Q_RESET_ALIVE.1	(1,5) → (4,7)
W8 / GND	Q_RESET_ALIVE.1 → R_RST_RELEASE_OFF.2	(4,7) → (9,3)
W9 / GND	R_RST_RELEASE_OFF.2 → R_STBA_OFF.2	(9,3) → (13,7)
W10 / GND	R_STBA_OFF.2 → R_STBB_OFF.2	(13,7) → (20,7)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP24 — matrix-switches

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

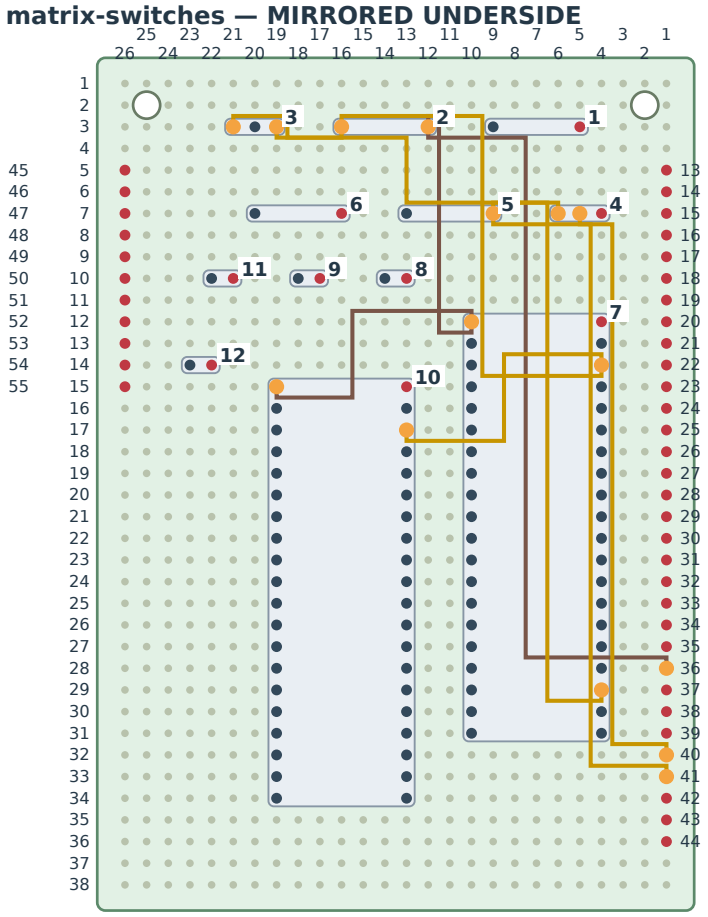
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	R_STBB_OFF.2 → U_MT_A.16	(20,7) → (4,27)
W12 / GND	U_MT_A.16 → U_MT_A.20	(4,27) → (4,31)
W13 / GND	U_MT_A.20 → U_MT_A.35	(4,31) → (10,17)
W14 / GND	U_MT_A.35 → U_MT_B.16	(10,17) → (13,30)
W15 / GND	U_MT_B.16 → U_MT_B.20	(13,30) → (13,34)
W16 / GND	U_MT_B.20 → U_MT_B.35	(13,34) → (19,20)
W17 / 12V_SYS	C_MT0_BULK.1 → C_MT0.1	(17,10) → (13,10)
W18 / 12V_SYS	C_MT0.1 → C_MT1_BULK.1	(13,10) → (22,14)
W19 / 12V_SYS	C_MT1_BULK.1 → C_MT1.1	(22,14) → (21,10)
W20 / 12V_SYS	C_MT1.1 → J_LINK_24.1	(21,10) → (1,28)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP24 — matrix-switches

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

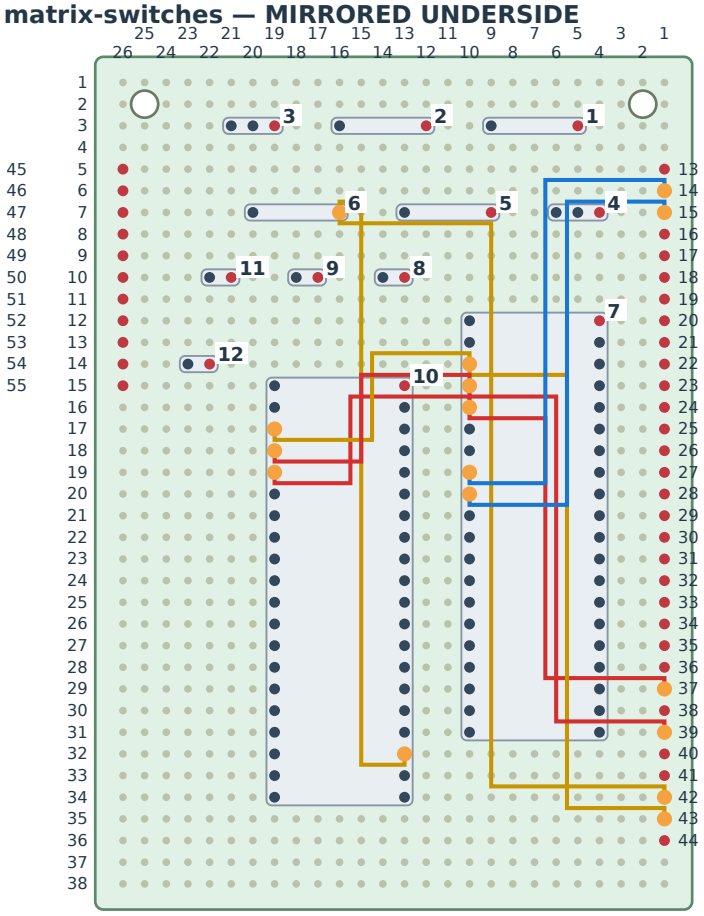
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / 12V_SYS	J_LINK_24.1 → R_RESET_ON.1	(1,28) → (12,3)
W22 / 12V_SYS	R_RESET_ON.1 → U_MT_A.40	(12,3) → (10,12)
W23 / 12V_SYS	U_MT_A.40 → U_MT_B.40	(10,12) → (19,15)
W24 / XBAR_RESET	Q_RESET_RELEASE.3 → R_RESET_ON.2	(21,3) → (16,3)
W25 / XBAR_RESET	R_RESET_ON.2 → U_MT_A.3	(16,3) → (4,14)
W26 / XBAR_RESET	U_MT_A.3 → U_MT_B.3	(4,14) → (13,17)
W27 / RESET_MID	Q_RESET_ALIVE.3 → Q_RESET_RELEASE.1	(6,7) → (19,3)
W28 / ROUTER_ALIVE	J_LINK_28.1 → Q_RESET_ALIVE.2	(1,32) → (5,7)
W29 / STROBE_A	J_LINK_29.1 → R_STBA_OFF.1	(1,33) → (9,7)
W30 / STROBE_A	R_STBA_OFF.1 → U_MT_A.18	(9,7) → (4,29)

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CP24 — matrix-switches

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

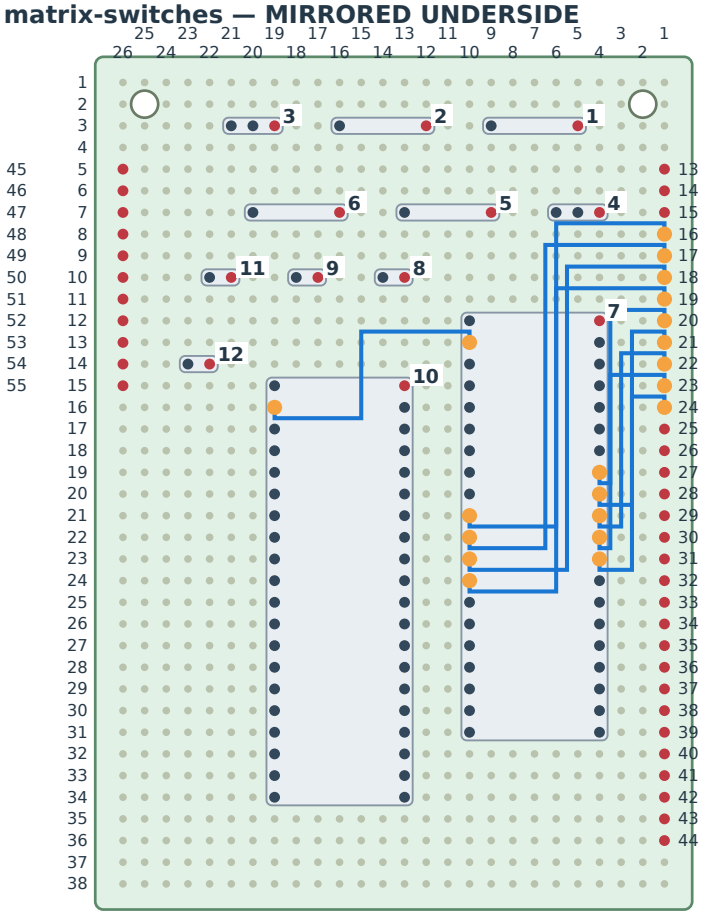
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / STROBE_B	J_LINK_30.1 → R_STBB_OFF.1	(1,34) → (16,7)
W32 / STROBE_B	R_STBB_OFF.1 → U_MT_B.18	(16,7) → (13,32)
W33 / 5V_SYS	J_LINK_25.1 → U_MT_A.36	(1,29) → (10,16)
W34 / 5V_SYS	U_MT_A.36 → U_MT_B.36	(10,16) → (19,19)
W35 / XBAR_DATA	J_LINK_31.1 → U_MT_A.38	(1,35) → (10,14)
W36 / XBAR_DATA	U_MT_A.38 → U_MT_B.38	(10,14) → (19,17)
W37 / 5V_EDU	J_LINK_27.1 → U_MT_A.37	(1,31) → (10,15)
W38 / 5V_EDU	U_MT_A.37 → U_MT_B.37	(10,15) → (19,18)
W39 / NODE00	J_LINK_2.1 → U_MT_A.33	(1,6) → (10,19)
W40 / NODE01	J_LINK_3.1 → U_MT_A.32	(1,7) → (10,20)

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CP24 — matrix-switches

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

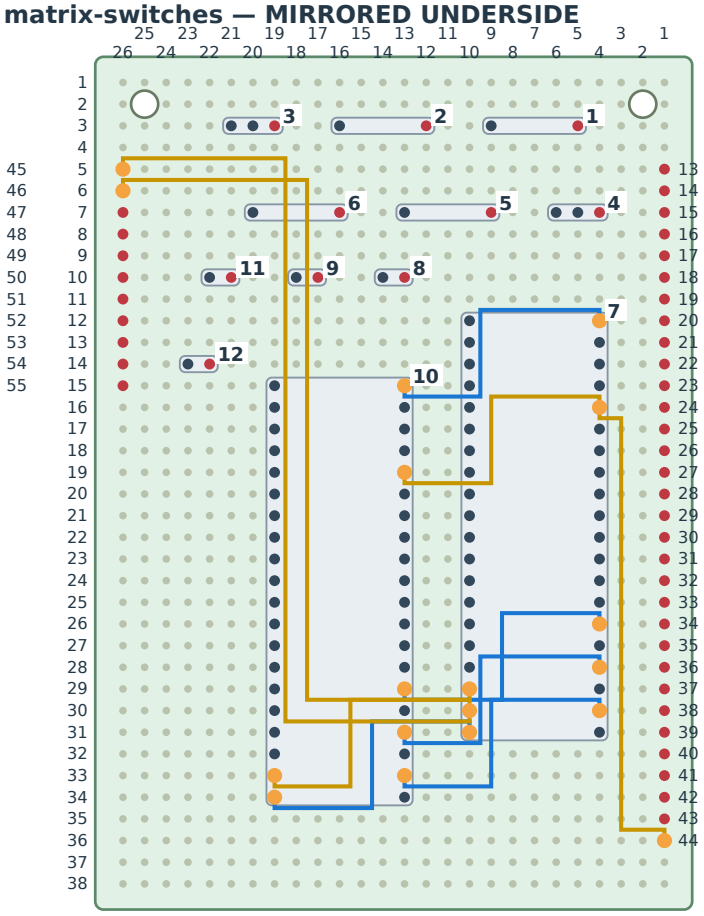
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / NODE02	J_LINK_4.1 → U_MT_A.31	(1,8) → (10,21)
W42 / NODE03	J_LINK_5.1 → U_MT_A.30	(1,9) → (10,22)
W43 / NODE04	J_LINK_6.1 → U_MT_A.29	(1,10) → (10,23)
W44 / NODE05	J_LINK_7.1 → U_MT_A.28	(1,11) → (10,24)
W45 / NODE06	J_LINK_8.1 → U_MT_A.8	(1,12) → (4,19)
W46 / NODE07	J_LINK_9.1 → U_MT_A.9	(1,13) → (4,20)
W47 / NODE08	J_LINK_10.1 → U_MT_A.10	(1,14) → (4,21)
W48 / NODE09	J_LINK_11.1 → U_MT_A.11	(1,15) → (4,22)
W49 / NODE10	J_LINK_12.1 → U_MT_A.12	(1,16) → (4,23)
W50 / BUS2	U_MT_A.39 → U_MT_B.39	(10,13) → (19,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP24 — matrix-switches

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

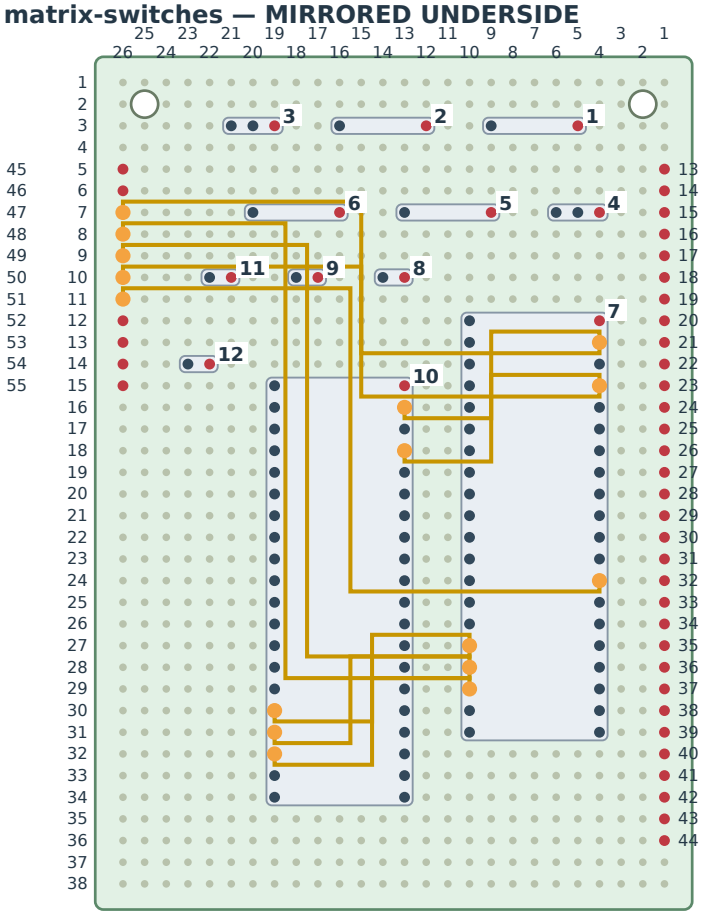
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W51 / BUS3	U_MT_A.1 → U_MT_B.1	(4,12) → (13,15)
W52 / BUS4	U_MT_A.21 → U_MT_B.21	(10,31) → (19,34)
W53 / BUS5	U_MT_A.19 → U_MT_B.19	(4,30) → (13,33)
W54 / BUS6	U_MT_A.17 → U_MT_B.17	(4,28) → (13,31)
W55 / BUS7	U_MT_A.15 → U_MT_B.15	(4,26) → (13,29)
W56 / AX0	J_LINK_32.1 → U_MT_A.5	(1,36) → (4,16)
W57 / AX0	U_MT_A.5 → U_MT_B.5	(4,16) → (13,19)
W58 / AX1	J_LINK_33.1 → U_MT_A.22	(26,5) → (10,30)
W59 / AX1	U_MT_A.22 → U_MT_B.22	(10,30) → (19,33)
W60 / AX2	J_LINK_34.1 → U_MT_A.23	(26,6) → (10,29)

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CP24 — matrix-switches

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

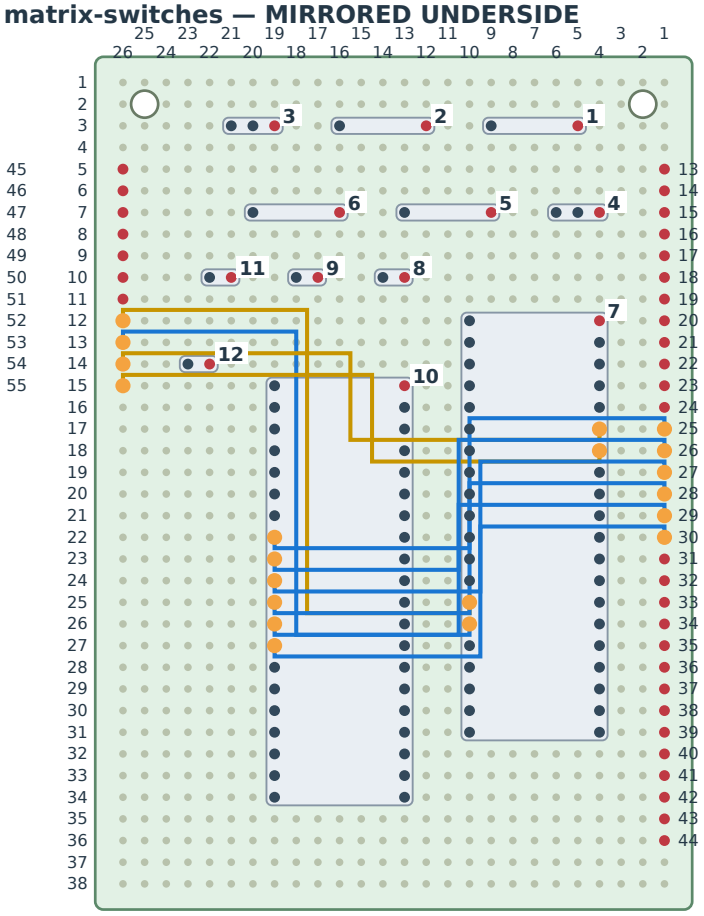
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W61 / AX2	U_MT_A.23 → U_MT_B.23	(10,29) → (19,32)
W62 / AX3	J_LINK_35.1 → U_MT_A.4	(26,7) → (4,15)
W63 / AX3	U_MT_A.4 → U_MT_B.4	(4,15) → (13,18)
W64 / AY0	J_LINK_36.1 → U_MT_A.24	(26,8) → (10,28)
W65 / AY0	U_MT_A.24 → U_MT_B.24	(10,28) → (19,31)
W66 / AY1	J_LINK_37.1 → U_MT_A.25	(26,9) → (10,27)
W67 / AY1	U_MT_A.25 → U_MT_B.25	(10,27) → (19,30)
W68 / AY2	J_LINK_38.1 → U_MT_A.2	(26,10) → (4,13)
W69 / AY2	U_MT_A.2 → U_MT_B.2	(4,13) → (13,16)
W70 / SOURCE_HI	J_LINK_39.1 → U_MT_A.13	(26,11) → (4,24)

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CP24 — matrix-switches

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

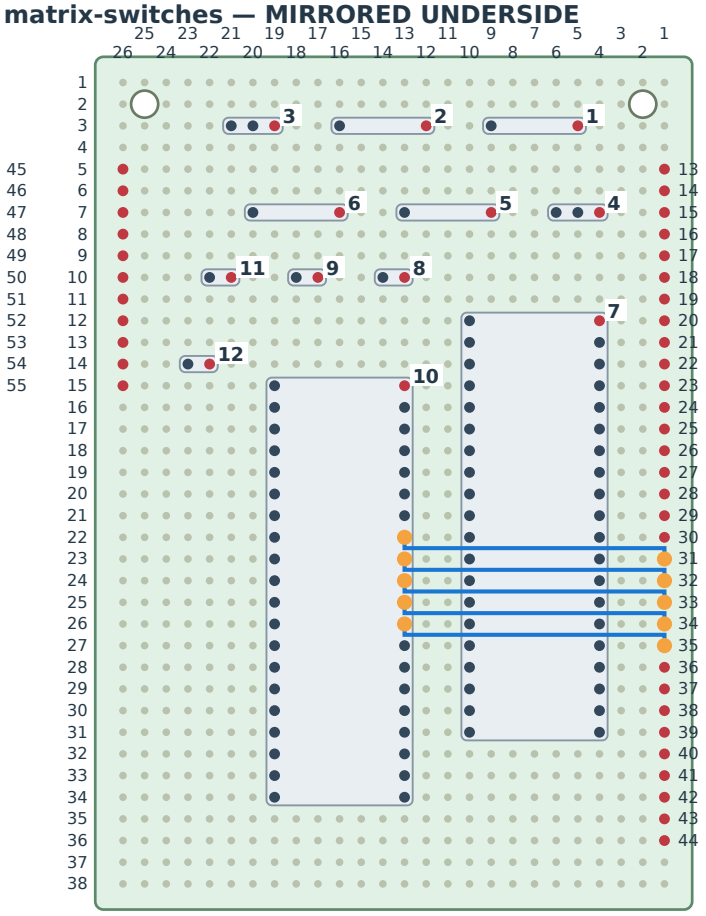
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W71 / SOURCE_LO	J_LINK_40.1 → U_MT_A.27	(26,12) → (10,25)
W72 / DIAG_NODE	J_LINK_41.1 → U_MT_A.26	(26,13) → (10,26)
W73 / DISCHARGE	J_LINK_42.1 → U_MT_A.6	(26,14) → (4,17)
W74 / CAL_LOAD	J_LINK_43.1 → U_MT_A.7	(26,15) → (4,18)
W75 / NODE11	J_LINK_13.1 → U_MT_B.33	(1,17) → (19,22)
W76 / NODE12	J_LINK_14.1 → U_MT_B.32	(1,18) → (19,23)
W77 / NODE13	J_LINK_15.1 → U_MT_B.31	(1,19) → (19,24)
W78 / NODE14	J_LINK_16.1 → U_MT_B.30	(1,20) → (19,25)
W79 / NODE15	J_LINK_17.1 → U_MT_B.29	(1,21) → (19,26)
W80 / NODE16	J_LINK_18.1 → U_MT_B.28	(1,22) → (19,27)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP24 — matrix-switches

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

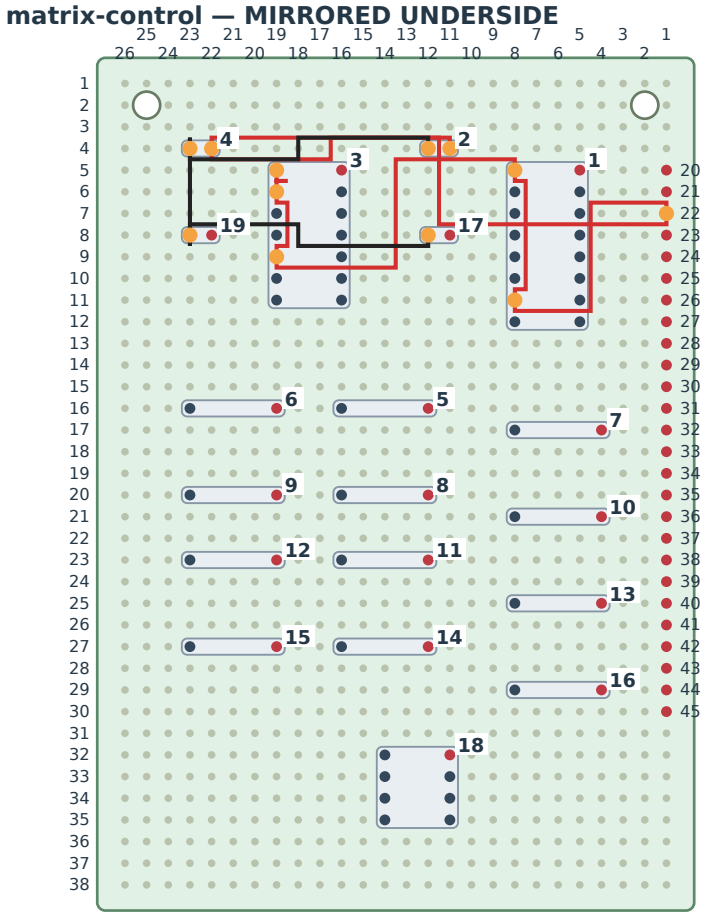
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W81 / NODE17	J_LINK_19.1 → U_MT_B.8	(1,23) → (13,22)
W82 / NODE18	J_LINK_20.1 → U_MT_B.9	(1,24) → (13,23)
W83 / NODE19	J_LINK_21.1 → U_MT_B.10	(1,25) → (13,24)
W84 / NODE20	J_LINK_22.1 → U_MT_B.11	(1,26) → (13,25)
W85 / NODE21	J_LINK_23.1 → U_MT_B.12	(1,27) → (13,26)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP24 — matrix-control

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

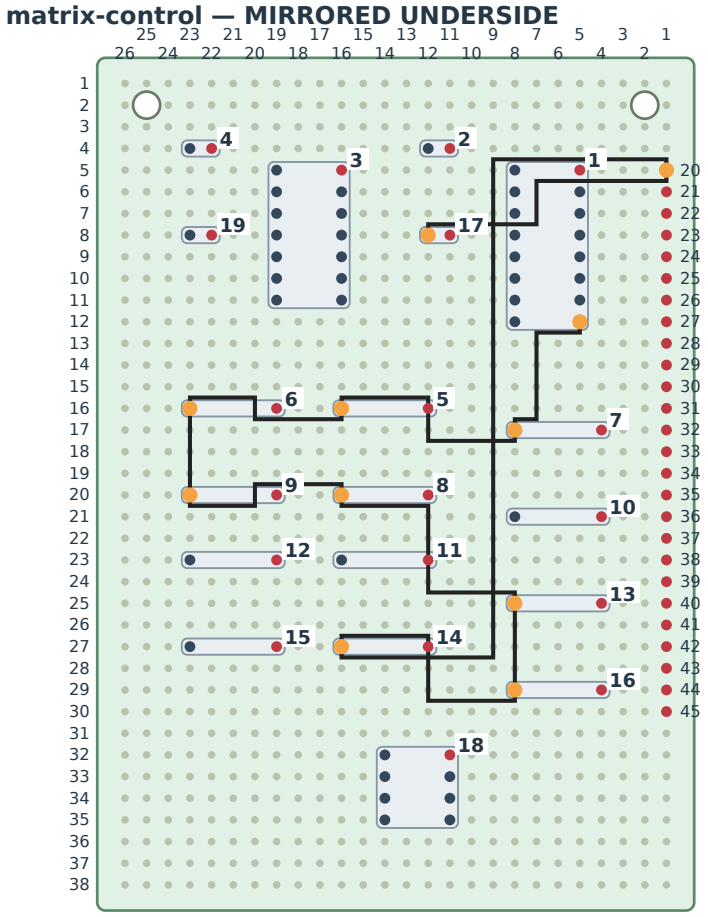
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / 5V_SYS	C_ADDR.1 → C_BUFFER.1	(11,4) → (22,4)
W2 / 5V_SYS	C_BUFFER.1 → J_LINK_3.1	(22,4) → (1,7)
W3 / 5V_SYS	J_LINK_3.1 → U_ADDR.10	(1,7) → (8,11)
W4 / 5V_SYS	U_ADDR.10 → U_ADDR.16	(8,11) → (8,5)
W5 / 5V_SYS	U_ADDR.16 → U_BUFFER.10	(8,5) → (19,9)
W6 / 5V_SYS	U_BUFFER.10 → U_BUFFER.13	(19,9) → (19,6)
W7 / 5V_SYS	U_BUFFER.13 → U_BUFFER.14	(19,6) → (19,5)
W8 / GND	C_ADDR.2 → C_BUFFER.2	(12,4) → (23,4)
W9 / GND	C_BUFFER.2 → C_DIAG_VCC.2	(23,4) → (23,8)
W10 / GND	C_DIAG_VCC.2 → C_DIAG.2	(23,8) → (12,8)

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CP24 — matrix-control

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

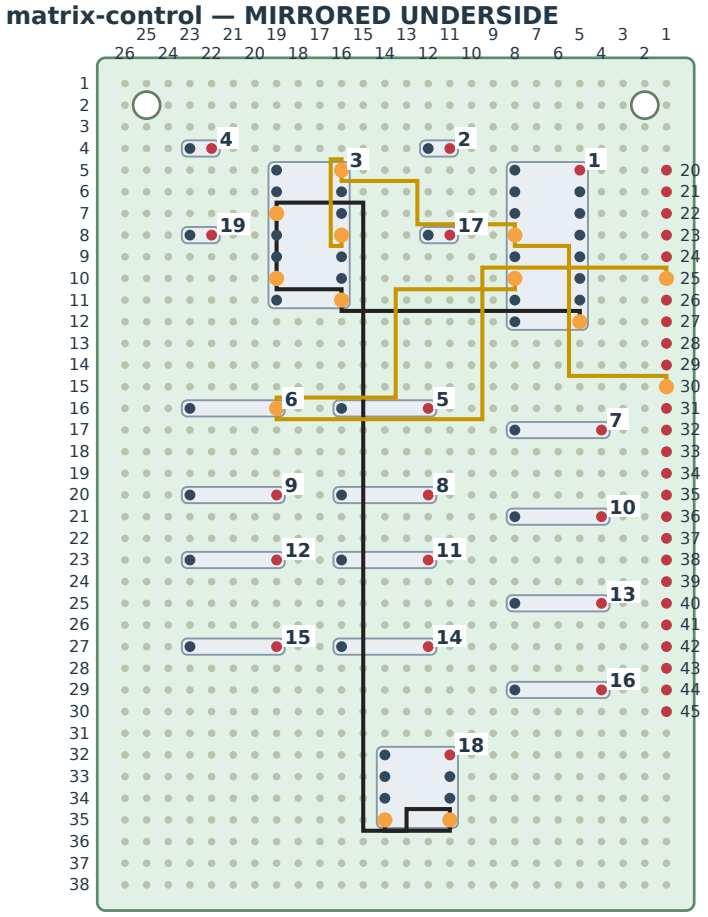
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	C_DIAG.2 → J_LINK_1.1	(12,8) → (1,5)
W12 / GND	J_LINK_1.1 → R_CAL.2	(1,5) → (16,27)
W13 / GND	R_CAL.2 → R_DIAG_BOTTOM.2	(16,27) → (8,29)
W14 / GND	R_DIAG_BOTTOM.2 → R_DISCHARGE.2	(8,29) → (8,25)
W15 / GND	R_DISCHARGE.2 → R_STB_A_IN_OFF.2	(8,25) → (16,20)
W16 / GND	R_STB_A_IN_OFF.2 → R_STB_B_IN_OFF.2	(16,20) → (23,20)
W17 / GND	R_STB_B_IN_OFF.2 → R_XBAR_CLK_OFF.2	(23,20) → (23,16)
W18 / GND	R_XBAR_CLK_OFF.2 → R_XBAR_DAT_OFF.2	(23,16) → (16,16)
W19 / GND	R_XBAR_DAT_OFF.2 → R_XBAR_LATCH_OFF.2	(16,16) → (8,17)
W20 / GND	R_XBAR_LATCH_OFF.2 → U_ADDR.8	(8,17) → (5,12)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP24 — matrix-control

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

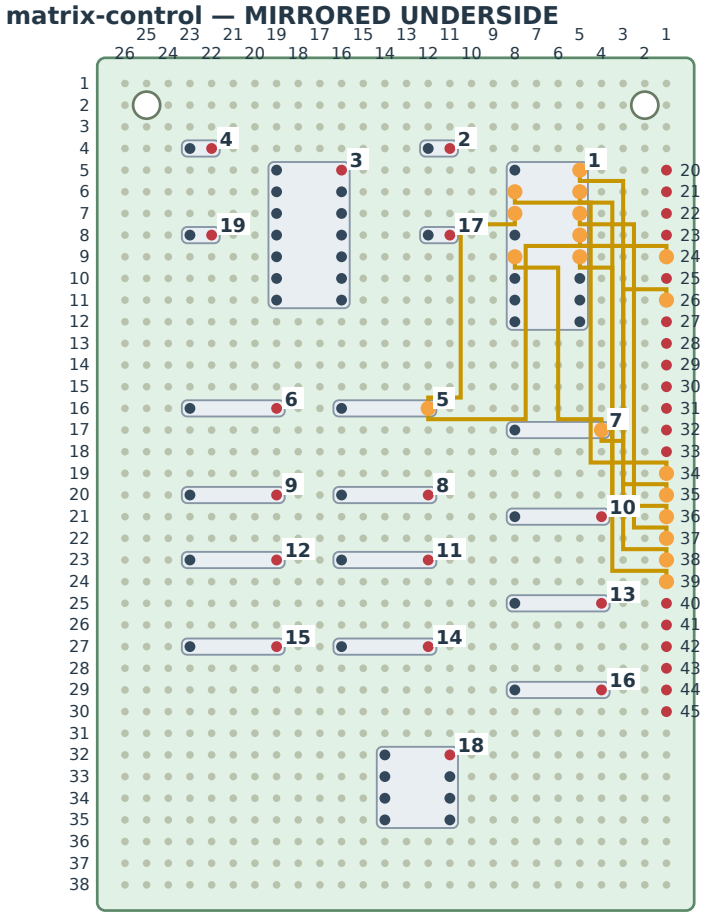
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / GND	U_ADDR.8 → U_BUFFER.7	(5,12) → (16,11)
W22 / GND	U_BUFFER.7 → U_BUFFER.9	(16,11) → (19,10)
W23 / GND	U_BUFFER.9 → U_BUFFER.12	(19,10) → (19,7)
W24 / GND	U_BUFFER.12 → U_DIAG.4	(19,7) → (11,35)
W25 / GND	U_DIAG.4 → U_DIAG.5	(11,35) → (14,35)
W26 / XBAR_OE_N	J_LINK_11.1 → U_ADDR.13	(1,15) → (8,8)
W27 / XBAR_OE_N	U_ADDR.13 → U_BUFFER.1	(8,8) → (16,5)
W28 / XBAR_OE_N	U_BUFFER.1 → U_BUFFER.4	(16,5) → (16,8)
W29 / XBAR_CLK	J_LINK_6.1 → R_XBAR_CLK_OFF.1	(1,10) → (19,16)
W30 / XBAR_CLK	R_XBAR_CLK_OFF.1 → U_ADDR.11	(19,16) → (8,10)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP24 — matrix-control

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

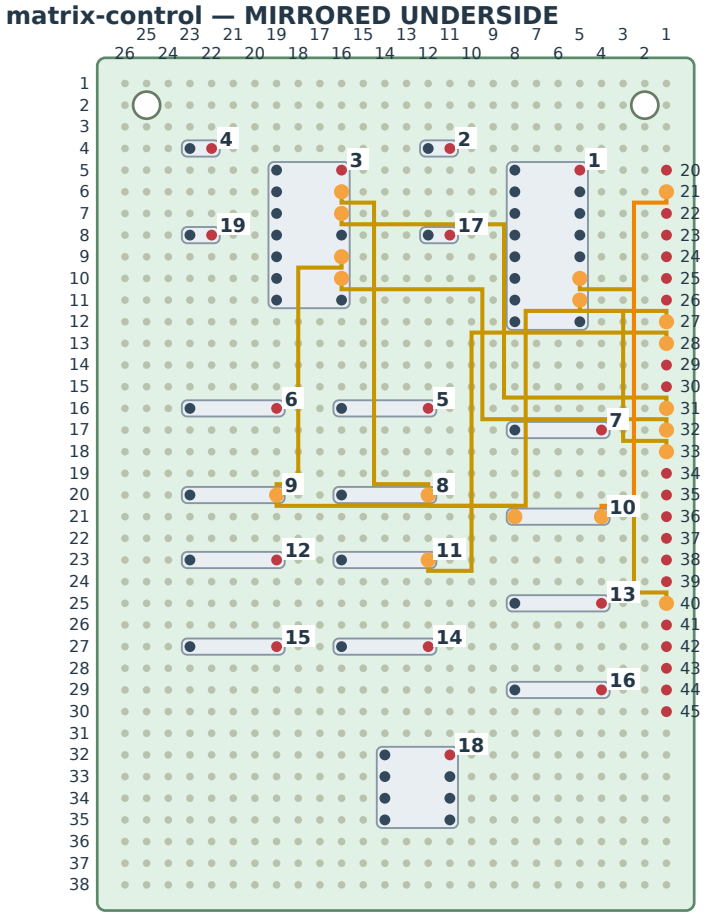
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / XBAR_LATCH	J_LINK_7.1 → R_XBAR_LATCH_OFF.1	(1,11) → (4,17)
W32 / XBAR_LATCH	R_XBAR_LATCH_OFF.1 → U_ADDR.12	(4,17) → (8,9)
W33 / XBAR_DAT	J_LINK_5.1 → R_XBAR_DAT_OFF.1	(1,9) → (12,16)
W34 / XBAR_DAT	R_XBAR_DAT_OFF.1 → U_ADDR.14	(12,16) → (8,7)
W35 / AX0	J_LINK_15.1 → U_ADDR.15	(1,19) → (8,6)
W36 / AX1	J_LINK_16.1 → U_ADDR.1	(1,20) → (5,5)
W37 / AX2	J_LINK_17.1 → U_ADDR.2	(1,21) → (5,6)
W38 / AX3	J_LINK_18.1 → U_ADDR.3	(1,22) → (5,7)
W39 / AY0	J_LINK_19.1 → U_ADDR.4	(1,23) → (5,8)
W40 / AY1	J_LINK_20.1 → U_ADDR.5	(1,24) → (5,9)

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CP24 — matrix-control

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

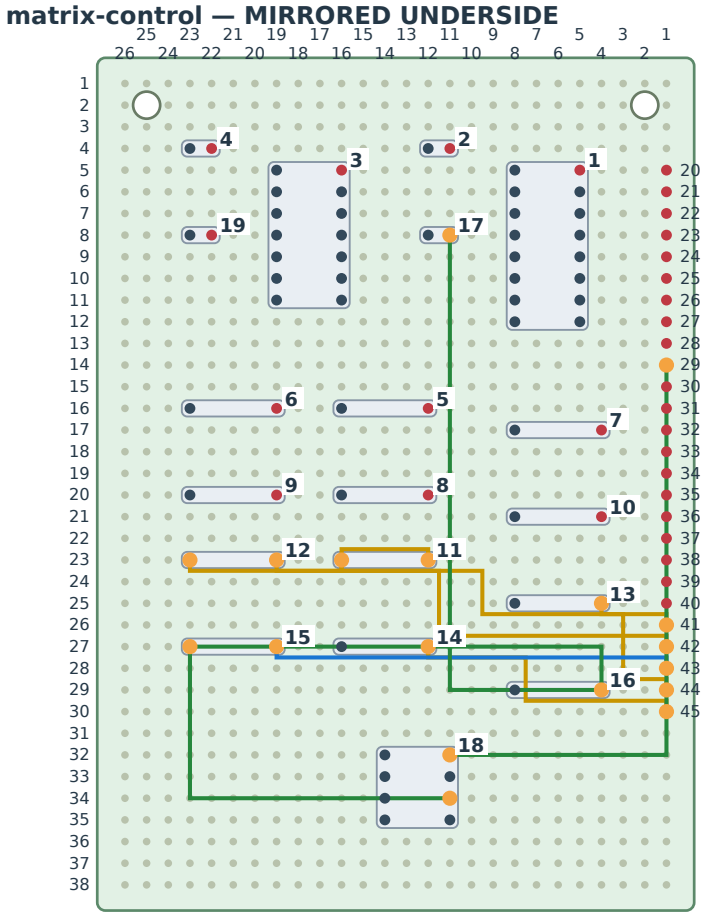
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / AY2	J_LINK_21.1 → U_ADDR.6	(1,25) → (5,10)
W42 / XBAR_DATA	J_LINK_14.1 → U_ADDR.7	(1,18) → (5,11)
W43 / STB_A_IN	J_LINK_8.1 → R_STB_A_IN_OFF.1	(1,12) → (12,20)
W44 / STB_A_IN	R_STB_A_IN_OFF.1 → U_BUFFER.2	(12,20) → (16,6)
W45 / STROBE_A	J_LINK_12.1 → U_BUFFER.3	(1,16) → (16,7)
W46 / STB_B_IN	J_LINK_9.1 → R_STB_B_IN_OFF.1	(1,13) → (19,20)
W47 / STB_B_IN	R_STB_B_IN_OFF.1 → U_BUFFER.5	(19,20) → (16,9)
W48 / STROBE_B	J_LINK_13.1 → U_BUFFER.6	(1,17) → (16,10)
W49 / 3V3_SYS	D_SOURCE.1 → J_LINK_2.1	(4,21) → (1,6)
W50 / SOURCE_FEED	D_SOURCE.2 → R_SOURCE_HI.1	(8,21) → (12,23)

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CP24 — matrix-control

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

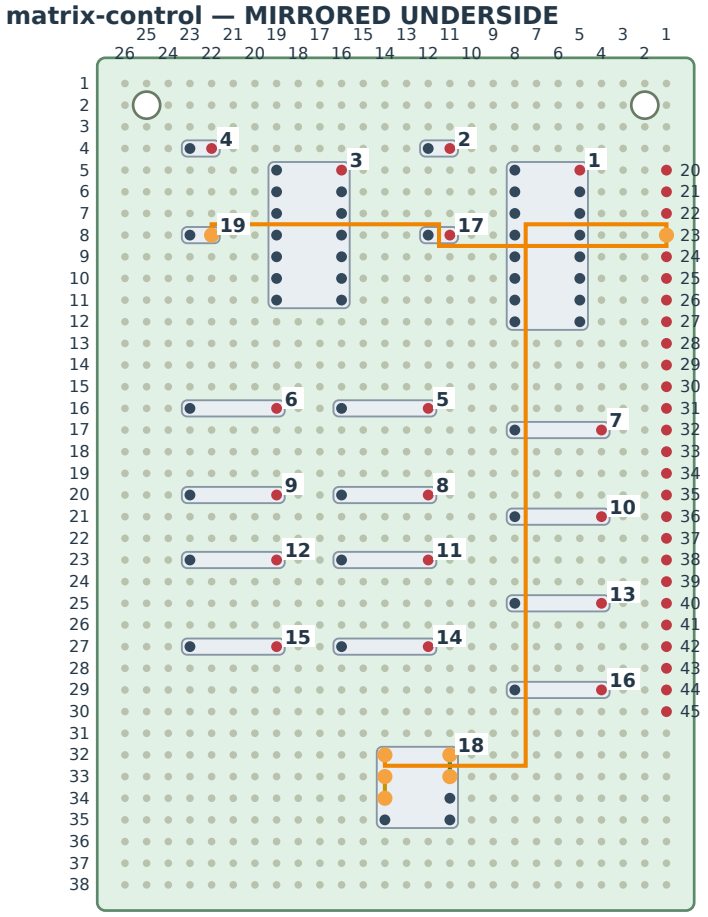
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W51 / SOURCE_FEED	R_SOURCE_HI.1 → R_SOURCE_LO.1	(12,23) → (19,23)
W52 / SOURCE_HI	J_LINK_22.1 → R_SOURCE_HI.2	(1,26) → (16,23)
W53 / SOURCE_LO	J_LINK_23.1 → R_SOURCE_LO.2	(1,27) → (23,23)
W54 / DISCHARGE	J_LINK_25.1 → R_DISCHARGE.1	(1,29) → (4,25)
W55 / CAL_LOAD	J_LINK_26.1 → R_CAL.1	(1,30) → (12,27)
W56 / DIAG_NODE	J_LINK_24.1 → R_DIAG_TOP.1	(1,28) → (19,27)
W57 / DIAG_DIV	C_DIAG.1 → R_DIAG_BOTTOM.1	(11,8) → (4,29)
W58 / DIAG_DIV	R_DIAG_BOTTOM.1 → R_DIAG_TOP.2	(4,29) → (23,27)
W59 / DIAG_DIV	R_DIAG_TOP.2 → U_DIAG.3	(23,27) → (11,34)
W60 / DIAG_ADC	J_LINK_10.1 → U_DIAG.1	(1,14) → (11,32)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP24 — matrix-control

Install these wires after their endpoint components; inspect each net with power isolated.



ID / net	From → to	TOP holes
W61 / DIAG_ADC	U_DIAG.1 → U_DIAG.2	(11,32) → (11,33)
W62 / USB_ROUTER_3V3	C_DIAG_VCC.1 → J_LINK_4.1	(22,8) → (1,8)
W63 / USB_ROUTER_3V3	J_LINK_4.1 → U_DIAG.8	(1,8) → (14,32)
W64 / DIAG_UNUSED	U_DIAG.6 → U_DIAG.7	(14,34) → (14,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

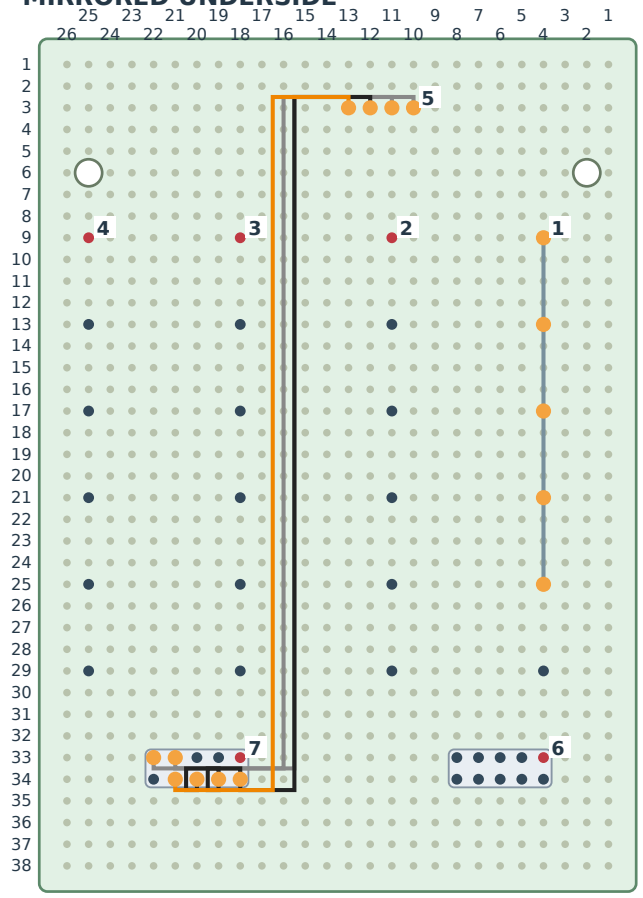
TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP27 — a2

Install these wires after their endpoint components; inspect each net with power isolated.

a2 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

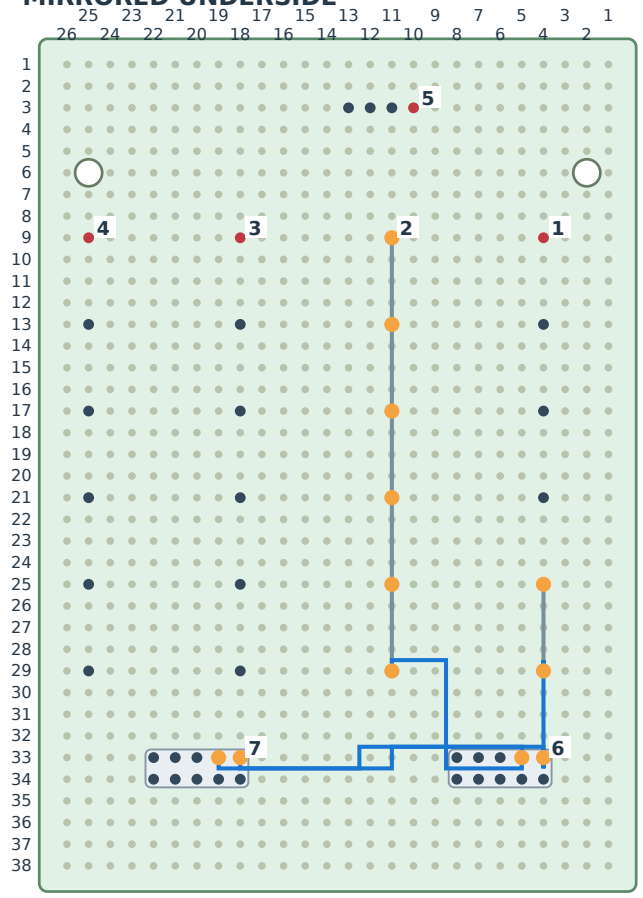
ID / net	From → to	TOP holes
W1 / SCL_POSITION_1	J_SERVICE.1 → J_SMART.9	(10,3) → (22,33)
W2 / SDA_POSITION_1	J_SERVICE.2 → J_SMART.7	(11,3) → (21,33)
W3 / GND	J_SERVICE.3 → J_SMART.2	(12,3) → (18,34)
W4 / GND	J_SMART.2 → J_SMART.4	(18,34) → (19,34)
W5 / GND	J_SMART.4 → J_SMART.6	(19,34) → (20,34)
W6 / 3V3_SYS	J_SERVICE.4 → J_SMART.8	(13,3) → (21,34)
W7 / NODE04	J_NODE04.1 → J_NODE04.2	(4,9) → (4,13)
W8 / NODE04	J_NODE04.2 → J_NODE04.3	(4,13) → (4,17)
W9 / NODE04	J_NODE04.3 → J_NODE04.4	(4,17) → (4,21)
W10 / NODE04	J_NODE04.4 → J_NODE04.5	(4,21) → (4,25)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — a2

Install these wires after their endpoint components; inspect each net with power isolated.

a2 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

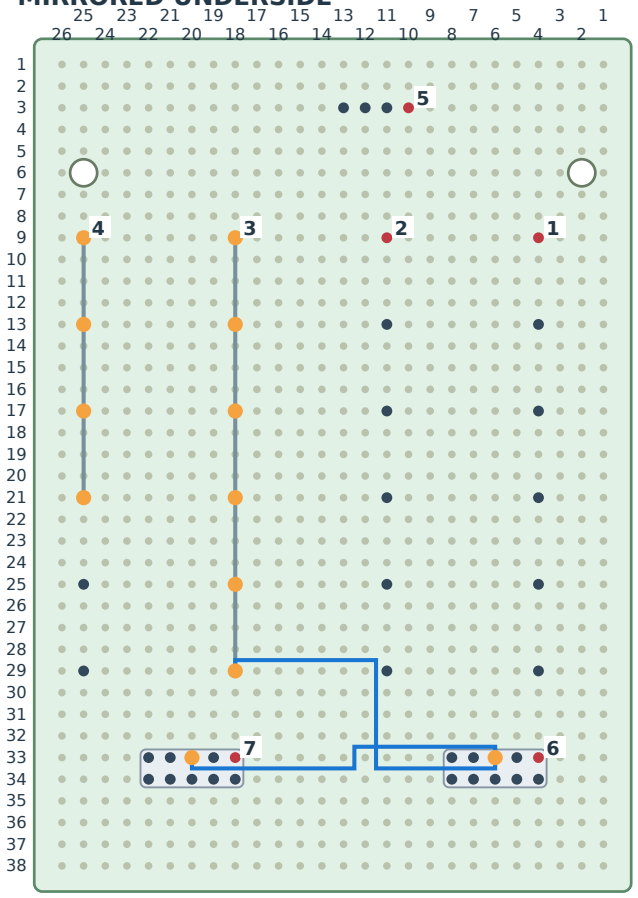
ID / net	From → to	TOP holes
W11 / NODE04	J_NODE04.5 → J_NODE04.6	(4,25) → (4,29)
W12 / NODE04	J_NODE04.6 → J_NODE.1	(4,29) → (4,33)
W13 / NODE04	J_NODE.1 → J_SMART.1	(4,33) → (18,33)
W14 / NODE05	J_NODE05.1 → J_NODE05.2	(11,9) → (11,13)
W15 / NODE05	J_NODE05.2 → J_NODE05.3	(11,13) → (11,17)
W16 / NODE05	J_NODE05.3 → J_NODE05.4	(11,17) → (11,21)
W17 / NODE05	J_NODE05.4 → J_NODE05.5	(11,21) → (11,25)
W18 / NODE05	J_NODE05.5 → J_NODE05.6	(11,25) → (11,29)
W19 / NODE05	J_NODE05.6 → J_NODE.3	(11,29) → (5,33)
W20 / NODE05	J_NODE.3 → J_SMART.3	(5,33) → (19,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — a2

Install these wires after their endpoint components; inspect each net with power isolated.

a2 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

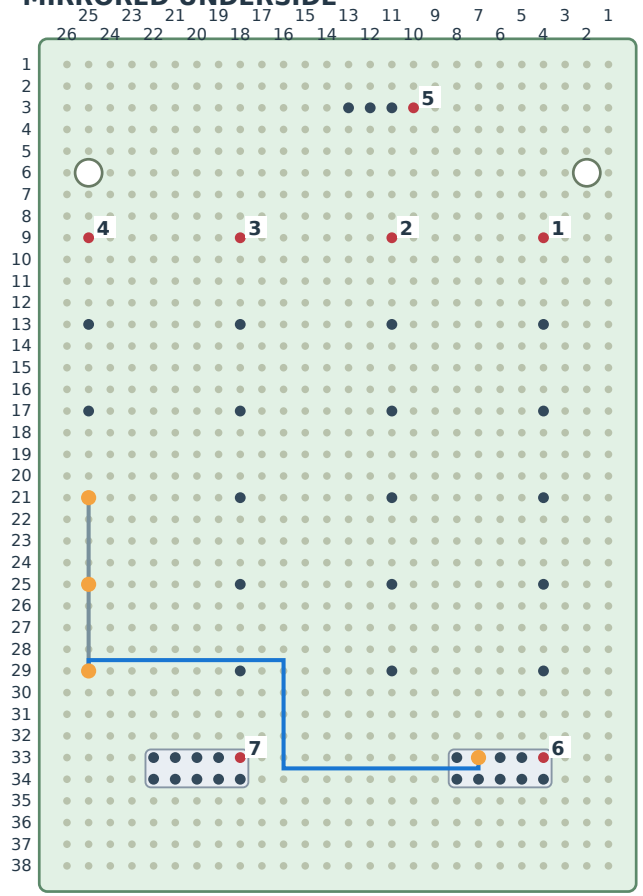
ID / net	From → to	TOP holes
W21 / NODE06	J_NODE06.1 → J_NODE06.2	(18,9) → (18,13)
W22 / NODE06	J_NODE06.2 → J_NODE06.3	(18,13) → (18,17)
W23 / NODE06	J_NODE06.3 → J_NODE06.4	(18,17) → (18,21)
W24 / NODE06	J_NODE06.4 → J_NODE06.5	(18,21) → (18,25)
W25 / NODE06	J_NODE06.5 → J_NODE06.6	(18,25) → (18,29)
W26 / NODE06	J_NODE06.6 → J_NODE.5	(18,29) → (6,33)
W27 / NODE06	J_NODE.5 → J_SMART.5	(6,33) → (20,33)
W28 / NODE07	J_NODE07.1 → J_NODE07.2	(25,9) → (25,13)
W29 / NODE07	J_NODE07.2 → J_NODE07.3	(25,13) → (25,17)
W30 / NODE07	J_NODE07.3 → J_NODE07.4	(25,17) → (25,21)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — a2

Install these wires after their endpoint components; inspect each net with power isolated.

a2 — MIRRORED UNDERSIDE



ID / net	From → to	TOP holes
W31 / NODE07	J_NODE07.4 → J_NODE07.5	(25,21) → (25,25)
W32 / NODE07	J_NODE07.5 → J_NODE07.6	(25,25) → (25,29)
W33 / NODE07	J_NODE07.6 → J_NODE.7	(25,29) → (7,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

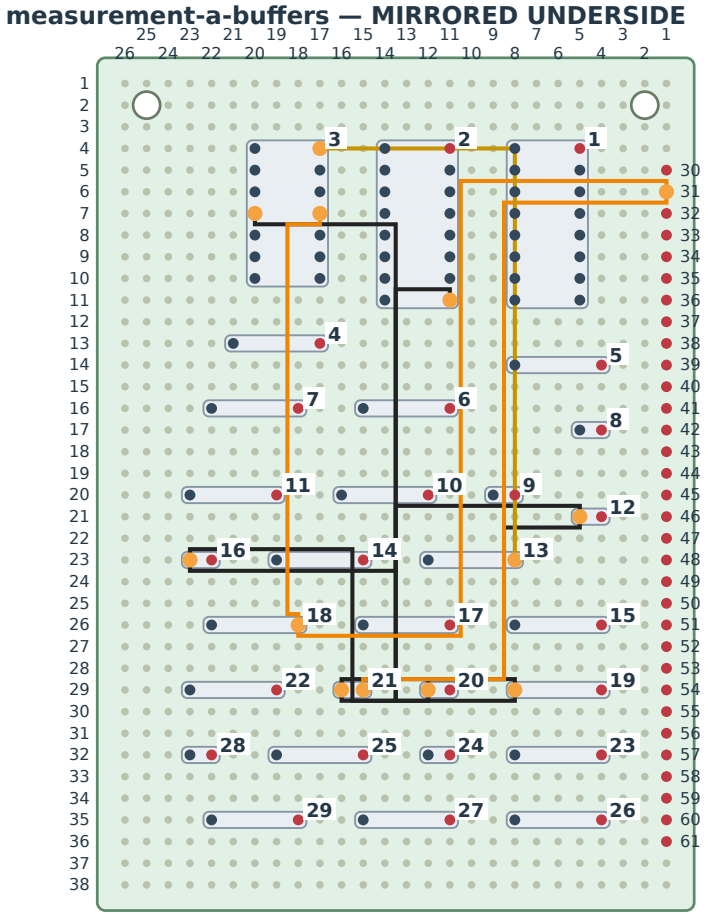
70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP27 — measurement-a-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

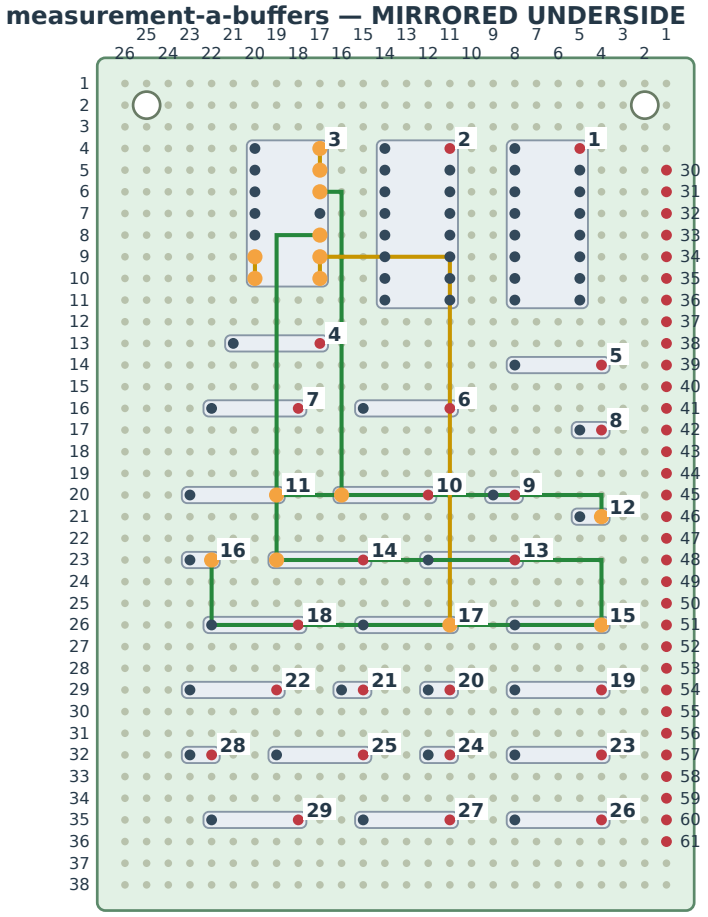
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W12 / GND	U_A_LED_DRIVER_1.8 → C_A_BUFFERS.2	(11,11) → (16,29)
W13 / GND	C_A_BUFFERS.2 → C_A_GENERAL_BIAS.2	(16,29) → (12,29)
W14 / GND	C_A_GENERAL_BIAS.2 → C_P1_1.2	(12,29) → (5,21)
W15 / GND	C_P1_1.2 → C_P1_2.2	(5,21) → (23,23)
W16 / GND	C_P1_2.2 → R_A_GENERAL_BIAS_BOTTOM.2	(23,23) → (8,29)
W17 / GND	R_A_GENERAL_BIAS_BOTTOM.2 → U_A_BUFFERS.11	(8,29) → (20,7)
W40 / 3V3_SYS	C_A_BUFFERS.1 → J_LINK_2.1	(15,29) → (1,6)
W41 / 3V3_SYS	J_LINK_2.1 → R_A_GENERAL_BIAS_TOP.1	(1,6) → (18,26)
W42 / 3V3_SYS	R_A_GENERAL_BIAS_TOP.1 → U_A_BUFFERS.4	(18,26) → (17,7)
W43 / P1_1_BUF	R_P1_1_ADC.1 → U_A_BUFFERS.1	(8,23) → (17,4)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — measurement-a-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

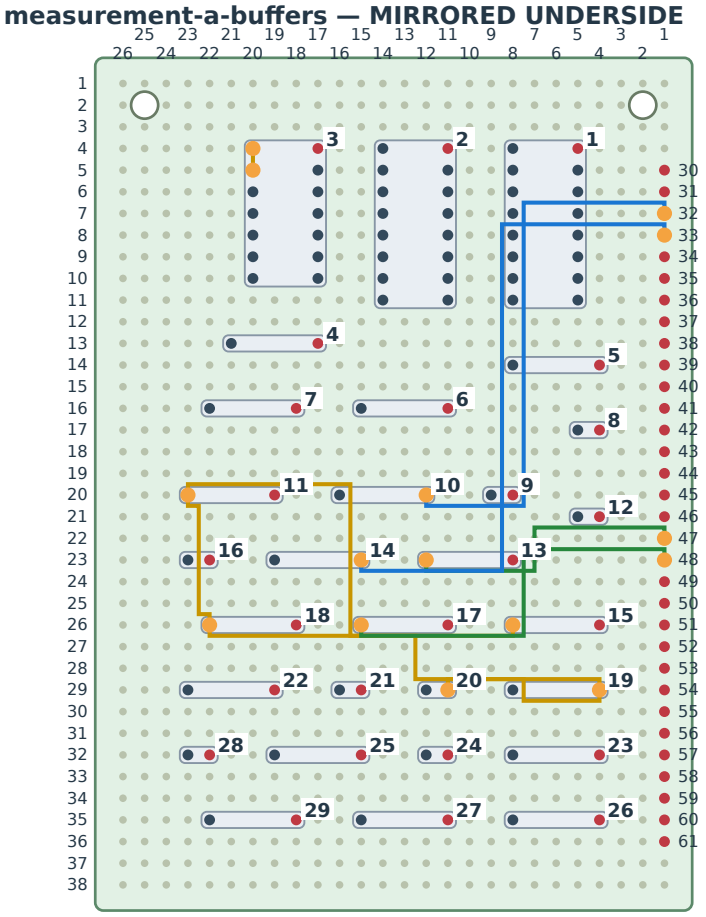
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W44 / P1_1_BUF	U_A_BUFFERS.1 → U_A_BUFFERS.2	(17,4) → (17,5)
W45 / P1_1_DIV	C_P1_1.1 → R_P1_1_BOTTOM.1	(4,21) → (19,20)
W46 / P1_1_DIV	R_P1_1_BOTTOM.1 → R_P1_1_TOP.2	(19,20) → (16,20)
W47 / P1_1_DIV	R_P1_1_TOP.2 → U_A_BUFFERS.3	(16,20) → (17,6)
W48 / P1_2_BUF	R_P1_2_ADC.1 → U_A_BUFFERS.6	(11,26) → (17,9)
W49 / P1_2_BUF	U_A_BUFFERS.6 → U_A_BUFFERS.7	(17,9) → (17,10)
W50 / P1_2_DIV	C_P1_2.1 → R_P1_2_BOTTOM.1	(22,23) → (4,26)
W51 / P1_2_DIV	R_P1_2_BOTTOM.1 → R_P1_2_TOP.2	(4,26) → (19,23)
W52 / P1_2_DIV	R_P1_2_TOP.2 → U_A_BUFFERS.5	(19,23) → (17,8)
W53 / P2_1_BUF	U_A_BUFFERS.8 → U_A_BUFFERS.9	(20,10) → (20,9)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — measurement-a-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

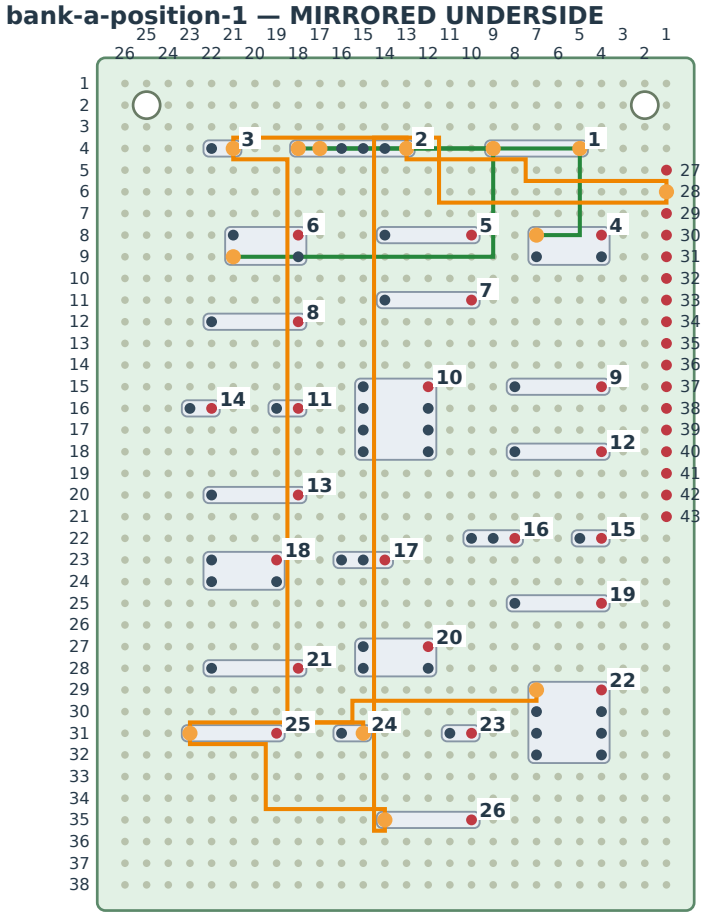
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W58 / P2_2_BUF	U_A_BUFFERS.13 → U_A_BUFFERS.14	(20,5) → (20,4)
W71 / NODE04	J_LINK_3.1 → R_P1_1_TOP.1	(1,7) → (12,20)
W72 / A_GENERAL_BIAS	C_A_GENERAL_BIAS.1 → R_A_GENERAL_BIAS_BOTTOM.1	(11,29) → (4,29)
W73 / A_GENERAL_BIAS	R_A_GENERAL_BIAS_BOTTOM.1 → R_A_GENERAL_BIAS_TOP.2	(4,29) → (22,26)
W74 / A_GENERAL_BIAS	R_A_GENERAL_BIAS_TOP.2 → R_P1_1_BOTTOM.2	(22,26) → (23,20)
W75 / A_GENERAL_BIAS	R_P1_1_BOTTOM.2 → R_P1_2_BOTTOM.2	(23,20) → (8,26)
W78 / P1_1_MUX	J_LINK_18.1 → R_P1_1_ADC.2	(1,22) → (12,23)
W79 / NODE05	J_LINK_4.1 → R_P1_2_TOP.1	(1,8) → (15,23)
W80 / P1_2_MUX	J_LINK_19.1 → R_P1_2_ADC.2	(1,23) → (15,26)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — bank-a-position-1

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

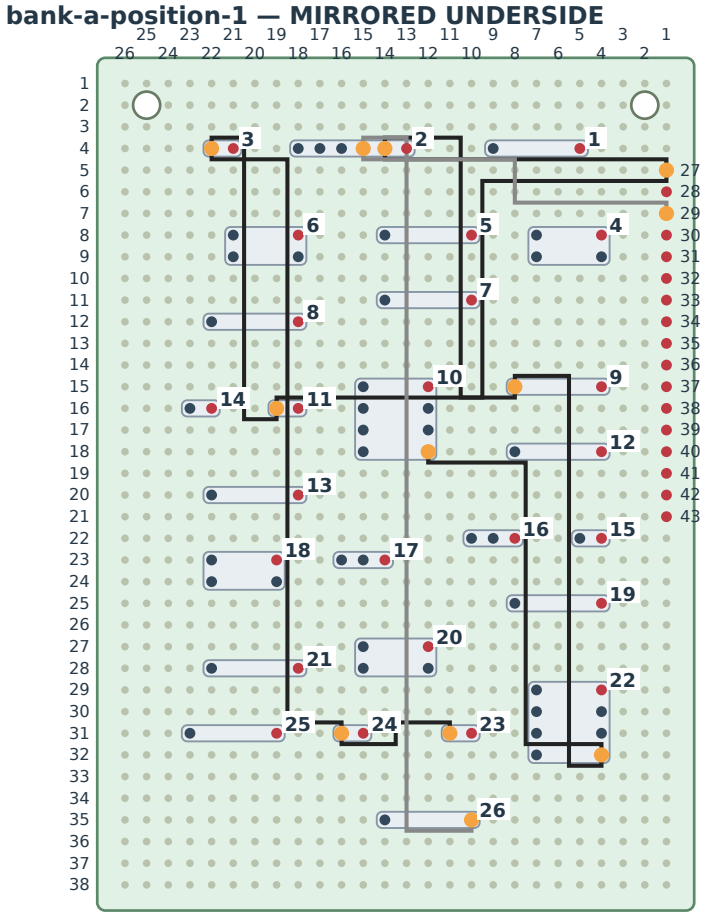
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / P1_SHUNT_P	R_P1_SHUNT.1 → J_P1_CURRENT_MODULE.5	(5,4) → (17,4)
W2 / P1_SHUNT_P	R_P1_SHUNT.1 → K_P1_AMM_P.4	(5,4) → (7,8)
W3 / P1_SHUNT_N	R_P1_SHUNT.2 → J_P1_CURRENT_MODULE.6	(9,4) → (18,4)
W4 / P1_SHUNT_N	R_P1_SHUNT.2 → K_P1_AMM_N.3	(9,4) → (21,9)
W5 / 3V3_SYS	C_P1_BUS_B.1 → C_P1_CURRENT.1	(15,31) → (21,4)
W6 / 3V3_SYS	C_P1_CURRENT.1 → J_LINK_2.1	(21,4) → (1,6)
W7 / 3V3_SYS	J_LINK_2.1 → J_P1_CURRENT_MODULE.1	(1,6) → (13,4)
W8 / 3V3_SYS	J_P1_CURRENT_MODULE.1 → R_P1_SCL.2	(13,4) → (14,35)
W9 / 3V3_SYS	R_P1_SCL.2 → R_P1_SDA.2	(14,35) → (23,31)
W10 / 3V3_SYS	R_P1_SDA.2 → U_P1_BUS.8	(23,31) → (7,29)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — bank-a-position-1

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

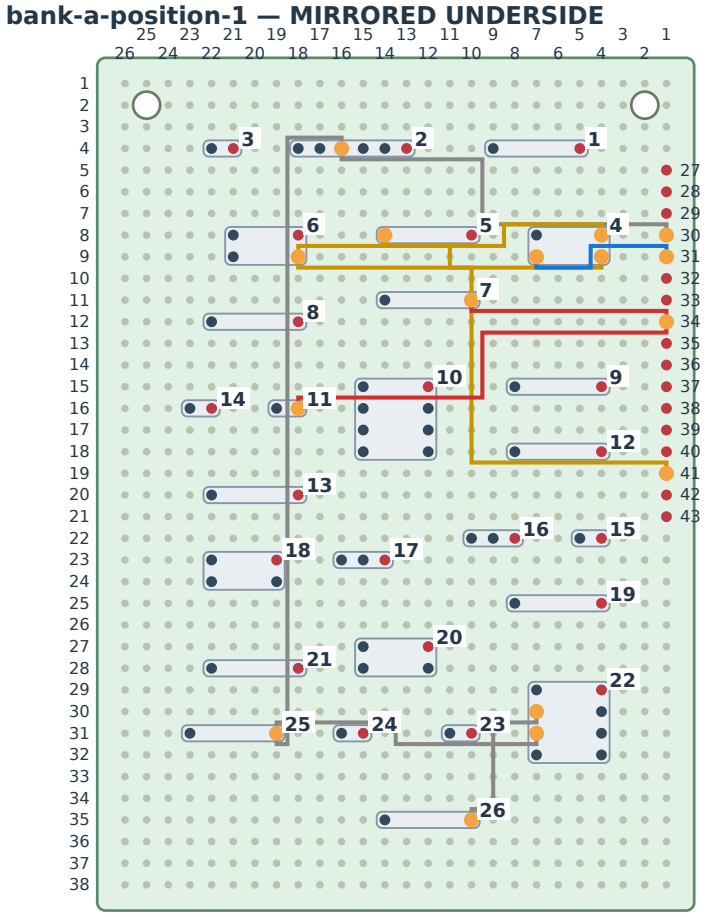
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	C_P1_BUS_A.2 → C_P1_BUS_B.2	(11,31) → (16,31)
W12 / GND	C_P1_BUS_B.2 → C_P1_CURRENT.2	(16,31) → (22,4)
W13 / GND	C_P1_CURRENT.2 → C_P1_LIMIT.2	(22,4) → (19,16)
W14 / GND	C_P1_LIMIT.2 → J_LINK_1.1	(19,16) → (1,5)
W15 / GND	J_LINK_1.1 → J_P1_CURRENT_MODULE.2	(1,5) → (14,4)
W16 / GND	J_P1_CURRENT_MODULE.2 → R_P1_SINK_SENSE.2	(14,4) → (8,15)
W17 / GND	R_P1_SINK_SENSE.2 → U_P1_BUS.4	(8,15) → (4,32)
W18 / GND	U_P1_BUS.4 → U_P1_LIMIT.4	(4,32) → (12,18)
W19 / SCL_POSITION_1	J_LINK_3.1 → J_P1_CURRENT_MODULE.3	(1,7) → (15,4)
W20 / SCL_POSITION_1	J_P1_CURRENT_MODULE.3 → R_P1_SCL.1	(15,4) → (10,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — bank-a-position-1

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

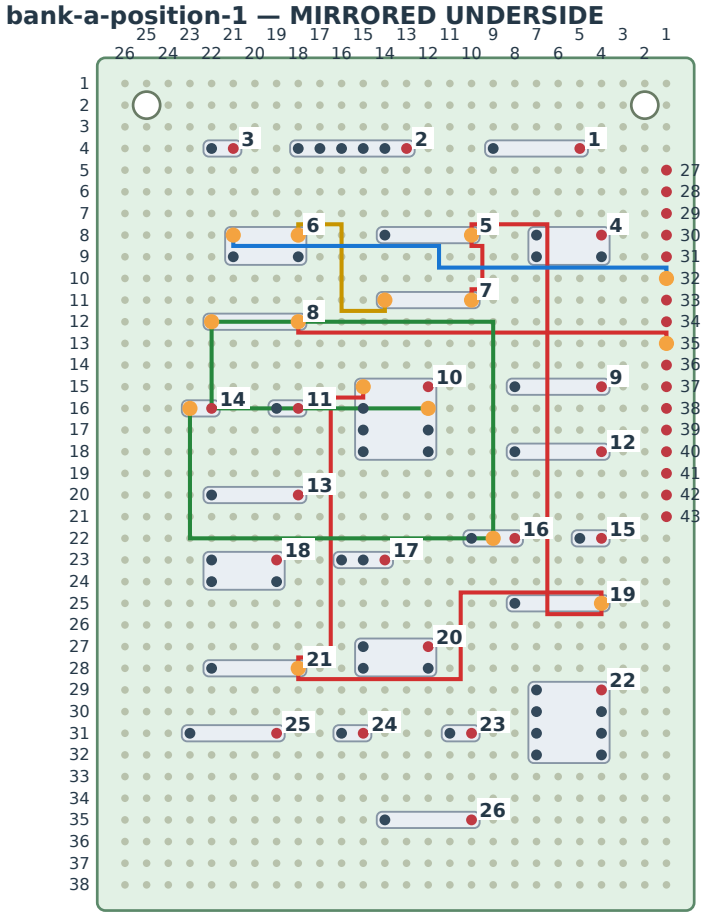
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / SCL_POSITION_1	R_P1_SCL.1 → U_P1_BUS.7	(10,35) → (7,30)
W22 / SDA_POSITION_1	J_LINK_4.1 → J_P1_CURRENT_MODULE.4	(1,8) → (16,4)
W23 / SDA_POSITION_1	J_P1_CURRENT_MODULE.4 → R_P1_SDA.1	(16,4) → (19,31)
W24 / SDA_POSITION_1	R_P1_SDA.1 → U_P1_BUS.6	(19,31) → (7,31)
W25 / P1_K_AMM_P_LED	K_P1_AMM_P.1 → R_P1_K_AMM_P_LED.2	(4,8) → (14,8)
W26 / P1_AMM_CATHODE	J_LINK_15.1 → K_P1_AMM_N.2	(1,19) → (18,9)
W27 / P1_AMM_CATHODE	K_P1_AMM_N.2 → K_P1_AMM_P.2	(18,9) → (4,9)
W28 / NODE04	J_LINK_5.1 → K_P1_AMM_P.3	(1,9) → (7,9)
W29 / 5V_SYS	C_P1_LIMIT.1 → J_LINK_8.1	(18,16) → (1,12)
W30 / 5V_SYS	J_LINK_8.1 → R_P1_K_AMM_N_LED.1	(1,12) → (10,11)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — bank-a-position-1

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

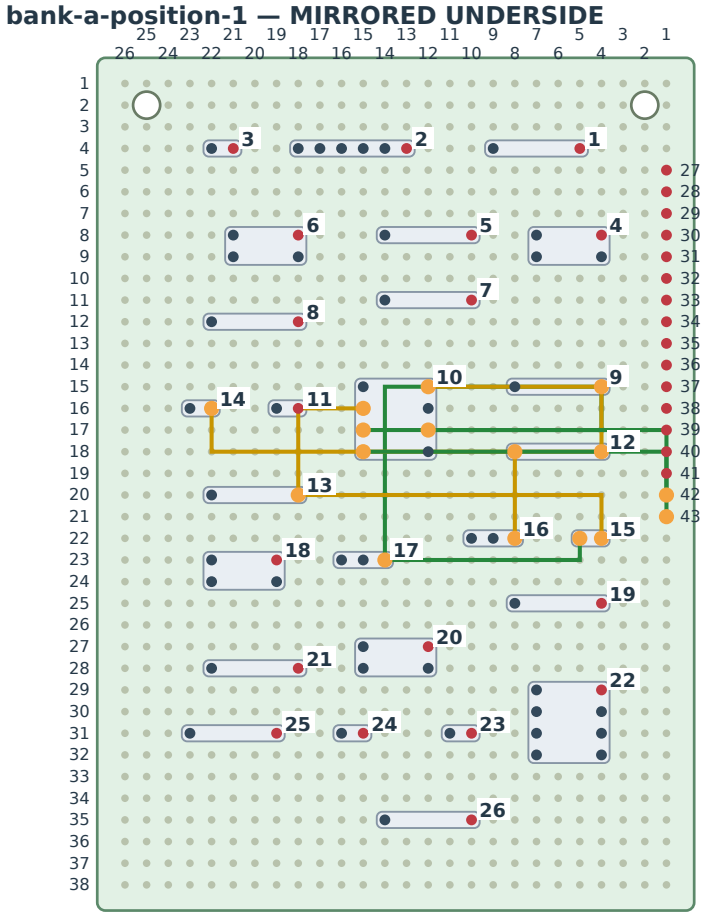
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / 5V_SYS	R_P1_K_AMM_N_LED.1 → R_P1_K_AMM_P_LED.1	(10,11) → (10,8)
W32 / 5V_SYS	R_P1_K_AMM_P_LED.1 → R_P1_K_HIGH_LED.1	(10,8) → (4,25)
W33 / 5V_SYS	R_P1_K_HIGH_LED.1 → R_P1_K_LOW_LED.1	(4,25) → (18,28)
W34 / 5V_SYS	R_P1_K_LOW_LED.1 → U_P1_LIMIT.8	(18,28) → (15,15)
W35 / P1_K_AMM_N_LED	K_P1_AMM_N.1 → R_P1_K_AMM_N_LED.2	(18,8) → (14,11)
W36 / NODE05	J_LINK_6.1 → K_P1_AMM_N.4	(1,10) → (21,8)
W37 / 5V_EDU	J_LINK_9.1 → R_P1_SOURCE_SENSE.1	(1,13) → (18,12)
W38 / P1_SOURCE_SENSE	C_P1_HI_COMP.2 → Q_P1_HIGH.2	(23,16) → (9,22)
W39 / P1_SOURCE_SENSE	Q_P1_HIGH.2 → R_P1_SOURCE_SENSE.2	(9,22) → (22,12)
W40 / P1_SOURCE_SENSE	R_P1_SOURCE_SENSE.2 → U_P1_LIMIT.2	(22,12) → (12,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — bank-a-position-1

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

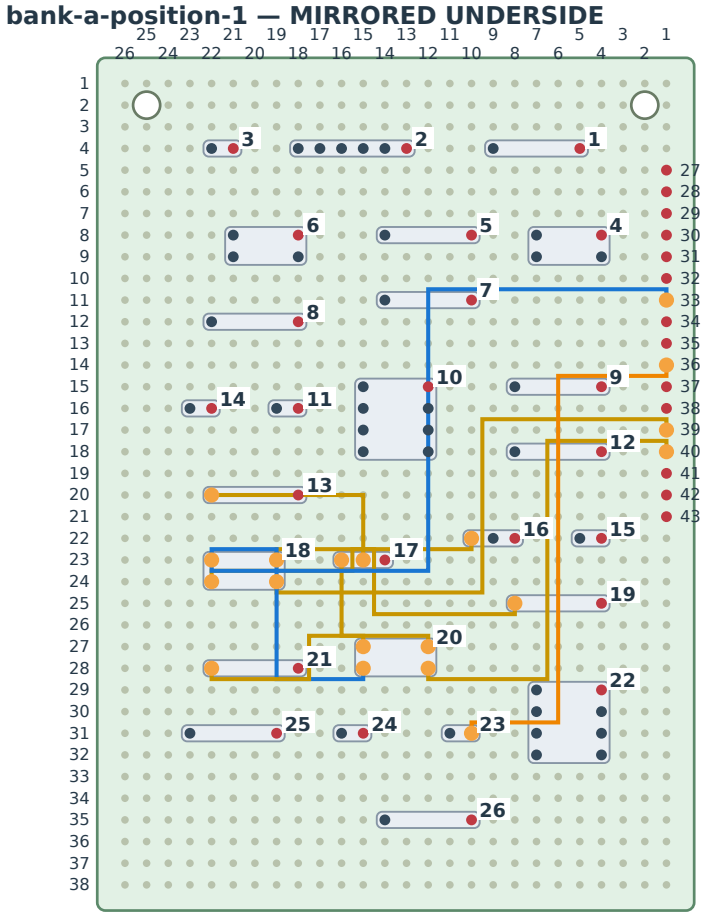
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / P1_SINK_SENSE	C_P1_LO_COMP.2 → Q_P1_LOW.1	(5,22) → (14,23)
W42 / P1_SINK_SENSE	Q_P1_LOW.1 → R_P1_SINK_SENSE.1	(14,23) → (4,15)
W43 / P1_SINK_SENSE	R_P1_SINK_SENSE.1 → U_P1_LIMIT.6	(4,15) → (15,17)
W44 / P1_HI_GATE_DRIVE	C_P1_HI_COMP.1 → R_P1_HI_GATE.1	(22,16) → (4,18)
W45 / P1_HI_GATE_DRIVE	R_P1_HI_GATE.1 → U_P1_LIMIT.1	(4,18) → (12,15)
W46 / A_REF_HI	J_LINK_16.1 → U_P1_LIMIT.3	(1,20) → (12,17)
W47 / A_REF_LO	J_LINK_17.1 → U_P1_LIMIT.5	(1,21) → (15,18)
W48 / P1_LO_GATE_DRIVE	C_P1_LO_COMP.1 → R_P1_LO_GATE.1	(4,22) → (18,20)
W49 / P1_LO_GATE_DRIVE	R_P1_LO_GATE.1 → U_P1_LIMIT.7	(18,20) → (15,16)
W50 / P1_HI_GATE	Q_P1_HIGH.1 → R_P1_HI_GATE.2	(8,22) → (8,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — bank-a-position-1

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

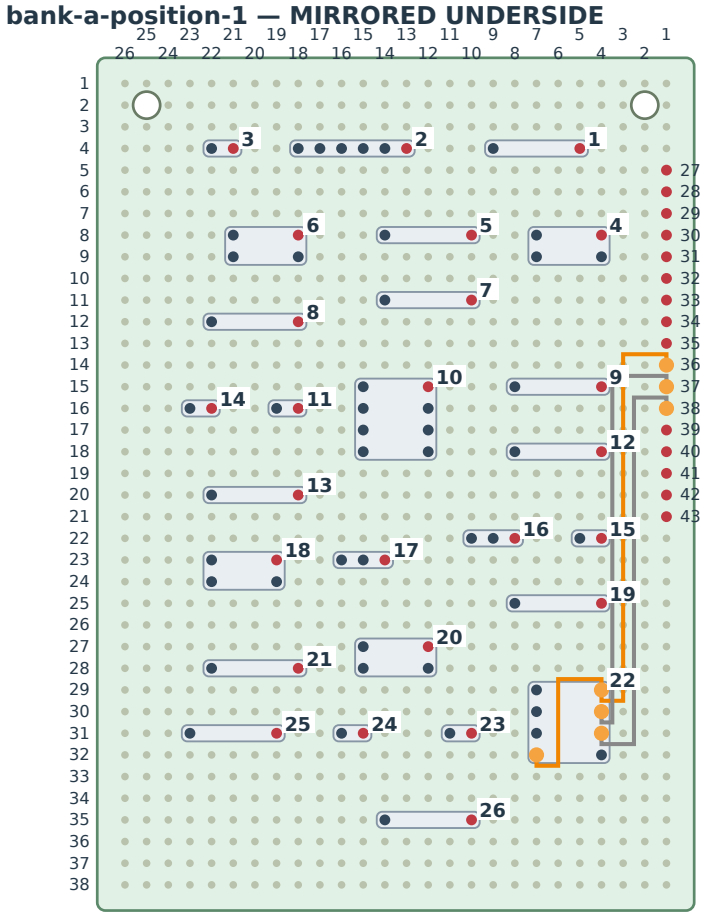
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W51 / P1_LO_GATE	Q_P1_LOW.2 → R_P1_LO_GATE.2	(15,23) → (22,20)
W52 / P1_SOURCE_OUT	K_P1_HIGH.3 → Q_P1_HIGH.3	(22,24) → (10,22)
W53 / P1_SINK_IN	K_P1_LOW.4 → Q_P1_LOW.3	(15,27) → (16,23)
W54 / P1_K_HIGH_LED	K_P1_HIGH.1 → R_P1_K_HIGH_LED.2	(19,23) → (8,25)
W55 / P1_K_HIGH_CATHODE	J_LINK_13.1 → K_P1_HIGH.2	(1,17) → (19,24)
W56 / NODE06	J_LINK_7.1 → K_P1_HIGH.4	(1,11) → (22,23)
W57 / NODE06	K_P1_HIGH.4 → K_P1_LOW.3	(22,23) → (15,28)
W58 / P1_K_LOW_LED	K_P1_LOW.1 → R_P1_K_LOW_LED.2	(12,27) → (22,28)
W59 / P1_K_LOW_CATHODE	J_LINK_14.1 → K_P1_LOW.2	(1,18) → (12,28)
W60 / USB_SMART_3V3	C_P1_BUS_A.1 → J_LINK_10.1	(10,31) → (1,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP27 — bank-a-position-1

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

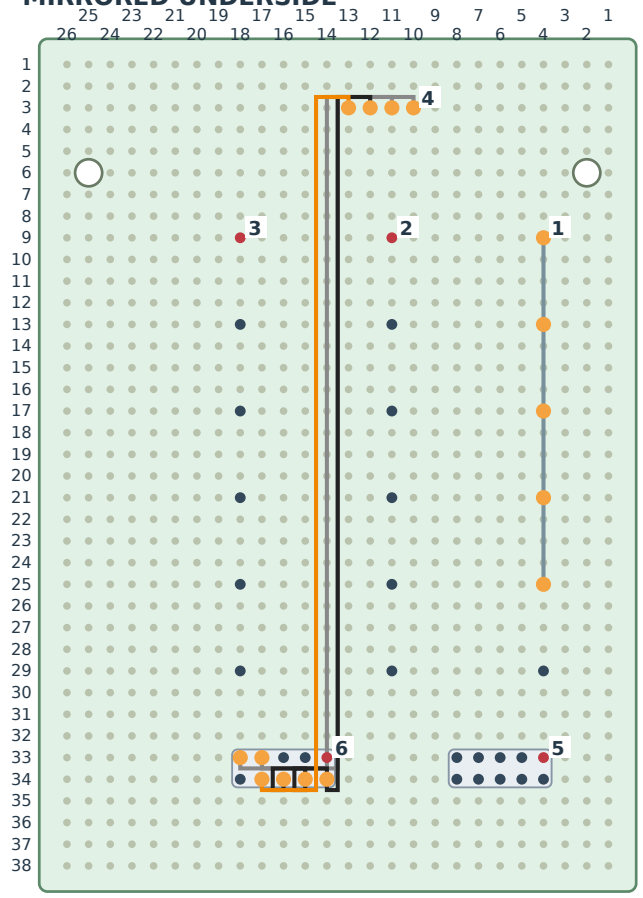
ID / net	From → to	TOP holes
W61 / USB_SMART_3V3	J_LINK_10.1 → U_P1_BUS.1	(1,14) → (4,29)
W62 / USB_SMART_3V3	U_P1_BUS.1 → U_P1_BUS.5	(4,29) → (7,32)
W63 / SCL_A	J_LINK_11.1 → U_P1_BUS.2	(1,15) → (4,30)
W64 / SDA_1	J_LINK_12.1 → U_P1_BUS.3	(1,16) → (4,31)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — a3

Install these wires after their endpoint components; inspect each net with power isolated.

a3 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

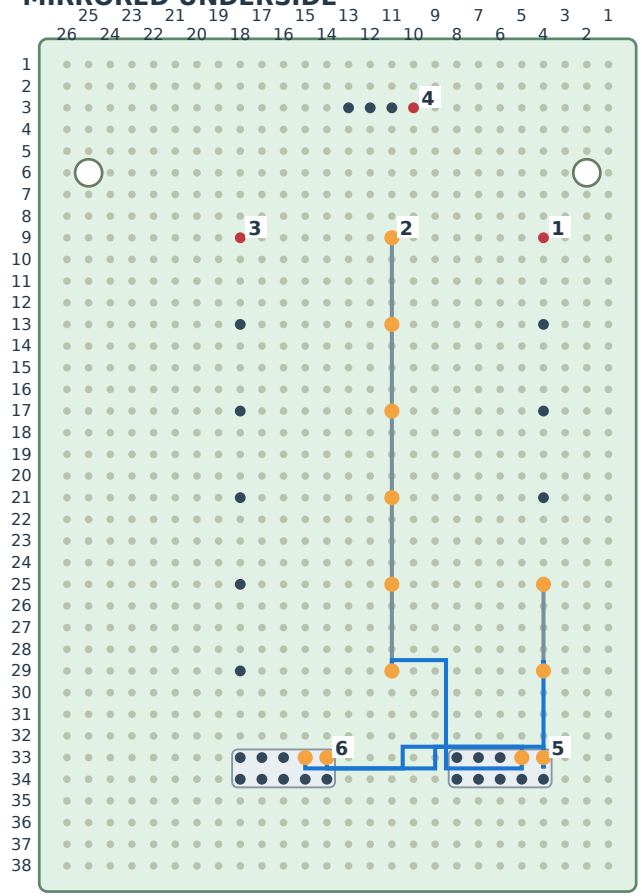
ID / net	From → to	TOP holes
W1 / SCL_POSITION_2	J_SERVICE.1 → J_SMART.9	(10,3) → (18,33)
W2 / SDA_POSITION_2	J_SERVICE.2 → J_SMART.7	(11,3) → (17,33)
W3 / GND	J_SERVICE.3 → J_SMART.2	(12,3) → (14,34)
W4 / GND	J_SMART.2 → J_SMART.4	(14,34) → (15,34)
W5 / GND	J_SMART.4 → J_SMART.6	(15,34) → (16,34)
W6 / 3V3_SYS	J_SERVICE.4 → J_SMART.8	(13,3) → (17,34)
W7 / NODE08	J_NODE08.1 → J_NODE08.2	(4,9) → (4,13)
W8 / NODE08	J_NODE08.2 → J_NODE08.3	(4,13) → (4,17)
W9 / NODE08	J_NODE08.3 → J_NODE08.4	(4,17) → (4,21)
W10 / NODE08	J_NODE08.4 → J_NODE08.5	(4,21) → (4,25)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — a3

Install these wires after their endpoint components; inspect each net with power isolated.

a3 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

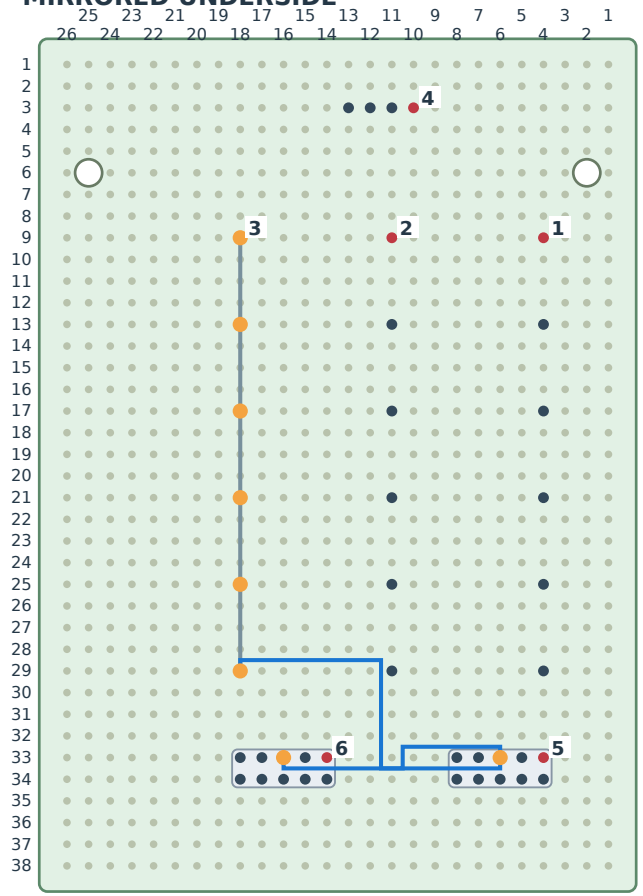
ID / net	From → to	TOP holes
W11 / NODE08	J_NODE08.5 → J_NODE08.6	(4,25) → (4,29)
W12 / NODE08	J_NODE08.6 → J_NODE.1	(4,29) → (4,33)
W13 / NODE08	J_NODE.1 → J_SMART.1	(4,33) → (14,33)
W14 / NODE09	J_NODE09.1 → J_NODE09.2	(11,9) → (11,13)
W15 / NODE09	J_NODE09.2 → J_NODE09.3	(11,13) → (11,17)
W16 / NODE09	J_NODE09.3 → J_NODE09.4	(11,17) → (11,21)
W17 / NODE09	J_NODE09.4 → J_NODE09.5	(11,21) → (11,25)
W18 / NODE09	J_NODE09.5 → J_NODE09.6	(11,25) → (11,29)
W19 / NODE09	J_NODE09.6 → J_NODE.3	(11,29) → (5,33)
W20 / NODE09	J_NODE.3 → J_SMART.3	(5,33) → (15,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — a3

Install these wires after their endpoint components; inspect each net with power isolated.

a3 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

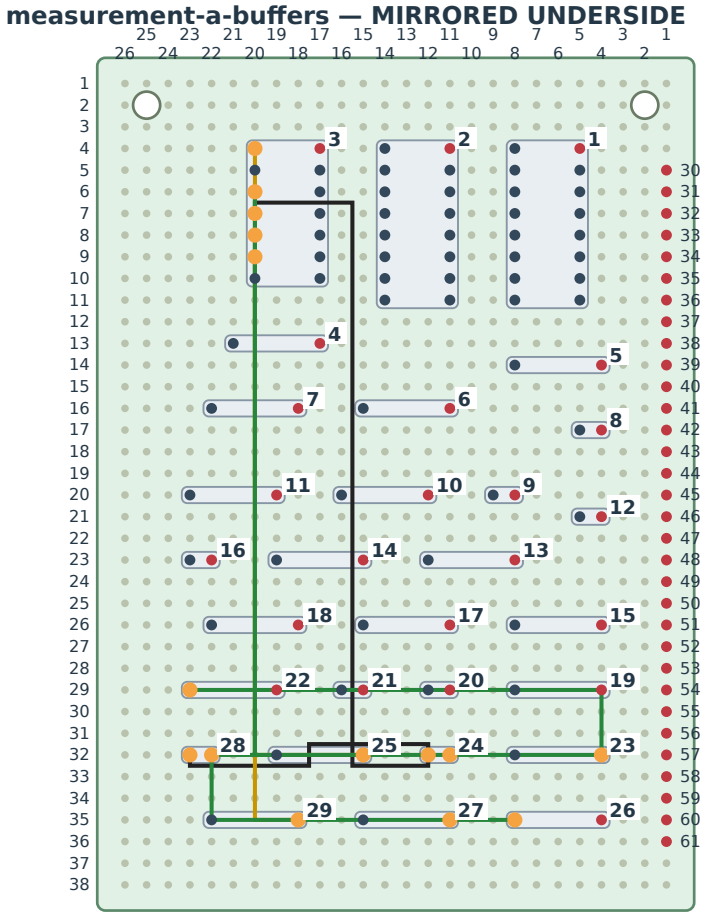
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / NODE10	J_NODE10.1 → J_NODE10.2	(18,9) → (18,13)
W22 / NODE10	J_NODE10.2 → J_NODE10.3	(18,13) → (18,17)
W23 / NODE10	J_NODE10.3 → J_NODE10.4	(18,17) → (18,21)
W24 / NODE10	J_NODE10.4 → J_NODE10.5	(18,21) → (18,25)
W25 / NODE10	J_NODE10.5 → J_NODE10.6	(18,25) → (18,29)
W26 / NODE10	J_NODE10.6 → J_NODE.5	(18,29) → (6,33)
W27 / NODE10	J_NODE.5 → J_SMART.5	(6,33) → (16,33)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — measurement-a-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

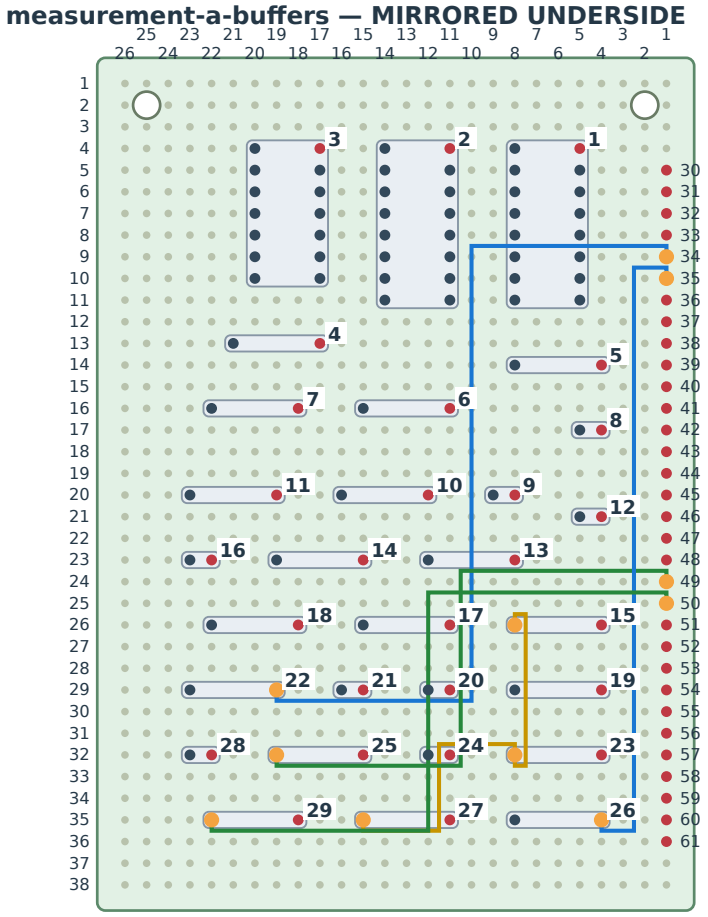
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W18 / GND	U_A_BUFFERS.11 → C_P2_1.2	(20,7) → (12,32)
W19 / GND	C_P2_1.2 → C_P2_2.2	(12,32) → (23,32)
W54 / P2_1_BUF	U_A_BUFFERS.9 → R_P2_1_ADC.1	(20,9) → (15,32)
W55 / P2_1_DIV	U_A_BUFFERS.10 → C_P2_1.1	(20,8) → (11,32)
W56 / P2_1_DIV	C_P2_1.1 → R_P2_1_BOTTOM.1	(11,32) → (4,32)
W57 / P2_1_DIV	R_P2_1_BOTTOM.1 → R_P2_1_TOP.2	(4,32) → (23,29)
W59 / P2_2_BUF	U_A_BUFFERS.14 → R_P2_2_ADC.1	(20,4) → (18,35)
W60 / P2_2_DIV	U_A_BUFFERS.12 → C_P2_2.1	(20,6) → (22,32)
W61 / P2_2_DIV	C_P2_2.1 → R_P2_2_BOTTOM.1	(22,32) → (11,35)
W62 / P2_2_DIV	R_P2_2_BOTTOM.1 → R_P2_2_TOP.2	(11,35) → (8,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — measurement-a-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

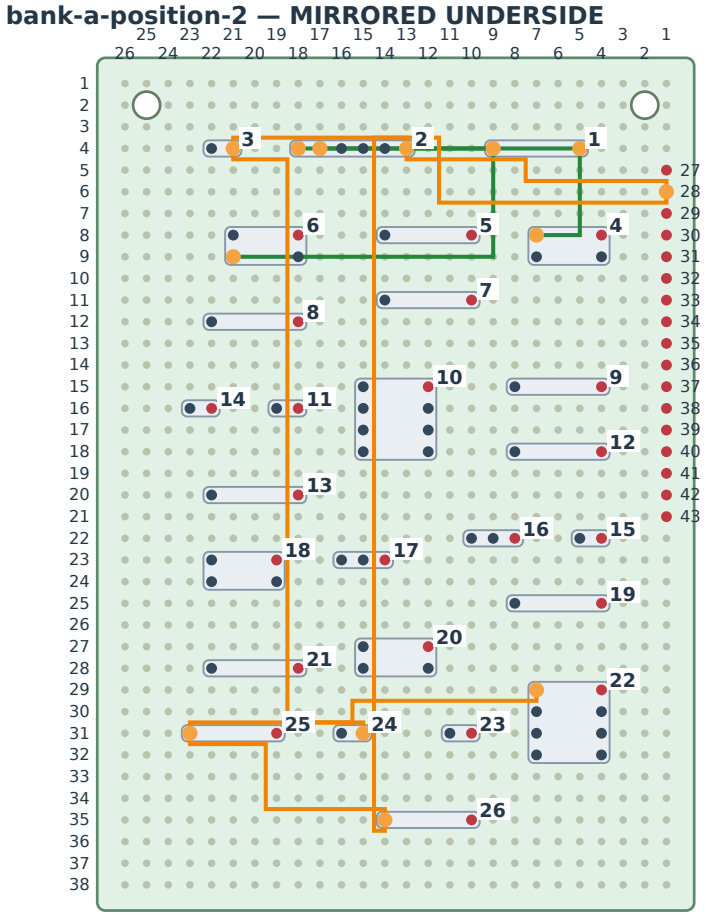
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W76 / A_GENERAL_BIAS	R_P1_2_BOTTOM.2 → R_P2_1_BOTTOM.2	(8,26) → (8,32)
W77 / A_GENERAL_BIAS	R_P2_1_BOTTOM.2 → R_P2_2_BOTTOM.2	(8,32) → (15,35)
W81 / NODE08	J_LINK_5.1 → R_P2_1_TOP.1	(1,9) → (19,29)
W82 / P2_1_MUX	J_LINK_20.1 → R_P2_1_ADC.2	(1,24) → (19,32)
W83 / NODE09	J_LINK_6.1 → R_P2_2_TOP.1	(1,10) → (4,35)
W84 / P2_2_MUX	J_LINK_21.1 → R_P2_2_ADC.2	(1,25) → (22,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — bank-a-position-2

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

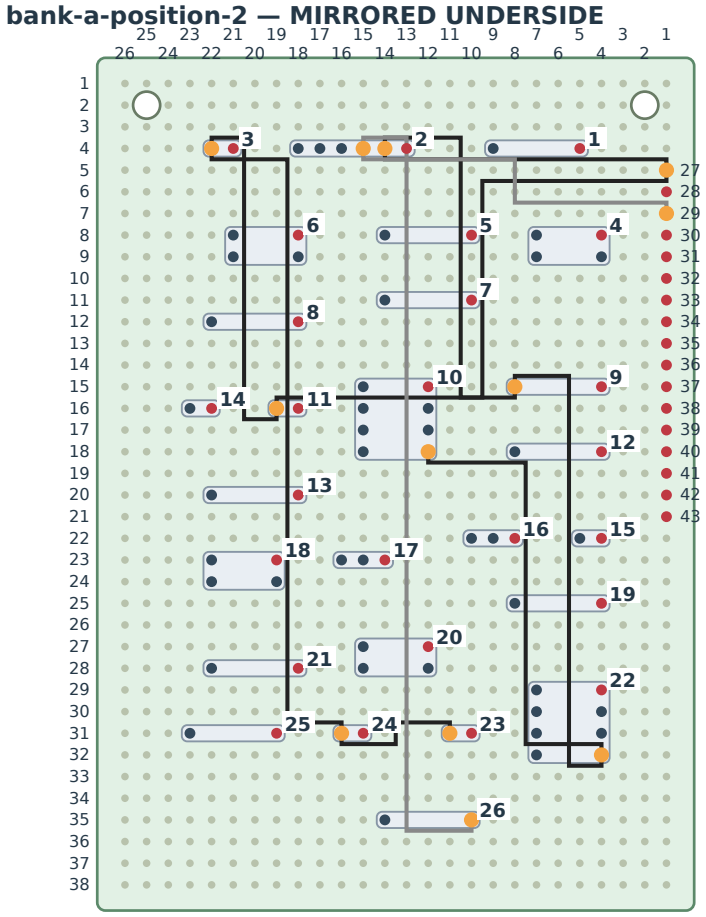
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / P2_SHUNT_P	R_P2_SHUNT.1 → J_P2_CURRENT_MODULE.5	(5,4) → (17,4)
W2 / P2_SHUNT_P	R_P2_SHUNT.1 → K_P2_AMM_P.4	(5,4) → (7,8)
W3 / P2_SHUNT_N	R_P2_SHUNT.2 → J_P2_CURRENT_MODULE.6	(9,4) → (18,4)
W4 / P2_SHUNT_N	R_P2_SHUNT.2 → K_P2_AMM_N.3	(9,4) → (21,9)
W5 / 3V3_SYS	C_P2_BUS_B.1 → C_P2_CURRENT.1	(15,31) → (21,4)
W6 / 3V3_SYS	C_P2_CURRENT.1 → J_LINK_2.1	(21,4) → (1,6)
W7 / 3V3_SYS	J_LINK_2.1 → J_P2_CURRENT_MODULE.1	(1,6) → (13,4)
W8 / 3V3_SYS	J_P2_CURRENT_MODULE.1 → R_P2_SCL.2	(13,4) → (14,35)
W9 / 3V3_SYS	R_P2_SCL.2 → R_P2_SDA.2	(14,35) → (23,31)
W10 / 3V3_SYS	R_P2_SDA.2 → U_P2_BUS.8	(23,31) → (7,29)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — bank-a-position-2

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

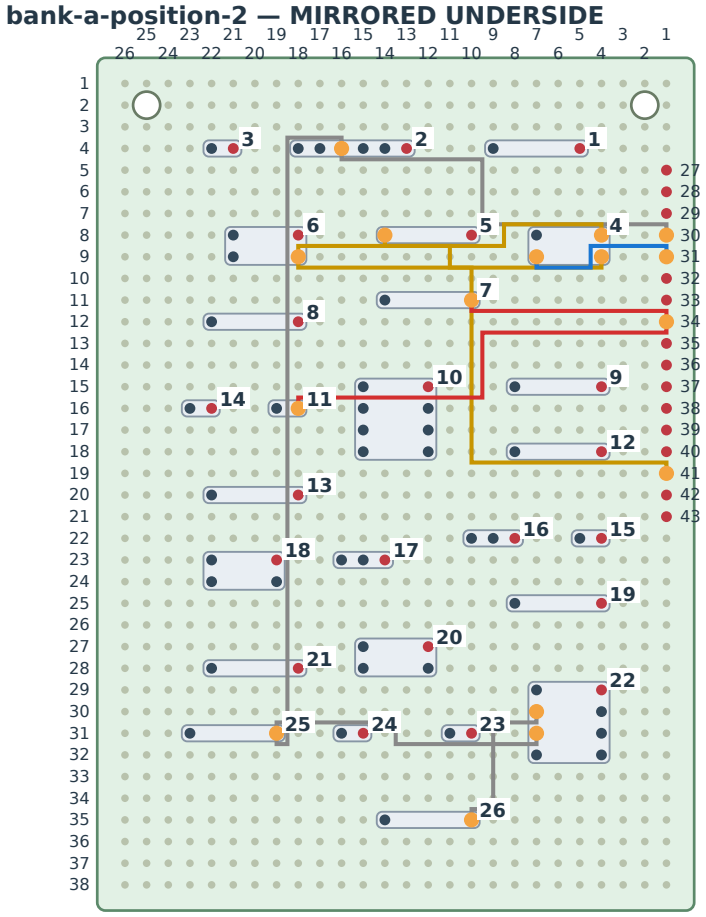
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	C_P2_BUS_A.2 → C_P2_BUS_B.2	(11,31) → (16,31)
W12 / GND	C_P2_BUS_B.2 → C_P2_CURRENT.2	(16,31) → (22,4)
W13 / GND	C_P2_CURRENT.2 → C_P2_LIMIT.2	(22,4) → (19,16)
W14 / GND	C_P2_LIMIT.2 → J_LINK_1.1	(19,16) → (1,5)
W15 / GND	J_LINK_1.1 → J_P2_CURRENT_MODULE.2	(1,5) → (14,4)
W16 / GND	J_P2_CURRENT_MODULE.2 → R_P2_SINK_SENSE.2	(14,4) → (8,15)
W17 / GND	R_P2_SINK_SENSE.2 → U_P2_BUS.4	(8,15) → (4,32)
W18 / GND	U_P2_BUS.4 → U_P2_LIMIT.4	(4,32) → (12,18)
W19 / SCL_POSITION_2	J_LINK_3.1 → J_P2_CURRENT_MODULE.3	(1,7) → (15,4)
W20 / SCL_POSITION_2	J_P2_CURRENT_MODULE.3 → R_P2_SCL.1	(15,4) → (10,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — bank-a-position-2

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

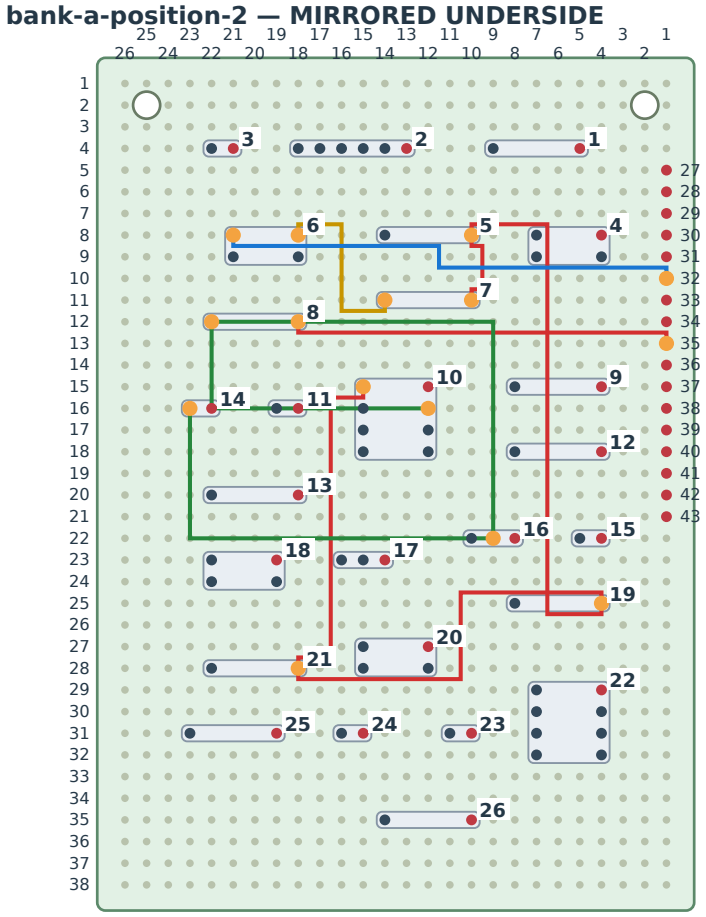
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / SCL_POSITION_2	R_P2_SCL.1 → U_P2_BUS.7	(10,35) → (7,30)
W22 / SDA_POSITION_2	J_LINK_4.1 → J_P2_CURRENT_MODULE.4	(1,8) → (16,4)
W23 / SDA_POSITION_2	J_P2_CURRENT_MODULE.4 → R_P2_SDA.1	(16,4) → (19,31)
W24 / SDA_POSITION_2	R_P2_SDA.1 → U_P2_BUS.6	(19,31) → (7,31)
W25 / P2_K_AMM_P_LED	K_P2_AMM_P.1 → R_P2_K_AMM_P_LED.2	(4,8) → (14,8)
W26 / P2_AMM_CATHODE	J_LINK_15.1 → K_P2_AMM_N.2	(1,19) → (18,9)
W27 / P2_AMM_CATHODE	K_P2_AMM_N.2 → K_P2_AMM_P.2	(18,9) → (4,9)
W28 / NODE08	J_LINK_5.1 → K_P2_AMM_P.3	(1,9) → (7,9)
W29 / 5V_SYS	C_P2_LIMIT.1 → J_LINK_8.1	(18,16) → (1,12)
W30 / 5V_SYS	J_LINK_8.1 → R_P2_K_AMM_N_LED.1	(1,12) → (10,11)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — bank-a-position-2

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

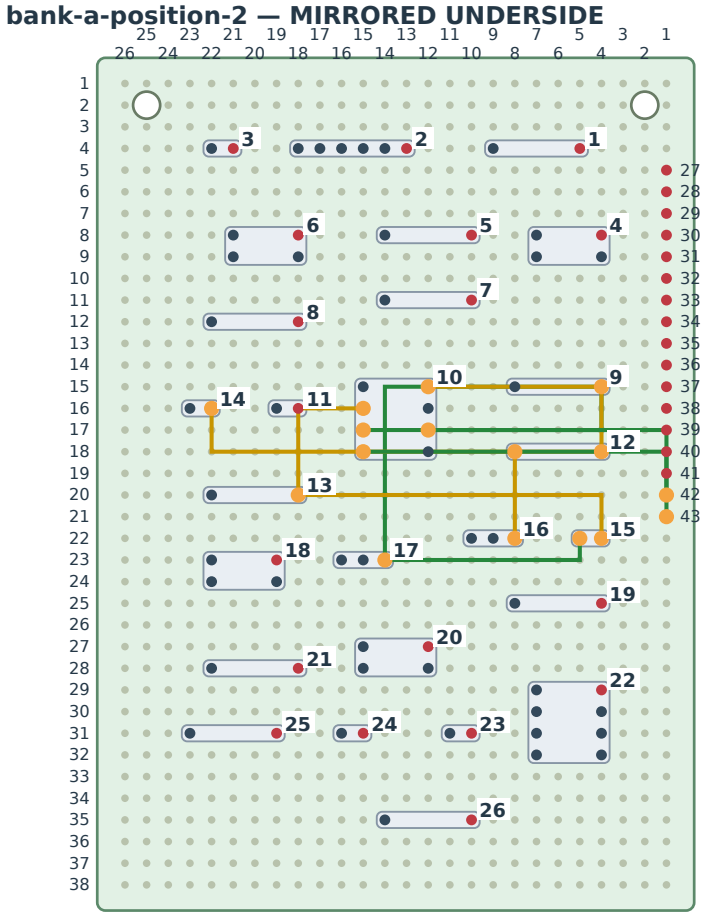
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / 5V_SYS	R_P2_K_AMM_N_LED.1 → R_P2_K_AMM_P_LED.1	(10,11) → (10,8)
W32 / 5V_SYS	R_P2_K_AMM_P_LED.1 → R_P2_K_HIGH_LED.1	(10,8) → (4,25)
W33 / 5V_SYS	R_P2_K_HIGH_LED.1 → R_P2_K_LOW_LED.1	(4,25) → (18,28)
W34 / 5V_SYS	R_P2_K_LOW_LED.1 → U_P2_LIMIT.8	(18,28) → (15,15)
W35 / P2_K_AMM_N_LED	K_P2_AMM_N.1 → R_P2_K_AMM_N_LED.2	(18,8) → (14,11)
W36 / NODE09	J_LINK_6.1 → K_P2_AMM_N.4	(1,10) → (21,8)
W37 / 5V_EDU	J_LINK_9.1 → R_P2_SOURCE_SENSE.1	(1,13) → (18,12)
W38 / P2_SOURCE_SENSE	C_P2_HI_COMP.2 → Q_P2_HIGH.2	(23,16) → (9,22)
W39 / P2_SOURCE_SENSE	Q_P2_HIGH.2 → R_P2_SOURCE_SENSE.2	(9,22) → (22,12)
W40 / P2_SOURCE_SENSE	R_P2_SOURCE_SENSE.2 → U_P2_LIMIT.2	(22,12) → (12,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — bank-a-position-2

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

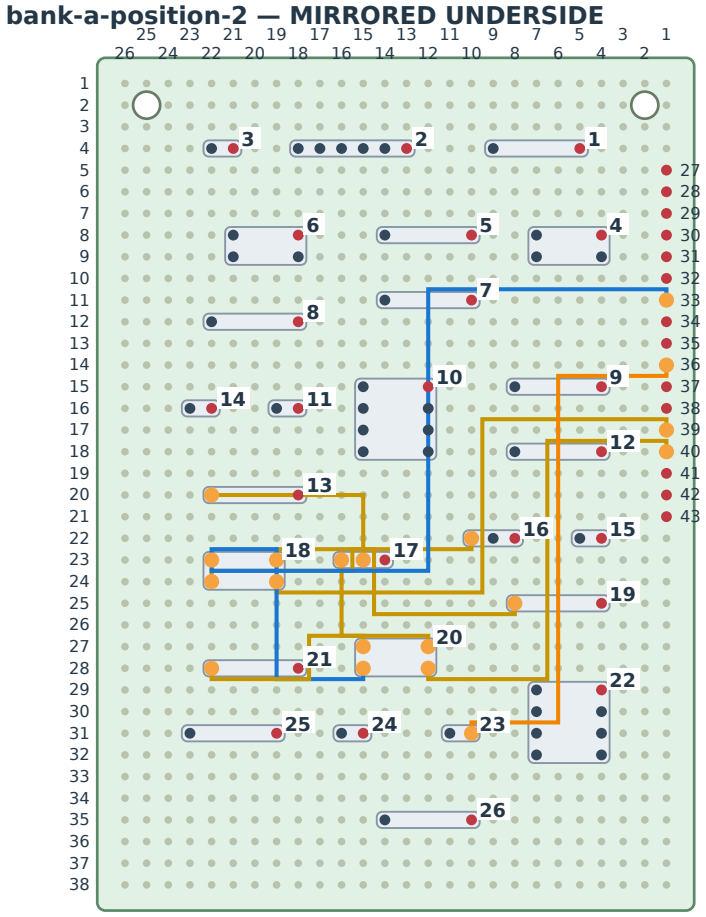
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / P2_SINK_SENSE	C_P2_LO_COMP.2 → Q_P2_LOW.1	(5,22) → (14,23)
W42 / P2_SINK_SENSE	Q_P2_LOW.1 → R_P2_SINK_SENSE.1	(14,23) → (4,15)
W43 / P2_SINK_SENSE	R_P2_SINK_SENSE.1 → U_P2_LIMIT.6	(4,15) → (15,17)
W44 / P2_HI_GATE_DRIVE	C_P2_HI_COMP.1 → R_P2_HI_GATE.1	(22,16) → (4,18)
W45 / P2_HI_GATE_DRIVE	R_P2_HI_GATE.1 → U_P2_LIMIT.1	(4,18) → (12,15)
W46 / A_REF_HI	J_LINK_16.1 → U_P2_LIMIT.3	(1,20) → (12,17)
W47 / A_REF_LO	J_LINK_17.1 → U_P2_LIMIT.5	(1,21) → (15,18)
W48 / P2_LO_GATE_DRIVE	C_P2_LO_COMP.1 → R_P2_LO_GATE.1	(4,22) → (18,20)
W49 / P2_LO_GATE_DRIVE	R_P2_LO_GATE.1 → U_P2_LIMIT.7	(18,20) → (15,16)
W50 / P2_HI_GATE	Q_P2_HIGH.1 → R_P2_HI_GATE.2	(8,22) → (8,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — bank-a-position-2

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

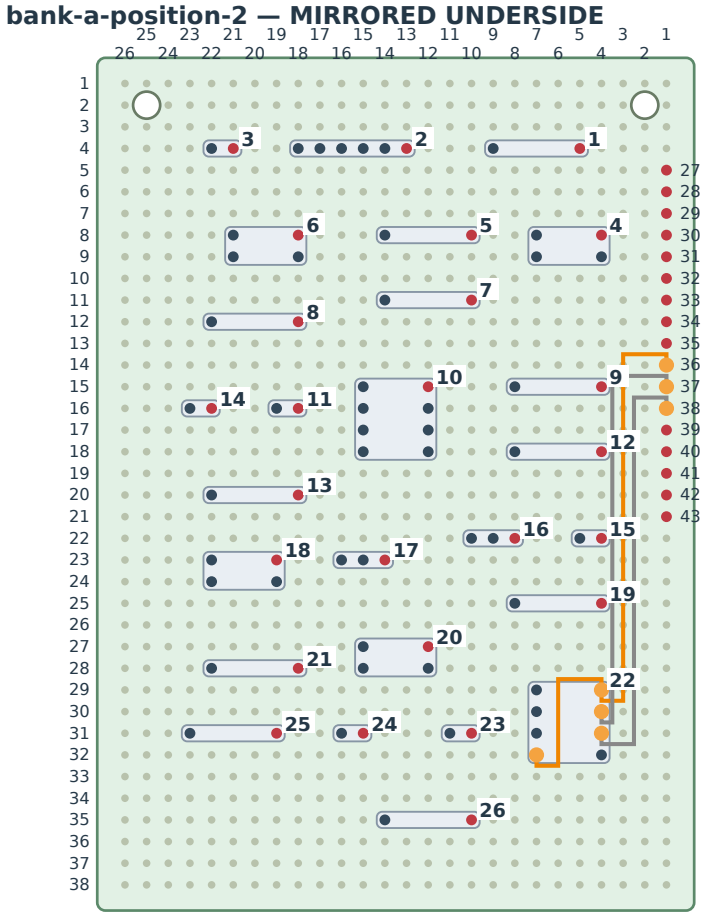
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W51 / P2_LO_GATE	Q_P2_LOW.2 → R_P2_LO_GATE.2	(15,23) → (22,20)
W52 / P2_SOURCE_OUT	K_P2_HIGH.3 → Q_P2_HIGH.3	(22,24) → (10,22)
W53 / P2_SINK_IN	K_P2_LOW.4 → Q_P2_LOW.3	(15,27) → (16,23)
W54 / P2_K_HIGH_LED	K_P2_HIGH.1 → R_P2_K_HIGH_LED.2	(19,23) → (8,25)
W55 / P2_K_HIGH_CATHODE	J_LINK_13.1 → K_P2_HIGH.2	(1,17) → (19,24)
W56 / NODE10	J_LINK_7.1 → K_P2_HIGH.4	(1,11) → (22,23)
W57 / NODE10	K_P2_HIGH.4 → K_P2_LOW.3	(22,23) → (15,28)
W58 / P2_K_LOW_LED	K_P2_LOW.1 → R_P2_K_LOW_LED.2	(12,27) → (22,28)
W59 / P2_K_LOW_CATHODE	J_LINK_14.1 → K_P2_LOW.2	(1,18) → (12,28)
W60 / USB_SMART_3V3	C_P2_BUS_A.1 → J_LINK_10.1	(10,31) → (1,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP28 — bank-a-position-2

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

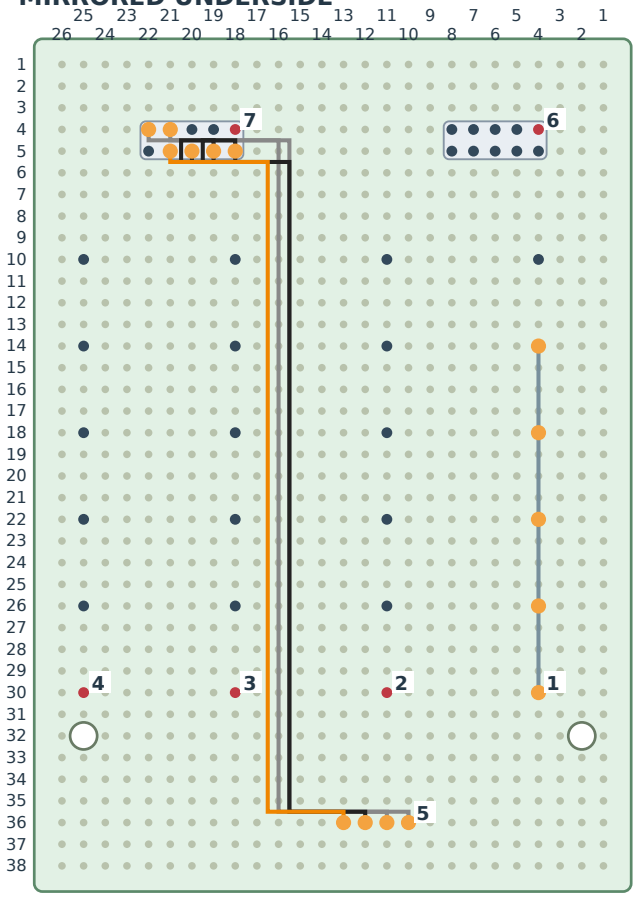
ID / net	From → to	TOP holes
W61 / USB_SMART_3V3	J_LINK_10.1 → U_P2_BUS.1	(1,14) → (4,29)
W62 / USB_SMART_3V3	U_P2_BUS.1 → U_P2_BUS.5	(4,29) → (7,32)
W63 / SCL_A	J_LINK_11.1 → U_P2_BUS.2	(1,15) → (4,30)
W64 / SDA_2	J_LINK_12.1 → U_P2_BUS.3	(1,16) → (4,31)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — b1

Install these wires after their endpoint components; inspect each net with power isolated.

b1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

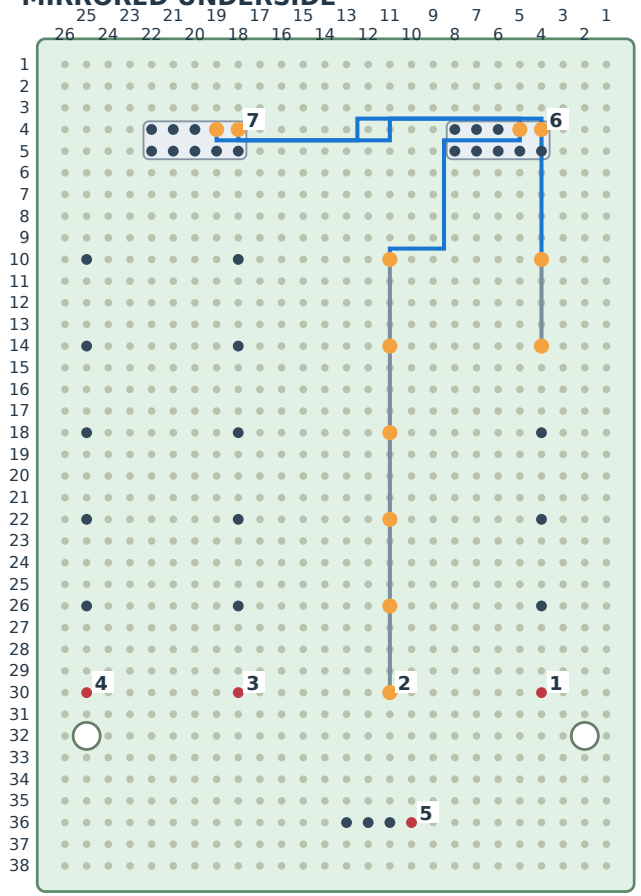
ID / net	From → to	TOP holes
W1 / SCL_POSITION_3	J_SERVICE.1 → J_SMART.9	(10,36) → (22,4)
W2 / SDA_POSITION_3	J_SERVICE.2 → J_SMART.7	(11,36) → (21,4)
W3 / GND	J_SERVICE.3 → J_SMART.2	(12,36) → (18,5)
W4 / GND	J_SMART.2 → J_SMART.4	(18,5) → (19,5)
W5 / GND	J_SMART.4 → J_SMART.6	(19,5) → (20,5)
W6 / 3V3_SYS	J_SERVICE.4 → J_SMART.8	(13,36) → (21,5)
W7 / NODE11	J_NODE11.1 → J_NODE11.2	(4,30) → (4,26)
W8 / NODE11	J_NODE11.2 → J_NODE11.3	(4,26) → (4,22)
W9 / NODE11	J_NODE11.3 → J_NODE11.4	(4,22) → (4,18)
W10 / NODE11	J_NODE11.4 → J_NODE11.5	(4,18) → (4,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — b1

Install these wires after their endpoint components; inspect each net with power isolated.

b1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

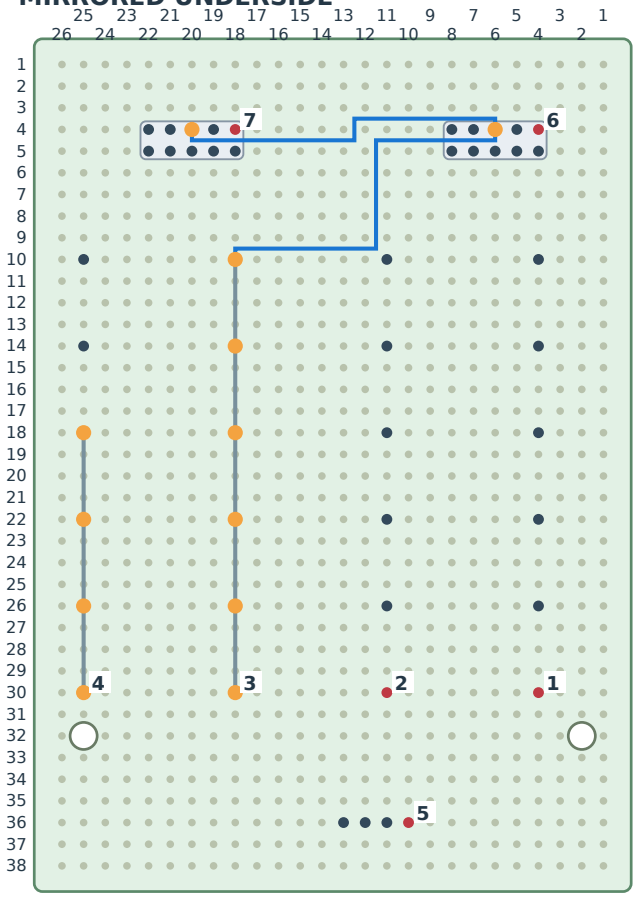
ID / net	From → to	TOP holes
W11 / NODE11	J_NODE11.5 → J_NODE11.6	(4,14) → (4,10)
W12 / NODE11	J_NODE11.6 → J_NODE.1	(4,10) → (4,4)
W13 / NODE11	J_NODE.1 → J_SMART.1	(4,4) → (18,4)
W14 / NODE12	J_NODE12.1 → J_NODE12.2	(11,30) → (11,26)
W15 / NODE12	J_NODE12.2 → J_NODE12.3	(11,26) → (11,22)
W16 / NODE12	J_NODE12.3 → J_NODE12.4	(11,22) → (11,18)
W17 / NODE12	J_NODE12.4 → J_NODE12.5	(11,18) → (11,14)
W18 / NODE12	J_NODE12.5 → J_NODE12.6	(11,14) → (11,10)
W19 / NODE12	J_NODE12.6 → J_NODE.3	(11,10) → (5,4)
W20 / NODE12	J_NODE.3 → J_SMART.3	(5,4) → (19,4)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — b1

Install these wires after their endpoint components; inspect each net with power isolated.

b1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

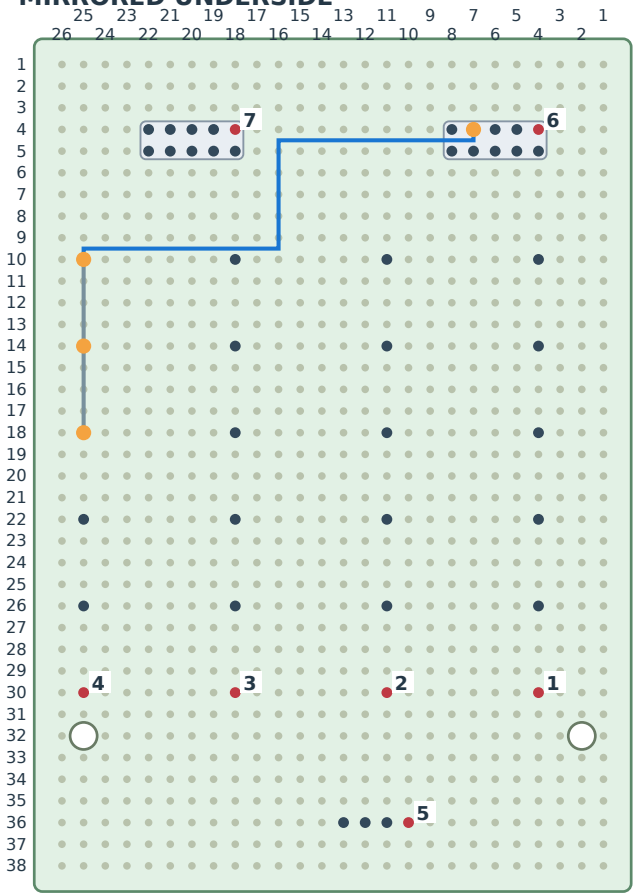
ID / net	From → to	TOP holes
W21 / NODE13	J_NODE13.1 → J_NODE13.2	(18,30) → (18,26)
W22 / NODE13	J_NODE13.2 → J_NODE13.3	(18,26) → (18,22)
W23 / NODE13	J_NODE13.3 → J_NODE13.4	(18,22) → (18,18)
W24 / NODE13	J_NODE13.4 → J_NODE13.5	(18,18) → (18,14)
W25 / NODE13	J_NODE13.5 → J_NODE13.6	(18,14) → (18,10)
W26 / NODE13	J_NODE13.6 → J_NODE.5	(18,10) → (6,4)
W27 / NODE13	J_NODE.5 → J_SMART.5	(6,4) → (20,4)
W28 / NODE14	J_NODE14.1 → J_NODE14.2	(25,30) → (25,26)
W29 / NODE14	J_NODE14.2 → J_NODE14.3	(25,26) → (25,22)
W30 / NODE14	J_NODE14.3 → J_NODE14.4	(25,22) → (25,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — b1

Install these wires after their endpoint components; inspect each net with power isolated.

b1 — MIRRORED UNDERSIDE



ID / net	From → to	TOP holes
W31 / NODE14	J_NODE14.4 → J_NODE14.5	(25,18) → (25,14)
W32 / NODE14	J_NODE14.5 → J_NODE14.6	(25,14) → (25,10)
W33 / NODE14	J_NODE14.6 → J_NODE.7	(25,10) → (7,4)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

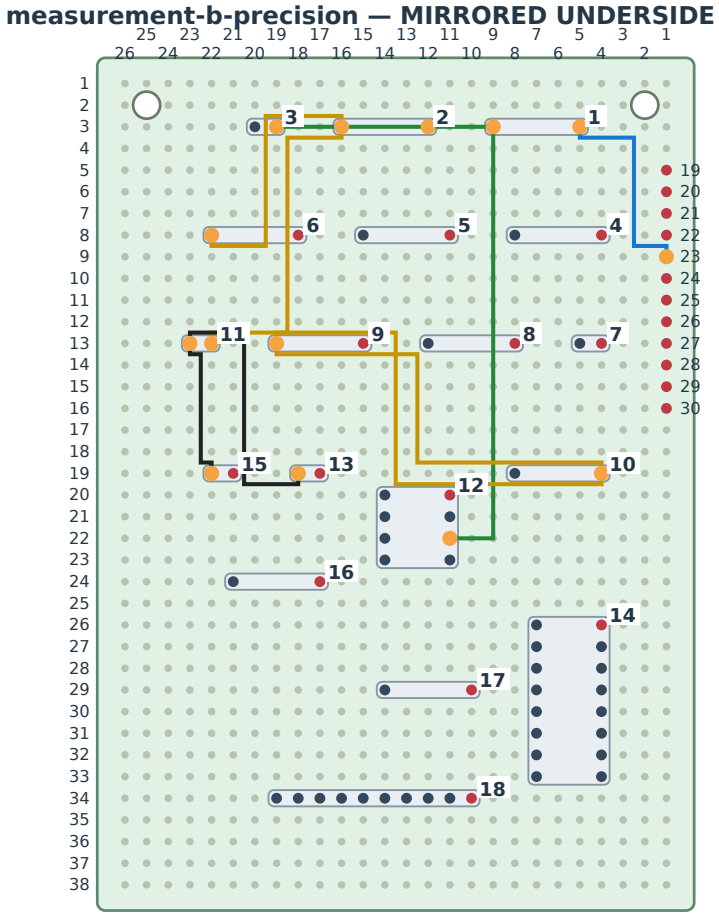
70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP29 — measurement-b-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

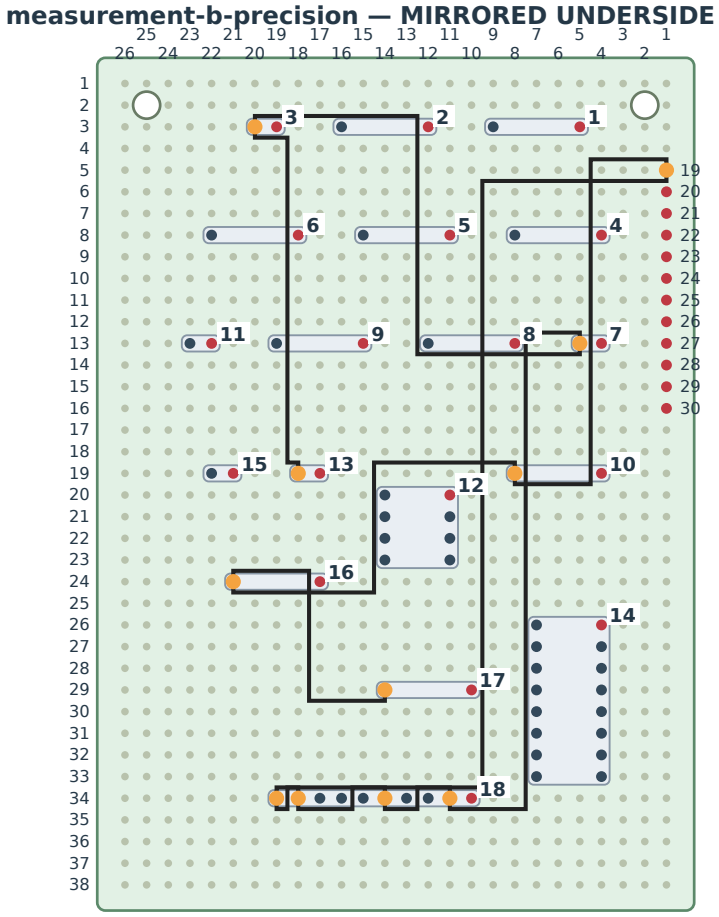
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / NODE11	J_LINK_5.1 → R_P3_1_TOP.1	(1,9) → (5,3)
W2 / P3_1_DIV	C_P3_1.1 → R_P3_1_BOTTOM.1	(19,3) → (12,3)
W3 / P3_1_DIV	R_P3_1_BOTTOM.1 → R_P3_1_TOP.2	(12,3) → (9,3)
W4 / P3_1_DIV	R_P3_1_TOP.2 → U_B_PRECISION.3	(9,3) → (11,22)
W5 / B_PREC_BIAS	C_B_PREC_BIAS.1 → R_B_PREC_BIAS_BOTTOM.1	(22,13) → (4,19)
W6 / B_PREC_BIAS	R_B_PREC_BIAS_BOTTOM.1 → R_B_PREC_BIAS_TOP.2	(4,19) → (19,13)
W7 / B_PREC_BIAS	R_B_PREC_BIAS_TOP.2 → R_P3_1_BOTTOM.2	(19,13) → (16,3)
W8 / B_PREC_BIAS	R_P3_1_BOTTOM.2 → R_P3_2_BOTTOM.2	(16,3) → (22,8)
W9 / GND	C_B_MUX.2 → C_B_PREC_BIAS.2	(22,19) → (23,13)
W10 / GND	C_B_PREC_BIAS.2 → C_B_PRECISION.2	(23,13) → (18,19)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — measurement-b-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

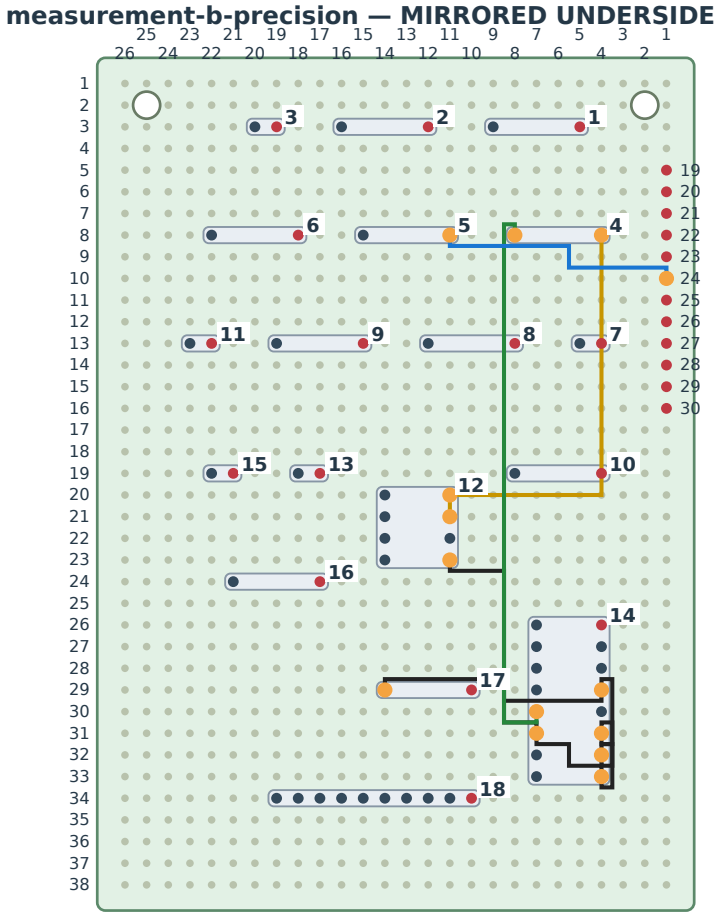
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	C_B_PRECISION.2 → C_P3_1.2	(18,19) → (20,3)
W12 / GND	C_P3_1.2 → C_P3_2.2	(20,3) → (5,13)
W13 / GND	C_P3_2.2 → J_B_ADC_MODULE.2	(5,13) → (11,34)
W14 / GND	J_B_ADC_MODULE.2 → J_B_ADC_MODULE.5	(11,34) → (14,34)
W15 / GND	J_B_ADC_MODULE.5 → J_B_ADC_MODULE.9	(14,34) → (18,34)
W16 / GND	J_B_ADC_MODULE.9 → J_B_ADC_MODULE.10	(18,34) → (19,34)
W17 / GND	J_B_ADC_MODULE.10 → J_LINK_1.1	(19,34) → (1,5)
W18 / GND	J_LINK_1.1 → R_B_PREC_BIAS_BOTTOM.2	(1,5) → (8,19)
W19 / GND	R_B_PREC_BIAS_BOTTOM.2 → R_B_SEL_A.2	(8,19) → (21,24)
W20 / GND	R_B_SEL_A.2 → R_B_SEL_B.2	(21,24) → (14,29)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — measurement-b-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

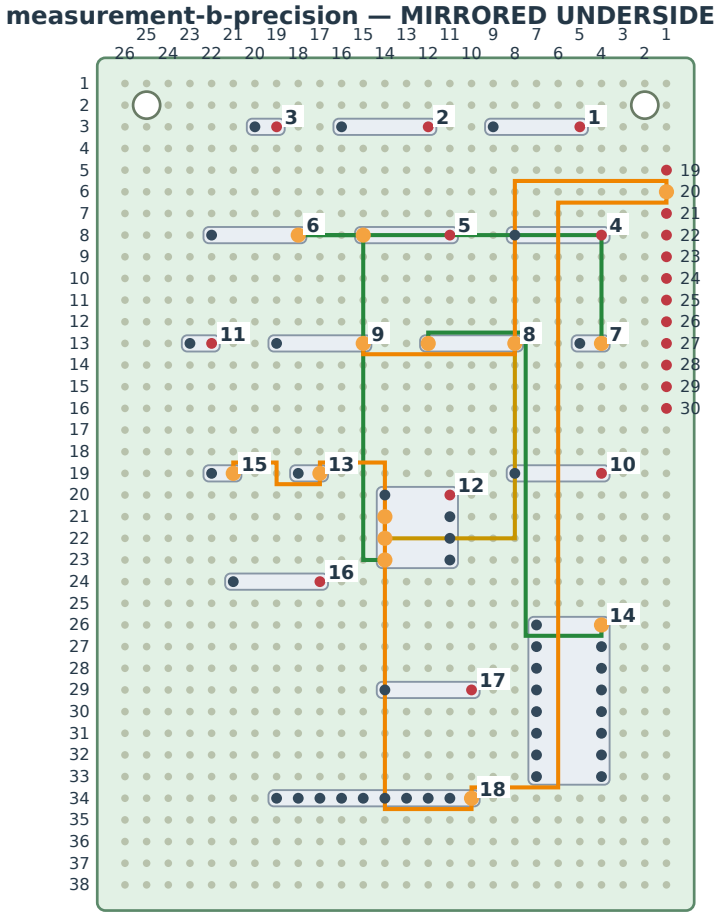
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / GND	R_B_SEL_B.2 → U_B_MUX.4	(14,29) → (4,29)
W22 / GND	U_B_MUX.4 → U_B_MUX.6	(4,29) → (4,31)
W23 / GND	U_B_MUX.6 → U_B_MUX.7	(4,31) → (4,32)
W24 / GND	U_B_MUX.7 → U_B_MUX.8	(4,32) → (4,33)
W25 / GND	U_B_MUX.8 → U_B_MUX.11	(4,33) → (7,31)
W26 / GND	U_B_MUX.11 → U_B_PRECISION.4	(7,31) → (11,23)
W27 / P3_1_BUF	R_P3_1_ADC.1 → U_B_PRECISION.1	(4,8) → (11,20)
W28 / P3_1_BUF	U_B_PRECISION.1 → U_B_PRECISION.2	(11,20) → (11,21)
W29 / P3_1_MUX	R_P3_1_ADC.2 → U_B_MUX.12	(8,8) → (7,30)
W30 / NODE12	J_LINK_6.1 → R_P3_2_TOP.1	(1,10) → (11,8)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — measurement-b-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

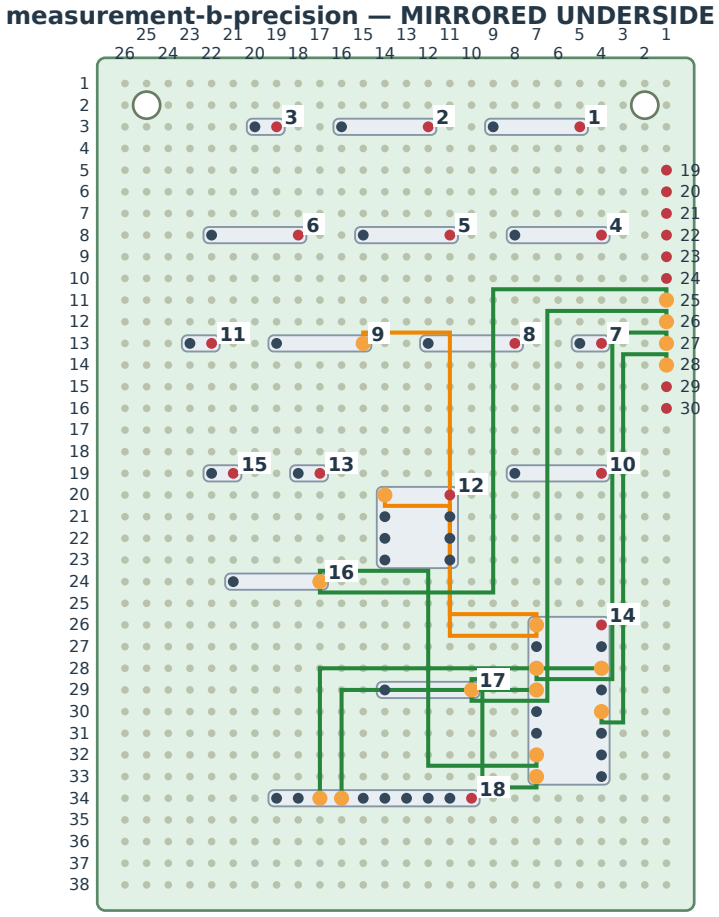
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / P3_2_DIV	C_P3_2.1 → R_P3_2_BOTTOM.1	(4,13) → (18,8)
W32 / P3_2_DIV	R_P3_2_BOTTOM.1 → R_P3_2_TOP.2	(18,8) → (15,8)
W33 / P3_2_DIV	R_P3_2_TOP.2 → U_B_PRECISION.5	(15,8) → (14,23)
W34 / P3_2_BUF	R_P3_2_ADC.1 → U_B_PRECISION.6	(8,13) → (14,22)
W35 / P3_2_BUF	U_B_PRECISION.6 → U_B_PRECISION.7	(14,22) → (14,21)
W36 / P3_2_MUX	R_P3_2_ADC.2 → U_B_MUX.1	(12,13) → (4,26)
W37 / 3V3_SYS	C_B_MUX.1 → C_B_PRECISION.1	(21,19) → (17,19)
W38 / 3V3_SYS	C_B_PRECISION.1 → J_B_ADC_MODULE.1	(17,19) → (10,34)
W39 / 3V3_SYS	J_B_ADC_MODULE.1 → J_LINK_2.1	(10,34) → (1,6)
W40 / 3V3_SYS	J_LINK_2.1 → R_B_PREC_BIAS_TOP.1	(1,6) → (15,13)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — measurement-b-precision

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

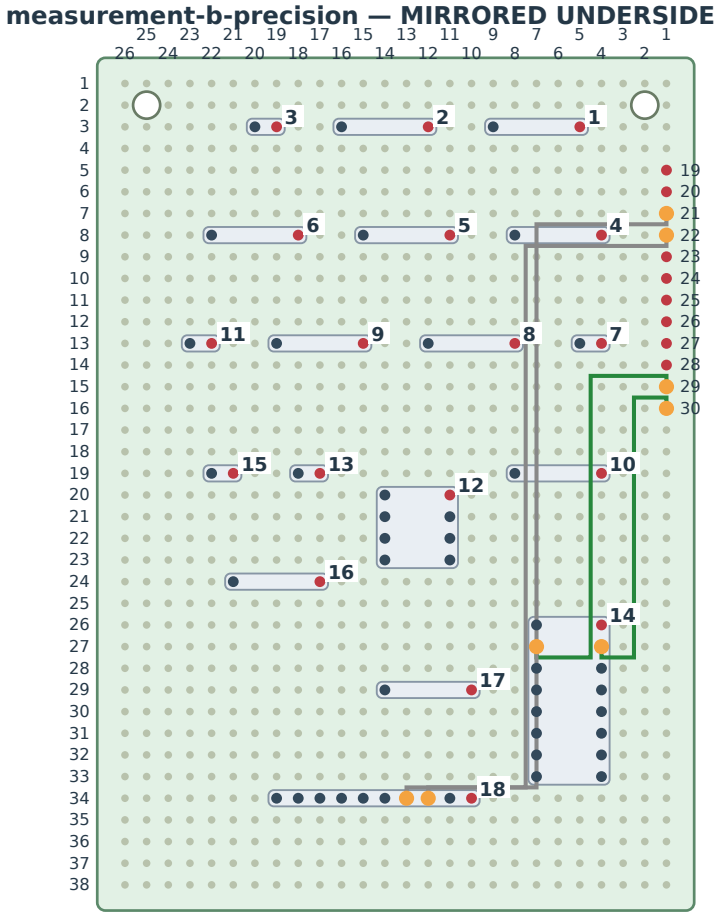
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / 3V3_SYS	R_B_PREC_BIAS_TOP.1 → U_B_MUX.16	(15,13) → (7,26)
W42 / 3V3_SYS	U_B_MUX.16 → U_B_PRECISION.8	(7,26) → (14,20)
W43 / ADC_B_P2	J_B_ADC_MODULE.8 → U_B_MUX.3	(17,34) → (4,28)
W44 / ADC_B_P1	J_B_ADC_MODULE.7 → U_B_MUX.13	(16,34) → (7,29)
W45 / MUX_B_A	J_LINK_7.1 → R_B_SEL_A.1	(1,11) → (17,24)
W46 / MUX_B_A	R_B_SEL_A.1 → U_B_MUX.10	(17,24) → (7,32)
W47 / MUX_B_B	J_LINK_8.1 → R_B_SEL_B.1	(1,12) → (10,29)
W48 / MUX_B_B	R_B_SEL_B.1 → U_B_MUX.9	(10,29) → (7,33)
W49 / P4_1_MUX	J_LINK_9.1 → U_B_MUX.14	(1,13) → (7,28)
W50 / P4_2_MUX	J_LINK_10.1 → U_B_MUX.5	(1,14) → (4,30)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — measurement-b-precision

Install these wires after their endpoint components; inspect each net with power isolated.



ID / net	From → to	TOP holes
W51 / P5_1_MUX	J_LINK_11.1 → U_B_MUX.15	(1,15) → (7,27)
W52 / P5_2_MUX	J_LINK_12.1 → U_B_MUX.2	(1,16) → (4,27)
W53 / SCL_POSITION_3	J_B_ADC_MODULE.3 → J_LINK_3.1	(12,34) → (1,7)
W54 / SDA_POSITION_3	J_B_ADC_MODULE.4 → J_LINK_4.1	(13,34) → (1,8)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

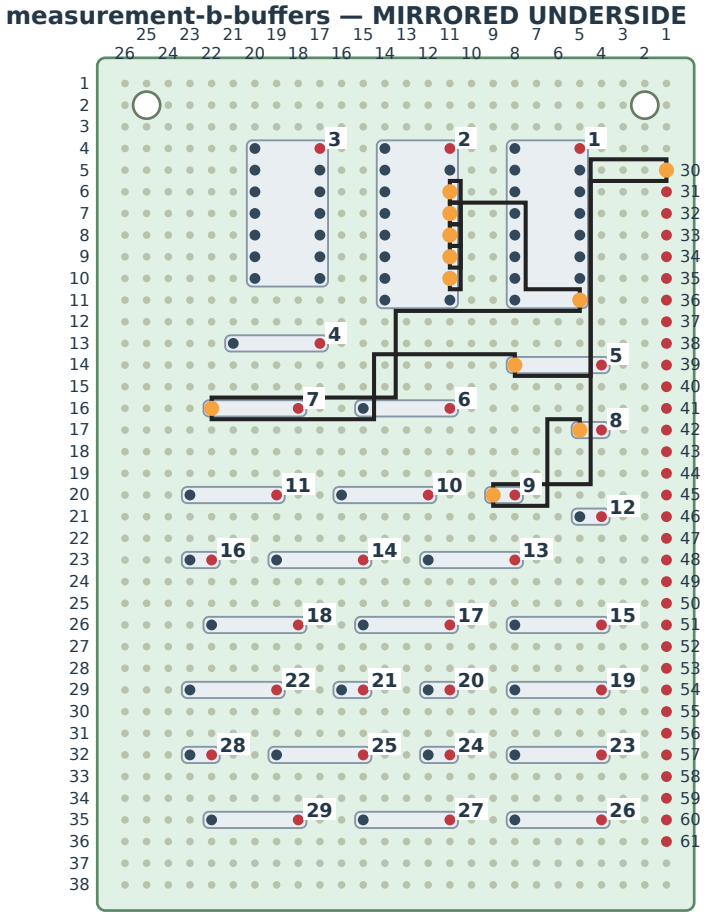
70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP29 — measurement-b-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

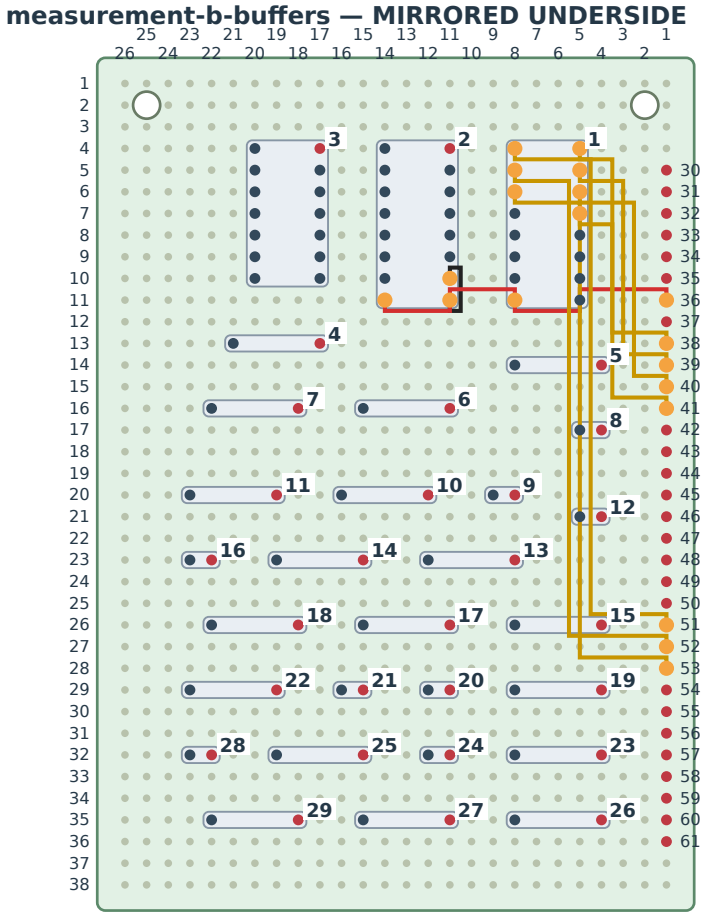
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / GND	C_B_REFHI.2 → C_B_REFLO.2	(5,17) → (9,20)
W2 / GND	C_B_REFLO.2 → J_LINK_1.1	(9,20) → (1,5)
W3 / GND	J_LINK_1.1 → R_B_HI_BOTTOM.2	(1,5) → (8,14)
W4 / GND	R_B_HI_BOTTOM.2 → R_B_LO_BOTTOM.2	(8,14) → (22,16)
W5 / GND	R_B_LO_BOTTOM.2 → U_B_LED_DRIVER_0.8	(22,16) → (5,11)
W6 / GND	U_B_LED_DRIVER_0.8 → U_B_LED_DRIVER_1.3	(5,11) → (11,6)
W7 / GND	U_B_LED_DRIVER_1.3 → U_B_LED_DRIVER_1.4	(11,6) → (11,7)
W8 / GND	U_B_LED_DRIVER_1.4 → U_B_LED_DRIVER_1.5	(11,7) → (11,8)
W9 / GND	U_B_LED_DRIVER_1.5 → U_B_LED_DRIVER_1.6	(11,8) → (11,9)
W10 / GND	U_B_LED_DRIVER_1.6 → U_B_LED_DRIVER_1.7	(11,9) → (11,10)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — measurement-b-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

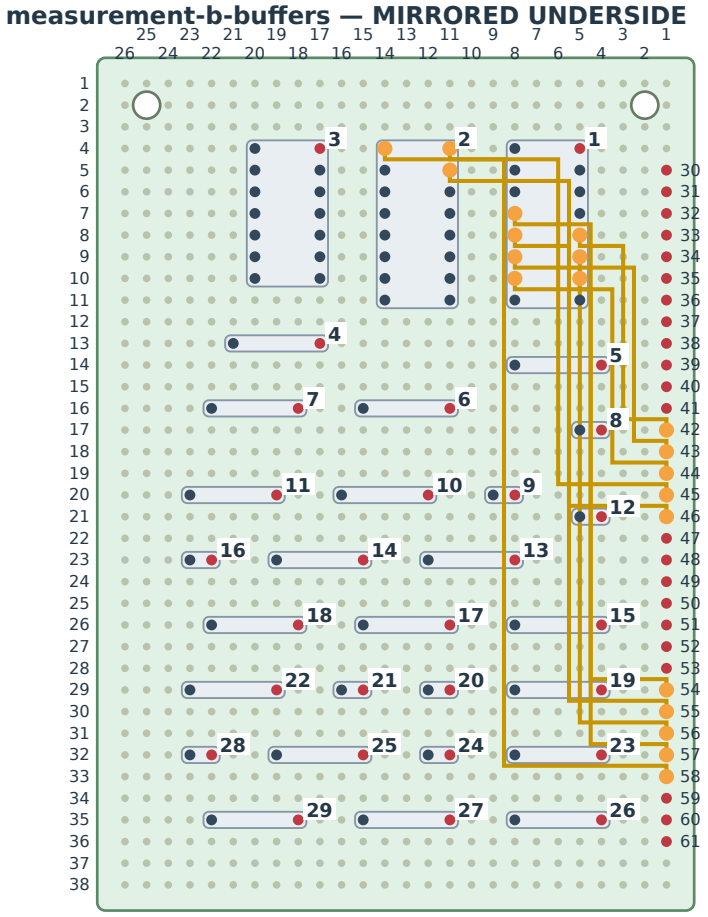
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	U_B_LED_DRIVER_1.7 → U_B_LED_DRIVER_1.8	(11,10) → (11,11)
W20 / 5V_SYS	J_LINK_7.1 → U_B_LED_DRIVER_0.9	(1,11) → (8,11)
W21 / 5V_SYS	U_B_LED_DRIVER_0.9 → U_B_LED_DRIVER_1.9	(8,11) → (14,11)
W22 / RELAY_9	J_LINK_9.1 → U_B_LED_DRIVER_0.1	(1,13) → (5,4)
W23 / P3_K_HIGH_CATHODE	J_LINK_22.1 → U_B_LED_DRIVER_0.16	(1,26) → (8,4)
W24 / RELAY_10	J_LINK_10.1 → U_B_LED_DRIVER_0.2	(1,14) → (5,5)
W25 / P3_K_LOW_CATHODE	J_LINK_23.1 → U_B_LED_DRIVER_0.15	(1,27) → (8,5)
W26 / RELAY_11	J_LINK_11.1 → U_B_LED_DRIVER_0.3	(1,15) → (5,6)
W27 / P3_AMM_CATHODE	J_LINK_24.1 → U_B_LED_DRIVER_0.14	(1,28) → (8,6)
W28 / RELAY_12	J_LINK_12.1 → U_B_LED_DRIVER_0.4	(1,16) → (5,7)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — measurement-b-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

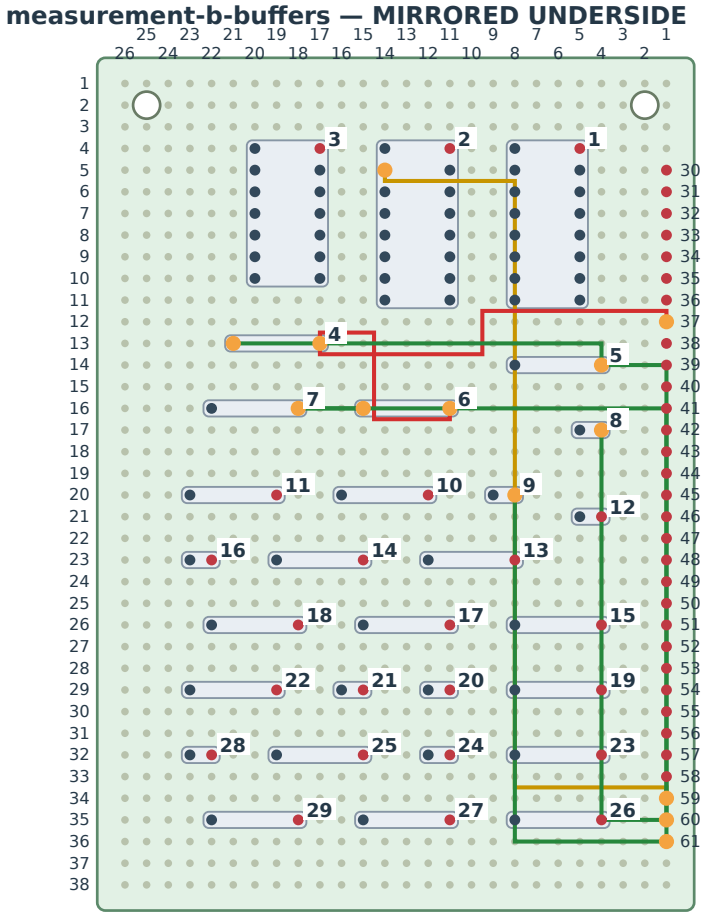
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W29 / P4_K_HIGH_CATHODE	J_LINK_25.1 → U_B_LED_DRIVER_0.13	(1,29) → (8,7)
W30 / RELAY_13	J_LINK_13.1 → U_B_LED_DRIVER_0.5	(1,17) → (5,8)
W31 / P4_K_LOW_CATHODE	J_LINK_26.1 → U_B_LED_DRIVER_0.12	(1,30) → (8,8)
W32 / RELAY_14	J_LINK_14.1 → U_B_LED_DRIVER_0.6	(1,18) → (5,9)
W33 / P4_AMM_CATHODE	J_LINK_27.1 → U_B_LED_DRIVER_0.11	(1,31) → (8,9)
W34 / RELAY_15	J_LINK_15.1 → U_B_LED_DRIVER_0.7	(1,19) → (5,10)
W35 / P5_K_HIGH_CATHODE	J_LINK_28.1 → U_B_LED_DRIVER_0.10	(1,32) → (8,10)
W36 / RELAY_16	J_LINK_16.1 → U_B_LED_DRIVER_1.1	(1,20) → (11,4)
W37 / P5_K_LOW_CATHODE	J_LINK_29.1 → U_B_LED_DRIVER_1.16	(1,33) → (14,4)
W38 / RELAY_17	J_LINK_17.1 → U_B_LED_DRIVER_1.2	(1,21) → (11,5)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — measurement-b-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

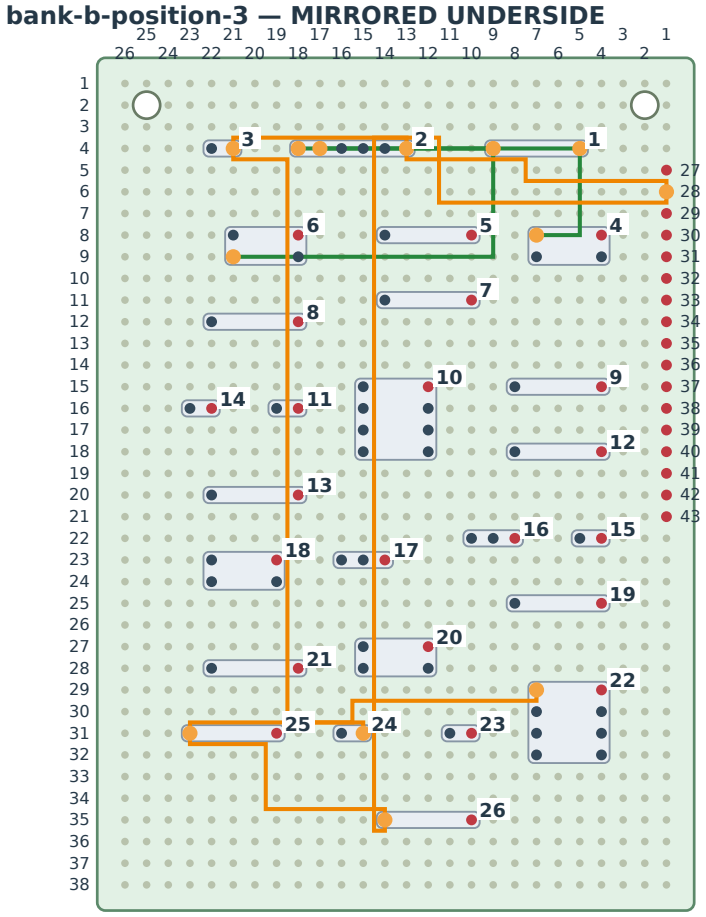
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W39 / P5_AMM_CATHODE	J_LINK_30.1 → U_B_LED_DRIVER_1.15	(1,34) → (14,5)
W63 / 5V_EDU	J_LINK_8.1 → R_B_HI_TOP.1	(1,12) → (17,13)
W64 / 5V_EDU	R_B_HI_TOP.1 → R_B_LO_TOP.1	(17,13) → (11,16)
W65 / B_REF_HI	C_B_REFHI.1 → J_LINK_31.1	(4,17) → (1,35)
W66 / B_REF_HI	J_LINK_31.1 → R_B_HI_BOTTOM.1	(1,35) → (4,14)
W67 / B_REF_HI	R_B_HI_BOTTOM.1 → R_B_HI_TOP.2	(4,14) → (21,13)
W68 / B_REF_LO	C_B_REFLO.1 → J_LINK_32.1	(8,20) → (1,36)
W69 / B_REF_LO	J_LINK_32.1 → R_B_LO_BOTTOM.1	(1,36) → (18,16)
W70 / B_REF_LO	R_B_LO_BOTTOM.1 → R_B_LO_TOP.2	(18,16) → (15,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — bank-b-position-3

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

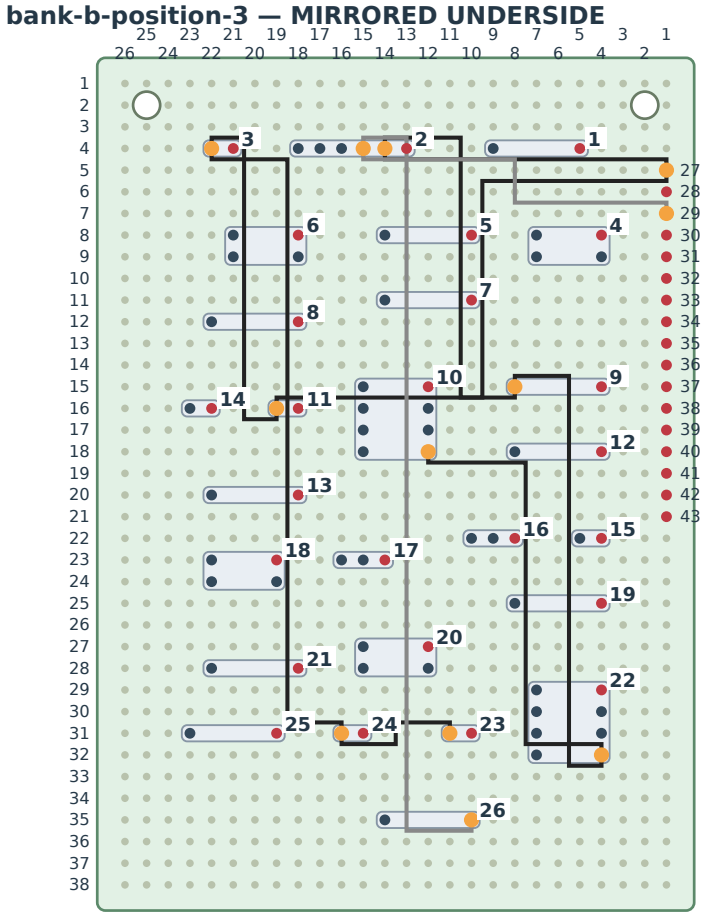
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / P3_SHUNT_P	R_P3_SHUNT.1 → J_P3_CURRENT_MODULE.5	(5,4) → (17,4)
W2 / P3_SHUNT_P	R_P3_SHUNT.1 → K_P3_AMM_P.4	(5,4) → (7,8)
W3 / P3_SHUNT_N	R_P3_SHUNT.2 → J_P3_CURRENT_MODULE.6	(9,4) → (18,4)
W4 / P3_SHUNT_N	R_P3_SHUNT.2 → K_P3_AMM_N.3	(9,4) → (21,9)
W5 / 3V3_SYS	C_P3_BUS_B.1 → C_P3_CURRENT.1	(15,31) → (21,4)
W6 / 3V3_SYS	C_P3_CURRENT.1 → J_LINK_2.1	(21,4) → (1,6)
W7 / 3V3_SYS	J_LINK_2.1 → J_P3_CURRENT_MODULE.1	(1,6) → (13,4)
W8 / 3V3_SYS	J_P3_CURRENT_MODULE.1 → R_P3_SCL.2	(13,4) → (14,35)
W9 / 3V3_SYS	R_P3_SCL.2 → R_P3_SDA.2	(14,35) → (23,31)
W10 / 3V3_SYS	R_P3_SDA.2 → U_P3_BUS.8	(23,31) → (7,29)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — bank-b-position-3

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

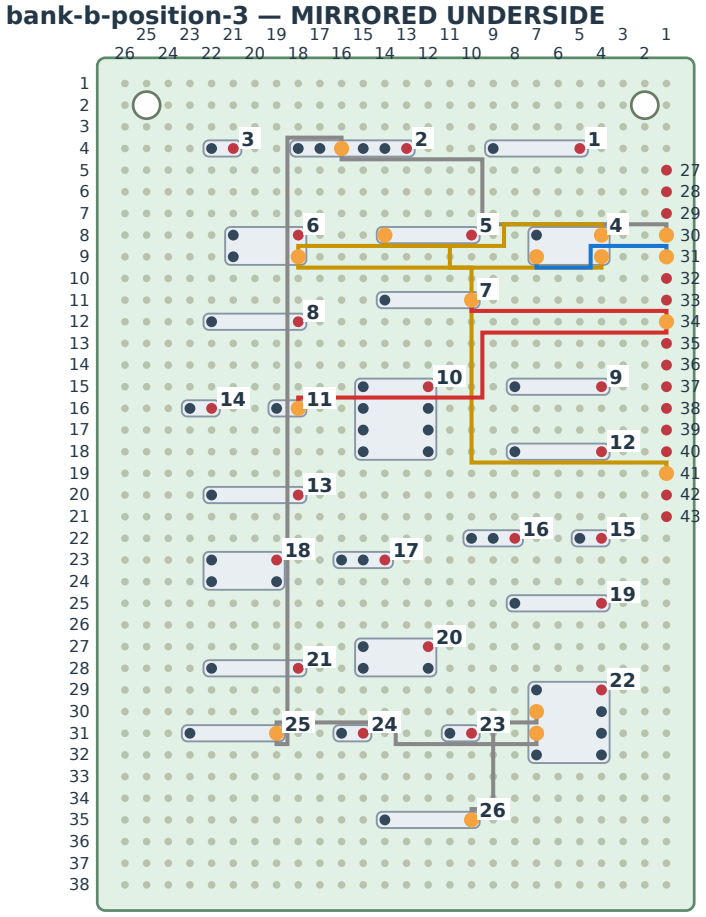
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	C_P3_BUS_A.2 → C_P3_BUS_B.2	(11,31) → (16,31)
W12 / GND	C_P3_BUS_B.2 → C_P3_CURRENT.2	(16,31) → (22,4)
W13 / GND	C_P3_CURRENT.2 → C_P3_LIMIT.2	(22,4) → (19,16)
W14 / GND	C_P3_LIMIT.2 → J_LINK_1.1	(19,16) → (1,5)
W15 / GND	J_LINK_1.1 → J_P3_CURRENT_MODULE.2	(1,5) → (14,4)
W16 / GND	J_P3_CURRENT_MODULE.2 → R_P3_SINK_SENSE.2	(14,4) → (8,15)
W17 / GND	R_P3_SINK_SENSE.2 → U_P3_BUS.4	(8,15) → (4,32)
W18 / GND	U_P3_BUS.4 → U_P3_LIMIT.4	(4,32) → (12,18)
W19 / SCL_POSITION_3	J_LINK_3.1 → J_P3_CURRENT_MODULE.3	(1,7) → (15,4)
W20 / SCL_POSITION_3	J_P3_CURRENT_MODULE.3 → R_P3_SCL.1	(15,4) → (10,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — bank-b-position-3

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

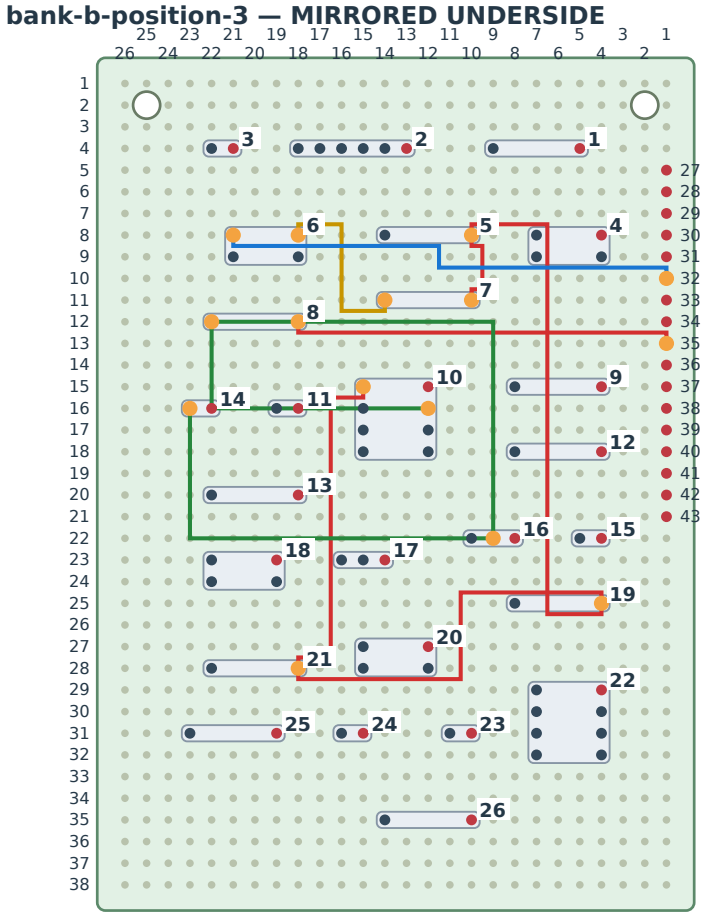
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / SCL_POSITION_3	R_P3_SCL.1 → U_P3_BUS.7	(10,35) → (7,30)
W22 / SDA_POSITION_3	J_LINK_4.1 → J_P3_CURRENT_MODULE.4	(1,8) → (16,4)
W23 / SDA_POSITION_3	J_P3_CURRENT_MODULE.4 → R_P3_SDA.1	(16,4) → (19,31)
W24 / SDA_POSITION_3	R_P3_SDA.1 → U_P3_BUS.6	(19,31) → (7,31)
W25 / P3_K_AMM_P_LED	K_P3_AMM_P.1 → R_P3_K_AMM_P_LED.2	(4,8) → (14,8)
W26 / P3_AMM_CATHODE	J_LINK_15.1 → K_P3_AMM_N.2	(1,19) → (18,9)
W27 / P3_AMM_CATHODE	K_P3_AMM_N.2 → K_P3_AMM_P.2	(18,9) → (4,9)
W28 / NODE11	J_LINK_5.1 → K_P3_AMM_P.3	(1,9) → (7,9)
W29 / 5V_SYS	C_P3_LIMIT.1 → J_LINK_8.1	(18,16) → (1,12)
W30 / 5V_SYS	J_LINK_8.1 → R_P3_K_AMM_N_LED.1	(1,12) → (10,11)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — bank-b-position-3

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

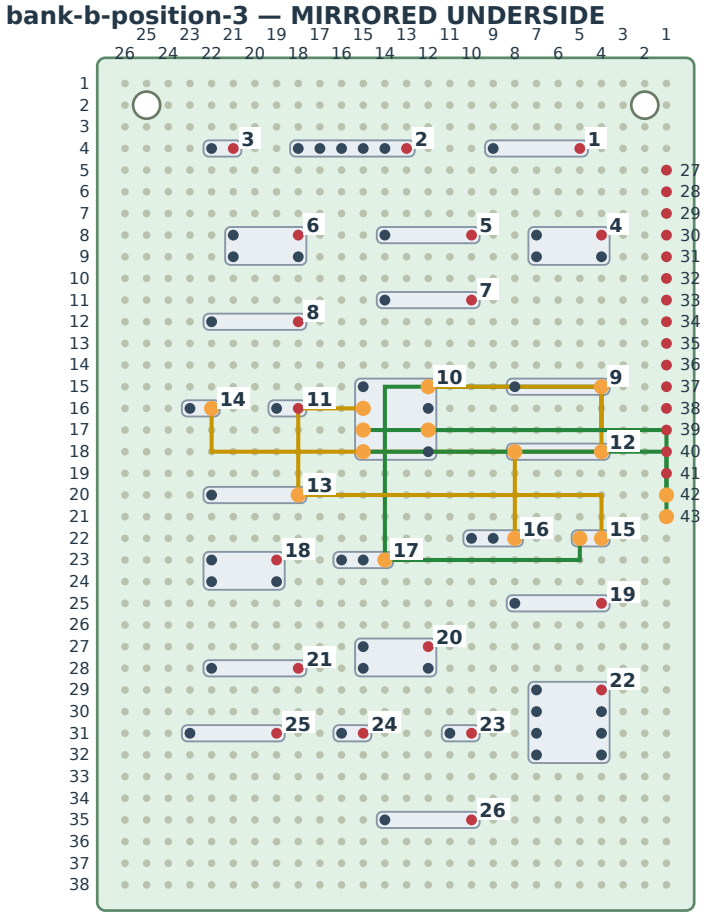
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / 5V_SYS	R_P3_K_AMM_N_LED.1 → R_P3_K_AMM_P_LED.1	(10,11) → (10,8)
W32 / 5V_SYS	R_P3_K_AMM_P_LED.1 → R_P3_K_HIGH_LED.1	(10,8) → (4,25)
W33 / 5V_SYS	R_P3_K_HIGH_LED.1 → R_P3_K_LOW_LED.1	(4,25) → (18,28)
W34 / 5V_SYS	R_P3_K_LOW_LED.1 → U_P3_LIMIT.8	(18,28) → (15,15)
W35 / P3_K_AMM_N_LED	K_P3_AMM_N.1 → R_P3_K_AMM_N_LED.2	(18,8) → (14,11)
W36 / NODE12	J_LINK_6.1 → K_P3_AMM_N.4	(1,10) → (21,8)
W37 / 5V_EDU	J_LINK_9.1 → R_P3_SOURCE_SENSE.1	(1,13) → (18,12)
W38 / P3_SOURCE_SENSE	C_P3_HI_COMP.2 → Q_P3_HIGH.2	(23,16) → (9,22)
W39 / P3_SOURCE_SENSE	Q_P3_HIGH.2 → R_P3_SOURCE_SENSE.2	(9,22) → (22,12)
W40 / P3_SOURCE_SENSE	R_P3_SOURCE_SENSE.2 → U_P3_LIMIT.2	(22,12) → (12,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — bank-b-position-3

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

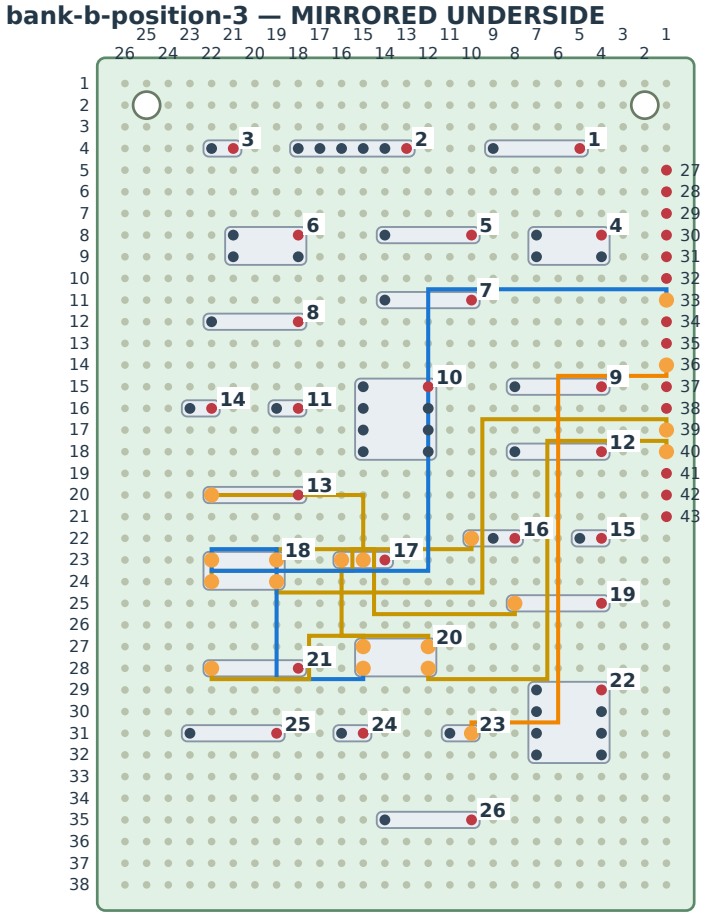
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / P3_SINK_SENSE	C_P3_LO_COMP.2 → Q_P3_LOW.1	(5,22) → (14,23)
W42 / P3_SINK_SENSE	Q_P3_LOW.1 → R_P3_SINK_SENSE.1	(14,23) → (4,15)
W43 / P3_SINK_SENSE	R_P3_SINK_SENSE.1 → U_P3_LIMIT.6	(4,15) → (15,17)
W44 / P3_HI_GATE_DRIVE	C_P3_HI_COMP.1 → R_P3_HI_GATE.1	(22,16) → (4,18)
W45 / P3_HI_GATE_DRIVE	R_P3_HI_GATE.1 → U_P3_LIMIT.1	(4,18) → (12,15)
W46 / B_REF_HI	J_LINK_16.1 → U_P3_LIMIT.3	(1,20) → (12,17)
W47 / B_REF_LO	J_LINK_17.1 → U_P3_LIMIT.5	(1,21) → (15,18)
W48 / P3_LO_GATE_DRIVE	C_P3_LO_COMP.1 → R_P3_LO_GATE.1	(4,22) → (18,20)
W49 / P3_LO_GATE_DRIVE	R_P3_LO_GATE.1 → U_P3_LIMIT.7	(18,20) → (15,16)
W50 / P3_HI_GATE	Q_P3_HIGH.1 → R_P3_HI_GATE.2	(8,22) → (8,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — bank-b-position-3

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

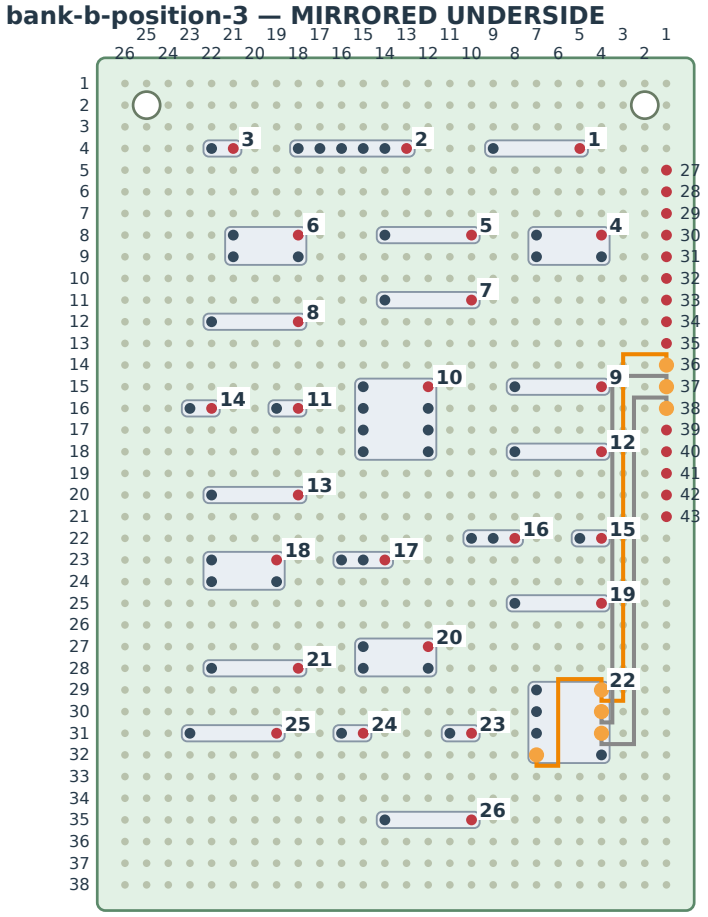
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W51 / P3_LO_GATE	Q_P3_LOW.2 → R_P3_LO_GATE.2	(15,23) → (22,20)
W52 / P3_SOURCE_OUT	K_P3_HIGH.3 → Q_P3_HIGH.3	(22,24) → (10,22)
W53 / P3_SINK_IN	K_P3_LOW.4 → Q_P3_LOW.3	(15,27) → (16,23)
W54 / P3_K_HIGH_LED	K_P3_HIGH.1 → R_P3_K_HIGH_LED.2	(19,23) → (8,25)
W55 / P3_K_HIGH_CATHODE	J_LINK_13.1 → K_P3_HIGH.2	(1,17) → (19,24)
W56 / NODE13	J_LINK_7.1 → K_P3_HIGH.4	(1,11) → (22,23)
W57 / NODE13	K_P3_HIGH.4 → K_P3_LOW.3	(22,23) → (15,28)
W58 / P3_K_LOW_LED	K_P3_LOW.1 → R_P3_K_LOW_LED.2	(12,27) → (22,28)
W59 / P3_K_LOW_CATHODE	J_LINK_14.1 → K_P3_LOW.2	(1,18) → (12,28)
W60 / USB_SMART_3V3	C_P3_BUS_A.1 → J_LINK_10.1	(10,31) → (1,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP29 — bank-b-position-3

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

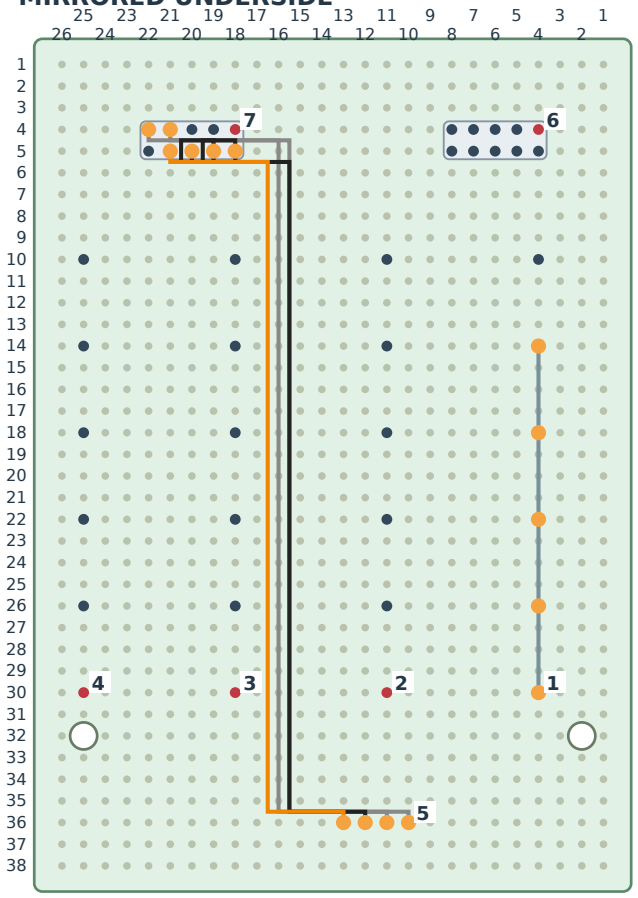
ID / net	From → to	TOP holes
W61 / USB_SMART_3V3	J_LINK_10.1 → U_P3_BUS.1	(1,14) → (4,29)
W62 / USB_SMART_3V3	U_P3_BUS.1 → U_P3_BUS.5	(4,29) → (7,32)
W63 / SCL_A	J_LINK_11.1 → U_P3_BUS.2	(1,15) → (4,30)
W64 / SDA_3	J_LINK_12.1 → U_P3_BUS.3	(1,16) → (4,31)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — b2

Install these wires after their endpoint components; inspect each net with power isolated.

b2 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

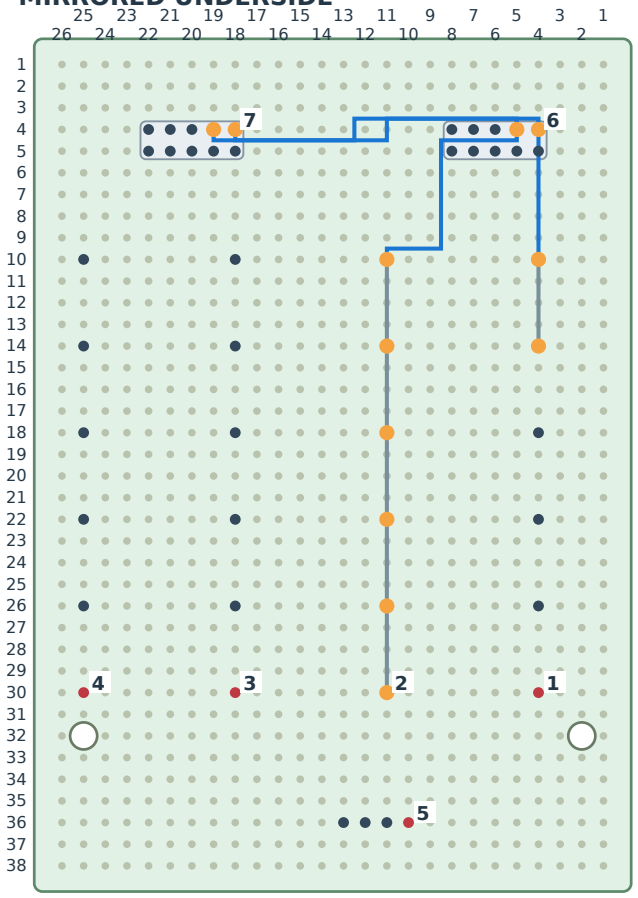
ID / net	From → to	TOP holes
W1 / SCL_POSITION_4	J_SERVICE.1 → J_SMART.9	(10,36) → (22,4)
W2 / SDA_POSITION_4	J_SERVICE.2 → J_SMART.7	(11,36) → (21,4)
W3 / GND	J_SERVICE.3 → J_SMART.2	(12,36) → (18,5)
W4 / GND	J_SMART.2 → J_SMART.4	(18,5) → (19,5)
W5 / GND	J_SMART.4 → J_SMART.6	(19,5) → (20,5)
W6 / 3V3_SYS	J_SERVICE.4 → J_SMART.8	(13,36) → (21,5)
W7 / NODE15	J_NODE15.1 → J_NODE15.2	(4,30) → (4,26)
W8 / NODE15	J_NODE15.2 → J_NODE15.3	(4,26) → (4,22)
W9 / NODE15	J_NODE15.3 → J_NODE15.4	(4,22) → (4,18)
W10 / NODE15	J_NODE15.4 → J_NODE15.5	(4,18) → (4,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — b2

Install these wires after their endpoint components; inspect each net with power isolated.

b2 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

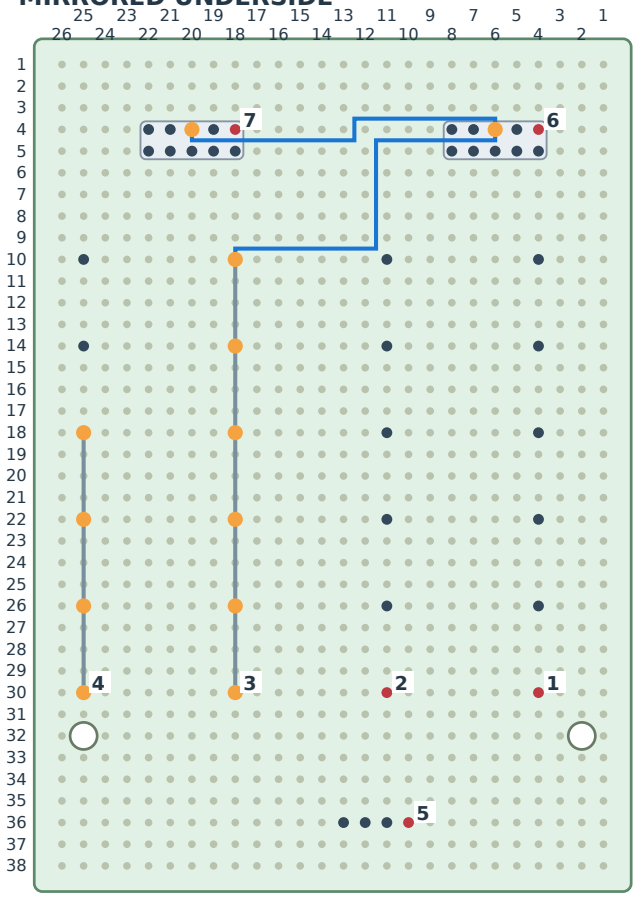
ID / net	From → to	TOP holes
W11 / NODE15	J_NODE15.5 → J_NODE15.6	(4,14) → (4,10)
W12 / NODE15	J_NODE15.6 → J_NODE.1	(4,10) → (4,4)
W13 / NODE15	J_NODE.1 → J_SMART.1	(4,4) → (18,4)
W14 / NODE16	J_NODE16.1 → J_NODE16.2	(11,30) → (11,26)
W15 / NODE16	J_NODE16.2 → J_NODE16.3	(11,26) → (11,22)
W16 / NODE16	J_NODE16.3 → J_NODE16.4	(11,22) → (11,18)
W17 / NODE16	J_NODE16.4 → J_NODE16.5	(11,18) → (11,14)
W18 / NODE16	J_NODE16.5 → J_NODE16.6	(11,14) → (11,10)
W19 / NODE16	J_NODE16.6 → J_NODE.3	(11,10) → (5,4)
W20 / NODE16	J_NODE.3 → J_SMART.3	(5,4) → (19,4)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — b2

Install these wires after their endpoint components; inspect each net with power isolated.

b2 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

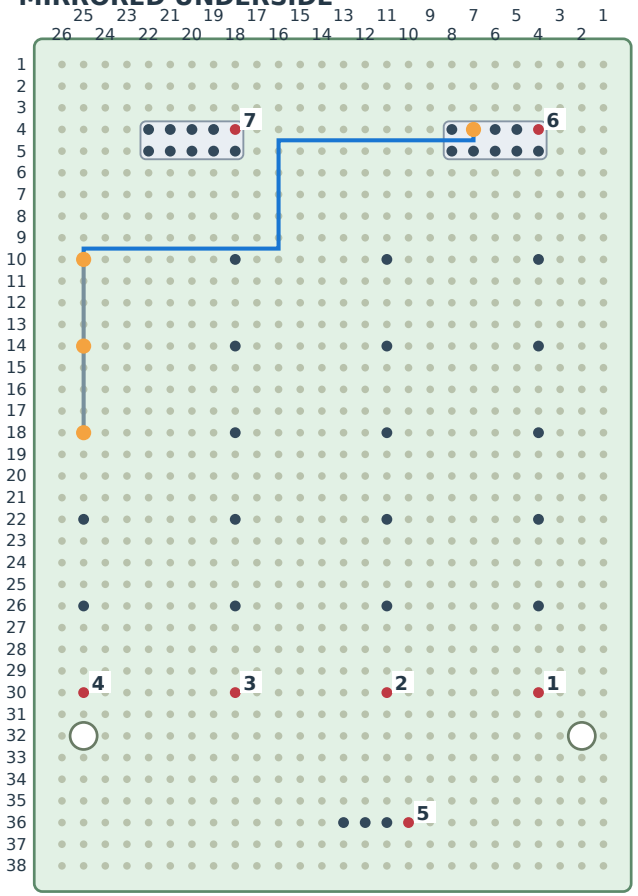
ID / net	From → to	TOP holes
W21 / NODE17	J_NODE17.1 → J_NODE17.2	(18,30) → (18,26)
W22 / NODE17	J_NODE17.2 → J_NODE17.3	(18,26) → (18,22)
W23 / NODE17	J_NODE17.3 → J_NODE17.4	(18,22) → (18,18)
W24 / NODE17	J_NODE17.4 → J_NODE17.5	(18,18) → (18,14)
W25 / NODE17	J_NODE17.5 → J_NODE17.6	(18,14) → (18,10)
W26 / NODE17	J_NODE17.6 → J_NODE.5	(18,10) → (6,4)
W27 / NODE17	J_NODE.5 → J_SMART.5	(6,4) → (20,4)
W28 / NODE18	J_NODE18.1 → J_NODE18.2	(25,30) → (25,26)
W29 / NODE18	J_NODE18.2 → J_NODE18.3	(25,26) → (25,22)
W30 / NODE18	J_NODE18.3 → J_NODE18.4	(25,22) → (25,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — b2

Install these wires after their endpoint components; inspect each net with power isolated.

b2 — MIRRORED UNDERSIDE



ID / net	From → to	TOP holes
W31 / NODE18	J_NODE18.4 → J_NODE18.5	(25,18) → (25,14)
W32 / NODE18	J_NODE18.5 → J_NODE18.6	(25,14) → (25,10)
W33 / NODE18	J_NODE18.6 → J_NODE.7	(25,10) → (7,4)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

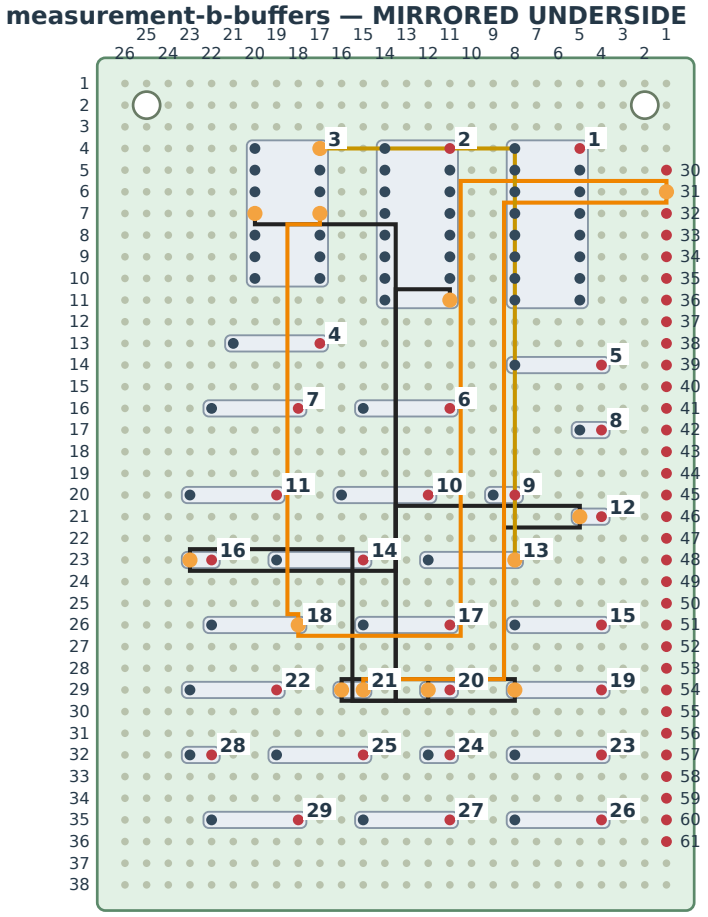
70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP30 — measurement-b-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

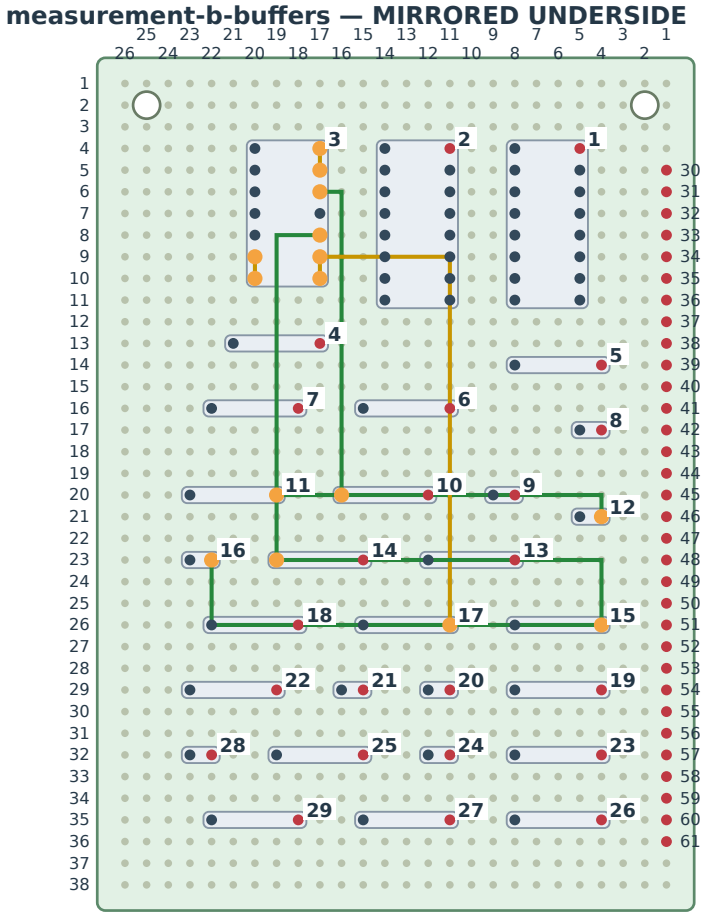
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W12 / GND	U_B_LED_DRIVER_1.8 → C_B_BUFFERS.2	(11,11) → (16,29)
W13 / GND	C_B_BUFFERS.2 → C_B_GENERAL_BIAS.2	(16,29) → (12,29)
W14 / GND	C_B_GENERAL_BIAS.2 → C_P4_1.2	(12,29) → (5,21)
W15 / GND	C_P4_1.2 → C_P4_2.2	(5,21) → (23,23)
W16 / GND	C_P4_2.2 → R_B_GENERAL_BIAS_BOTTOM.2	(23,23) → (8,29)
W17 / GND	R_B_GENERAL_BIAS_BOTTOM.2 → U_B_BUFFERS.11	(8,29) → (20,7)
W40 / 3V3_SYS	C_B_BUFFERS.1 → J_LINK_2.1	(15,29) → (1,6)
W41 / 3V3_SYS	J_LINK_2.1 → R_B_GENERAL_BIAS_TOP.1	(1,6) → (18,26)
W42 / 3V3_SYS	R_B_GENERAL_BIAS_TOP.1 → U_B_BUFFERS.4	(18,26) → (17,7)
W43 / P4_1_BUF	R_P4_1_ADC.1 → U_B_BUFFERS.1	(8,23) → (17,4)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — measurement-b-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

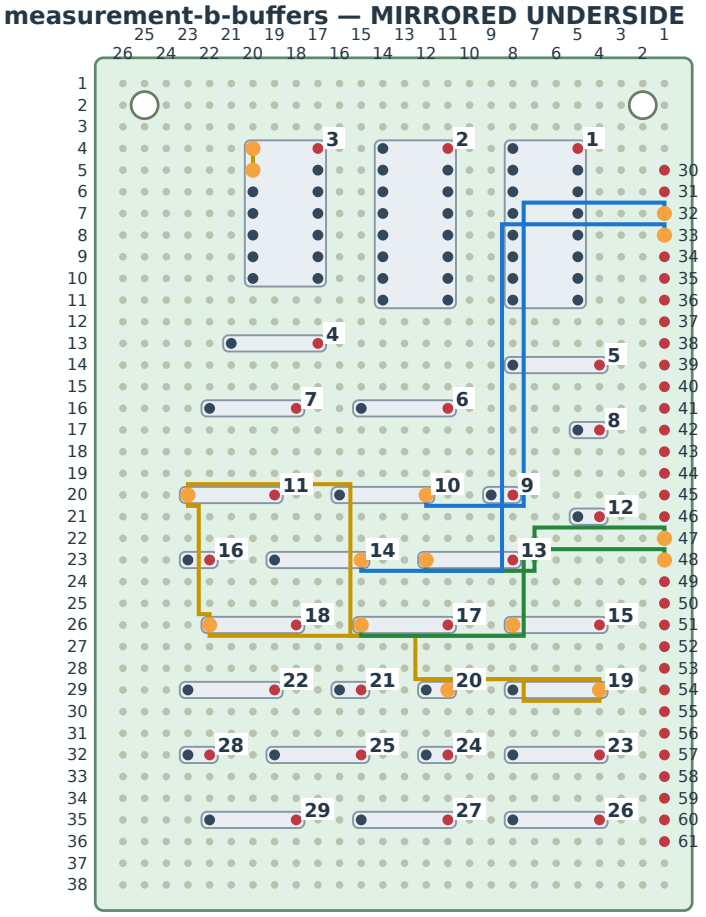
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W44 / P4_1_BUF	U_B_BUFFERS.1 → U_B_BUFFERS.2	(17,4) → (17,5)
W45 / P4_1_DIV	C_P4_1.1 → R_P4_1_BOTTOM.1	(4,21) → (19,20)
W46 / P4_1_DIV	R_P4_1_BOTTOM.1 → R_P4_1_TOP.2	(19,20) → (16,20)
W47 / P4_1_DIV	R_P4_1_TOP.2 → U_B_BUFFERS.3	(16,20) → (17,6)
W48 / P4_2_BUF	R_P4_2_ADC.1 → U_B_BUFFERS.6	(11,26) → (17,9)
W49 / P4_2_BUF	U_B_BUFFERS.6 → U_B_BUFFERS.7	(17,9) → (17,10)
W50 / P4_2_DIV	C_P4_2.1 → R_P4_2_BOTTOM.1	(22,23) → (4,26)
W51 / P4_2_DIV	R_P4_2_BOTTOM.1 → R_P4_2_TOP.2	(4,26) → (19,23)
W52 / P4_2_DIV	R_P4_2_TOP.2 → U_B_BUFFERS.5	(19,23) → (17,8)
W53 / P5_1_BUF	U_B_BUFFERS.8 → U_B_BUFFERS.9	(20,10) → (20,9)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — measurement-b-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

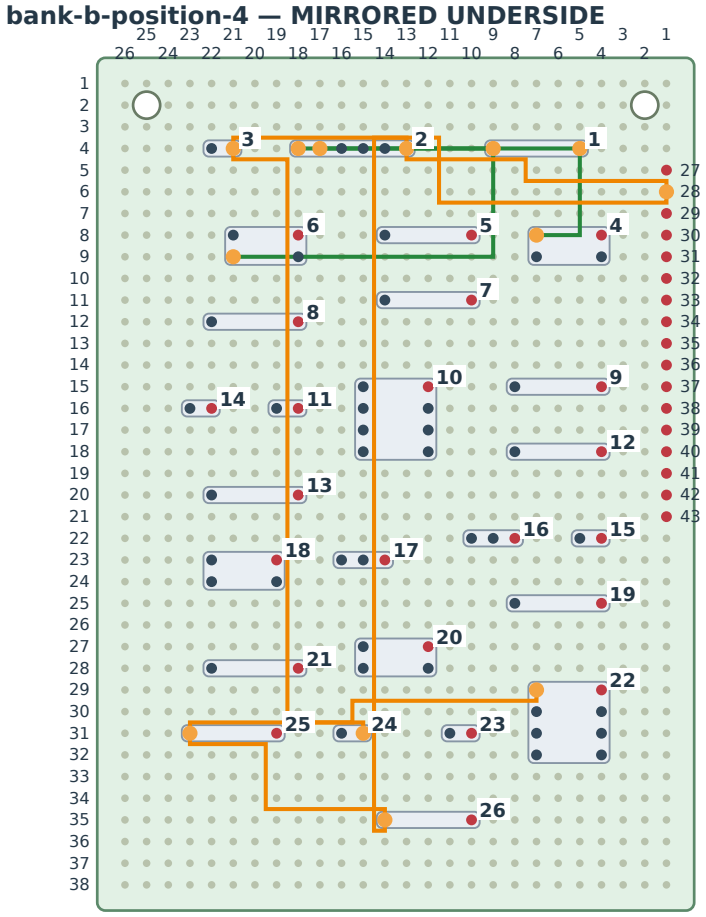
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W58 / P5_2_BUF	U_B_BUFFERS.13 → U_B_BUFFERS.14	(20,5) → (20,4)
W71 / NODE15	J_LINK_3.1 → R_P4_1_TOP.1	(1,7) → (12,20)
W72 / B_GENERAL_BIAS	C_B_GENERAL_BIAS.1 → R_B_GENERAL_BIAS_BOTTOM.1	(11,29) → (4,29)
W73 / B_GENERAL_BIAS	R_B_GENERAL_BIAS_BOTTOM.1 → R_B_GENERAL_BIAS_TOP.2	(4,29) → (22,26)
W74 / B_GENERAL_BIAS	R_B_GENERAL_BIAS_TOP.2 → R_P4_1_BOTTOM.2	(22,26) → (23,20)
W75 / B_GENERAL_BIAS	R_P4_1_BOTTOM.2 → R_P4_2_BOTTOM.2	(23,20) → (8,26)
W78 / P4_1_MUX	J_LINK_18.1 → R_P4_1_ADC.2	(1,22) → (12,23)
W79 / NODE16	J_LINK_4.1 → R_P4_2_TOP.1	(1,8) → (15,23)
W80 / P4_2_MUX	J_LINK_19.1 → R_P4_2_ADC.2	(1,23) → (15,26)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — bank-b-position-4

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

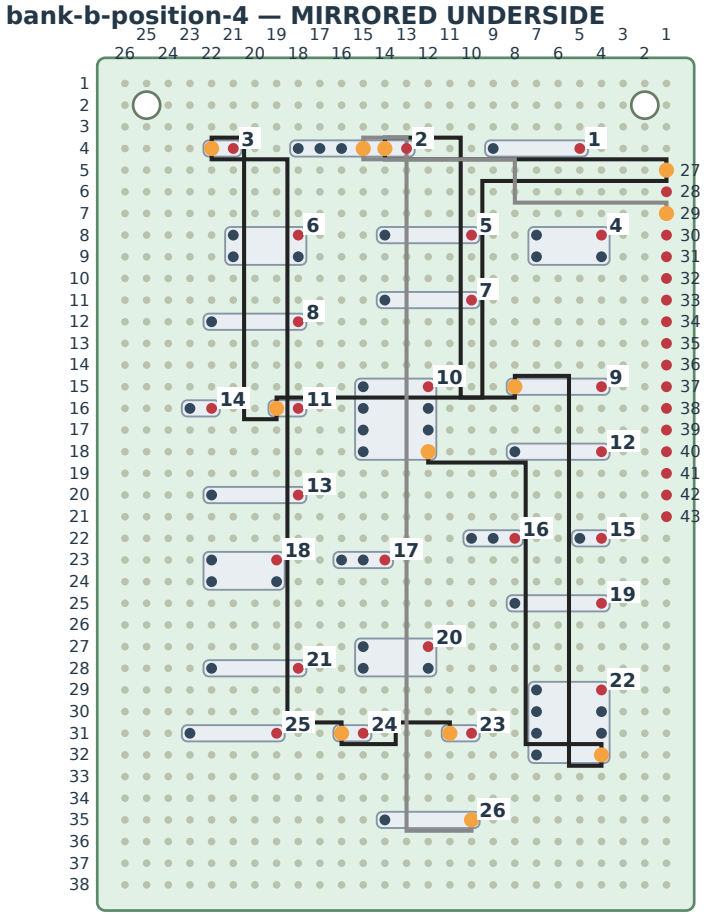
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / P4_SHUNT_P	R_P4_SHUNT.1 → J_P4_CURRENT_MODULE.5	(5,4) → (17,4)
W2 / P4_SHUNT_P	R_P4_SHUNT.1 → K_P4_AMM_P.4	(5,4) → (7,8)
W3 / P4_SHUNT_N	R_P4_SHUNT.2 → J_P4_CURRENT_MODULE.6	(9,4) → (18,4)
W4 / P4_SHUNT_N	R_P4_SHUNT.2 → K_P4_AMM_N.3	(9,4) → (21,9)
W5 / 3V3_SYS	C_P4_BUS_B.1 → C_P4_CURRENT.1	(15,31) → (21,4)
W6 / 3V3_SYS	C_P4_CURRENT.1 → J_LINK_2.1	(21,4) → (1,6)
W7 / 3V3_SYS	J_LINK_2.1 → J_P4_CURRENT_MODULE.1	(1,6) → (13,4)
W8 / 3V3_SYS	J_P4_CURRENT_MODULE.1 → R_P4_SCL.2	(13,4) → (14,35)
W9 / 3V3_SYS	R_P4_SCL.2 → R_P4_SDA.2	(14,35) → (23,31)
W10 / 3V3_SYS	R_P4_SDA.2 → U_P4_BUS.8	(23,31) → (7,29)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — bank-b-position-4

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

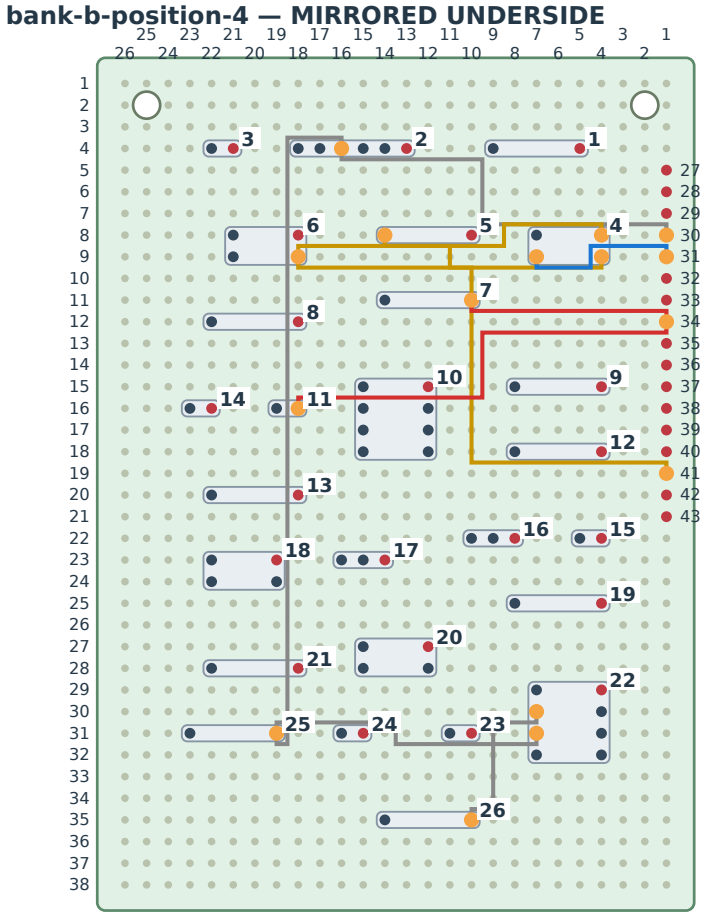
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	C_P4_BUS_A.2 → C_P4_BUS_B.2	(11,31) → (16,31)
W12 / GND	C_P4_BUS_B.2 → C_P4_CURRENT.2	(16,31) → (22,4)
W13 / GND	C_P4_CURRENT.2 → C_P4_LIMIT.2	(22,4) → (19,16)
W14 / GND	C_P4_LIMIT.2 → J_LINK_1.1	(19,16) → (1,5)
W15 / GND	J_LINK_1.1 → J_P4_CURRENT_MODULE.2	(1,5) → (14,4)
W16 / GND	J_P4_CURRENT_MODULE.2 → R_P4_SINK_SENSE.2	(14,4) → (8,15)
W17 / GND	R_P4_SINK_SENSE.2 → U_P4_BUS.4	(8,15) → (4,32)
W18 / GND	U_P4_BUS.4 → U_P4_LIMIT.4	(4,32) → (12,18)
W19 / SCL_POSITION_4	J_LINK_3.1 → J_P4_CURRENT_MODULE.3	(1,7) → (15,4)
W20 / SCL_POSITION_4	J_P4_CURRENT_MODULE.3 → R_P4_SCL.1	(15,4) → (10,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — bank-b-position-4

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

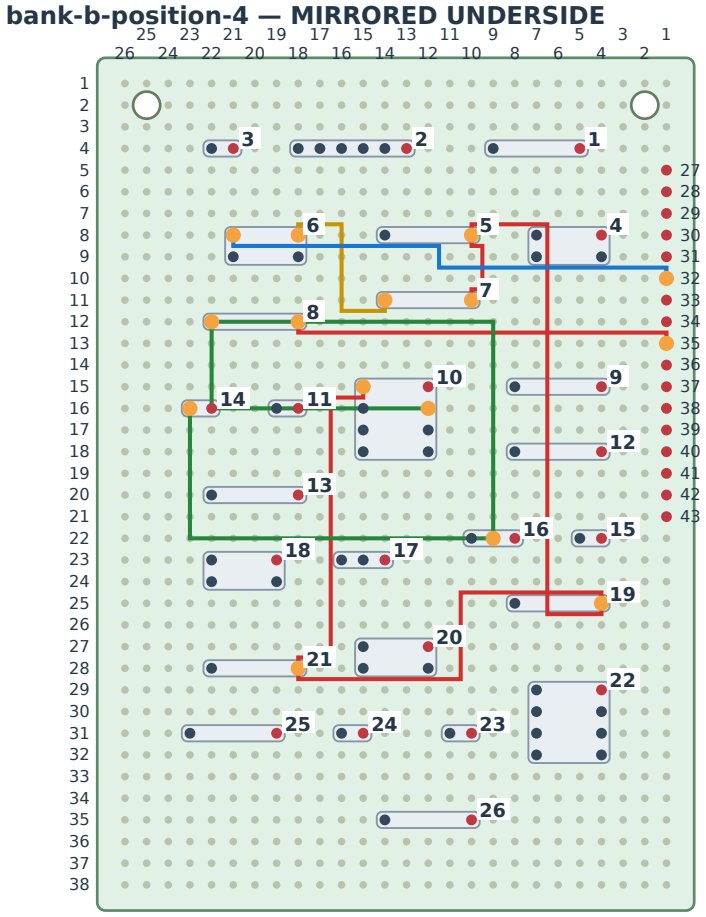
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / SCL_POSITION_4	R_P4_SCL.1 → U_P4_BUS.7	(10,35) → (7,30)
W22 / SDA_POSITION_4	J_LINK_4.1 → J_P4_CURRENT_MODULE.4	(1,8) → (16,4)
W23 / SDA_POSITION_4	J_P4_CURRENT_MODULE.4 → R_P4_SDA.1	(16,4) → (19,31)
W24 / SDA_POSITION_4	R_P4_SDA.1 → U_P4_BUS.6	(19,31) → (7,31)
W25 / P4_K_AMM_P_LED	K_P4_AMM_P.1 → R_P4_K_AMM_P_LED.2	(4,8) → (14,8)
W26 / P4_AMM_CATHODE	J_LINK_15.1 → K_P4_AMM_N.2	(1,19) → (18,9)
W27 / P4_AMM_CATHODE	K_P4_AMM_N.2 → K_P4_AMM_P.2	(18,9) → (4,9)
W28 / NODE15	J_LINK_5.1 → K_P4_AMM_P.3	(1,9) → (7,9)
W29 / 5V_SYS	C_P4_LIMIT.1 → J_LINK_8.1	(18,16) → (1,12)
W30 / 5V_SYS	J_LINK_8.1 → R_P4_K_AMM_N_LED.1	(1,12) → (10,11)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — bank-b-position-4

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

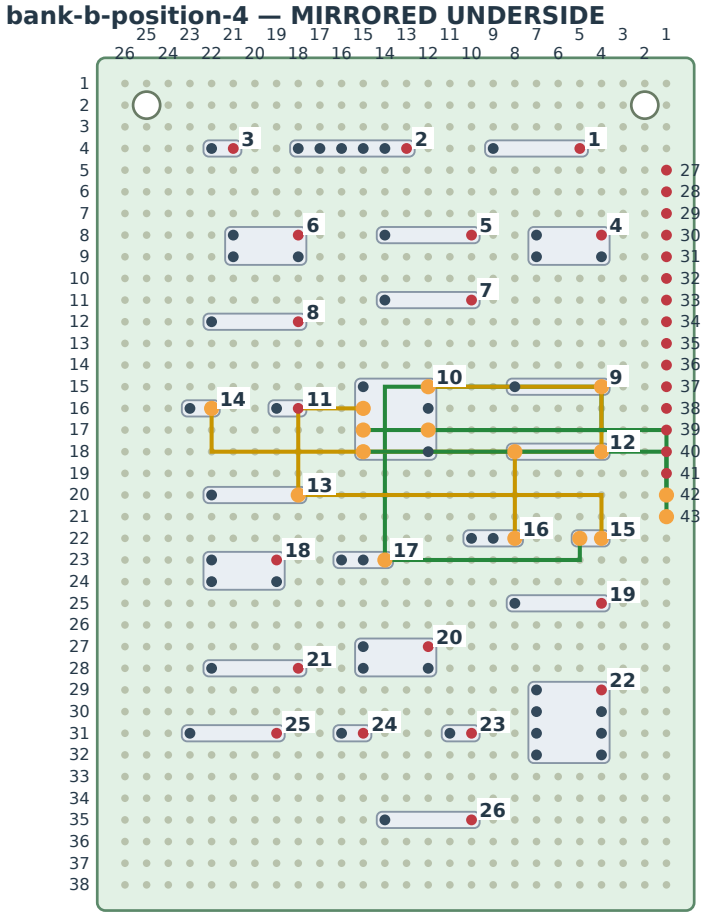
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / 5V_SYS	R_P4_K_AMM_N_LED.1 → R_P4_K_AMM_P_LED.1	(10,11) → (10,8)
W32 / 5V_SYS	R_P4_K_AMM_P_LED.1 → R_P4_K_HIGH_LED.1	(10,8) → (4,25)
W33 / 5V_SYS	R_P4_K_HIGH_LED.1 → R_P4_K_LOW_LED.1	(4,25) → (18,28)
W34 / 5V_SYS	R_P4_K_LOW_LED.1 → U_P4_LIMIT.8	(18,28) → (15,15)
W35 / P4_K_AMM_N_LED	K_P4_AMM_N.1 → R_P4_K_AMM_N_LED.2	(18,8) → (14,11)
W36 / NODE16	J_LINK_6.1 → K_P4_AMM_N.4	(1,10) → (21,8)
W37 / 5V_EDU	J_LINK_9.1 → R_P4_SOURCE_SENSE.1	(1,13) → (18,12)
W38 / P4_SOURCE_SENSE	C_P4_HI_COMP.2 → Q_P4_HIGH.2	(23,16) → (9,22)
W39 / P4_SOURCE_SENSE	Q_P4_HIGH.2 → R_P4_SOURCE_SENSE.2	(9,22) → (22,12)
W40 / P4_SOURCE_SENSE	R_P4_SOURCE_SENSE.2 → U_P4_LIMIT.2	(22,12) → (12,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — bank-b-position-4

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

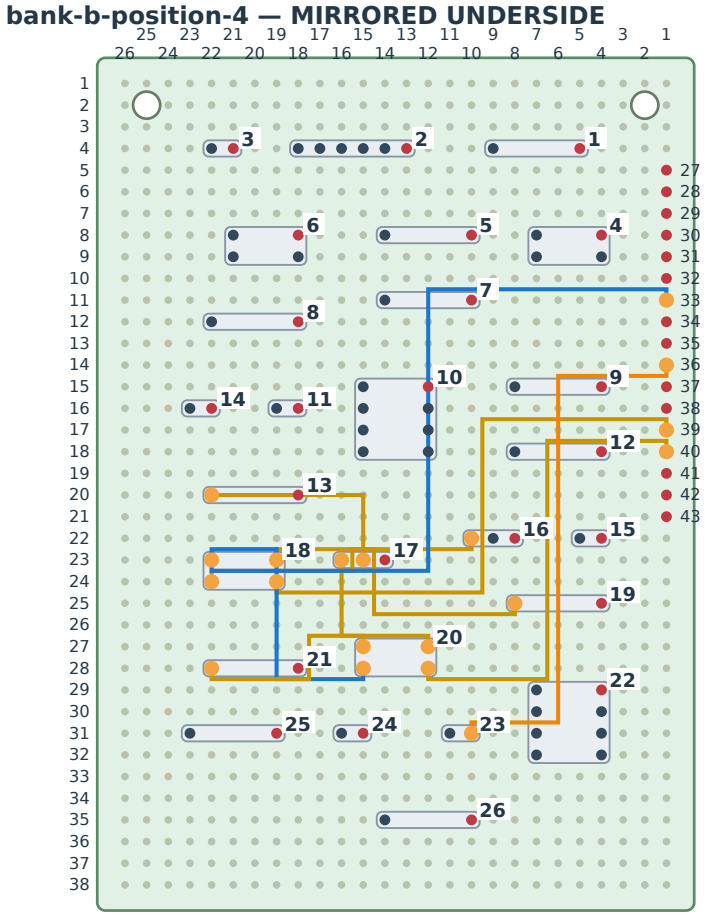
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / P4_SINK_SENSE	C_P4_LO_COMP.2 → Q_P4_LOW.1	(5,22) → (14,23)
W42 / P4_SINK_SENSE	Q_P4_LOW.1 → R_P4_SINK_SENSE.1	(14,23) → (4,15)
W43 / P4_SINK_SENSE	R_P4_SINK_SENSE.1 → U_P4_LIMIT.6	(4,15) → (15,17)
W44 / P4_HI_GATE_DRIVE	C_P4_HI_COMP.1 → R_P4_HI_GATE.1	(22,16) → (4,18)
W45 / P4_HI_GATE_DRIVE	R_P4_HI_GATE.1 → U_P4_LIMIT.1	(4,18) → (12,15)
W46 / B_REF_HI	J_LINK_16.1 → U_P4_LIMIT.3	(1,20) → (12,17)
W47 / B_REF_LO	J_LINK_17.1 → U_P4_LIMIT.5	(1,21) → (15,18)
W48 / P4_LO_GATE_DRIVE	C_P4_LO_COMP.1 → R_P4_LO_GATE.1	(4,22) → (18,20)
W49 / P4_LO_GATE_DRIVE	R_P4_LO_GATE.1 → U_P4_LIMIT.7	(18,20) → (15,16)
W50 / P4_HI_GATE	Q_P4_HIGH.1 → R_P4_HI_GATE.2	(8,22) → (8,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — bank-b-position-4

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

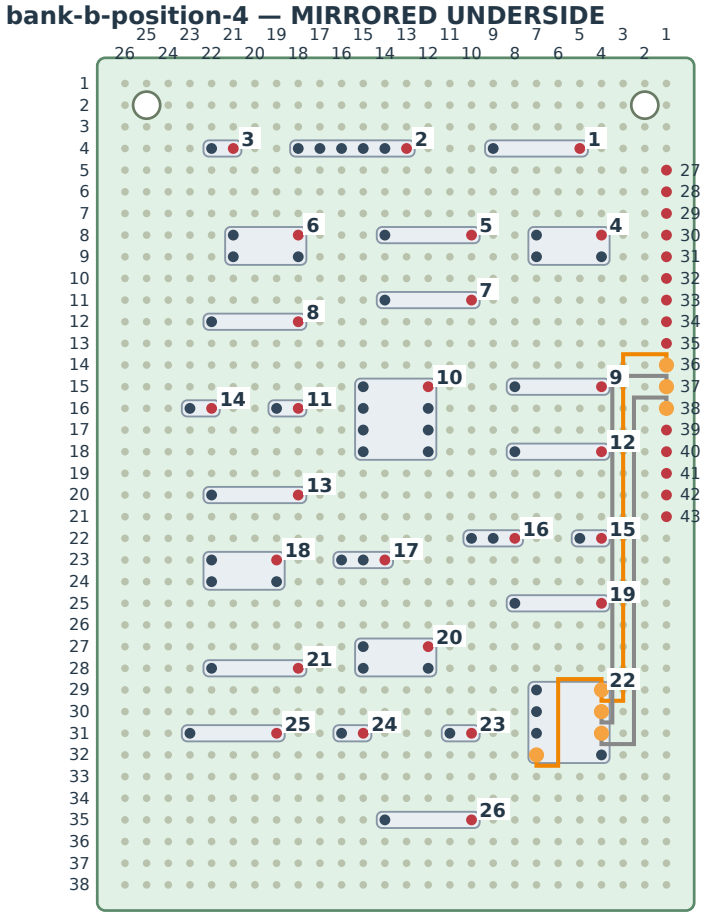
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W51 / P4_LO_GATE	Q_P4_LOW.2 → R_P4_LO_GATE.2	(15,23) → (22,20)
W52 / P4_SOURCE_OUT	K_P4_HIGH.3 → Q_P4_HIGH.3	(22,24) → (10,22)
W53 / P4_SINK_IN	K_P4_LOW.4 → Q_P4_LOW.3	(15,27) → (16,23)
W54 / P4_K_HIGH_LED	K_P4_HIGH.1 → R_P4_K_HIGH_LED.2	(19,23) → (8,25)
W55 / P4_K_HIGH_CATHODE	J_LINK_13.1 → K_P4_HIGH.2	(1,17) → (19,24)
W56 / NODE17	J_LINK_7.1 → K_P4_HIGH.4	(1,11) → (22,23)
W57 / NODE17	K_P4_HIGH.4 → K_P4_LOW.3	(22,23) → (15,28)
W58 / P4_K_LOW_LED	K_P4_LOW.1 → R_P4_K_LOW_LED.2	(12,27) → (22,28)
W59 / P4_K_LOW_CATHODE	J_LINK_14.1 → K_P4_LOW.2	(1,18) → (12,28)
W60 / USB_SMART_3V3	C_P4_BUS_A.1 → J_LINK_10.1	(10,31) → (1,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP30 — bank-b-position-4

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

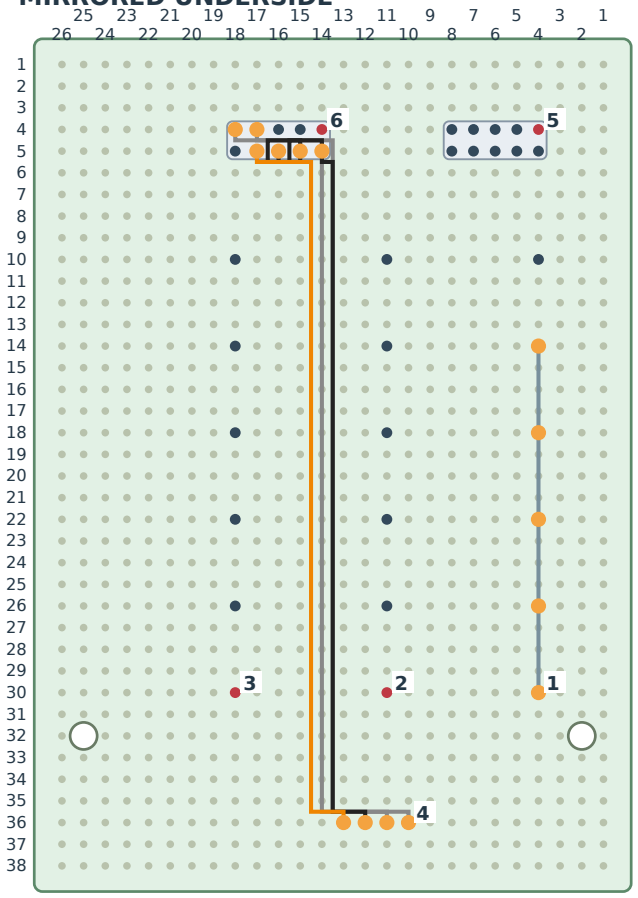
ID / net	From → to	TOP holes
W61 / USB_SMART_3V3	J_LINK_10.1 → U_P4_BUS.1	(1,14) → (4,29)
W62 / USB_SMART_3V3	U_P4_BUS.1 → U_P4_BUS.5	(4,29) → (7,32)
W63 / SCL_A	J_LINK_11.1 → U_P4_BUS.2	(1,15) → (4,30)
W64 / SDA_4	J_LINK_12.1 → U_P4_BUS.3	(1,16) → (4,31)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — b3

Install these wires after their endpoint components; inspect each net with power isolated.

b3 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

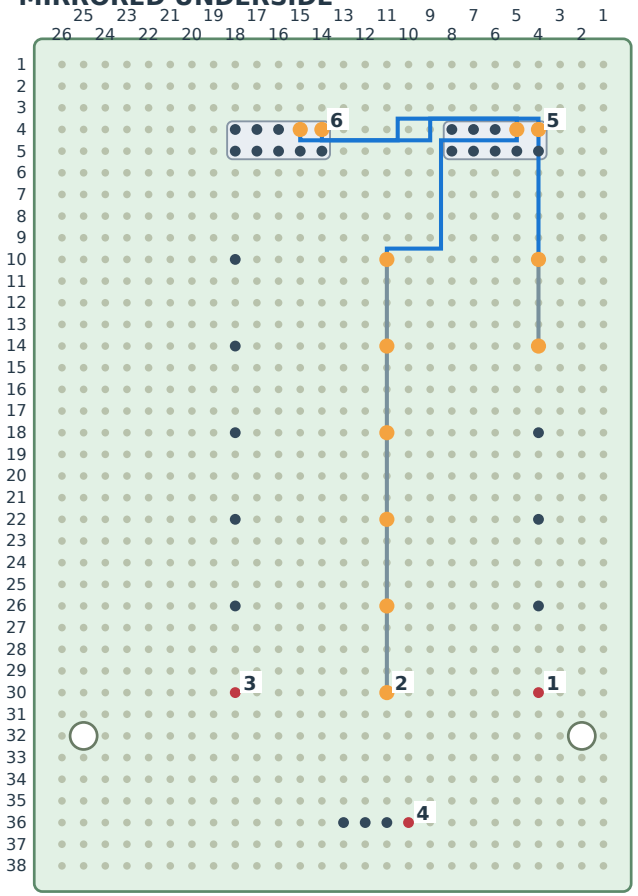
ID / net	From → to	TOP holes
W1 / SCL_POSITION_5	J_SERVICE.1 → J_SMART.9	(10,36) → (18,4)
W2 / SDA_POSITION_5	J_SERVICE.2 → J_SMART.7	(11,36) → (17,4)
W3 / GND	J_SERVICE.3 → J_SMART.2	(12,36) → (14,5)
W4 / GND	J_SMART.2 → J_SMART.4	(14,5) → (15,5)
W5 / GND	J_SMART.4 → J_SMART.6	(15,5) → (16,5)
W6 / 3V3_SYS	J_SERVICE.4 → J_SMART.8	(13,36) → (17,5)
W7 / NODE19	J_NODE19.1 → J_NODE19.2	(4,30) → (4,26)
W8 / NODE19	J_NODE19.2 → J_NODE19.3	(4,26) → (4,22)
W9 / NODE19	J_NODE19.3 → J_NODE19.4	(4,22) → (4,18)
W10 / NODE19	J_NODE19.4 → J_NODE19.5	(4,18) → (4,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — b3

Install these wires after their endpoint components; inspect each net with power isolated.

b3 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

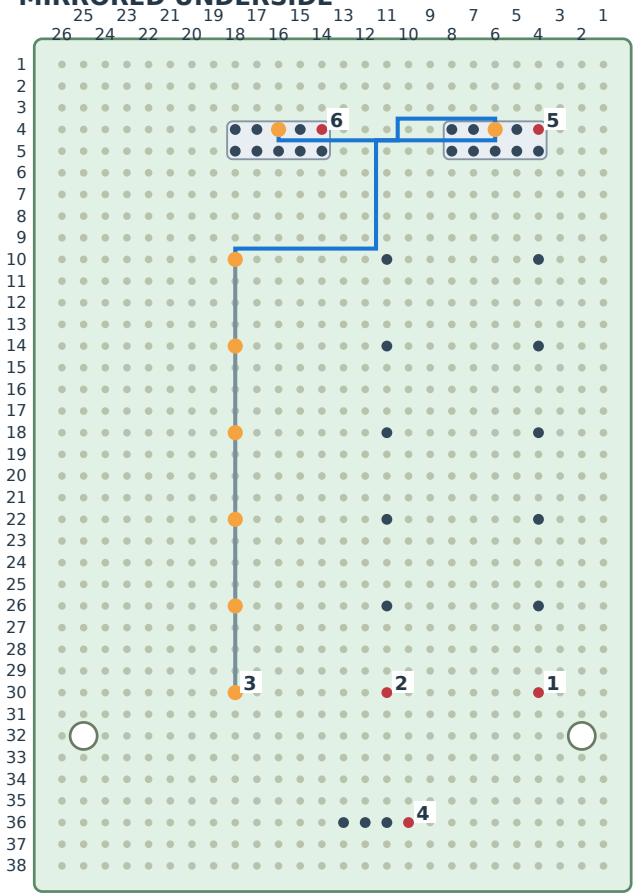
ID / net	From → to	TOP holes
W11 / NODE19	J_NODE19.5 → J_NODE19.6	(4,14) → (4,10)
W12 / NODE19	J_NODE19.6 → J_NODE.1	(4,10) → (4,4)
W13 / NODE19	J_NODE.1 → J_SMART.1	(4,4) → (14,4)
W14 / NODE20	J_NODE20.1 → J_NODE20.2	(11,30) → (11,26)
W15 / NODE20	J_NODE20.2 → J_NODE20.3	(11,26) → (11,22)
W16 / NODE20	J_NODE20.3 → J_NODE20.4	(11,22) → (11,18)
W17 / NODE20	J_NODE20.4 → J_NODE20.5	(11,18) → (11,14)
W18 / NODE20	J_NODE20.5 → J_NODE20.6	(11,14) → (11,10)
W19 / NODE20	J_NODE20.6 → J_NODE.3	(11,10) → (5,4)
W20 / NODE20	J_NODE.3 → J_SMART.3	(5,4) → (15,4)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — b3

Install these wires after their endpoint components; inspect each net with power isolated.

b3 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

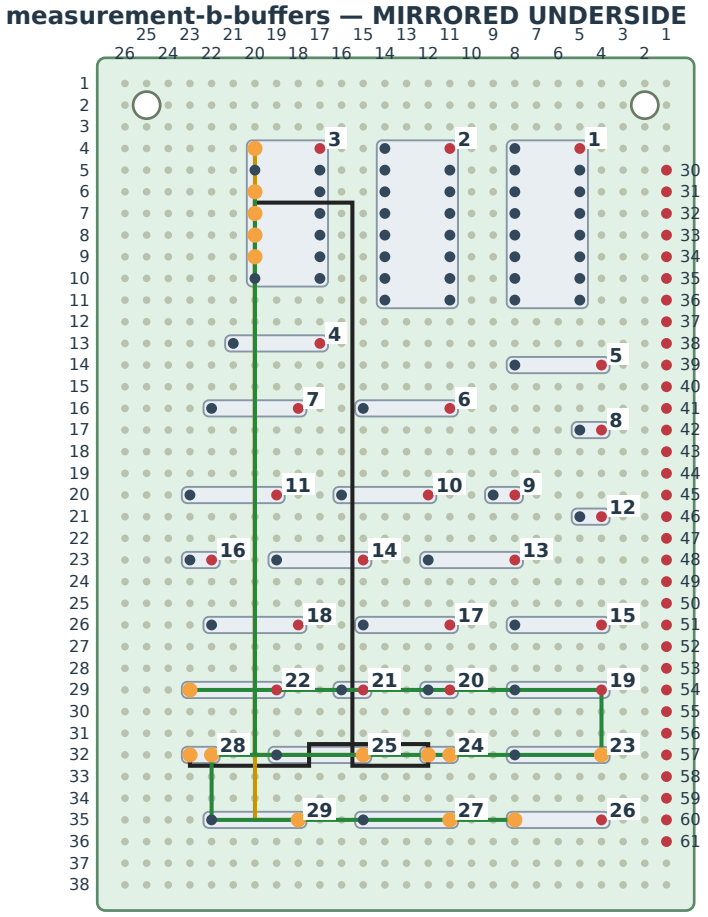
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / NODE21	J_NODE21.1 → J_NODE21.2	(18,30) → (18,26)
W22 / NODE21	J_NODE21.2 → J_NODE21.3	(18,26) → (18,22)
W23 / NODE21	J_NODE21.3 → J_NODE21.4	(18,22) → (18,18)
W24 / NODE21	J_NODE21.4 → J_NODE21.5	(18,18) → (18,14)
W25 / NODE21	J_NODE21.5 → J_NODE21.6	(18,14) → (18,10)
W26 / NODE21	J_NODE21.6 → J_NODE.5	(18,10) → (6,4)
W27 / NODE21	J_NODE.5 → J_SMART.5	(6,4) → (16,4)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — measurement-b-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

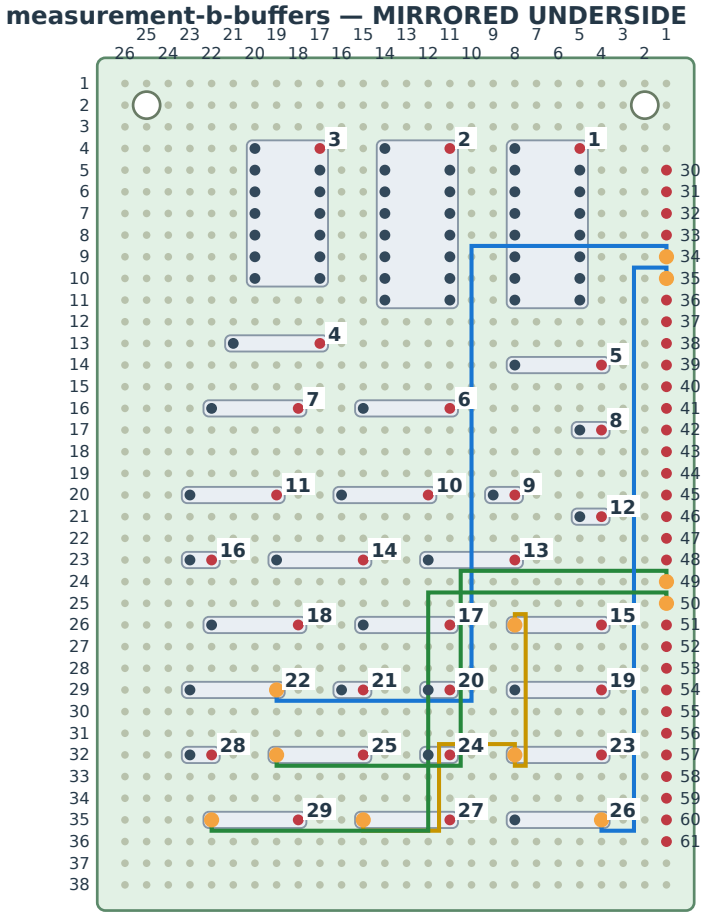
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W18 / GND	U_B_BUFFERS.11 → C_P5_1.2	(20,7) → (12,32)
W19 / GND	C_P5_1.2 → C_P5_2.2	(12,32) → (23,32)
W54 / P5_1_BUF	U_B_BUFFERS.9 → R_P5_1_ADC.1	(20,9) → (15,32)
W55 / P5_1_DIV	U_B_BUFFERS.10 → C_P5_1.1	(20,8) → (11,32)
W56 / P5_1_DIV	C_P5_1.1 → R_P5_1_BOTTOM.1	(11,32) → (4,32)
W57 / P5_1_DIV	R_P5_1_BOTTOM.1 → R_P5_1_TOP.2	(4,32) → (23,29)
W59 / P5_2_BUF	U_B_BUFFERS.14 → R_P5_2_ADC.1	(20,4) → (18,35)
W60 / P5_2_DIV	U_B_BUFFERS.12 → C_P5_2.1	(20,6) → (22,32)
W61 / P5_2_DIV	C_P5_2.1 → R_P5_2_BOTTOM.1	(22,32) → (11,35)
W62 / P5_2_DIV	R_P5_2_BOTTOM.1 → R_P5_2_TOP.2	(11,35) → (8,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — measurement-b-buffers

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

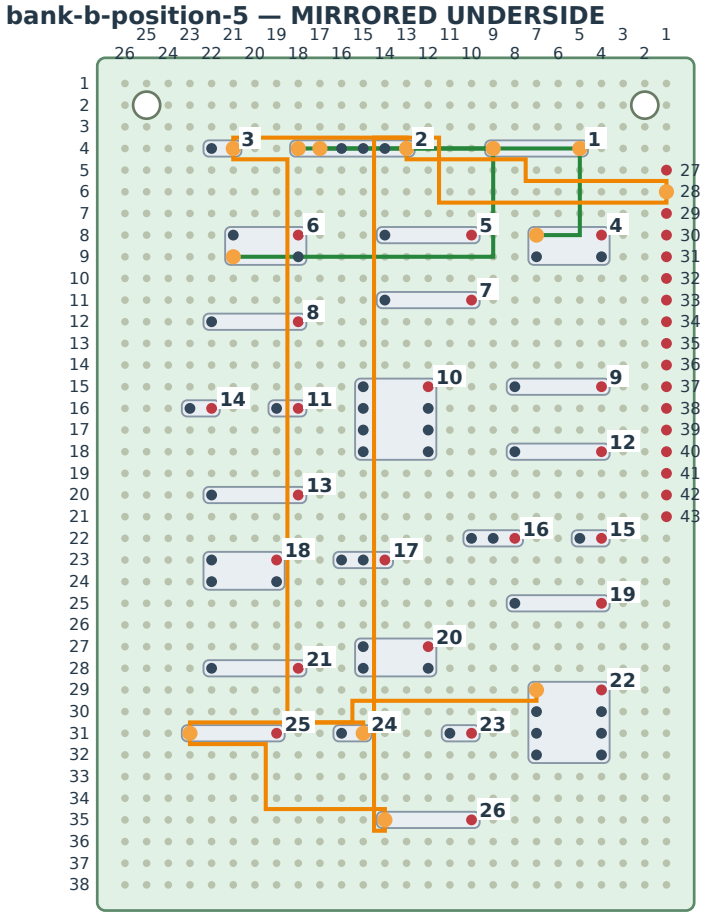
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W76 / B_GENERAL_BIAS	R_P4_2_BOTTOM.2 → R_P5_1_BOTTOM.2	(8,26) → (8,32)
W77 / B_GENERAL_BIAS	R_P5_1_BOTTOM.2 → R_P5_2_BOTTOM.2	(8,32) → (15,35)
W81 / NODE19	J_LINK_5.1 → R_P5_1_TOP.1	(1,9) → (19,29)
W82 / P5_1_MUX	J_LINK_20.1 → R_P5_1_ADC.2	(1,24) → (19,32)
W83 / NODE20	J_LINK_6.1 → R_P5_2_TOP.1	(1,10) → (4,35)
W84 / P5_2_MUX	J_LINK_21.1 → R_P5_2_ADC.2	(1,25) → (22,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — bank-b-position-5

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

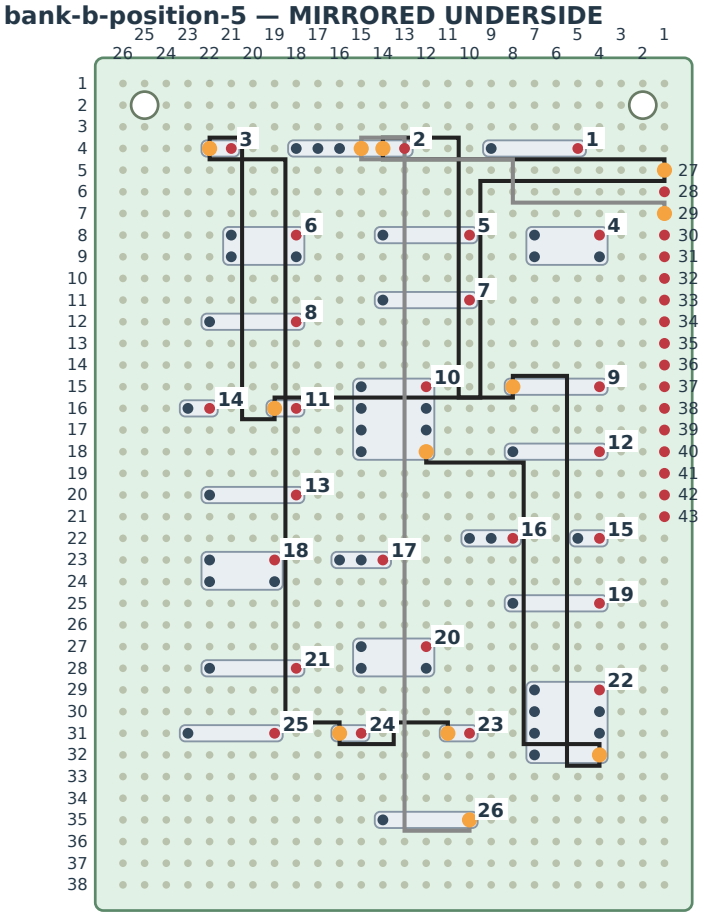
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W1 / P5_SHUNT_P	R_P5_SHUNT.1 → J_P5_CURRENT_MODULE.5	(5,4) → (17,4)
W2 / P5_SHUNT_P	R_P5_SHUNT.1 → K_P5_AMM_P.4	(5,4) → (7,8)
W3 / P5_SHUNT_N	R_P5_SHUNT.2 → J_P5_CURRENT_MODULE.6	(9,4) → (18,4)
W4 / P5_SHUNT_N	R_P5_SHUNT.2 → K_P5_AMM_N.3	(9,4) → (21,9)
W5 / 3V3_SYS	C_P5_BUS_B.1 → C_P5_CURRENT.1	(15,31) → (21,4)
W6 / 3V3_SYS	C_P5_CURRENT.1 → J_LINK_2.1	(21,4) → (1,6)
W7 / 3V3_SYS	J_LINK_2.1 → J_P5_CURRENT_MODULE.1	(1,6) → (13,4)
W8 / 3V3_SYS	J_P5_CURRENT_MODULE.1 → R_P5_SCL.2	(13,4) → (14,35)
W9 / 3V3_SYS	R_P5_SCL.2 → R_P5_SDA.2	(14,35) → (23,31)
W10 / 3V3_SYS	R_P5_SDA.2 → U_P5_BUS.8	(23,31) → (7,29)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — bank-b-position-5

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

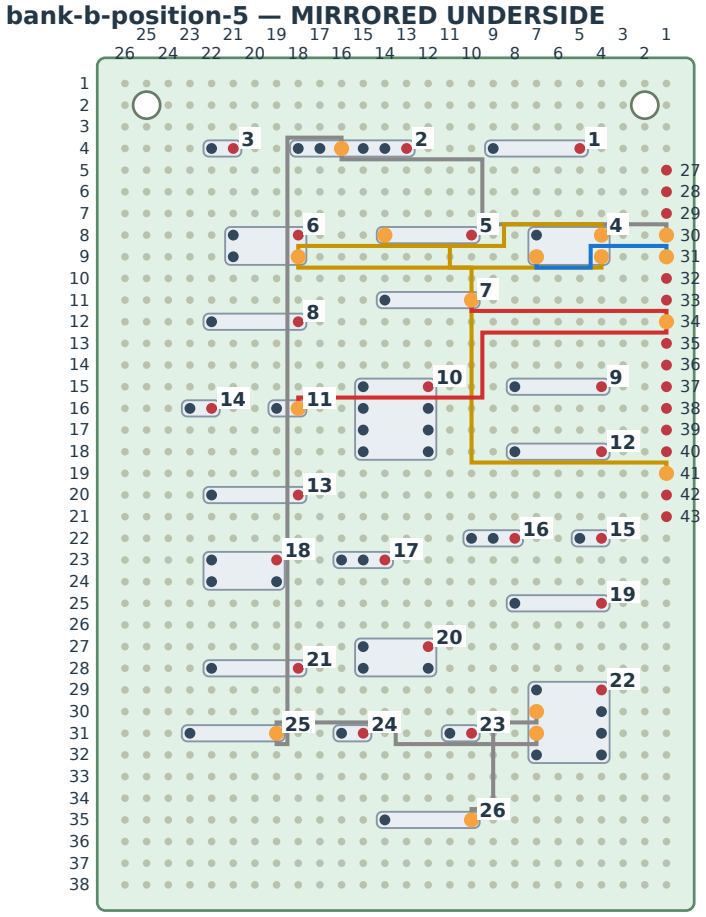
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W11 / GND	C_P5_BUS_A.2 → C_P5_BUS_B.2	(11,31) → (16,31)
W12 / GND	C_P5_BUS_B.2 → C_P5_CURRENT.2	(16,31) → (22,4)
W13 / GND	C_P5_CURRENT.2 → C_P5_LIMIT.2	(22,4) → (19,16)
W14 / GND	C_P5_LIMIT.2 → J_LINK_1.1	(19,16) → (1,5)
W15 / GND	J_LINK_1.1 → J_P5_CURRENT_MODULE.2	(1,5) → (14,4)
W16 / GND	J_P5_CURRENT_MODULE.2 → R_P5_SINK_SENSE.2	(14,4) → (8,15)
W17 / GND	R_P5_SINK_SENSE.2 → U_P5_BUS.4	(8,15) → (4,32)
W18 / GND	U_P5_BUS.4 → U_P5_LIMIT.4	(4,32) → (12,18)
W19 / SCL_POSITION_5	J_LINK_3.1 → J_P5_CURRENT_MODULE.3	(1,7) → (15,4)
W20 / SCL_POSITION_5	J_P5_CURRENT_MODULE.3 → R_P5_SCL.1	(15,4) → (10,35)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — bank-b-position-5

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

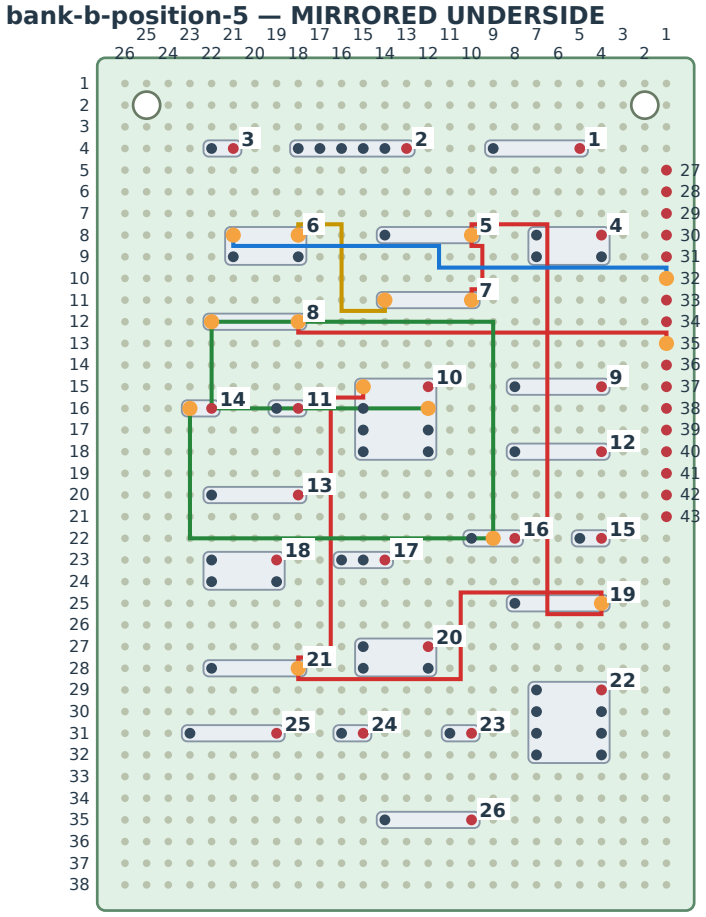
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W21 / SCL_POSITION_5	R_P5_SCL.1 → U_P5_BUS.7	(10,35) → (7,30)
W22 / SDA_POSITION_5	J_LINK_4.1 → J_P5_CURRENT_MODULE.4	(1,8) → (16,4)
W23 / SDA_POSITION_5	J_P5_CURRENT_MODULE.4 → R_P5_SDA.1	(16,4) → (19,31)
W24 / SDA_POSITION_5	R_P5_SDA.1 → U_P5_BUS.6	(19,31) → (7,31)
W25 / P5_K_AMM_P_LED	K_P5_AMM_P.1 → R_P5_K_AMM_P_LED.2	(4,8) → (14,8)
W26 / P5_AMM_CATHODE	J_LINK_15.1 → K_P5_AMM_N.2	(1,19) → (18,9)
W27 / P5_AMM_CATHODE	K_P5_AMM_N.2 → K_P5_AMM_P.2	(18,9) → (4,9)
W28 / NODE19	J_LINK_5.1 → K_P5_AMM_P.3	(1,9) → (7,9)
W29 / 5V_SYS	C_P5_LIMIT.1 → J_LINK_8.1	(18,16) → (1,12)
W30 / 5V_SYS	J_LINK_8.1 → R_P5_K_AMM_N_LED.1	(1,12) → (10,11)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — bank-b-position-5

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

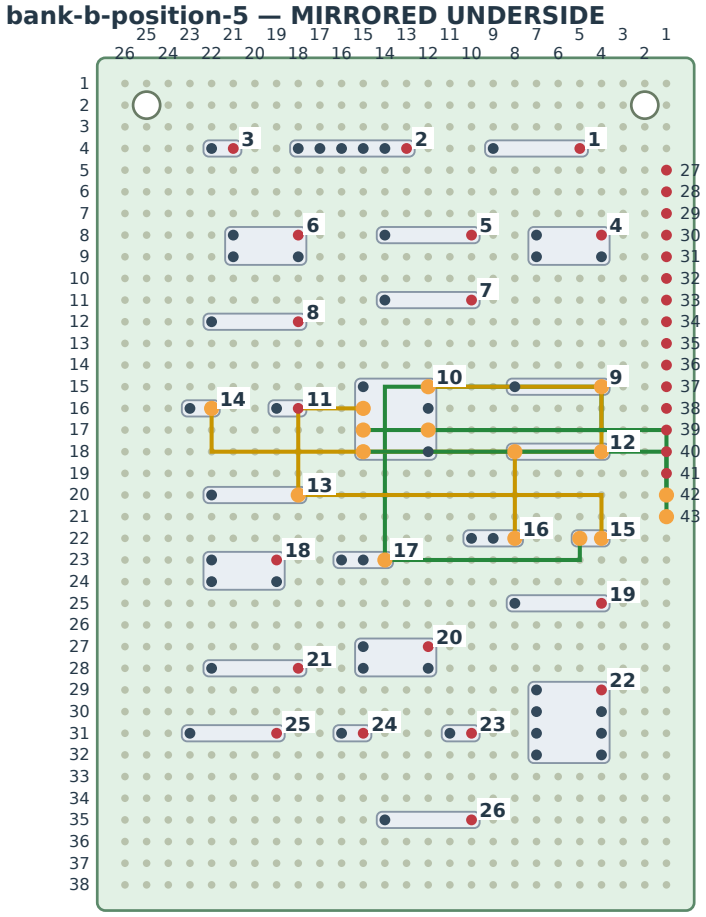
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W31 / 5V_SYS	R_P5_K_AMM_N_LED.1 → R_P5_K_AMM_P_LED.1	(10,11) → (10,8)
W32 / 5V_SYS	R_P5_K_AMM_P_LED.1 → R_P5_K_HIGH_LED.1	(10,8) → (4,25)
W33 / 5V_SYS	R_P5_K_HIGH_LED.1 → R_P5_K_LOW_LED.1	(4,25) → (18,28)
W34 / 5V_SYS	R_P5_K_LOW_LED.1 → U_P5_LIMIT.8	(18,28) → (15,15)
W35 / P5_K_AMM_N_LED	K_P5_AMM_N.1 → R_P5_K_AMM_N_LED.2	(18,8) → (14,11)
W36 / NODE20	J_LINK_6.1 → K_P5_AMM_N.4	(1,10) → (21,8)
W37 / 5V_EDU	J_LINK_9.1 → R_P5_SOURCE_SENSE.1	(1,13) → (18,12)
W38 / P5_SOURCE_SENSE	C_P5_HI_COMP.2 → Q_P5_HIGH.2	(23,16) → (9,22)
W39 / P5_SOURCE_SENSE	Q_P5_HIGH.2 → R_P5_SOURCE_SENSE.2	(9,22) → (22,12)
W40 / P5_SOURCE_SENSE	R_P5_SOURCE_SENSE.2 → U_P5_LIMIT.2	(22,12) → (12,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — bank-b-position-5

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

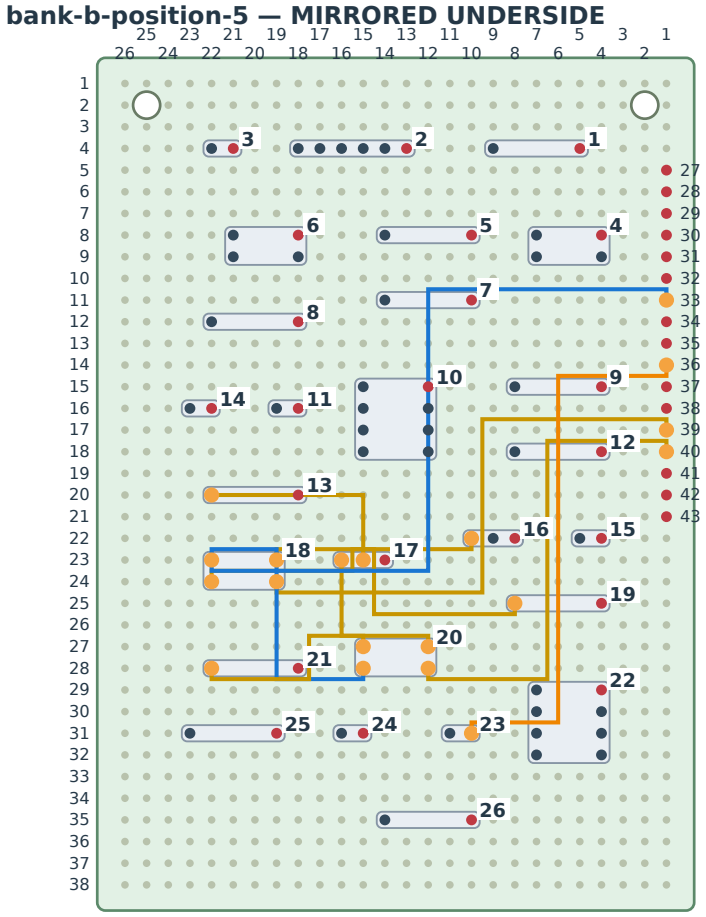
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W41 / P5_SINK_SENSE	C_P5_LO_COMP.2 → Q_P5_LOW.1	(5,22) → (14,23)
W42 / P5_SINK_SENSE	Q_P5_LOW.1 → R_P5_SINK_SENSE.1	(14,23) → (4,15)
W43 / P5_SINK_SENSE	R_P5_SINK_SENSE.1 → U_P5_LIMIT.6	(4,15) → (15,17)
W44 / P5_HI_GATE_DRIVE	C_P5_HI_COMP.1 → R_P5_HI_GATE.1	(22,16) → (4,18)
W45 / P5_HI_GATE_DRIVE	R_P5_HI_GATE.1 → U_P5_LIMIT.1	(4,18) → (12,15)
W46 / B_REF_HI	J_LINK_16.1 → U_P5_LIMIT.3	(1,20) → (12,17)
W47 / B_REF_LO	J_LINK_17.1 → U_P5_LIMIT.5	(1,21) → (15,18)
W48 / P5_LO_GATE_DRIVE	C_P5_LO_COMP.1 → R_P5_LO_GATE.1	(4,22) → (18,20)
W49 / P5_LO_GATE_DRIVE	R_P5_LO_GATE.1 → U_P5_LIMIT.7	(18,20) → (15,16)
W50 / P5_HI_GATE	Q_P5_HIGH.1 → R_P5_HI_GATE.2	(8,22) → (8,18)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — bank-b-position-5

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

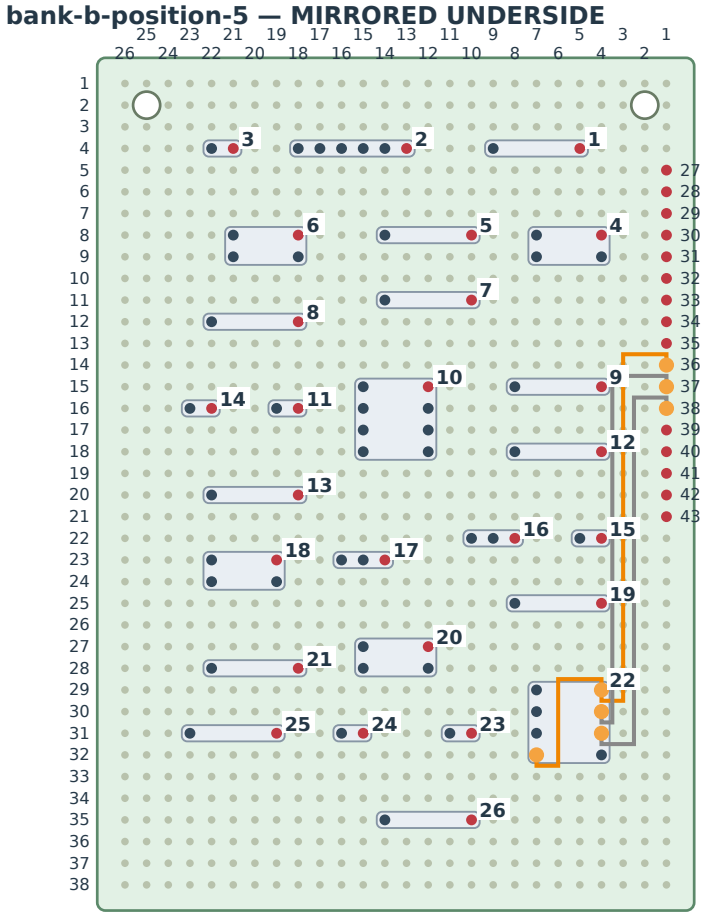
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W51 / P5_LO_GATE	Q_P5_LOW.2 → R_P5_LO_GATE.2	(15,23) → (22,20)
W52 / P5_SOURCE_OUT	K_P5_HIGH.3 → Q_P5_HIGH.3	(22,24) → (10,22)
W53 / P5_SINK_IN	K_P5_LOW.4 → Q_P5_LOW.3	(15,27) → (16,23)
W54 / P5_K_HIGH_LED	K_P5_HIGH.1 → R_P5_K_HIGH_LED.2	(19,23) → (8,25)
W55 / P5_K_HIGH_CATHODE	J_LINK_13.1 → K_P5_HIGH.2	(1,17) → (19,24)
W56 / NODE21	J_LINK_7.1 → K_P5_HIGH.4	(1,11) → (22,23)
W57 / NODE21	K_P5_HIGH.4 → K_P5_LOW.3	(22,23) → (15,28)
W58 / P5_K_LOW_LED	K_P5_LOW.1 → R_P5_K_LOW_LED.2	(12,27) → (22,28)
W59 / P5_K_LOW_CATHODE	J_LINK_14.1 → K_P5_LOW.2	(1,18) → (12,28)
W60 / USB_SMART_3V3	C_P5_BUS_A.1 → J_LINK_10.1	(10,31) → (1,14)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP31 — bank-b-position-5

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

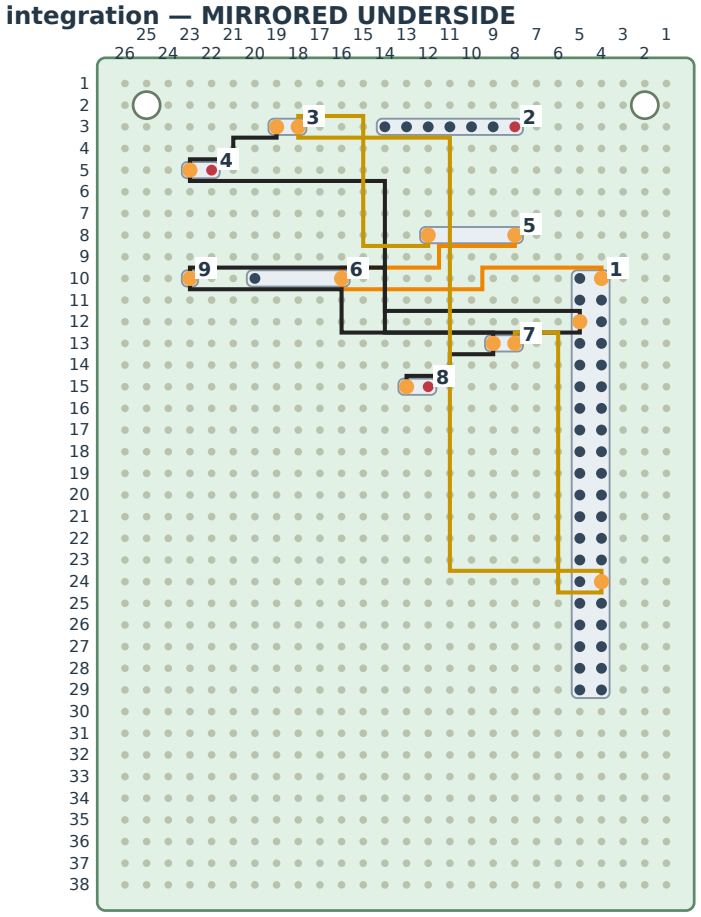
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W61 / USB_SMART_3V3	J_LINK_10.1 → U_P5_BUS.1	(1,14) → (4,29)
W62 / USB_SMART_3V3	U_P5_BUS.1 → U_P5_BUS.5	(4,29) → (7,32)
W63 / SCL_A	J_LINK_11.1 → U_P5_BUS.2	(1,15) → (4,30)
W64 / SDA_5	J_LINK_12.1 → U_P5_BUS.3	(1,16) → (4,31)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP32 — integration

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

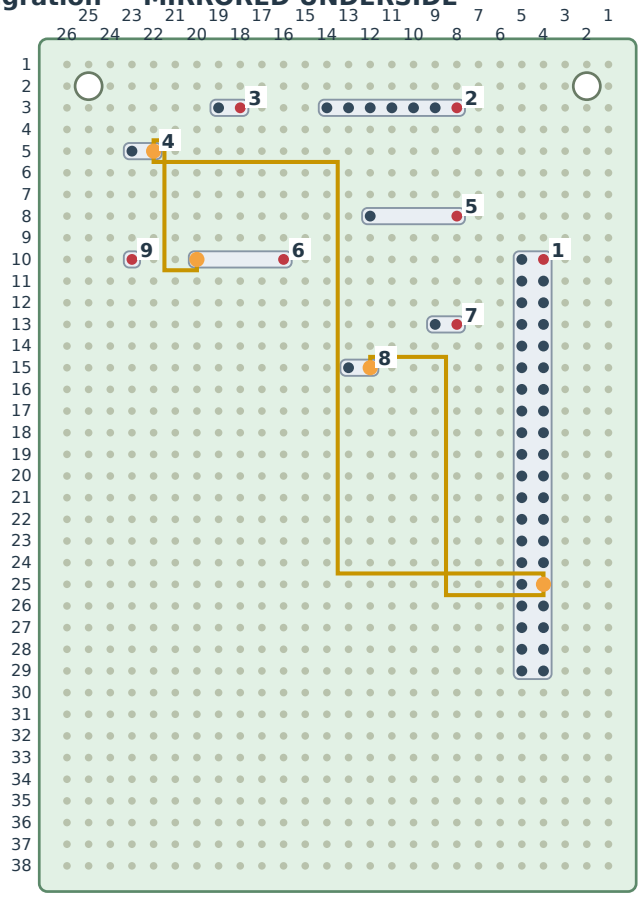
ID / net	From → to	TOP holes
W1 / PI_3V3	J_PI.1 → R_BACK.1	(4,10) → (16,10)
W2 / PI_3V3	R_BACK.1 → R_ENERGIZE.1	(16,10) → (8,8)
W4 / GND	C_BACK.2 → C_ENERGIZE.2	(13,15) → (9,13)
W5 / GND	C_ENERGIZE.2 → J_GROUND.1	(9,13) → (23,10)
W6 / GND	J_GROUND.1 → J_PI.6	(23,10) → (5,12)
W7 / GND	J_PI.6 → J_SW_BACK.2	(5,12) → (23,5)
W8 / GND	J_SW_BACK.2 → J_SW_ENERGIZE.2	(23,5) → (19,3)
W15 / ENERGIZE_N	C_ENERGIZE.1 → J_PI.29	(8,13) → (4,24)
W16 / ENERGIZE_N	J_PI.29 → J_SW_ENERGIZE.1	(4,24) → (18,3)
W17 / ENERGIZE_N	J_SW_ENERGIZE.1 → R_ENERGIZE.2	(18,3) → (12,8)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP32 — integration

Install these wires after their endpoint components; inspect each net with power isolated.

integration — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

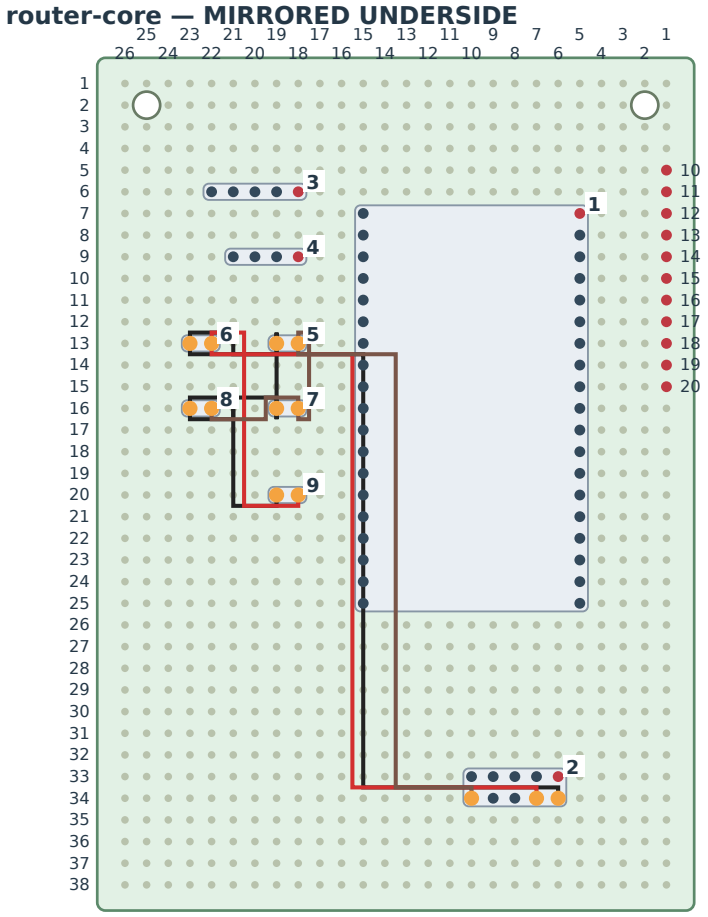
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W18 / BACK_N	C_BACK.1 → J_PI.31	(12,15) → (4,25)
W19 / BACK_N	J_PI.31 → J_SW_BACK.1	(4,25) → (22,5)
W20 / BACK_N	J_SW_BACK.1 → R_BACK.2	(22,5) → (20,10)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP32 — router-core

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

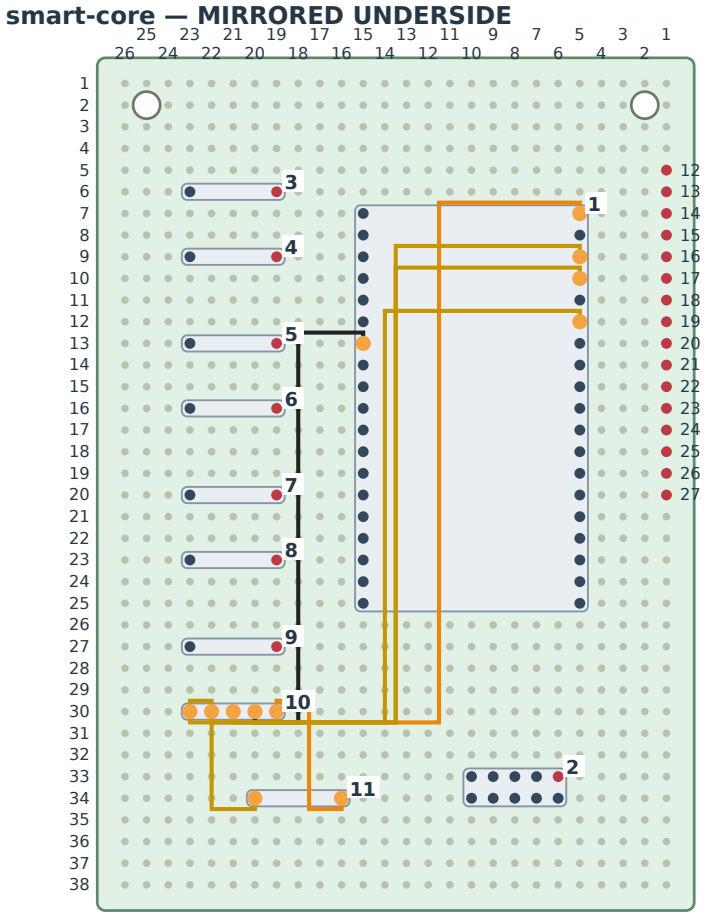
PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W8 / GND	J_POWER.2 → J_5V.2	(6,34) → (23,13)
W9 / GND	J_5V.2 → J_12V.2	(23,13) → (19,13)
W10 / GND	J_12V.2 → J_FAN1.2	(19,13) → (19,16)
W11 / GND	J_FAN1.2 → J_FAN2.2	(19,16) → (23,16)
W12 / GND	J_FAN2.2 → J_LIGHT.2	(23,16) → (19,20)
W31 / 5V_SYS	J_POWER.4 → J_5V.1	(7,34) → (22,13)
W32 / 5V_SYS	J_5V.1 → J_LIGHT.1	(22,13) → (18,20)
W33 / 12V_SYS	J_POWER.10 → J_12V.1	(10,34) → (18,13)
W34 / 12V_SYS	J_12V.1 → J_FAN1.1	(18,13) → (18,16)
W35 / 12V_SYS	J_FAN1.1 → J_FAN2.1	(18,16) → (22,16)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP32 — smart-core

Install these wires after their endpoint components; inspect each net with power isolated.



ID / net	From → to	TOP holes
W9 / USB_SMART_3V3	U_ESP2.1 → J_JOYSTICK.1	(5,7) → (19,30)
W10 / USB_SMART_3V3	J_JOYSTICK.1 → R_JOY_SW.1	(19,30) → (16,34)
W16 / GND	U_ESP2.26 → J_JOYSTICK.2	(15,13) → (20,30)
W39 / JOY_X	U_ESP2.3 → J_JOYSTICK.3	(5,9) → (21,30)
W40 / JOY_Y	U_ESP2.4 → J_JOYSTICK.4	(5,10) → (22,30)
W41 / JOY_SW_N	U_ESP2.6 → J_JOYSTICK.5	(5,12) → (23,30)
W42 / JOY_SW_N	J_JOYSTICK.5 → R_JOY_SW.2	(23,30) → (20,34)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

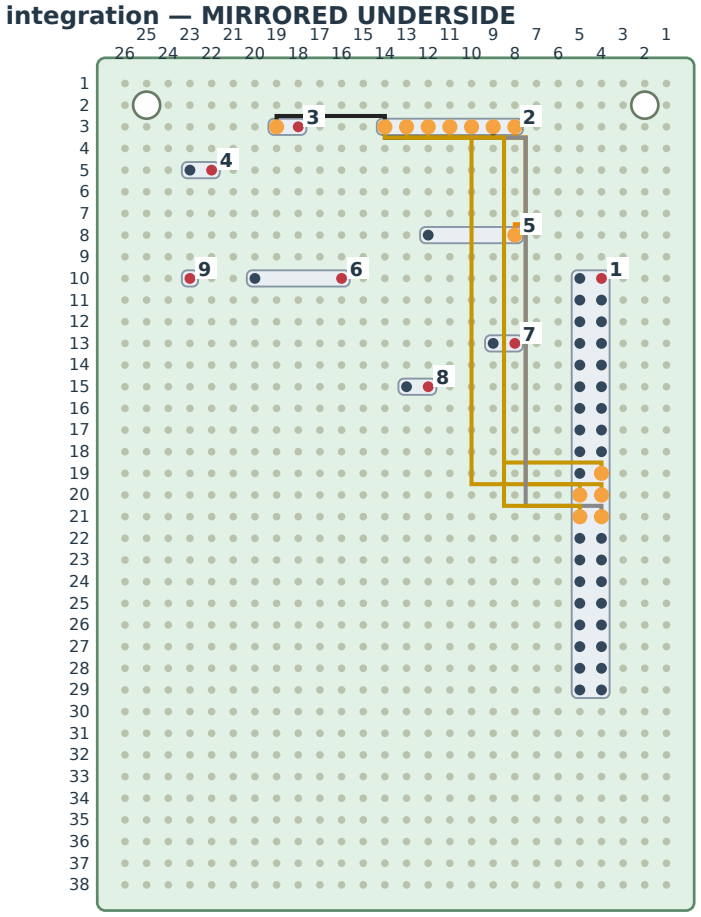
70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

CP34 — integration

Install these wires after their endpoint components; inspect each net with power isolated.



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

ID / net	From → to	TOP holes
W3 / PI_3V3	R_ENERGIZE.1 → J_RFID.1	(8,8) → (8,3)
W9 / GND	J_SW_ENERGIZE.2 → J_RFID.2	(19,3) → (9,3)
W10 / MOSI	J_PI.19 → J_RFID.4	(4,19) → (11,3)
W11 / MISO	J_PI.21 → J_RFID.5	(4,20) → (12,3)
W12 / RFID_RESET	J_PI.22 → J_RFID.7	(5,20) → (14,3)
W13 / SCLK	J_PI.23 → J_RFID.3	(4,21) → (10,3)
W14 / CE0_N	J_PI.24 → J_RFID.6	(5,21) → (13,3)

Insulated underside wire except the explicitly bare node straps. Strip only endpoints; sleeve crossings. A bend is not a solder joint. The SVG shows the existing prescribed paths, not manufactured PCB copper.

CP01 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H5 / GND	router-core / J_FIXTURE.2 TOP (19,6)	measurement-a-precision / JP_RETURN.2 TOP (5,35)
H155 / FIX_STIM	router-core / J_FIXTURE.5 TOP (22,6)	measurement-a-precision / J_LINK_6.1 TOP (1,10)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP06 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H1 / SCL_POSITION_0	measurement-a-precision / J_LINK_1.1 TOP (1,5)	a1 / J_SMART.9 TOP (22,33)
H2 / SCL_POSITION_0	a1 / J_SMART.9 TOP (22,33)	bank-a-position-0 / J_LINK_1.1 TOP (1,5)
H3 / SDA_POSITION_0	measurement-a-precision / J_LINK_2.1 TOP (1,6)	a1 / J_SMART.7 TOP (21,33)
H4 / SDA_POSITION_0	a1 / J_SMART.7 TOP (21,33)	bank-a-position-0 / J_LINK_2.1 TOP (1,6)
H6 / GND	measurement-a-precision / JP_RETURN.2 TOP (5,35)	a1 / J_SMART.2 TOP (18,34)
H7 / GND	a1 / J_SMART.2 TOP (18,34)	bank-a-position-0 / J_LINK_3.1 TOP (1,7)
H31 / 3V3_SYS	measurement-a-precision / J_LINK_3.1 TOP (1,7)	a1 / J_SMART.8 TOP (21,34)
H32 / 3V3_SYS	a1 / J_SMART.8 TOP (21,34)	bank-a-position-0 / J_LINK_4.1 TOP (1,8)
H52 / NODE00	a1 / J_NODE.1 TOP (4,33)	measurement-a-precision / J_LINK_4.1 TOP (1,8)
H55 / NODE01	a1 / J_NODE.3 TOP (5,33)	measurement-a-precision / J_LINK_5.1 TOP (1,9)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP09 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H8 / GND	bank-a-position-0 / J_LINK_3.1 TOP (1,7)	power-input / J_INPUT.2 TOP (6,11)
H9 / GND	power-input / J_INPUT.2 TOP (6,11)	power-interlock / J_MATRIX.2 TOP (6,34)
H33 / 3V3_SYS	bank-a-position-0 / J_LINK_4.1 TOP (1,8)	power-input / JP_3V3.2 TOP (9,29)
H34 / 3V3_SYS	power-input / JP_3V3.2 TOP (9,29)	power-interlock / J_MATRIX.3 TOP (7,33)
H116 / 12V_SYS	power-input / J_LINK_1.1 TOP (1,5)	power-interlock / J_MATRIX.10 TOP (10,34)
H122 / 5V_SYS	power-input / JP_LOGIC.2 TOP (5,29)	power-interlock / J_MATRIX.4 TOP (7,34)
H150 / EDU_PWR_EN	router-core / J_PROTOTYPE.4 TOP (21,9)	power-interlock / J_MATRIX.5 TOP (8,33)
H152 / EDU_FAULT_RESET	router-core / J_LINK_9.1 TOP (1,13)	power-interlock / J_MATRIX.7 TOP (9,33)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP10 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H10 / GND	power-interlock / J_MATRIX.2 TOP (6,34)	power-guards / J_LINK_1.1 TOP (1,5)
H119 / BUCK_5V	power-input / J_LINK_2.1 TOP (1,6)	power-guards / J_LINK_2.1 TOP (1,6)
H120 / BUCK_3V3	power-input / J_LINK_3.1 TOP (1,7)	power-guards / J_LINK_3.1 TOP (1,7)
H121 / SAFE_5V	power-input / JP_LOGIC.1 TOP (4,29)	power-guards / J_LINK_4.1 TOP (1,8)
H138 / SAFE_3V3	power-input / JP_3V3.1 TOP (8,29)	power-guards / J_LINK_5.1 TOP (1,9)
H139 / PRESET_N	power-interlock / J_TEST.5 TOP (20,27)	power-guards / J_LINK_6.1 TOP (1,10)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP11 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H11 / GND	power-guards / J_LINK_1.1 TOP (1,5)	power-analog / J_LINK_1.1 TOP (1,5)
H12 / GND	power-analog / J_LINK_1.1 TOP (1,5)	router-safety / J_LINK_1.1 TOP (1,5)
H35 / 3V3_SYS	power-interlock / J_MATRIX.3 TOP (7,33)	power-analog / J_LINK_2.1 TOP (1,6)
H36 / 3V3_SYS	power-analog / J_LINK_2.1 TOP (1,6)	router-core / J_POWER.3 TOP (7,33)
H117 / 12V_SYS	power-interlock / J_MATRIX.10 TOP (10,34)	router-core / J_POWER.10 TOP (10,34)
H123 / 5V_SYS	power-interlock / J_MATRIX.4 TOP (7,34)	power-analog / J_LINK_3.1 TOP (1,7)
H124 / 5V_SYS	power-analog / J_LINK_3.1 TOP (1,7)	router-core / J_POWER.4 TOP (7,34)
H140 / PRESET_N	power-guards / J_LINK_6.1 TOP (1,10)	power-analog / J_LINK_4.1 TOP (1,8)
H141 / USB_ROUTER_3V3	router-core / J_FIXTURE.1 TOP (18,6)	router-safety / J_LINK_3.1 TOP (1,7)
H151 / FAULT_USB	router-core / J_LINK_8.1 TOP (1,12)	router-safety / J_LINK_5.1 TOP (1,9)
H153 / CURRENT_USB	router-core / J_LINK_10.1 TOP (1,14)	router-safety / J_LINK_6.1 TOP (1,10)
H156 / 5V_EDU	power-interlock / J_MATRIX.1 TOP (6,33)	power-analog / J_LINK_5.1 TOP (1,9)
H157 / 5V_EDU	power-analog / J_LINK_5.1 TOP (1,9)	router-core / J_POWER.1 TOP (6,33)
H168 / EDU_FAULT	power-interlock / J_MATRIX.6 TOP (8,34)	router-core / J_POWER.6 TOP (8,34)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP11 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H169 / EDU_FAULT	router-core / J_POWER.6 TOP (8,34)	router-safety / J_LINK_7.1 TOP (1,11)
H170 / CURRENT_SENSE	power-interlock / J_MATRIX.8 TOP (9,34)	power-analog / J_LINK_6.1 TOP (1,10)
H171 / CURRENT_SENSE	power-analog / J_LINK_6.1 TOP (1,10)	router-core / J_POWER.8 TOP (9,34)
H172 / CURRENT_SENSE	router-core / J_POWER.8 TOP (9,34)	router-safety / J_LINK_8.1 TOP (1,12)
H173 / ROUTER_ALIVE	power-interlock / J_MATRIX.9 TOP (10,33)	router-core / J_POWER.9 TOP (10,33)
H204 / EDU_ON	power-interlock / J_SMART.7 TOP (19,33)	power-analog / J_LINK_7.1 TOP (1,11)
H240 / ALLOW_RAIL	power-analog / J_LINK_8.1 TOP (1,12)	power-interlock / J_LINK_1.1 TOP (1,5)
H241 / REF_TRIP	power-interlock / J_TEST.3 TOP (18,27)	power-analog / J_LINK_9.1 TOP (1,13)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP13 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H125 / 5V_SYS	router-core / J_POWER.4 TOP (7,34)	router-safety / J_LINK_2.1 TOP (1,6)
H149 / HEARTBEAT	router-core / J_LINK_7.1 TOP (1,11)	router-safety / J_LINK_4.1 TOP (1,8)
H174 / ROUTER_ALIVE	router-core / J_POWER.9 TOP (10,33)	router-safety / J_LINK_9.1 TOP (1,13)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP15 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H13 / GND	router-safety / J_LINK_1.1 TOP (1,5)	measurement-a-buffers / J_LINK_1.1 TOP (1,5)
H14 / GND	measurement-a-buffers / J_LINK_1.1 TOP (1,5)	smart-core / J_POWER.2 TOP (6,34)
H15 / GND	smart-core / J_POWER.2 TOP (6,34)	smart-safety / J_LINK_1.1 TOP (1,5)
H37 / 3V3_SYS	router-core / J_POWER.3 TOP (7,33)	smart-core / J_POWER.5 TOP (8,33)
H58 / NODE02	a1 / J_NODE.5 TOP (6,33)	bank-a-position-0 / J_LINK_7.1 TOP (1,11)
H126 / 5V_SYS	router-safety / J_LINK_2.1 TOP (1,6)	bank-a-position-0 / J_LINK_8.1 TOP (1,12)
H127 / 5V_SYS	bank-a-position-0 / J_LINK_8.1 TOP (1,12)	measurement-a-buffers / J_LINK_7.1 TOP (1,11)
H128 / 5V_SYS	measurement-a-buffers / J_LINK_7.1 TOP (1,11)	smart-core / J_POWER.1 TOP (6,33)
H129 / 5V_SYS	smart-core / J_POWER.1 TOP (6,33)	smart-safety / J_LINK_2.1 TOP (1,6)
H158 / 5V_EDU	router-core / J_POWER.1 TOP (6,33)	bank-a-position-0 / J_LINK_9.1 TOP (1,13)
H159 / 5V_EDU	bank-a-position-0 / J_LINK_9.1 TOP (1,13)	measurement-a-buffers / J_LINK_8.1 TOP (1,12)
H160 / 5V_EDU	measurement-a-buffers / J_LINK_8.1 TOP (1,12)	smart-core / J_POWER.3 TOP (7,33)
H177 / USB_SMART_3V3	bank-a-position-0 / J_LINK_10.1 TOP (1,14)	smart-core / J_LINK_1.1 TOP (1,5)
H178 / USB_SMART_3V3	smart-core / J_LINK_1.1 TOP (1,5)	smart-safety / J_LINK_3.1 TOP (1,7)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP15 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H184 / SR_SRCLK	smart-core / J_LINK_2.1 TOP (1,6)	smart-safety / J_LINK_4.1 TOP (1,8)
H185 / SR_RCLK	smart-core / J_LINK_3.1 TOP (1,7)	smart-safety / J_LINK_5.1 TOP (1,9)
H186 / SR_SER	smart-core / J_LINK_4.1 TOP (1,8)	smart-safety / J_LINK_6.1 TOP (1,10)
H187 / SMART_HEARTBEAT	smart-core / J_LINK_5.1 TOP (1,9)	smart-safety / J_LINK_7.1 TOP (1,11)
H188 / SCL_A	bank-a-position-0 / J_LINK_11.1 TOP (1,15)	smart-core / J_LINK_6.1 TOP (1,10)
H194 / SDA_0	bank-a-position-0 / J_LINK_12.1 TOP (1,16)	smart-core / J_LINK_7.1 TOP (1,11)
H200 / MUX_A_A	measurement-a-precision / J_LINK_7.1 TOP (1,11)	smart-core / J_LINK_13.1 TOP (1,17)
H201 / MUX_A_B	measurement-a-precision / J_LINK_8.1 TOP (1,12)	smart-core / J_LINK_14.1 TOP (1,18)
H205 / EDU_ON	power-analog / J_LINK_7.1 TOP (1,11)	smart-core / J_POWER.7 TOP (9,33)
H206 / EDU_ON	smart-core / J_POWER.7 TOP (9,33)	smart-safety / J_LINK_8.1 TOP (1,12)
H207 / RELAY_0	measurement-a-buffers / J_LINK_9.1 TOP (1,13)	smart-safety / J_LINK_9.1 TOP (1,13)
H208 / RELAY_1	measurement-a-buffers / J_LINK_10.1 TOP (1,14)	smart-safety / J_LINK_10.1 TOP (1,14)
H209 / RELAY_2	measurement-a-buffers / J_LINK_11.1 TOP (1,15)	smart-safety / J_LINK_11.1 TOP (1,15)
H210 / RELAY_3	measurement-a-buffers / J_LINK_12.1 TOP (1,16)	smart-safety / J_LINK_12.1 TOP (1,16)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP15 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H211 / RELAY_4	measurement-a-buffers / J_LINK_13.1 TOP (1,17)	smart-safety / J_LINK_13.1 TOP (1,17)
H212 / RELAY_5	measurement-a-buffers / J_LINK_14.1 TOP (1,18)	smart-safety / J_LINK_14.1 TOP (1,18)
H213 / RELAY_6	measurement-a-buffers / J_LINK_15.1 TOP (1,19)	smart-safety / J_LINK_15.1 TOP (1,19)
H214 / RELAY_7	measurement-a-buffers / J_LINK_16.1 TOP (1,20)	smart-safety / J_LINK_16.1 TOP (1,20)
H215 / RELAY_8	measurement-a-buffers / J_LINK_17.1 TOP (1,21)	smart-safety / J_LINK_17.1 TOP (1,21)
H246 / P0_K_HIGH_CATHODE	bank-a-position-0 / J_LINK_13.1 TOP (1,17)	measurement-a-buffers / J_LINK_22.1 TOP (1,26)
H247 / P0_K_LOW_CATHODE	bank-a-position-0 / J_LINK_14.1 TOP (1,18)	measurement-a-buffers / J_LINK_23.1 TOP (1,27)
H255 / A_REF_HI	bank-a-position-0 / J_LINK_16.1 TOP (1,20)	measurement-a-buffers / J_LINK_31.1 TOP (1,35)
H258 / A_REF_LO	bank-a-position-0 / J_LINK_17.1 TOP (1,21)	measurement-a-buffers / J_LINK_32.1 TOP (1,36)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP22 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H53 / NODE00	measurement-a-precision / J_LINK_4.1 TOP (1,8)	bank-a-position-0 / J_LINK_5.1 TOP (1,9)
H56 / NODE01	measurement-a-precision / J_LINK_5.1 TOP (1,9)	bank-a-position-0 / J_LINK_6.1 TOP (1,10)
H248 / P0_AMM_CATHODE	measurement-a-buffers / J_LINK_24.1 TOP (1,28)	bank-a-position-0 / J_LINK_15.1 TOP (1,19)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP24 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H16 / GND	smart-safety / J_LINK_1.1 TOP (1,5)	matrix-control / J_LINK_1.1 TOP (1,5)
H17 / GND	matrix-control / J_LINK_1.1 TOP (1,5)	matrix-switches / J_LINK_1.1 TOP (1,5)
H38 / 3V3_SYS	smart-core / J_POWER.5 TOP (8,33)	matrix-control / J_LINK_2.1 TOP (1,6)
H54 / NODE00	bank-a-position-0 / J_LINK_5.1 TOP (1,9)	matrix-switches / J_LINK_2.1 TOP (1,6)
H57 / NODE01	bank-a-position-0 / J_LINK_6.1 TOP (1,10)	matrix-switches / J_LINK_3.1 TOP (1,7)
H59 / NODE02	bank-a-position-0 / J_LINK_7.1 TOP (1,11)	matrix-switches / J_LINK_4.1 TOP (1,8)
H60 / NODE03	a1 / J_NODE.7 TOP (7,33)	matrix-switches / J_LINK_5.1 TOP (1,9)
H118 / 12V_SYS	router-core / J_POWER.10 TOP (10,34)	matrix-switches / J_LINK_24.1 TOP (1,28)
H130 / 5V_SYS	smart-safety / J_LINK_2.1 TOP (1,6)	matrix-control / J_LINK_3.1 TOP (1,7)
H131 / 5V_SYS	matrix-control / J_LINK_3.1 TOP (1,7)	matrix-switches / J_LINK_25.1 TOP (1,29)
H142 / USB_ROUTER_3V3	router-safety / J_LINK_3.1 TOP (1,7)	matrix-control / J_LINK_4.1 TOP (1,8)
H143 / XBAR_DAT	router-core / J_LINK_1.1 TOP (1,5)	matrix-control / J_LINK_5.1 TOP (1,9)
H144 / XBAR_CLK	router-core / J_LINK_2.1 TOP (1,6)	matrix-control / J_LINK_6.1 TOP (1,10)
H145 / XBAR_LATCH	router-core / J_LINK_3.1 TOP (1,7)	matrix-control / J_LINK_7.1 TOP (1,11)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP24 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H146 / STB_A_IN	router-core / J_LINK_4.1 TOP (1,8)	matrix-control / J_LINK_8.1 TOP (1,12)
H147 / STB_B_IN	router-core / J_LINK_5.1 TOP (1,9)	matrix-control / J_LINK_9.1 TOP (1,13)
H148 / RST_RELEASE	router-core / J_LINK_6.1 TOP (1,10)	matrix-switches / J_LINK_26.1 TOP (1,30)
H154 / DIAG_ADC	router-core / J_LINK_11.1 TOP (1,15)	matrix-control / J_LINK_10.1 TOP (1,14)
H161 / 5V_EDU	smart-core / J_POWER.3 TOP (7,33)	matrix-switches / J_LINK_27.1 TOP (1,31)
H175 / ROUTER_ALIVE	router-safety / J_LINK_9.1 TOP (1,13)	matrix-switches / J_LINK_28.1 TOP (1,32)
H176 / XBAR_OE_N	router-safety / J_LINK_10.1 TOP (1,14)	matrix-control / J_LINK_11.1 TOP (1,15)
H225 / STROBE_A	matrix-control / J_LINK_12.1 TOP (1,16)	matrix-switches / J_LINK_29.1 TOP (1,33)
H226 / STROBE_B	matrix-control / J_LINK_13.1 TOP (1,17)	matrix-switches / J_LINK_30.1 TOP (1,34)
H227 / XBAR_DATA	matrix-control / J_LINK_14.1 TOP (1,18)	matrix-switches / J_LINK_31.1 TOP (1,35)
H228 / AX0	matrix-control / J_LINK_15.1 TOP (1,19)	matrix-switches / J_LINK_32.1 TOP (1,36)
H229 / AX1	matrix-control / J_LINK_16.1 TOP (1,20)	matrix-switches / J_LINK_33.1 TOP (26,5)
H230 / AX2	matrix-control / J_LINK_17.1 TOP (1,21)	matrix-switches / J_LINK_34.1 TOP (26,6)
H231 / AX3	matrix-control / J_LINK_18.1 TOP (1,22)	matrix-switches / J_LINK_35.1 TOP (26,7)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP24 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H232 / AY0	matrix-control / J_LINK_19.1 TOP (1,23)	matrix-switches / J_LINK_36.1 TOP (26,8)
H233 / AY1	matrix-control / J_LINK_20.1 TOP (1,24)	matrix-switches / J_LINK_37.1 TOP (26,9)
H234 / AY2	matrix-control / J_LINK_21.1 TOP (1,25)	matrix-switches / J_LINK_38.1 TOP (26,10)
H235 / SOURCE_HI	matrix-control / J_LINK_22.1 TOP (1,26)	matrix-switches / J_LINK_39.1 TOP (26,11)
H236 / SOURCE_LO	matrix-control / J_LINK_23.1 TOP (1,27)	matrix-switches / J_LINK_40.1 TOP (26,12)
H237 / DIAG_NODE	matrix-control / J_LINK_24.1 TOP (1,28)	matrix-switches / J_LINK_41.1 TOP (26,13)
H238 / DISCHARGE	matrix-control / J_LINK_25.1 TOP (1,29)	matrix-switches / J_LINK_42.1 TOP (26,14)
H239 / CAL_LOAD	matrix-control / J_LINK_26.1 TOP (1,30)	matrix-switches / J_LINK_43.1 TOP (26,15)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP27 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H18 / GND	matrix-switches / J_LINK_1.1 TOP (1,5)	a2 / J_SMART.2 TOP (18,34)
H19 / GND	a2 / J_SMART.2 TOP (18,34)	bank-a-position-1 / J_LINK_1.1 TOP (1,5)
H39 / 3V3_SYS	matrix-control / J_LINK_2.1 TOP (1,6)	a2 / J_SMART.8 TOP (21,34)
H40 / 3V3_SYS	a2 / J_SMART.8 TOP (21,34)	bank-a-position-1 / J_LINK_2.1 TOP (1,6)
H41 / 3V3_SYS	bank-a-position-1 / J_LINK_2.1 TOP (1,6)	measurement-a-buffers / J_LINK_2.1 TOP (1,6)
H61 / SCL_POSITION_1	a2 / J_SMART.9 TOP (22,33)	bank-a-position-1 / J_LINK_3.1 TOP (1,7)
H62 / SDA_POSITION_1	a2 / J_SMART.7 TOP (21,33)	bank-a-position-1 / J_LINK_4.1 TOP (1,8)
H63 / NODE04	matrix-switches / J_LINK_6.1 TOP (1,10)	a2 / J_NODE.1 TOP (4,33)
H64 / NODE04	a2 / J_NODE.1 TOP (4,33)	bank-a-position-1 / J_LINK_5.1 TOP (1,9)
H65 / NODE04	bank-a-position-1 / J_LINK_5.1 TOP (1,9)	measurement-a-buffers / J_LINK_3.1 TOP (1,7)
H66 / NODE05	matrix-switches / J_LINK_7.1 TOP (1,11)	a2 / J_NODE.3 TOP (5,33)
H67 / NODE05	a2 / J_NODE.3 TOP (5,33)	bank-a-position-1 / J_LINK_6.1 TOP (1,10)
H68 / NODE05	bank-a-position-1 / J_LINK_6.1 TOP (1,10)	measurement-a-buffers / J_LINK_4.1 TOP (1,8)
H69 / NODE06	matrix-switches / J_LINK_8.1 TOP (1,12)	a2 / J_NODE.5 TOP (6,33)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP27 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H70 / NODE06	a2 / J_NODE.5 TOP (6,33)	bank-a-position-1 / J_LINK_7.1 TOP (1,11)
H71 / NODE07	matrix-switches / J_LINK_9.1 TOP (1,13)	a2 / J_NODE.7 TOP (7,33)
H132 / 5V_SYS	matrix-switches / J_LINK_25.1 TOP (1,29)	bank-a-position-1 / J_LINK_8.1 TOP (1,12)
H162 / 5V_EDU	matrix-switches / J_LINK_27.1 TOP (1,31)	bank-a-position-1 / J_LINK_9.1 TOP (1,13)
H179 / USB_SMART_3V3	smart-safety / J_LINK_3.1 TOP (1,7)	bank-a-position-1 / J_LINK_10.1 TOP (1,14)
H189 / SCL_A	smart-core / J_LINK_6.1 TOP (1,10)	bank-a-position-1 / J_LINK_11.1 TOP (1,15)
H195 / SDA_1	smart-core / J_LINK_8.1 TOP (1,12)	bank-a-position-1 / J_LINK_12.1 TOP (1,16)
H242 / P1_1_MUX	measurement-a-precision / J_LINK_9.1 TOP (1,13)	measurement-a-buffers / J_LINK_18.1 TOP (1,22)
H243 / P1_2_MUX	measurement-a-precision / J_LINK_10.1 TOP (1,14)	measurement-a-buffers / J_LINK_19.1 TOP (1,23)
H249 / P1_K_HIGH_CATHODE	measurement-a-buffers / J_LINK_25.1 TOP (1,29)	bank-a-position-1 / J_LINK_13.1 TOP (1,17)
H250 / P1_K_LOW_CATHODE	measurement-a-buffers / J_LINK_26.1 TOP (1,30)	bank-a-position-1 / J_LINK_14.1 TOP (1,18)
H251 / P1_AMM_CATHODE	measurement-a-buffers / J_LINK_27.1 TOP (1,31)	bank-a-position-1 / J_LINK_15.1 TOP (1,19)
H256 / A_REF_HI	measurement-a-buffers / J_LINK_31.1 TOP (1,35)	bank-a-position-1 / J_LINK_16.1 TOP (1,20)
H259 / A_REF_LO	measurement-a-buffers / J_LINK_32.1 TOP (1,36)	bank-a-position-1 / J_LINK_17.1 TOP (1,21)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP28 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H20 / GND	bank-a-position-1 / J_LINK_1.1 TOP (1,5)	a3 / J_SMART.2 TOP (14,34)
H21 / GND	a3 / J_SMART.2 TOP (14,34)	bank-a-position-2 / J_LINK_1.1 TOP (1,5)
H42 / 3V3_SYS	measurement-a-buffers / J_LINK_2.1 TOP (1,6)	a3 / J_SMART.8 TOP (17,34)
H43 / 3V3_SYS	a3 / J_SMART.8 TOP (17,34)	bank-a-position-2 / J_LINK_2.1 TOP (1,6)
H72 / SCL_POSITION_2	a3 / J_SMART.9 TOP (18,33)	bank-a-position-2 / J_LINK_3.1 TOP (1,7)
H73 / SDA_POSITION_2	a3 / J_SMART.7 TOP (17,33)	bank-a-position-2 / J_LINK_4.1 TOP (1,8)
H74 / NODE08	matrix-switches / J_LINK_10.1 TOP (1,14)	a3 / J_NODE.1 TOP (4,33)
H75 / NODE08	a3 / J_NODE.1 TOP (4,33)	bank-a-position-2 / J_LINK_5.1 TOP (1,9)
H76 / NODE08	bank-a-position-2 / J_LINK_5.1 TOP (1,9)	measurement-a-buffers / J_LINK_5.1 TOP (1,9)
H77 / NODE09	matrix-switches / J_LINK_11.1 TOP (1,15)	a3 / J_NODE.3 TOP (5,33)
H78 / NODE09	a3 / J_NODE.3 TOP (5,33)	bank-a-position-2 / J_LINK_6.1 TOP (1,10)
H79 / NODE09	bank-a-position-2 / J_LINK_6.1 TOP (1,10)	measurement-a-buffers / J_LINK_6.1 TOP (1,10)
H80 / NODE10	matrix-switches / J_LINK_12.1 TOP (1,16)	a3 / J_NODE.5 TOP (6,33)
H81 / NODE10	a3 / J_NODE.5 TOP (6,33)	bank-a-position-2 / J_LINK_7.1 TOP (1,11)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP28 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H133 / 5V_SYS	bank-a-position-1 / J_LINK_8.1 TOP (1,12)	bank-a-position-2 / J_LINK_8.1 TOP (1,12)
H163 / 5V_EDU	bank-a-position-1 / J_LINK_9.1 TOP (1,13)	bank-a-position-2 / J_LINK_9.1 TOP (1,13)
H180 / USB_SMART_3V3	bank-a-position-1 / J_LINK_10.1 TOP (1,14)	bank-a-position-2 / J_LINK_10.1 TOP (1,14)
H190 / SCL_A	bank-a-position-1 / J_LINK_11.1 TOP (1,15)	bank-a-position-2 / J_LINK_11.1 TOP (1,15)
H196 / SDA_2	smart-core / J_LINK_9.1 TOP (1,13)	bank-a-position-2 / J_LINK_12.1 TOP (1,16)
H244 / P2_1_MUX	measurement-a-precision / J_LINK_11.1 TOP (1,15)	measurement-a-buffers / J_LINK_20.1 TOP (1,24)
H245 / P2_2_MUX	measurement-a-precision / J_LINK_12.1 TOP (1,16)	measurement-a-buffers / J_LINK_21.1 TOP (1,25)
H252 / P2_K_HIGH_CATHODE	measurement-a-buffers / J_LINK_28.1 TOP (1,32)	bank-a-position-2 / J_LINK_13.1 TOP (1,17)
H253 / P2_K_LOW_CATHODE	measurement-a-buffers / J_LINK_29.1 TOP (1,33)	bank-a-position-2 / J_LINK_14.1 TOP (1,18)
H254 / P2_AMM_CATHODE	measurement-a-buffers / J_LINK_30.1 TOP (1,34)	bank-a-position-2 / J_LINK_15.1 TOP (1,19)
H257 / A_REF_HI	bank-a-position-1 / J_LINK_16.1 TOP (1,20)	bank-a-position-2 / J_LINK_16.1 TOP (1,20)
H260 / A_REF_LO	bank-a-position-1 / J_LINK_17.1 TOP (1,21)	bank-a-position-2 / J_LINK_17.1 TOP (1,21)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP29 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H22 / GND	bank-a-position-2 / J_LINK_1.1 TOP (1,5)	b1 / J_SMART.2 TOP (18,5)
H23 / GND	b1 / J_SMART.2 TOP (18,5)	bank-b-position-3 / J_LINK_1.1 TOP (1,5)
H24 / GND	bank-b-position-3 / J_LINK_1.1 TOP (1,5)	measurement-b-buffers / J_LINK_1.1 TOP (1,5)
H25 / GND	measurement-b-buffers / J_LINK_1.1 TOP (1,5)	measurement-b-precision / J_LINK_1.1 TOP (1,5)
H44 / 3V3_SYS	bank-a-position-2 / J_LINK_2.1 TOP (1,6)	b1 / J_SMART.8 TOP (21,5)
H45 / 3V3_SYS	b1 / J_SMART.8 TOP (21,5)	bank-b-position-3 / J_LINK_2.1 TOP (1,6)
H46 / 3V3_SYS	bank-b-position-3 / J_LINK_2.1 TOP (1,6)	measurement-b-precision / J_LINK_2.1 TOP (1,6)
H82 / SCL_POSITION_3	b1 / J_SMART.9 TOP (22,4)	bank-b-position-3 / J_LINK_3.1 TOP (1,7)
H83 / SCL_POSITION_3	bank-b-position-3 / J_LINK_3.1 TOP (1,7)	measurement-b-precision / J_LINK_3.1 TOP (1,7)
H84 / SDA_POSITION_3	b1 / J_SMART.7 TOP (21,4)	bank-b-position-3 / J_LINK_4.1 TOP (1,8)
H85 / SDA_POSITION_3	bank-b-position-3 / J_LINK_4.1 TOP (1,8)	measurement-b-precision / J_LINK_4.1 TOP (1,8)
H86 / NODE11	matrix-switches / J_LINK_13.1 TOP (1,17)	b1 / J_NODE.1 TOP (4,4)
H87 / NODE11	b1 / J_NODE.1 TOP (4,4)	bank-b-position-3 / J_LINK_5.1 TOP (1,9)
H88 / NODE11	bank-b-position-3 / J_LINK_5.1 TOP (1,9)	measurement-b-precision / J_LINK_5.1 TOP (1,9)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP29 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H89 / NODE12	matrix-switches / J_LINK_14.1 TOP (1,18)	b1 / J_NODE.3 TOP (5,4)
H90 / NODE12	b1 / J_NODE.3 TOP (5,4)	bank-b-position-3 / J_LINK_6.1 TOP (1,10)
H91 / NODE12	bank-b-position-3 / J_LINK_6.1 TOP (1,10)	measurement-b-precision / J_LINK_6.1 TOP (1,10)
H92 / NODE13	matrix-switches / J_LINK_15.1 TOP (1,19)	b1 / J_NODE.5 TOP (6,4)
H93 / NODE13	b1 / J_NODE.5 TOP (6,4)	bank-b-position-3 / J_LINK_7.1 TOP (1,11)
H94 / NODE14	matrix-switches / J_LINK_16.1 TOP (1,20)	b1 / J_NODE.7 TOP (7,4)
H134 / 5V_SYS	bank-a-position-2 / J_LINK_8.1 TOP (1,12)	bank-b-position-3 / J_LINK_8.1 TOP (1,12)
H135 / 5V_SYS	bank-b-position-3 / J_LINK_8.1 TOP (1,12)	measurement-b-buffers / J_LINK_7.1 TOP (1,11)
H164 / 5V_EDU	bank-a-position-2 / J_LINK_9.1 TOP (1,13)	bank-b-position-3 / J_LINK_9.1 TOP (1,13)
H165 / 5V_EDU	bank-b-position-3 / J_LINK_9.1 TOP (1,13)	measurement-b-buffers / J_LINK_8.1 TOP (1,12)
H181 / USB_SMART_3V3	bank-a-position-2 / J_LINK_10.1 TOP (1,14)	bank-b-position-3 / J_LINK_10.1 TOP (1,14)
H191 / SCL_A	bank-a-position-2 / J_LINK_11.1 TOP (1,15)	bank-b-position-3 / J_LINK_11.1 TOP (1,15)
H197 / SDA_3	smart-core / J_LINK_10.1 TOP (1,14)	bank-b-position-3 / J_LINK_12.1 TOP (1,16)
H202 / MUX_B_A	smart-core / J_LINK_15.1 TOP (1,19)	measurement-b-precision / J_LINK_7.1 TOP (1,11)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP29 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H203 / MUX_B_B	smart-core / J_LINK_16.1 TOP (1,20)	measurement-b-precision / J_LINK_8.1 TOP (1,12)
H216 / RELAY_9	smart-safety / J_LINK_18.1 TOP (1,22)	measurement-b-buffers / J_LINK_9.1 TOP (1,13)
H217 / RELAY_10	smart-safety / J_LINK_19.1 TOP (1,23)	measurement-b-buffers / J_LINK_10.1 TOP (1,14)
H218 / RELAY_11	smart-safety / J_LINK_20.1 TOP (1,24)	measurement-b-buffers / J_LINK_11.1 TOP (1,15)
H219 / RELAY_12	smart-safety / J_LINK_21.1 TOP (1,25)	measurement-b-buffers / J_LINK_12.1 TOP (1,16)
H220 / RELAY_13	smart-safety / J_LINK_22.1 TOP (1,26)	measurement-b-buffers / J_LINK_13.1 TOP (1,17)
H221 / RELAY_14	smart-safety / J_LINK_23.1 TOP (1,27)	measurement-b-buffers / J_LINK_14.1 TOP (1,18)
H222 / RELAY_15	smart-safety / J_LINK_24.1 TOP (1,28)	measurement-b-buffers / J_LINK_15.1 TOP (1,19)
H223 / RELAY_16	smart-safety / J_LINK_25.1 TOP (1,29)	measurement-b-buffers / J_LINK_16.1 TOP (1,20)
H224 / RELAY_17	smart-safety / J_LINK_26.1 TOP (1,30)	measurement-b-buffers / J_LINK_17.1 TOP (1,21)
H265 / P3_K_HIGH_CATHODE	bank-b-position-3 / J_LINK_13.1 TOP (1,17)	measurement-b-buffers / J_LINK_22.1 TOP (1,26)
H266 / P3_K_LOW_CATHODE	bank-b-position-3 / J_LINK_14.1 TOP (1,18)	measurement-b-buffers / J_LINK_23.1 TOP (1,27)
H267 / P3_AMM_CATHODE	bank-b-position-3 / J_LINK_15.1 TOP (1,19)	measurement-b-buffers / J_LINK_24.1 TOP (1,28)
H274 / B_REF_HI	bank-b-position-3 / J_LINK_16.1 TOP (1,20)	measurement-b-buffers / J_LINK_31.1 TOP (1,35)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP29 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H277 / B_REF_LO	bank-b-position-3 / J_LINK_17.1 TOP (1,21)	measurement-b-buffers / J_LINK_32.1 TOP (1,36)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP30 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H26 / GND	measurement-b-precision / J_LINK_1.1 TOP (1,5)	b2 / J_SMART.2 TOP (18,5)
H27 / GND	b2 / J_SMART.2 TOP (18,5)	bank-b-position-4 / J_LINK_1.1 TOP (1,5)
H47 / 3V3_SYS	measurement-b-precision / J_LINK_2.1 TOP (1,6)	b2 / J_SMART.8 TOP (21,5)
H48 / 3V3_SYS	b2 / J_SMART.8 TOP (21,5)	bank-b-position-4 / J_LINK_2.1 TOP (1,6)
H49 / 3V3_SYS	bank-b-position-4 / J_LINK_2.1 TOP (1,6)	measurement-b-buffers / J_LINK_2.1 TOP (1,6)
H95 / SCL_POSITION_4	b2 / J_SMART.9 TOP (22,4)	bank-b-position-4 / J_LINK_3.1 TOP (1,7)
H96 / SDA_POSITION_4	b2 / J_SMART.7 TOP (21,4)	bank-b-position-4 / J_LINK_4.1 TOP (1,8)
H97 / NODE15	matrix-switches / J_LINK_17.1 TOP (1,21)	b2 / J_NODE.1 TOP (4,4)
H98 / NODE15	b2 / J_NODE.1 TOP (4,4)	bank-b-position-4 / J_LINK_5.1 TOP (1,9)
H99 / NODE15	bank-b-position-4 / J_LINK_5.1 TOP (1,9)	measurement-b-buffers / J_LINK_3.1 TOP (1,7)
H100 / NODE16	matrix-switches / J_LINK_18.1 TOP (1,22)	b2 / J_NODE.3 TOP (5,4)
H101 / NODE16	b2 / J_NODE.3 TOP (5,4)	bank-b-position-4 / J_LINK_6.1 TOP (1,10)
H102 / NODE16	bank-b-position-4 / J_LINK_6.1 TOP (1,10)	measurement-b-buffers / J_LINK_4.1 TOP (1,8)
H103 / NODE17	matrix-switches / J_LINK_19.1 TOP (1,23)	b2 / J_NODE.5 TOP (6,4)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP30 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H104 / NODE17	b2 / J_NODE.5 TOP (6,4)	bank-b-position-4 / J_LINK_7.1 TOP (1,11)
H105 / NODE18	matrix-switches / J_LINK_20.1 TOP (1,24)	b2 / J_NODE.7 TOP (7,4)
H136 / 5V_SYS	measurement-b-buffers / J_LINK_7.1 TOP (1,11)	bank-b-position-4 / J_LINK_8.1 TOP (1,12)
H166 / 5V_EDU	measurement-b-buffers / J_LINK_8.1 TOP (1,12)	bank-b-position-4 / J_LINK_9.1 TOP (1,13)
H182 / USB_SMART_3V3	bank-b-position-3 / J_LINK_10.1 TOP (1,14)	bank-b-position-4 / J_LINK_10.1 TOP (1,14)
H192 / SCL_A	bank-b-position-3 / J_LINK_11.1 TOP (1,15)	bank-b-position-4 / J_LINK_11.1 TOP (1,15)
H198 / SDA_4	smart-core / J_LINK_11.1 TOP (1,15)	bank-b-position-4 / J_LINK_12.1 TOP (1,16)
H261 / P4_1_MUX	measurement-b-precision / J_LINK_9.1 TOP (1,13)	measurement-b-buffers / J_LINK_18.1 TOP (1,22)
H262 / P4_2_MUX	measurement-b-precision / J_LINK_10.1 TOP (1,14)	measurement-b-buffers / J_LINK_19.1 TOP (1,23)
H268 / P4_K_HIGH_CATHODE	measurement-b-buffers / J_LINK_25.1 TOP (1,29)	bank-b-position-4 / J_LINK_13.1 TOP (1,17)
H269 / P4_K_LOW_CATHODE	measurement-b-buffers / J_LINK_26.1 TOP (1,30)	bank-b-position-4 / J_LINK_14.1 TOP (1,18)
H270 / P4_AMM_CATHODE	measurement-b-buffers / J_LINK_27.1 TOP (1,31)	bank-b-position-4 / J_LINK_15.1 TOP (1,19)
H275 / B_REF_HI	measurement-b-buffers / J_LINK_31.1 TOP (1,35)	bank-b-position-4 / J_LINK_16.1 TOP (1,20)
H278 / B_REF_LO	measurement-b-buffers / J_LINK_32.1 TOP (1,36)	bank-b-position-4 / J_LINK_17.1 TOP (1,21)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP31 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H28 / GND	bank-b-position-4 / J_LINK_1.1 TOP (1,5)	b3 / J_SMART.2 TOP (14,5)
H29 / GND	b3 / J_SMART.2 TOP (14,5)	bank-b-position-5 / J_LINK_1.1 TOP (1,5)
H50 / 3V3_SYS	measurement-b-buffers / J_LINK_2.1 TOP (1,6)	b3 / J_SMART.8 TOP (17,5)
H51 / 3V3_SYS	b3 / J_SMART.8 TOP (17,5)	bank-b-position-5 / J_LINK_2.1 TOP (1,6)
H106 / SCL_POSITION_5	b3 / J_SMART.9 TOP (18,4)	bank-b-position-5 / J_LINK_3.1 TOP (1,7)
H107 / SDA_POSITION_5	b3 / J_SMART.7 TOP (17,4)	bank-b-position-5 / J_LINK_4.1 TOP (1,8)
H108 / NODE19	matrix-switches / J_LINK_21.1 TOP (1,25)	b3 / J_NODE.1 TOP (4,4)
H109 / NODE19	b3 / J_NODE.1 TOP (4,4)	bank-b-position-5 / J_LINK_5.1 TOP (1,9)
H110 / NODE19	bank-b-position-5 / J_LINK_5.1 TOP (1,9)	measurement-b-buffers / J_LINK_5.1 TOP (1,9)
H111 / NODE20	matrix-switches / J_LINK_22.1 TOP (1,26)	b3 / J_NODE.3 TOP (5,4)
H112 / NODE20	b3 / J_NODE.3 TOP (5,4)	bank-b-position-5 / J_LINK_6.1 TOP (1,10)
H113 / NODE20	bank-b-position-5 / J_LINK_6.1 TOP (1,10)	measurement-b-buffers / J_LINK_6.1 TOP (1,10)
H114 / NODE21	matrix-switches / J_LINK_23.1 TOP (1,27)	b3 / J_NODE.5 TOP (6,4)
H115 / NODE21	b3 / J_NODE.5 TOP (6,4)	bank-b-position-5 / J_LINK_7.1 TOP (1,11)

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CP31 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H137 / 5V_SYS	bank-b-position-4 / J_LINK_8.1 TOP (1,12)	bank-b-position-5 / J_LINK_8.1 TOP (1,12)
H167 / 5V_EDU	bank-b-position-4 / J_LINK_9.1 TOP (1,13)	bank-b-position-5 / J_LINK_9.1 TOP (1,13)
H183 / USB_SMART_3V3	bank-b-position-4 / J_LINK_10.1 TOP (1,14)	bank-b-position-5 / J_LINK_10.1 TOP (1,14)
H193 / SCL_A	bank-b-position-4 / J_LINK_11.1 TOP (1,15)	bank-b-position-5 / J_LINK_11.1 TOP (1,15)
H199 / SDA_5	smart-core / J_LINK_12.1 TOP (1,16)	bank-b-position-5 / J_LINK_12.1 TOP (1,16)
H263 / P5_1_MUX	measurement-b-precision / J_LINK_11.1 TOP (1,15)	measurement-b-buffers / J_LINK_20.1 TOP (1,24)
H264 / P5_2_MUX	measurement-b-precision / J_LINK_12.1 TOP (1,16)	measurement-b-buffers / J_LINK_21.1 TOP (1,25)
H271 / P5_K_HIGH_CATHODE	measurement-b-buffers / J_LINK_28.1 TOP (1,32)	bank-b-position-5 / J_LINK_13.1 TOP (1,17)
H272 / P5_K_LOW_CATHODE	measurement-b-buffers / J_LINK_29.1 TOP (1,33)	bank-b-position-5 / J_LINK_14.1 TOP (1,18)
H273 / P5_AMM_CATHODE	measurement-b-buffers / J_LINK_30.1 TOP (1,34)	bank-b-position-5 / J_LINK_15.1 TOP (1,19)
H276 / B_REF_HI	bank-b-position-4 / J_LINK_16.1 TOP (1,20)	bank-b-position-5 / J_LINK_16.1 TOP (1,20)
H279 / B_REF_LO	bank-b-position-4 / J_LINK_17.1 TOP (1,21)	bank-b-position-5 / J_LINK_17.1 TOP (1,21)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.

CP32 — inter-board harnesses

Install after local wires; power isolated. Confirm every conductor end to end before plugging in.

ID / purpose	From terminal	To terminal
H30 / GND	bank-b-position-5 / J_LINK_1.1 TOP (1,5)	integration / J_PI.6 TOP (5,12)

TOP hole numbering remains the same when viewed from the mirrored underside. Use insulated wire and strain relief. Lengths follow the measured enclosure layout; the cross-board topology is specified here, but no measured harness length or purchased connector body is implied. A named net without its conductor is not connected.